ELECTRONICS ENGINEERING (ECE) BOARD EXAM

Complete Solutions, Comprehensive Theory & High-Yield Reviewer

Source: PB Exams (25) — 100 Solved Board-Style Examination Questions

Communications Engineering • Transmission Lines, Antennas & Wave Propagation • Digital Signal Processing • Data Networks • Industrial Automation & Instrumentation

PART I — Complete Step-by-Step Solutions (Questions 1–100)

Question 1: According to Bessel functions for FM, what happens to the carrier power when the modulation index $\beta = 2.4$?

Topic: Angle Modulation / FM Bessel Functions
Concept & Explanation: In Frequency Modulation, the carrier component is modulated by the ordinary zero-order Bessel function of the first kind, $J_0(\beta)$. The first zero of $J_0(\beta)$ occurs at approximately $\beta \approx 2.4048$. At this modulation index, the amplitude of the carrier frequency component becomes zero ($J_0(2.4) \approx 0$), meaning all carrier power is transferred entirely into the frequency sidebands.
Board Exam Tip: Carrier nulls occur at $\beta = 2.4, 5.52, 8.65, 11.79$. The condition $\beta = 2.4$ is frequently exploited in the laboratory to calibrate FM frequency deviation.
Answer: The first carrier null occurs, and carrier power drops to zero.

Question 2: According to Parseval's Theorem, what is the relationship between the energy of a signal in the time domain and its energy in the frequency domain?

Topic: Signals & Systems / Spectral Energy Analysis
Concept & Explanation: Parseval's theorem (or Rayleigh's Energy Theorem) states that the total energy of a continuous-time finite-energy signal computed by integrating the squared magnitude over all time equals the integral of its energy spectral density across all frequencies:

$$E = \int_{-\infty}^{\infty} |x(t)|^2 \, dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(\omega)|^2 \, d\omega = \int_{-\infty}^{\infty} |X(f)|^2 \, df$$

Thus, energy is conserved across domain transformations.
Board Exam Tip: Time-domain energy equals frequency-domain energy; Fourier transformation is a unitary, energy-preserving mapping.
Answer: The time-domain energy and frequency-domain energy are equal.

Question 3: When is the phenomenon of "granular noise" in a Delta Modulator most prominent?

Topic: Digital Communications / Delta Modulation
Concept & Explanation: Delta Modulation uses a 1-bit quantizer with a fixed step size Δ. When the input signal is flat (DC) or varies very slowly, the step size Δ is larger than the signal change per sample period. The feedback approximation hunts above and below the input signal, producing high-frequency quantization ripple termed granular noise. (Conversely, when the signal slope exceeds Δ/T_s, slope overload distortion occurs.)
Board Exam Tip: Granular noise $\rightarrow$ flat/slow signal (step size too large). Slope overload $\rightarrow$ rapidly rising signal (step size too small).
Answer: When the input signal remains perfectly flat (DC) or changes very slowly, making the quantizer output oscillate around the input and producing granular noise.

Question 4: If you are designing an application that streams real-time live video and cannot tolerate high latency, but can tolerate dropped pixels, which Transport Layer protocol should you select?

Topic: Data Communications / Transport Layer Protocols
Concept & Explanation: Live video and VoIP demand minimal latency and jitter. TCP uses connection setup handshakes, windowed flow control, and retransmissions, creating unpredictable queuing delays. UDP (User Datagram Protocol) is connectionless and unacknowledged; dropped datagrams cause minor visual artifacts but preserve strict temporal real-time continuity.
Board Exam Tip: Real-time, loss-tolerant $\rightarrow$ UDP. Reliable, loss-intolerant (files, web, banking) $\rightarrow$ TCP.
Answer: UDP

Question 5: Find the equivalent noise temperature of a receiver with a noise figure of 3 dB (assume $T_0 = 290$ K).

Topic: Noise Analysis / Noise Temperature
Given: $NF = 3$ dB, reference temperature $T_0 = 290$ K. | **Find:** Equivalent noise temperature T_e in Kelvin.
Formula & Solution:
1. Convert Noise Figure from dB to linear noise factor F:

$$F = 10^{NF/10} = 10^{3/10} = 10^{0.3} \approx 1.9953 \approx 2.0$$

2. Compute equivalent noise temperature:

$$T_e = T_0(F - 1) = 290 \text{ K} \times (2.0 - 1) = 290 \text{ K} \times 1 = 290 \text{ K}$$

Board Exam Tip: A 3 dB noise figure indicates that the receiver adds noise equal to the thermal noise of a source at room temperature (290 K).
Answer: 290 K

Question 6: Determine the Noise Figure (NF) in dB of an amplifier that has an input SNR of 40 dB and an output SNR of 34 dB.

Topic: RF Amplifiers / Noise Figure Analysis
Given: $\text{SNR}_{in} = 40$ dB, $\text{SNR}_{out} = 34$ dB. | **Find:** Noise Figure NF in dB.
Formula & Solution:
By definition, Noise Figure is the decibel ratio of input SNR to output SNR:

$$NF \text{ (dB)} = \text{SNR}_{in} \text{ (dB)} - \text{SNR}_{out} \text{ (dB)} = 40 \text{ dB} - 34 \text{ dB} = 6 \text{ dB}$$

Board Exam Tip: An ideal noiseless amplifier has $NF = 0$ dB ($\text{SNR}_{out} = \text{SNR}_{in}$). Every real amplifier degrades SNR, making $NF > 0$ dB.
Answer: 6 dB

Question 7: Given a reflection coefficient of magnitude 0.20 on a transmission line, calculate the Voltage Standing Wave Ratio (VSWR).

Topic: Transmission Lines / Reflection Analysis
Given: Reflection coefficient magnitude $|\Gamma| = 0.20$. | **Find:** Voltage Standing Wave Ratio (VSWR).
Formula & Solution:
Using the standard VSWR formula:

$$\text{VSWR} = \frac{1 + |\Gamma|}{1 - |\Gamma|} = \frac{1 + 0.20}{1 - 0.20} = \frac{1.20}{0.80} = 1.50$$

Board Exam Tip: For a matched line $|\Gamma| = 0 \implies$ VSWR $= 1$. For total reflection (open or short) $|\Gamma| = 1 \implies$ VSWR $= \infty$.

Answer: 1.5

Question 8: What hardware device modulates digital data from a computer into analog audio tones for transmission over traditional telephone lines?

Topic: Data Communications / Hardware Interfaces

Concept & Explanation: The term "Modem" is a portmanteau of Modulator-Demodulator. It converts digital square pulses from terminal equipment into audio frequency-shift keyed or quadrature-modulated analog tones that match the 300–3400 Hz bandwidth of the Public Switched Telephone Network (PSTN).

Board Exam Tip: Modem $\rightarrow$ Digital to Voiceband Analog conversion and vice versa.

Answer: Modem

Question 9: A 4–20 mA temperature transmitter is scaled to a range of $0°$C to $200°$C. What temperature is indicated by a loop current of 12 mA?

Topic: Industrial Instrumentation / Current Loops

Given: Current range: $I_{min} = 4$ mA, $I_{max} = 20$ mA (span = 16 mA). Process range: $T_{min} = 0°$C, $T_{max} = 200°$C (span = $200°$C). Loop current $I = 12$ mA. | **Find:** Indicated temperature T.

Formula & Solution:

1. Calculate fractional span:

$$\text{Fraction} = \frac{I - I_{min}}{I_{max} - I_{min}} = \frac{12 - 4}{20 - 4} = \frac{8}{16} = 0.50 \quad (50\%)$$

2. Map to process temperature:

$$T = T_{min} + \text{Fraction} \times (T_{max} - T_{min}) = 0°\text{C} + 0.50 \times (200 - 0) = 100°\text{C}$$

Board Exam Tip: 12 mA is the exact midpoint of a 4–20 mA loop ($4 + 8 = 12$ mA), corresponding to 50% of any calibrated process range.

Answer: $100°$**C**

Question 10: A digital radio link requires a bit rate of 10 Mbps. If 64-QAM modulation is used, what is the theoretical minimum Nyquist bandwidth required?

Topic: Digital Communications / Spectral Efficiency

Given: Bit rate $R_b = 10$ Mbps, modulation order $M = 64$ (64-QAM). | **Find:** Minimum Nyquist bandwidth B_{min}.

Formula & Solution:

1. Bits per symbol k:

$$k = \log_2(M) = \log_2(64) = 6 \text{ bits/symbol}$$

2. Baud (symbol) rate R_s:

$$R_s = \frac{R_b}{k} = \frac{10 \text{ Mbps}}{6 \text{ bits/symbol}} = 1.6667 \text{ MBaud}$$

3. For ideal Nyquist filtering ($r = 0$), minimum double-sideband RF bandwidth equals the symbol rate:

$$B_{min} = R_s \approx 1.67 \text{ MHz}$$

Board Exam Tip: Spectral efficiency of M-QAM is $\eta = \log_2(M)$ bps/Hz. Thus, $B_{min} = R_b / \log_2(M) = 10/6 = 1.67$ MHz.

Answer: 1.67 MHz

Question 11: What is the fundamental working principle of the Cyclic Redundancy Check (CRC) error detection method?

Topic: Data Link Layer / Error Detection
Concept & Explanation: In CRC, data bits are treated as algebraic coefficients of a binary polynomial $D(x)$. The transmitter multiplies $D(x)$ by x^n and performs modulo-2 polynomial division using a standardized generator polynomial $G(x)$. The resulting remainder polynomial $R(x)$ is appended to the frame as the Frame Check Sequence (FCS). The receiver divides the entire sequence by $G(x)$; a non-zero remainder indicates an error.
Board Exam Tip: CRC uses modulo-2 arithmetic (XOR logic) and polynomial division; it catches all single-bit, double-bit, and burst errors $\leq$ generator degree.
Answer: It treats the data block as a binary polynomial, divides it by a generator polynomial using modulo-2 arithmetic, and appends the resulting remainder.

Question 12: [Original exam item omitted / mapped to Topic 13 in source PDF]: What is a key difference between a Local Area Network (LAN) and a Wide Area Network (WAN) at the OSI Data Link Layer?

Topic: Data Communications / OSI Architecture
Concept & Explanation: LANs operate over localized geographical domains utilizing multi-access shared media governed by IEEE 802 standards (e.g., Ethernet IEEE 802.3) with 48-bit physical MAC addresses. WANs bridge large distances over leased telecommunication lines using point-to-point Layer-2 encapsulation protocols such as PPP (Point-to-Point Protocol), HDLC, or Frame Relay.
Board Exam Tip: LAN Data Link = Ethernet + MAC addressing. WAN Data Link = Point-to-point framing (HDLC, PPP).
Answer: LANs commonly use protocols such as Ethernet with MAC addressing, while WANs may use protocols such as PPP or HDLC for point-to-point communication.

Question 14: Calculate the maximum quantization error (quantization noise) in a PCM system with a total input voltage range of 0 to 5 V and an 8-bit ADC.

Topic: Digital Communications / PCM Systems
Given: Voltage range $V_{range} = 5.0\ \text{V} - 0\ \text{V} = 5\ \text{V}$, ADC resolution $n = 8$ bits. | **Find:** Maximum quantization error $Q_{max} = \Delta/2$.
Formula & Solution:
1. Calculate quantization step size (resolution) Δ:

$$\Delta = \frac{V_{range}}{2^n} = \frac{5\ \text{V}}{2^8} = \frac{5}{256} \approx 0.01953125\ \text{V} = 19.53\ \text{mV}$$

2. The maximum quantization error for a midtread or midrise linear quantizer is half the step size:

$$Q_{max} = \pm\frac{\Delta}{2} = \pm\frac{19.53125\ \text{mV}}{2} \approx \pm 9.7656\ \text{mV} \approx 9.77\ \text{mV}$$

Board Exam Tip: Peak error is $\Delta/2$; RMS quantization noise voltage is $\Delta/\sqrt{12}$.
Answer: 9.77 mV

Question 15: Convert a power level of 2 watts into dBm.

Topic: RF Engineering / Decibel Calculations
Given: Power $P = 2$ Watts $= 2000$ mW. | **Find:** Power level in dBm.
Formula & Solution:
Using the standard dBm conversion formula referenced to 1 mW:

$$P\ (\text{dBm}) = 10\log_{10}\left(\frac{P\ \text{in mW}}{1\ \text{mW}}\right) = 10\log_{10}(2000)$$

$$P\,(\text{dBm}) = 10\left(\log_{10}(2) + \log_{10}(1000)\right) = 10(0.30103 + 3) = 10(3.30103) \approx 33.01\ \text{dBm} \approx 33\ \text{dBm}$$

Board Exam Shortcut: 1 W $= 30$ dBm. Doubling power adds $+3$ dB. Therefore, 2 W $= 30 + 3 = 33$ dBm.

Answer: 33 dBm

Question 16: What is the main purpose of the μ-law compression formula in North American telephone systems?

Topic: Telecommunications / Companding

Concept & Explanation: Speech signals contain high peak-to-average ratios: quiet phonemes occur often, while loud vowels occur rarely. Uniform quantization introduces high percentage error on low-amplitude sounds. The logarithmic μ-law compander compresses the dynamic range before encoding, allocating smaller step sizes to weak signals and wider steps to strong signals, ensuring a nearly constant SQNR over a 40 dB dynamic range.

Board Exam Tip: North America & Japan use μ-law ($\mu = 255$); Europe uses A-law ($A = 87.6$). Both provide non-uniform quantization.

Answer: It applies smaller quantization steps to low-amplitude and larger steps to high-amplitude signals, improving SQNR.

Question 17: In the TCP/IP protocol suite, what is the primary role of the Internet Protocol (IP)?

Topic: Data Networks / Internet Protocol

Concept & Explanation: The Internet Protocol operates at the Internet (Network) Layer. Its core responsibility is host identification via logical hierarchical IP addressing, packet encapsulation, and best-effort, connectionless packet routing across intermediate router hops.

Board Exam Tip: IP provides logical addressing and routing; it does not guarantee delivery, packet sequencing, or error correction.

Answer: To provide best-effort logical addressing and routing of packets across interconnected networks.

Question 18: An ISDN Primary Rate Interface (PRI) in North America consists of how many channels?

Topic: Telecommunications / ISDN

Concept & Explanation: In North America and Japan, ISDN PRI is built on the 1.544 Mbps T1 carrier, which allocates 24 time slots: 23 Bearer (B) channels carrying 64 kbps user payload and 1 Data (D) channel carrying 64 kbps signaling ($23B + D$). In Europe, E1-based PRI provides $30B + 2D$ (or $30B + D$).

Board Exam Tip: North American PRI = $23B + D$ (T1, 1.544 Mbps). European PRI = $30B + D$ (E1, 2.048 Mbps).

Answer: 23 Bearer channels + 1 Data channel

Question 19: In a T1 carrier system, 24 voice channels are multiplexed. Each frame adds 1 framing bit. Calculate the total number of bits per frame.

Topic: Telecommunications / T-Carrier Systems

Given: Voice channels $N = 24$, resolution per channel $n = 8$ bits, framing bits $F = 1$ bit. | **Find:** Total bits per T1 frame.

Formula & Solution:

1. Bits from digitized voice channels:

$$\text{Voice bits} = 24\ \text{channels} \times 8\ \text{bits/channel} = 192\ \text{bits}$$

2. Total frame size including synchronization bit:

$$\text{Total Frame Bits} = 192 + 1 = 193 \text{ bits}$$

Board Exam Tip: 193 bits/frame $\times$ 8000 frames/s $= 1.544$ Mbps (Standard DS1 bit rate).

Answer: 193 bits

Question 20: Which type of noise is characterized by an equal distribution of power across all frequencies?

Topic: Noise Analysis / Random Processes

Concept & Explanation: White noise has a flat, constant power spectral density ($S_n(f) = N_0/2$ W/Hz) across all frequencies, directly analogous to white light containing equal intensities of all visible wavelengths. Thermal (Johnson-Nyquist) noise generated by random electron agitation is modeled as white noise over practical RF bands.

Board Exam Tip: Constant PSD across all frequencies = White Noise. $1/f$ PSD = Pink or Flicker Noise.

Answer: White Noise

Question 21: A communication channel has a bandwidth of 4 kHz and a signal-to-noise ratio of 15 dB. Calculate the Shannon channel capacity.

Topic: Information Theory / Shannon-Hartley Theorem

Given: Bandwidth $B = 4$ kHz $= 4000$ Hz, $\text{SNR}_{dB} = 15$ dB. | **Find:** Channel Capacity C in kbps.

Formula & Solution:

1. Convert SNR from dB to linear power ratio:

$$\text{SNR} = 10^{15/10} = 10^{1.5} \approx 31.6228$$

2. Apply the Shannon-Hartley capacity formula:

$$C = B \log_2(1 + \text{SNR}) = 4000 \times \log_2(1 + 31.6228) = 4000 \times \log_2(32.6228)$$

3. Using base change $\log_2(x) = \ln(x)/\ln(2)$:

$$\log_2(32.6228) = \frac{\ln(32.6228)}{\ln(2)} \approx \frac{3.48498}{0.693147} \approx 5.02776$$

$$C = 4000 \times 5.02776 = 20{,}111 \text{ bps} \approx 20.2 \text{ kbps}$$

Answer: 20.2 kbps

Question 22: What is the primary purpose of a stub matching network on a transmission line?

Topic: Transmission Lines / Impedance Matching

Concept & Explanation: When a load does not equal the line characteristic impedance ($Z_L \neq Z_0$), reflections create standing waves. A stub (a short or open section of transmission line) placed in parallel at a computed distance from the load presents a pure susceptance (jB) that cancels the load susceptance, making the total input admittance equal to the characteristic line conductance (Y_0), thus matching the line and eliminating standing waves.

Board Exam Tip: Stub matching introduces pure reactance/susceptance to cancel line reflections without dissipating resistive power.

Answer: To introduce a complementary reactance that cancels the load reactance and achieves impedance matching.

Question 23: What is the primary radiation pattern of a Marconi antenna, which is a quarter-wave vertical monopole mounted over a ground plane?

Topic: Antenna Engineering / Monopoles
Concept & Explanation: A Marconi antenna is a vertical quarter-wavelength ($\lambda/4$) monopole operated against a ground plane (or conducting earth). Because the radiator has rotational symmetry around its vertical axis, its radiation pattern in the azimuth (horizontal) plane is completely non-directional (omnidirectional, circle). In the elevation (vertical) plane, radiation forms a donut-like figure-eight clipped at the ground surface.
Board Exam Tip: Marconi antenna = $\lambda/4$ vertical; Horizontal pattern = Omnidirectional (circle); Elevation pattern = Figure-eight above ground.

Answer: An omnidirectional radiation pattern in the horizontal plane, with approximately uniform radiation in all azimuthal directions.

Question 24: Which digital modulation technique modifies both the amplitude and the phase of the carrier to define its constellation points?

Topic: Digital Communications / Modulation Schemes
Concept & Explanation: Quadrature Amplitude Modulation (QAM) modulates two orthogonal sinusoidal carriers (in-phase I and quadrature Q) with independent multilevel amplitudes. Each distinct constellation point represents a unique combination of carrier amplitude $\sqrt{I^2 + Q^2}$ and carrier phase angle $\arctan(Q/I)$.
Board Exam Tip: ASK = Amplitude only. PSK = Phase only. FSK = Frequency only. QAM = Both Amplitude and Phase.

Answer: QAM

Question 25: For an FM signal with a frequency deviation of 50 kHz and a modulating frequency of 10 kHz, calculate the required bandwidth using Carson's Rule.

Topic: Angle Modulation / FM Bandwidth
Given: Peak frequency deviation $\Delta f = 50$ kHz, modulating frequency $f_m = 10$ kHz. | **Find:** Transmission bandwidth B_{FM} using Carson's Rule.
Formula & Solution:
Carson's Rule states:

$$B_{FM} = 2(\Delta f + f_m)$$

$$B_{FM} = 2(50 \text{ kHz} + 10 \text{ kHz}) = 2(60 \text{ kHz}) = 120 \text{ kHz}$$

Board Exam Tip: Carson's Rule accounts for approximately 98% of total radiated FM power. In terms of modulation index: $B = 2f_m(\beta + 1)$.

Answer: 120 kHz

Question 26: What is the primary purpose of the Frame Check Sequence (FCS) field added to a data frame at the Data Link Layer (Layer 2)?

Topic: Data Link Layer / Framing
Concept & Explanation: The FCS field, located at the trailer of an Ethernet or HDLC frame, stores a 16-bit or 32-bit CRC checksum computed over the frame's address, control, and payload fields. The receiver re-evaluates this mathematical polynomial; if any bit has flipped during transmission, the checksum fails and the damaged frame is silently dropped.
Board Exam Tip: FCS provides Layer-2 bit error detection (not correction).

Answer: To carry a CRC-based error-detection value calculated from the frame contents to detect bit errors introduced during transmission.

Question 27: On a standard Smith Chart, what does the horizontal center axis represent?

Topic: Transmission Lines / Smith Chart
Concept & Explanation: The Smith Chart represents complex reflection coefficients Γ plotted on the normalized impedance plane $z = r + jx$. The straight horizontal diameter passing through the center represents zero reactance ($x = 0$), mapping all purely resistive normalized impedances from short-circuit ($r = 0$, leftmost) through matched load ($r = 1$, center) to open-circuit ($r = \infty$, rightmost).
Board Exam Tip: Horizontal center line $\rightarrow$ Pure resistance ($x = 0$). Upper half $\rightarrow$ Inductive ($+jx$). Lower half $\rightarrow$ Capacitive ($-jx$).
Answer: Purely resistive impedances (zero reactance).

Question 28: In the OSI model, which layer ensures reliable, error-free, end-to-end data delivery and handles flow control (e.g., TCP)?
Topic: OSI Reference Model / Layer 4
Concept & Explanation: Layer 4 (Transport Layer) provides transparent, end-to-end data transport between application processes on host devices. It manages segment sequencing, port addressing, flow control (sliding window), error recovery (retransmissions), and connection management (TCP three-way handshake).
Board Exam Tip: End-to-end reliable transmission and port numbers = Transport Layer (Layer 4). Hop-to-hop framing = Data Link (Layer 2).
Answer: Layer 4: Transport Layer

Question 29: A transmitter operates with an EIRP of 100 watts. Calculate the power density at a distance of 5 km in free space.
Topic: Antennas & Wave Propagation / Friis Transmission
Given: EIRP $= P_t G_t = 100$ W, distance $R = 5$ km $= 5000$ m. | **Find:** Power density S in W/m^2.
Formula & Solution:
Power density radiated isotropically over a spherical surface of radius R:

$$S = \frac{\text{EIRP}}{4\pi R^2} = \frac{100}{4\pi(5000)^2} = \frac{100}{4\pi \times 25 \times 10^6} = \frac{100}{100\pi \times 10^6} = \frac{1}{\pi \times 10^6}$$

$$S = \frac{1}{3.14159 \times 10^6} \approx 3.183 \times 10^{-7} \text{ W/m}^2$$

Answer: 3.18×10^{-7} **W/m^2**

Question 30: Analyze why baseband signals are typically not transmitted directly over wireless radio links without modulation.
Topic: Telecommunications Fundamentals / Antenna Physics
Concept & Explanation: An efficient antenna must have a physical size comparable to the wavelength: $L \approx \lambda/2 = c/(2f)$. For voice signals at 3 kHz, the required antenna length would be $L = 3 \times 10^8/(2 \times 3000) = 50{,}000$ meters $= 50$ km, which is physically impractical. Modulation shifts baseband energy to high carrier frequencies where antennas measure centimeters or inches, and enables frequency-division multiplexing.
Board Exam Tip: Direct baseband radiation fails due to: (1) enormous required antenna size, (2) inability to multiplex multiple channels, and (3) extreme signal attenuation.
Answer: Their low frequencies would require impractically large antennas.

Question 31: Analyze the total power distribution in an AM signal modulated at $m = 0.5$. What percentage of the total transmitted power is contained in both sidebands combined?
Topic: Analog Communications / AM Power Relations

Given: Modulation index $m = 0.5$. | **Find:** Sideband power percentage of total power.
Formula & Solution:
1. Total power formula for double-sideband full-carrier AM:

$$P_t = P_c\left(1 + \frac{m^2}{2}\right) = P_c\left(1 + \frac{0.5^2}{2}\right) = P_c\left(1 + \frac{0.25}{2}\right) = 1.125P_c$$

2. Total power in both sidebands:

$$P_{SB} = P_c\left(\frac{m^2}{2}\right) = 0.125P_c$$

3. Percentage sideband power:

$$\%P_{SB} = \frac{P_{SB}}{P_t} \times 100\% = \frac{0.125P_c}{1.125P_c} \times 100\% = \frac{0.125}{1.125} \times 100\% = \frac{1}{9} \times 100\% \approx 11.11\%$$

Board Exam Shortcut: Sideband fraction = $\frac{m^2}{2+m^2}$. For $m = 0.5 \implies \frac{0.25}{2.25} = \frac{1}{9} \approx 11.11\%$.
Answer: 11.11%

Question 32: What is the theoretical gain of an isotropic radiator?
Topic: Antenna Engineering / Gain Reference
Concept & Explanation: An isotropic radiator is a theoretical, loss-free point source that radiates power equally in all directions over a sphere. Because it serves as the universal reference standard for antenna directivity and gain, its gain is defined as unity (1.0 in linear ratio), which converts to $10\log_{10}(1.0) = 0$ dBi.
Board Exam Tip: Isotropic radiator gain = 1.0 (linear) $= 0$ dBi. Dipole gain = 1.64 (linear) $= 2.15$ dBi.
Answer: 0 dBi

Question 33: Which network topology uses a single backbone cable to which all nodes are tapped directly?
Topic: Data Networks / Topologies
Concept & Explanation: In a bus topology, a continuous linear transmission line (coaxial trunk) connects all communicating stations via vampire taps or BNC T-connectors. The ends of the trunk must be terminated with matching resistors (typically $50\,\Omega$) to prevent signal reflections.
Board Exam Tip: Single continuous trunk with taps = Bus Topology. Hub/switch in center = Star. Closed loop = Ring.
Answer: Bus Topology

Question 34: Calculate the thermal noise voltage developed across a $50\,\Omega$ resistor at a temperature of 290 K over a 1 MHz bandwidth ($k = 1.38 \times 10^{-23}$ J/K).
Topic: Noise Analysis / Johnson-Nyquist Noise
Given: $R = 50\,\Omega$, $T = 290$ K, $B = 1$ MHz $= 10^6$ Hz, $k = 1.38 \times 10^{-23}$ J/K. | **Find:** RMS thermal noise voltage V_n in μV.
Formula & Solution:
Johnson's noise voltage equation:

$$V_n = \sqrt{4kTBR}$$

$$4kTBR = 4 \times (1.38 \times 10^{-23}) \times 290 \times 10^6 \times 50 = 8.004 \times 10^{-13}\ \text{V}^2$$

$$V_n = \sqrt{8.004 \times 10^{-13}} \approx 8.9465 \times 10^{-7}\ \text{V} \approx 0.89\,\mu\text{V}$$

Board Exam Tip: Noise voltage scales with the square root of bandwidth and resistance: doubling R increases V_n by $\sqrt{2} \approx 1.414$.

Answer: $0.89\,\mu$**V**

Question 35: You need to design a digital filter that completely removes a DC offset from an incoming signal but passes high frequencies. Which zero placement on the Z-plane achieves this?

Topic: Digital Signal Processing / Z-Transform Filter Design
Concept & Explanation: The frequency response of a digital filter is evaluated by tracing the unit circle $z = e^{j\omega}$, where $\omega = 0$ corresponds to DC (zero frequency) and $\omega = \pi$ represents the half-sampling frequency (Nyquist). Placing a transmission zero directly at $z = 1$ ($e^{j0} = 1$) drives the system frequency response $|H(e^{j0})| = 0$, completely rejecting DC bias while passing higher alternating frequencies (high-pass filter).
Board Exam Tip: Zero at $z = 1 \implies$ Blocks DC (High-Pass). Zero at $z = -1 \implies$ Blocks Nyquist (Low-Pass).
Answer: Place a zero at z = 1

Question 36: Determine the velocity factor of a coaxial cable if the wave propagates through it at 2×10^8 m/s ($c = 3 \times 10^8$ m/s).

Topic: Transmission Lines / Wave Velocity
Given: Line phase velocity $v = 2 \times 10^8$ m/s, free-space velocity $c = 3 \times 10^8$ m/s. | **Find:** Velocity factor VF.
Formula & Solution:
By definition, velocity factor is the ratio of wave velocity in the medium to velocity in free space:

$$VF = \frac{v}{c} = \frac{2 \times 10^8 \text{ m/s}}{3 \times 10^8 \text{ m/s}} = \frac{2}{3} \approx 0.667 \approx 0.67$$

Board Exam Tip: $VF = 1/\sqrt{\epsilon_r}$. A velocity factor of 0.67 indicates a solid polyethylene dielectric ($\epsilon_r \approx 2.25$).
Answer: 0.67

Question 37: Which multiplexing technique is utilized extensively in fiber optic communications to send multiple light wavelengths through a single strand of fiber?

Topic: Optical Communications / WDM
Concept & Explanation: Wavelength Division Multiplexing (WDM) combines multiple optical carrier signals onto a single optical fiber strand by using different wavelengths (colors) of laser light (e.g., in the 1310 nm and 1550 nm low-loss optical windows). It represents the optical equivalent of Frequency Division Multiplexing (FDM).
Board Exam Tip: Fiber optic wavelength-based multiplexing = WDM (Dense WDM / Coarse WDM).
Answer: Wavelength Division Multiplexing (WDM)

Question 38: How is the half-power beamwidth (HPBW) of a directional antenna formally defined?

Topic: Antenna Engineering / Antenna Patterns
Concept & Explanation: In an antenna radiation pattern plot, the Half-Power Beamwidth (HPBW or $\theta_{3\text{dB}}$) is the angular span between the two symmetrical points on the major radiation lobe where the radiated power density drops to exactly half (50% or -3 dB) of its on-axis peak value ($E = E_{max}/\sqrt{2} \approx 0.707 E_{max}$).
Board Exam Tip: HPBW $= -3$ dB power points = 70.7% field strength points.
Answer: The angular separation between the two points on the main lobe where the radiated power density is one-half of its maximum value (-3 dB).

Question 39: In a SCADA system, what is the primary role of a Human-Machine Interface (HMI)?

Topic: Industrial Automation / SCADA Architecture

Concept & Explanation: The Human-Machine Interface (HMI) is the graphical operator interface in a supervisory control and data acquisition system. It renders telemetry, schematic mimics, alarms, trends, and diagnostic data collected from Remote Terminal Units (RTUs) and Programmable Logic Controllers (PLCs), allowing human operators to monitor and issue process commands.

Board Exam Tip: HMI = Graphic display, monitoring, alarming, and human supervisory input to PLC/SCADA networks.

Answer: To provide operators with graphical displays, real-time system information, alarms, and controls for interacting with the process.

Question 40: An analog signal is sampled at 8 kHz and quantized into 256 distinct levels. Calculate the PCM transmission bit rate.

Topic: Digital Communications / Pulse Code Modulation

Given: $f_s = 8$ kHz $= 8000$ samples/sec, Quantization levels $L = 256$. | **Find:** Bit rate R_b in kbps.

Formula & Solution:

1. Number of bits per sample n:

$$n = \log_2(L) = \log_2(256) = 8 \text{ bits/sample}$$

2. Transmission bit rate:

$$R_b = n \times f_s = 8 \text{ bits/sample} \times 8000 \text{ samples/s} = 64{,}000 \text{ bps} = 64 \text{ kbps}$$

Board Exam Tip: This 64 kbps channel forms the fundamental Digital Signal 0 (DS0) building block for global digital telephony.

Answer: 64 kbps

Question 41: In waveguide theory, what does the term "cutoff frequency" refer to?

Topic: Microwaves / Waveguides

Concept & Explanation: A hollow metallic waveguide operates as a high-pass filter. The cutoff frequency (f_c) is the absolute minimum operational frequency below which electromagnetic wave propagation cannot occur in that mode; electromagnetic fields decay exponentially as evanescent waves without transmitting real power.

Board Exam Tip: For rectangular waveguide TE_{10} dominant mode: $f_c = c/(2a)$, where a is the broad wall dimension.

Answer: The lowest frequency at which a particular mode can propagate through the waveguide.

Question 42: Calculate the free space path loss (FSPL) in dB for a 1 GHz signal traveling 10 km.

Topic: Microwave Engineering / Friis Propagation

Given: $f = 1$ GHz $= 1000$ MHz, $d = 10$ km. | **Find:** Free Space Path Loss in dB.

Formula & Solution:

Logarithmic FSPL formula:

$$\text{FSPL (dB)} = 32.44 + 20\log_{10}(f_{\text{MHz}}) + 20\log_{10}(d_{\text{km}})$$

$$\text{FSPL} = 32.44 + 20\log_{10}(1000) + 20\log_{10}(10)$$

$$20\log_{10}(1000) = 20 \times 3 = 60 \text{ dB}, \quad 20\log_{10}(10) = 20 \times 1 = 20 \text{ dB}$$

$$\text{FSPL} = 32.44 + 60 + 20 = 112.44 \text{ dB} \approx 112.4 \text{ dB}$$

Board Exam Shortcut: Standard baseline: At 1 GHz and 1 km, FSPL ≈ 92.4 dB. Increasing distance to 10 km ($10\times$) adds $20\log_{10}(10) = 20$ dB, yielding $92.4 + 20 = 112.4$ dB.

Answer: 112.4 dB

Question 43: In radio-wave propagation, what does the Maximum Usable Frequency (MUF) represent?

Topic: Antennas & Wave Propagation / Ionospheric Refraction

Concept & Explanation: Maximum Usable Frequency (MUF) is the highest radio frequency that can be reflected/refracted back to Earth by an ionospheric layer for a specified transmission path distance and angle of incidence. Signals above the MUF penetrate through the ionosphere into outer space.

Board Exam Tip: MUF = Critical Frequency $\times \sec(\theta_i) = f_c/\cos(\theta_i)$ (Secant Law).

Answer: The highest frequency that can be returned to Earth by the ionosphere for a specified propagation path.

Question 44: What is the effect of lowering the Intermediate Frequency (IF) in a superheterodyne receiver?

Topic: RF Receivers / Intermediate Frequency Choice

Concept & Explanation: Lowering the IF reduces the center frequency of the IF tuned stages ($BW = f_{IF}/Q$). This allows sharper, narrower passbands for a given filter Q, significantly improving selectivity between adjacent channels. However, the image frequency ($f_{img} = f_{RF} \pm 2f_{IF}$) moves much closer to the desired carrier, severely degrading image frequency rejection at the RF preselector.

Board Exam Tip: Lower IF $\rightarrow$ Better adjacent channel selectivity, worse image rejection. Higher IF $\rightarrow$ Better image rejection, worse adjacent selectivity.

Answer: Adjacent channel selectivity improves, but image frequency rejection degrades.

Question 45: A continuous-time signal is given as $x(t) = 4\cos(100\pi t) + 8\sin(300\pi t)$. Calculate the minimum Nyquist sampling frequency required to prevent aliasing.

Topic: Signals & Systems / Nyquist Sampling Theorem

Given: Signal terms: $\omega_1 = 100\pi$ rad/s, $\omega_2 = 300\pi$ rad/s. | **Find:** Nyquist sampling frequency f_s in Hz.

Formula & Solution:

1. Find component frequencies in Hertz ($\omega = 2\pi f$):

$$f_1 = \frac{100\pi}{2\pi} = 50 \text{ Hz}, \quad f_2 = \frac{300\pi}{2\pi} = 150 \text{ Hz}$$

2. The maximum signal frequency is $f_{max} = 150$ Hz. 3. Nyquist sampling rate requires:

$$f_s \geq 2f_{max} = 2 \times 150 \text{ Hz} = 300 \text{ Hz}$$

Board Exam Tip: Nyquist rate is always determined strictly by the highest frequency component present in the composite signal, regardless of amplitude.

Answer: 300 Hz

Question 46: Which of the following defines an energy signal?

Topic: Signals & Systems / Signal Classification

Concept & Explanation: An energy signal possesses finite, non-zero total energy ($0 < E < \infty$) evaluated over all time. Because its total energy is spread over infinite time, its normalized average

power calculated as $P = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt$ is identically zero ($P = 0$).
Board Exam Tip: A signal cannot be both an energy and a power signal. Aperiodic pulses are energy signals; periodic waveforms are power signals ($0 < P < \infty$, $E = \infty$).
Answer: Its total energy is finite and non-zero, and its average power is zero.

Question 47: What is a significant disadvantage of a standard Star topology compared with a Mesh topology?
Topic: Data Networks / Topologies
Concept & Explanation: In a Star network, every host node connects to a single central hub or switch. If that central device suffers hardware or power failure, communication across the entire local network ceases immediately. Conversely, a Mesh topology contains multiple redundant point-to-point links, routing traffic around link or node failures.
Board Exam Tip: Star topology has an inherent single point of failure (the central hub/switch).
Answer: If the central hub or switch fails in a Star topology, the entire network can become unavailable because the central device represents a single point of failure.

Question 48: Which pulse modulation scheme encodes analog amplitude into the precise timing of a constant-width pulse relative to a reference clock?
Topic: Pulse Modulation / Timing Schemes
Concept & Explanation: Pulse Position Modulation (PPM) maintains constant pulse amplitude and constant pulse duration, modulating the temporal position (displacement) of each pulse within its allocated time frame proportionally to the instantaneous analog sample amplitude.
Board Exam Tip: PAM = pulse height; PWM (PDM) = pulse width; PPM = pulse time position. PPM and PWM have high noise immunity because limiters clip amplitude noise.
Answer: Pulse Position Modulation (PPM)

Question 49: Calculate the physical length of a half-wave dipole operating at 150 MHz in free space (assume a velocity factor of 0.95 for the physical wire).
Topic: Antennas / Resonant Length
Given: $f = 150$ MHz $= 150 \times 10^6$ Hz, $c = 3 \times 10^8$ m/s, $k = 0.95$. | **Find:** Physical antenna length L in meters.
Formula & Solution:
1. Free-space wavelength:

$$\lambda_0 = \frac{c}{f} = \frac{3 \times 10^8 \text{ m/s}}{150 \times 10^6 \text{ Hz}} = 2.0 \text{ meters}$$

2. Physical half-wavelength accounting for velocity factor:

$$L = \frac{k\lambda_0}{2} = \frac{0.95 \times 2.0}{2} = 0.95 \text{ meters}$$

Board Exam Shortcut: L (meters) $\approx \frac{142.5 \times k}{f_{\text{MHz}}} = \frac{142.5 \times 0.95}{150} \approx 0.90$–$0.95$ m.
Answer: 0.95 m

Question 50: Calculate the beamwidth (in degrees) of a parabolic reflector antenna with a diameter of 3 m operating at 4 GHz. Use the approximation $\theta \approx 70\lambda/D$.
Topic: Antenna Engineering / Parabolic Dishes
Given: $D = 3$ m, $f = 4$ GHz $= 4 \times 10^9$ Hz. | **Find:** Beamwidth θ in degrees.
Formula & Solution:
1. Compute operating wavelength:

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8 \text{ m/s}}{4 \times 10^9 \text{ Hz}} = 0.075 \text{ meters}$$

2. Apply beamwidth approximation:

$$\theta \approx \frac{70\lambda}{D} = \frac{70 \times 0.075}{3} = \frac{5.25}{3} = 1.75^\circ$$

Board Exam Tip: Increasing dish diameter or frequency produces a narrower beamwidth and higher antenna gain.

Answer: 1.75°

Question 51: An AM transmitter radiates 50 kW of unmodulated carrier power. If it is modulated by a sinusoidal signal at 100% modulation, calculate the total radiated power.

Topic: Analog Communications / AM Power Relations

Given: Carrier power $P_c = 50$ kW, modulation index $m = 1.0$ (100%). | **Find:** Total radiated power P_t in kW.

Formula & Solution:

Total AM power equation:

$$P_t = P_c\left(1 + \frac{m^2}{2}\right) = 50 \text{ kW} \times \left(1 + \frac{1.0^2}{2}\right) = 50 \times 1.5 = 75 \text{ kW}$$

Board Exam Shortcut: At 100% sinusoidal AM modulation, total power always increases by exactly 50% over unmodulated carrier power ($P_t = 1.5P_c$).

Answer: 75 kW

Question 52: Why are 4-wire circuits commonly preferred over 2-wire circuits for long-distance data communication links?

Topic: Telephony / Long-Haul Transmission

Concept & Explanation: A 2-wire circuit carries bidirectional traffic over a single pair, creating impedance hybrid mismatches, singing, and near-end crosstalk when amplifiers are inserted. A 4-wire circuit establishes physically independent forward (transmit) and reverse (receive) paths, allowing unidirectional linear electronic repeaters to be inserted without feedback oscillation.

Board Exam Tip: 4-wire circuits allow true full-duplex communication and unidirectional amplification across long-haul trunks.

Answer: 4-wire circuits provide separate transmit and receive paths, enabling full-duplex operation and easier signal amplification.

Question 53: In a superheterodyne receiver, what is the primary purpose of the Intermediate Frequency (IF) amplifier stages?

Topic: RF Receivers / Receiver Architecture

Concept & Explanation: Tuning multi-stage RF filters synchronously across a wide frequency band is technically difficult and yields non-uniform selectivity. In a superheterodyne receiver, incoming RF signals are converted to a constant IF. The fixed-tuned IF amplifier stages deliver the vast majority of receiver gain (often > 60 dB) and achieve sharp adjacent-channel selectivity with high-Q fixed bandpass filters.

Board Exam Tip: In a superheterodyne receiver, sensitivity (gain) and selectivity (filtering) are primarily determined by the IF stages.

Answer: To provide the bulk of the receiver's gain and selectivity at a fixed frequency.

Question 54: In a typical PLC scan cycle, if a physical input changes from OFF to ON while the user program is being executed, when will the PLC normally register this change?

Topic: Industrial Automation / Programmable Logic Controllers

Concept & Explanation: A standard PLC executes a cyclical three-step routine: (1) Input Scan,

(2) Program Logic Execution, and (3) Output Scan. During the Program Execution phase, the CPU reads only from the static Input Image Table. If a physical field contact transitions after the Input Scan has completed, that change is ignored until the next Input Scan phase of the subsequent scan cycle.
Board Exam Tip: Field inputs are latched once per cycle during the Input Scan. Changes mid-cycle are recognized on the following cycle.

Answer: During the next Input Scan phase of the subsequent PLC scan cycle.

Question 55: What receiver circuit provides the function of varying the RF amplifier gain based on the strength of the received signal to prevent audio volume fluctuation?

Topic: RF Receivers / Automatic Gain Control
Concept & Explanation: The Automatic Gain Control (AGC) circuit rectifies and filters the IF or detector output to generate a DC control voltage proportional to the carrier signal strength. This bias voltage is fed back to the RF and IF amplifier stages, reducing amplifier gain for strong incoming signals and boosting gain for weak signals, preventing speaker overload and fading.
Board Exam Tip: AGC maintains a constant audio/data output volume regardless of incoming RF signal fading or strength variations.

Answer: Automatic Gain Control (AGC)

Question 56: A rectangular waveguide has a broad wall dimension $a = 2.5$ cm. Calculate the cutoff frequency for the dominant TE_{10} mode.

Topic: Microwave Engineering / Waveguides
Given: Broad wall dimension $a = 2.5$ cm $= 0.025$ m, mode = TE_{10}, $c = 3 \times 10^8$ m/s. | **Find:** Cutoff frequency f_c in GHz.
Formula & Solution:
For the dominant TE_{10} mode in a rectangular waveguide, the cutoff wavelength is $\lambda_c = 2a$. The cutoff frequency is:

$$f_c = \frac{c}{\lambda_c} = \frac{c}{2a} = \frac{3 \times 10^8 \text{ m/s}}{2 \times 0.025 \text{ m}} = \frac{3 \times 10^8}{0.05} = 6 \times 10^9 \text{ Hz} = 6 \text{ GHz}$$

Board Exam Tip: Waveguide propagation is impossible below f_c. Frequencies $f > 6$ GHz propagate freely down this guide.

Answer: 6 GHz

Question 57: What is the primary function of a Network Gateway compared with a standard Router?

Topic: Data Communications / Inter-Networking Devices
Concept & Explanation: A standard router interconnects networks at Layer 3 (Network Layer) using a common protocol suite (e.g., IPv4/IPv6). A gateway operates across higher layers (Layers 4 through 7) of the OSI model, performing protocol translation, message mapping, and syntax conversion between fundamentally incompatible networking architectures (such as TCP/IP to IBM SNA, or Modbus to Profinet).
Board Exam Tip: Repeater = L1; Bridge/Switch = L2; Router = L3; Gateway = L4 to L7 (protocol translation between dissimilar networks).

Answer: A gateway can operate at higher OSI layers to translate between fundamentally different network architectures or protocols, such as TCP/IP and IBM SNA.

Question 58: How do the theoretical bit error rates (BERs) of BPSK and QPSK compare when both are subjected to the same channel conditions and (E_b/N_0)?

Topic: Digital Communications / Error Probability Analysis

Concept & Explanation: In QPSK, the bit stream is demultiplexed into two orthogonal bit streams (I and Q), each modulated at half the overall symbol rate. Because the I and Q channels are completely orthogonal and do not interfere with each other, each axis behaves as an independent BPSK signal with energy per bit E_b. Consequently, both BPSK and QPSK share the exact same theoretical bit error probability curve: $P_b = Q\left(\sqrt{2E_b/N_0}\right)$. QPSK achieves twice the spectral efficiency of BPSK for the same BER.
Board Exam Tip: BPSK and QPSK have identical BER vs. E_b/N_0. QPSK doubles the bit rate in the same bandwidth.
Answer: BPSK and QPSK have exactly the same theoretical BER under the same (E_b/N_0) conditions.

Question 59: What type of wide-area network technology switches fixed-size 53-byte cells rather than variable-length packets?
Topic: Data Communications / ATM Networks
Concept & Explanation: Asynchronous Transfer Mode (ATM) is a connection-oriented high-speed WAN switching technology based on small, uniform, fixed-length packets termed cells. Each cell contains exactly 53 bytes: a 5-byte header and a 48-byte payload. Fixed cell sizing enables predictable hardware switching and low jitter, ideal for real-time voice and video traffic.
Board Exam Tip: Fixed 53-byte cells (5 header + 48 payload) = ATM.
Answer: Asynchronous Transfer Mode (ATM)

Question 60: A binary source generates zeroes with a probability of 0.25 and ones with a probability of 0.75. Calculate the entropy of the source in bits/symbol.
Topic: Information Theory / Source Entropy
Given: $P(0) = 0.25$, $P(1) = 0.75$. | **Find:** Source entropy $H(S)$ in bits/symbol.
Formula & Solution:
Shannon's source entropy formula:

$$H(S) = -\sum_i P_i \log_2(P_i) = -[0.25 \log_2(0.25) + 0.75 \log_2(0.75)]$$

Evaluating the logarithmic terms:

$$\log_2(0.25) = \log_2(2^{-2}) = -2$$

$$\log_2(0.75) = \log_2(3/4) = \log_2(3) - 2 \approx 1.58496 - 2 = -0.41504$$

Substituting into entropy equation:

$$H(S) = -[0.25(-2) + 0.75(-0.41504)] = -[-0.500 - 0.31128] = 0.81128 \approx 0.811 \text{ bits/symbol}$$

Board Exam Tip: Maximum entropy for a binary source occurs when symbols are equiprobable ($P = 0.5$), yielding $H_{max} = 1.0$ bit/symbol.
Answer: 0.811

Question 61: In cryptography, what defines an asymmetric encryption system (Public Key Infrastructure)?
Topic: Network Security / Cryptographic Systems
Concept & Explanation: Asymmetric cryptography utilizes a mathematically related key pair: a public key that can be distributed openly to the world, and a private key that must remain strictly secret with the owner. Data encrypted with the public key can only be decrypted by the matching private key, providing confidentiality without requiring pre-shared secret keys.
Board Exam Tip: Symmetric = single shared key. Asymmetric = key pair (public for encryption,

private for decryption).

Answer: It uses two mathematically linked keys: a public key for encrypting data and a private key for decrypting data.

Question 62: A strain gauge has a Gauge Factor (GF) of 2.0 and a nominal resistance of $120\,\Omega$. If it is subjected to a strain of 500 microstrain, calculate the change in resistance ΔR.

Topic: Industrial Instrumentation / Strain Gauges

Given: Gauge Factor $GF = 2.0$, nominal resistance $R = 120\,\Omega$, mechanical strain $\epsilon = 500 \times 10^{-6}$ ($500\,\mu\epsilon$). | **Find:** Change in resistance ΔR in Ω.

Formula & Solution:

The fundamental strain gauge governing equation is:

$$\frac{\Delta R}{R} = GF \times \epsilon$$

Solving for ΔR:

$$\Delta R = GF \times \epsilon \times R = 2.0 \times (500 \times 10^{-6}) \times 120\,\Omega$$

$$\Delta R = 1000 \times 10^{-6} \times 120 = 10^{-3} \times 120 = 0.120\,\Omega = 0.12\,\Omega$$

Board Exam Tip: Metallic foil strain gauges typically have $GF \approx 2.0$; piezoresistive semiconductor gauges have $GF > 100$.

Answer: $0.12\,\Omega$

Question 63: In a security alarm system, how does a normally closed (NC) magnetic door switch provide fail-safe operation when the door is opened or its wiring is interrupted?

Topic: Security Systems / Circuit Architecture

Concept & Explanation: A normally closed (NC) reed switch holds its internal contacts shut when the door magnet is adjacent, completing an electrical loop through which supervisory monitoring current flows. When an intruder opens the door, the magnet pulls away and the switch opens. Crucially, if an intruder cuts or severs the wiring, loop current drops to zero identically, immediately triggering an alarm. This closed-loop supervision ensures fail-safe operation.

Board Exam Tip: NC circuits trigger alarms upon loop interruption, protecting against severed wiring.

Answer: The closed door keeps the magnetic switch closed, maintaining circuit continuity; if the door opens or its wiring is cut, the circuit opens and triggers an alarm.

Question 64: Determine the output sequence $y[n]$ at $n = 2$ for an FIR filter described by $y[n] = x[n] - 0.5x[n-1]$, given the input sequence $x[n] = \{2, 4, 6, 8\}$ for $n = 0, 1, 2, 3$.

Topic: Digital Signal Processing / FIR Filters

Given: Difference equation: $y[n] = x[n] - 0.5x[n-1]$. Input: $x[0] = 2, x[1] = 4, x[2] = 6, x[3] = 8$. | **Find:** Filter output at $n = 2$, $y[2]$.

Formula & Solution:

Substitute $n = 2$ directly into the difference equation:

$$y[2] = x[2] - 0.5x[2-1] = x[2] - 0.5x[1]$$

From the given input sequence: $x[2] = 6$ and $x[1] = 4$.

$$y[2] = 6 - 0.5(4) = 6 - 2 = 4$$

Board Exam Tip: FIR filters have no feedback ($y[n-k]$ terms); outputs depend strictly on current and past inputs.

Answer: 4

Question 65: Which type of digital filter is unconditionally stable because it does not utilize feedback?

Topic: Digital Signal Processing / Filter Stability
Concept & Explanation: A Finite Impulse Response (FIR) filter's transfer function contains no feedback denominator poles other than trivial poles located at the origin ($z = 0$). Because all poles lie inside the unit circle on the Z-plane, the impulse response contains a finite number of samples and the system is inherently, unconditionally stable (BIBO stable).
Board Exam Tip: FIR filters are always unconditionally stable and can achieve exact linear phase; IIR filters use feedback and may become unstable.
Answer: Finite Impulse Response (FIR)

Question 66: Analyze the system function $H(z) = \frac{z}{z-1.5}$. Which of the following statements is true regarding its step response?

Topic: Digital Signal Processing / Z-Domain Stability
Concept & Explanation: The pole of this transfer function is located at $z = 1.5$. For a causal system to be Bounded-Input Bounded-Output (BIBO) stable, all poles must lie strictly inside the unit circle ($|z| < 1$). Because the pole $|z| = 1.5 > 1$, the inverse Z-transform yields an expanding geometric series $(1.5)^n u[n]$ whose step response diverges to infinity.
Board Exam Tip: Pole outside the unit circle ($|p| > 1$) for a causal system $\implies$ Unstable / diverges to infinity.
Answer: The system is unstable, and the step response diverges to infinity.

Question 67: Using High-Level Data Link Control (HDLC) bit-stuffing rules, what will the transmitter output if the data stream contains the sequence 111111?

Topic: Data Link Layer / HDLC Protocol
Given: Input data: `1 1 1 1 1 1` (six consecutive ones). | **Find:** Transmitted bit sequence after bit stuffing.
Formula & Solution:
The HDLC framing flag is `01111110`. To prevent user data from mimicking this flag, the transmitter inserts a '0' immediately after any five consecutive '1's, regardless of the following bit:

$$\text{Input: } \underbrace{1\,1\,1\,1\,1}_{5\text{ ones}}\,1 \quad \Longrightarrow \quad \text{Output: } \underbrace{1\,1\,1\,1\,1}_{5\text{ ones}}\,\mathbf{0}\,1$$

Transmitted stream: `1111101`.
Answer: 1111101

Question 68: A walkie-talkie communication link where only one party can speak at a time while holding a button is an example of what transmission mode?

Topic: Data Communications / Transmission Modes
Concept & Explanation: In half-duplex communication, data transmission is bidirectional across the shared channel, but transmission can occur in only one direction at any given instant. When one operator presses the Push-To-Talk (PTT) switch, that radio transmits while the other receives.
Board Exam Tip: Simplex = One-way only (Broadcast TV/Radio). Half-Duplex = Both ways, one at a time (Walkie-talkie). Full-Duplex = Both ways simultaneously (Telephone).
Answer: Half-Duplex

Question 69: An unmodulated AM transmitter draws an antenna current of 10 A. When the carrier is sinusoidally modulated, the current increases to 11.5 A. Calculate the modulation

index.

Topic: Analog Communications / AM Current Relations
Given: Unmodulated carrier current $I_c = 10$ A, modulated antenna current $I_t = 11.5$ A. | **Find:** Modulation index m.
Formula & Solution:
1. AM current formula:

$$I_t = I_c\sqrt{1 + \frac{m^2}{2}}$$

2. Solve for the ratio:

$$\frac{I_t}{I_c} = \frac{11.5}{10} = 1.15$$

3. Square both sides:

$$\left(\frac{I_t}{I_c}\right)^2 = (1.15)^2 = 1.3225 = 1 + \frac{m^2}{2}$$

4. Isolate m^2:

$$\frac{m^2}{2} = 1.3225 - 1 = 0.3225 \implies m^2 = 0.645$$

$$m = \sqrt{0.645} \approx 0.8031 \approx 0.81$$

Board Exam Tip: At 100% modulation ($m = 1.0$), antenna current rises to $I_t = I_c\sqrt{1.5} \approx 1.225I_c$ (22.5% increase).
Answer: 0.81

Question 70: How does the CSMA/CD protocol handle data collisions on a half-duplex Ethernet network?

Topic: Data Link Layer / Media Access Control
Concept & Explanation: Under Carrier Sense Multiple Access with Collision Detection (CSMA/CD), a node senses the physical wire before transmitting. If two nodes transmit simultaneously, their voltage levels superimpose, indicating a collision. Transmitting stations detect this abnormal signal, cease data transmission, broadcast a 32-bit jam signal to notify all stations, and execute a Truncated Binary Exponential Backoff algorithm before attempting retransmission.
Board Exam Tip: CSMA/CD steps: Listen $\rightarrow$ Transmit $\rightarrow$ Detect Collision $\rightarrow$ Send Jam Signal $\rightarrow$ Random Exponential Backoff $\rightarrow$ Retransmit.
Answer: When a collision is detected, transmitting devices stop, send a jam signal, wait for a randomly determined backoff period, and then attempt retransmission.

Question 71: Compute the linear convolution of two discrete sequences $x[n] = [1, 2]$ and $h[n] = [1, -1]$.

Topic: Digital Signal Processing / Convolution
Given: $x[n] = [1, 2]$ (length $L = 2$), $h[n] = [1, -1]$ (length $M = 2$). | **Find:** Linear convolution $y[n] = x[n] * h[n]$.
Formula & Solution:
Resulting sequence length is $N = L + M - 1 = 2 + 2 - 1 = 3$. Evaluating term by term:

$$y[0] = x[0] \cdot h[0] = 1 \times 1 = 1$$

$$y[1] = x[0] \cdot h[1] + x[1] \cdot h[0] = (1)(-1) + (2)(1) = -1 + 2 = 1$$

$$y[2] = x[1] \cdot h[1] = 2 \times (-1) = -2$$

Resulting sequence: $y[n] = [1, 1, -2]$.
Answer: [1, 1, -2]

Question 72: A QPSK modulator transmits at a baud rate of 2400 baud. Calculate the binary throughput (bit rate) of this system.

Topic: Digital Communications / Modulation

Given: $R_s = 2400$ Baud, modulation scheme: QPSK ($M = 4$). | **Find:** Bit rate R_b in bps.

Formula & Solution:

1. Number of bits per symbol in QPSK:

$$k = \log_2(M) = \log_2(4) = 2 \text{ bits/symbol}$$

2. Bit rate formula:

$$R_b = R_s \times k = 2400 \text{ Baud} \times 2 \text{ bits/symbol} = 4800 \text{ bps}$$

Board Exam Tip: In QPSK, bit rate is twice the baud rate ($R_b = 2R_s$). In BPSK, bit rate equals baud rate ($R_b = R_s$).

Answer: 4800 bps

Question 73: How does asynchronous transmission maintain bit timing without using a dedicated clock line?

Topic: Data Communications / Transmission Techniques

Concept & Explanation: Asynchronous transmission groups data into small characters (typically 8 bits). Each character is preceded by a Start bit (logic 0 transition from idle high), which synchronizes the receiver's local clock oscillator, and terminated by one or two Stop bits (logic 1). Because clock resynchronization occurs on every individual byte, long-term clock drift is prevented without a shared clock line.

Board Exam Tip: Asynchronous = Start/Stop bits per byte. Synchronous = Preamble/sync bytes, shared continuous clock.

Answer: Each byte is framed with a start bit that allows the receiver to synchronize its local clock for that byte.

Question 74: In a digital transmission system, under what condition does the Baud rate become equal to the Bit rate?

Topic: Digital Communications / Baud vs. Bit Rate

Concept & Explanation: Baud rate measures the number of signal symbol transitions per second (R_s), while bit rate measures transmitted binary information bits per second ($R_b = R_s \log_2 M$). When two-level binary signaling is used ($M = 2$, such as binary ASK, BPSK, or standard NRZ line codes), each symbol represents exactly one bit ($\log_2(2) = 1$), making Baud rate and Bit rate numerically equal.

Board Exam Tip: Baud rate = Bit rate only when $M = 2$ (binary signaling). When $M > 2$, Bit rate $>$ Baud rate.

Answer: When binary (two-level) signaling is used, transmitting exactly one bit per symbol.

Question 75: What is the primary effect of atmospheric ducting on VHF/UHF radio-wave propagation?

Topic: Wave Propagation / Tropospheric Ducting

Concept & Explanation: Atmospheric ducting occurs during temperature inversions when warm, dry air overlays cool, moist air, creating an abnormally steep negative refractive index gradient in the troposphere. This layer acts as a low-loss dielectric waveguide, trapping VHF, UHF, and microwave signals between atmospheric strata or between the layer and the Earth's surface, extending propagation hundreds of kilometers beyond the radio horizon.

Board Exam Tip: Tropospheric ducting channels VHF/UHF waves well beyond the normal radio horizon.

Answer: It acts as a dielectric waveguide, trapping radio waves and allowing them to propagate beyond the normal line-of-sight range.

Question 76: A 10-bit linear PCM system provides a certain dynamic range. If the system is upgraded to a 12-bit linear PCM system, how much does the dynamic range increase in dB?

Topic: Digital Communications / PCM Dynamic Range

Given: Initial bits $n_1 = 10$, upgraded bits $n_2 = 12$, change in bits $\Delta n = 2$. | **Find:** Dynamic range increase in dB.

Formula & Solution:

Dynamic range (and Signal-to-Quantization-Noise Ratio) in linear PCM increases by approximately 6.02 dB per additional quantization bit:

$$\Delta \text{DR} = 6.02 \times \Delta n = 6.02 \times (12 - 10) = 6.02 \times 2 = 12.04 \text{ dB} \approx 12 \text{ dB}$$

Board Exam Shortcut: Remember the "6 dB per bit rule" for PCM. Upgrading by 2 bits adds 12 dB.

Answer: 12 dB

Question 77: Determine the modulation index of a phase-modulated (PM) signal where the maximum phase deviation is 2.5 radians.

Topic: Angle Modulation / Phase Modulation

Given: Maximum phase deviation $\Delta\theta = 2.5$ radians. | **Find:** Phase modulation index m_p.

Formula & Solution:

By definition, the modulation index of a Phase Modulated (PM) wave is directly equal to its maximum peak phase deviation expressed in radians:

$$m_p = \Delta\theta = 2.5 \text{ radians} = 2.5$$

Board Exam Tip: In PM, modulation index $m_p = k_p V_m$ and is independent of modulating frequency f_m. In FM, $m_f = \Delta f / f_m$.

Answer: 2.5

Question 78: A signal with a frequency of 4 kHz is sampled at 6 kHz. Analyze the sampled signal to determine the principal aliased frequency.

Topic: Digital Signal Processing / Aliasing

Given: $f = 4$ kHz, sampling rate $f_s = 6$ kHz. | **Find:** Principal apparent aliased frequency f_a.

Formula & Solution:

1. Nyquist frequency is $f_N = f_s/2 = 6 \text{ kHz}/2 = 3 \text{ kHz}$. 2. Because $f = 4 \text{ kHz} > 3 \text{ kHz}$, aliasing occurs. The signal aliases across the folding frequency:

$$f_a = |f_s - f| = |6 \text{ kHz} - 4 \text{ kHz}| = 2 \text{ kHz}$$

Board Exam Tip: The apparent frequency under aliasing is $f_{alias} = |k f_s \pm f|$ falling within the fundamental Nyquist interval $[0, f_s/2]$.

Answer: 2 kHz

Question 79: What happens to the ground-wave propagation range as the transmitting frequency increases into the HF and VHF bands?

Topic: Antennas & Wave Propagation / Ground Waves

Concept & Explanation: Ground waves travel along the Earth's surface and induce alternating currents in the terrestrial soil. The Earth's finite conductivity and dielectric constant cause earth absorption to increase with the square of frequency. Beyond 2 MHz (HF and VHF), ground wave attenuation becomes severe, limiting practical ground-wave communication to VLF, LF, and MF

(AM broadcast) bands.
Board Exam Tip: Ground-wave attenuation increases sharply with frequency; it is unusable at VHF.
Answer: The ground-wave propagation range decreases significantly because attenuation over the Earth's surface increases at higher frequencies.

Question 80: Which layer of the atmosphere is primarily responsible for long-distance High Frequency (HF) skywave propagation?
Topic: Wave Propagation / Ionosphere
Concept & Explanation: The ionosphere, extending from approximately 50 km to 400 km above Earth, is ionized by solar ultraviolet and X-ray radiation into the D, E, F_1, and F_2 layers. The dense F_2 layer reflects HF radio waves (3–30 MHz) back to ground, enabling transcontinental skywave propagation.
Board Exam Tip: Skywave HF skip = Ionosphere (F_2 layer). Daytime MF attenuation = D layer. VHF ducting = Troposphere.
Answer: Ionosphere

Question 81: Calculate the total maximum user data throughput of an ISDN Basic Rate Interface (BRI) line if all B-channels are bonded together for a single data transfer.
Topic: Telecommunications / ISDN Architectures
Given: ISDN BRI configuration: $2B + D$. Bearer channels: 2×64 kbps, Data channel: 1×16 kbps. | **Find:** Bonded user payload throughput.
Formula & Solution:
ISDN Bearer (B) channels carry user payload, while the D channel is dedicated strictly to out-of-band signaling and control. When the two B-channels are bonded using Multilink PPP (MLPPP):

$$\text{Total User Throughput} = 2 \times 64 \text{ kbps} = 128 \text{ kbps}$$

Board Exam Tip: BRI total raw line rate is 192 kbps (including framing), user channel rate is 144 kbps ($2B + D$), and bonded user payload throughput is 128 kbps.
Answer: 128 kbps

Question 82: Which type of temperature sensor relies on the Seebeck effect to generate a small millivolt potential when two dissimilar metals are joined and subjected to a temperature gradient?
Topic: Industrial Instrumentation / Temperature Sensors
Concept & Explanation: A thermocouple consists of two dissimilar electrical conductors welded together to form a measurement junction and a reference junction. Under the Seebeck effect, a temperature differential between the junctions induces an open-circuit thermoelectric voltage proportional to the temperature gradient.
Board Exam Tip: Thermocouple = Seebeck effect (mV output). RTD = Platinum resistance increase with temperature. Thermistor = Non-linear semiconductor.
Answer: Thermocouple

Question 83: Why is Pulse Code Modulation (PCM) considered superior to Pulse Amplitude Modulation (PAM), Pulse Width Modulation (PWM), and Pulse Position Modulation (PPM) for long-distance transmission?
Topic: Digital Communications / Modulation Schemes
Concept & Explanation: PAM, PWM, and PPM are analog pulse modulation systems where noise and waveform distortions accumulate at every analog repeater along a transmission path. PCM converts analog samples into discrete binary codes. Digital regenerative repeaters detect and reconstruct clean binary pulses, stripping away channel noise at each repeater hop.

Board Exam Tip: PCM is digital and noise-immune because regenerative repeaters eliminate noise accumulation.

Answer: PCM is digital, enabling noise-free regeneration at repeaters.

Question 84: What is the standard reference impedance used for calculating dBm in telecommunication audio circuits?

Topic: Telephony / Decibel Standards

Concept & Explanation: In standard voice telecommunications and audio engineering, historical telephone line trunking and audio equipment are matched to a standard characteristic reference impedance of $600\,\Omega$. In RF systems, the standard is $50\,\Omega$ ($75\,\Omega$ for cable TV).

Board Exam Tip: Audio / Telecom standard impedance = $600\,\Omega$. RF standard = $50\,\Omega$. CATV = $75\,\Omega$. Across $600\,\Omega$, 0 dBm $= 1$ mW $\implies V = \sqrt{0.001 \times 600} \approx 0.775$ V.

Answer: 600 Ω

Question 85: What happens to the input impedance of a short-circuited transmission line that is exactly one-quarter wavelength ($\lambda/4$) long?

Topic: Transmission Lines / Impedance Inversion

Concept & Explanation: The input impedance equation for a lossless transmission line terminated in Z_L is:

$$Z_{in} = Z_0 \frac{Z_L + jZ_0 \tan(\beta l)}{Z_0 + jZ_L \tan(\beta l)}$$

For a quarter-wave line ($l = \lambda/4 \implies \beta l = \pi/2$), this reduces to the quarter-wave transformer relation:

$$Z_{in} = \frac{Z_0^2}{Z_L}$$

When terminated in a short circuit ($Z_L = 0$):

$$Z_{in} = \frac{Z_0^2}{0} \to \infty$$

The quarter-wave shorted stub acts as a parallel-resonant tank circuit, presenting an infinite input impedance (open circuit).

Board Exam Tip: $\lambda/4$ shorted stub $\to$ Open circuit. $\lambda/4$ open stub $\to$ Short circuit. $\lambda/2$ line $\to$ Repeats load impedance.

Answer: It appears as an open circuit (infinite impedance).

Question 86: What is the fundamental relationship between an antenna's directivity and its beamwidth?

Topic: Antenna Engineering / Directivity Relations

Concept & Explanation: Antenna directivity measures the concentration of radiated power in a specific direction relative to an isotropic source. If an antenna's beamwidth is made narrower, its radiated power is compressed into a smaller solid angle ($\Omega_A \approx \theta_{HP}\phi_{HP}$), increasing directivity:

$$D \approx \frac{41{,}253}{\theta_E^\circ \times \theta_H^\circ}$$

Thus, directivity and beamwidth are inversely related.

Board Exam Tip: Narrower beamwidth $\implies$ Higher directivity and higher gain.

Answer: They are inversely related, such that a narrower beamwidth generally results in higher antenna directivity.

Question 87: A transmitter outputs 20 W (43 dBm). The feedline has 3 dB of loss, and the antenna has a gain of 10 dBi. Calculate the Effective Isotropic Radiated Power (EIRP).

Topic: Antenna Systems / Link Budget Analysis

Given: Transmitter power $P_t = 43$ dBm (20 W), line loss $L_f = 3$ dB, antenna gain $G_t = 10$ dBi. | **Find:** EIRP in dBm and Watts.

Formula & Solution:

1. Calculate EIRP in logarithmic units:

$$\text{EIRP (dBm)} = P_t \text{ (dBm)} - L_f \text{ (dB)} + G_t \text{ (dBi)}$$

$$\text{EIRP} = 43 \text{ dBm} - 3 \text{ dB} + 10 \text{ dBi} = 50 \text{ dBm}$$

2. Convert 50 dBm to linear Watts:

$$P \text{ (W)} = 10^{(50-30)/10} = 10^{20/10} = 10^2 = 100 \text{ Watts}$$

Board Exam Shortcut: 30 dBm $= 1$ W, 40 dBm $= 10$ W, 50 dBm $= 100$ W.

Answer: 50 dBm (100 W)

Question 88: A transmission line has a characteristic impedance of $75\,\Omega$ and is terminated with a $50\,\Omega$ resistive load. Calculate the reflection coefficient.

Topic: Transmission Lines / Reflection Analysis

Given: Characteristic impedance $Z_0 = 75\,\Omega$, load impedance $Z_L = 50\,\Omega$. | **Find:** Voltage reflection coefficient Γ.

Formula & Solution:

Formula for voltage reflection coefficient:

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{50 - 75}{50 + 75} = \frac{-25}{125} = -\frac{1}{5} = -0.20$$

Board Exam Tip: The negative sign indicates a 180° phase inversion of the reflected voltage wave because $Z_L < Z_0$.

Answer: -0.20

Question 89: What is the main purpose of pseudo-random noise (PN) spreading codes in a Direct Sequence Spread Spectrum (DSSS) system?

Topic: Spread Spectrum / DSSS

Concept & Explanation: In DSSS, narrowband data bits are multiplied by a high-rate pseudo-random noise code (chipping sequence). This modulates and spreads signal energy across a much wider RF bandwidth. Spreading reduces power spectral density below the channel noise floor, providing high resistance to intentional jamming, multi-user code-division multiplexing, and low probability of intercept.

Board Exam Tip: DSSS spreads signal energy across bandwidth via PN code, providing jamming resistance and CDMA separation.

Answer: To spread the signal across a wide bandwidth, improving resistance to jamming and interference.

Question 90: Analyze why tight Closed Loop Power Control is an absolute necessity in a CDMA cellular network.

Topic: Cellular Systems / CDMA Fundamentals

Concept & Explanation: In CDMA, all mobiles transmit on the same carrier frequency simultaneously. If a mobile station near the base station transmits with excessive power, its signal overwhelms the weak signals of distant mobiles, blocking them from the receiver (the "Near-Far

Problem"). Closed-loop power control commands mobile transmitters to adjust output power up to 1500 times per second, ensuring all mobile transmissions arrive at the base station with approximately equal power.
Board Exam Tip: CDMA requires fast power control to eliminate the Near-Far problem and maximize system capacity.
Answer: To solve the Near-Far problem, ensuring all mobile signals arrive at the base station with equal power.

Question 91: In the Z-domain, what is the Region of Convergence (ROC) requirement for a causal and stable Linear Time-Invariant (LTI) system?
Topic: Digital Signal Processing / System Stability
Concept & Explanation: For a discrete-time LTI system to be BIBO stable, its unit impulse response must be absolutely summable, which requires the ROC of $H(z)$ to include the unit circle ($|z| = 1$). For the system to be causal, the ROC must be the exterior of a circle outside the outermost pole. Therefore, for a system that is both causal and stable, all poles must lie strictly inside the unit circle ($|p_k| < 1$).
Board Exam Tip: Causal + Stable $\iff$ All poles inside the unit circle, and ROC includes the unit circle.
Answer: The ROC must include the unit circle, and all poles must be inside the unit circle.

Question 92: In a Yagi-Uda antenna array, what is the typical physical length of the director element relative to the driven element?
Topic: Antenna Engineering / Parasitic Arrays
Concept & Explanation: A Yagi-Uda parasitic array uses a driven resonant half-wave element ($\lambda/2$). Directors placed in front of the driven element are cut approximately 5% shorter than the driven element, making their input impedance capacitive and steering the main radiation lobe forward. The reflector placed behind is cut approximately 5% longer, making it inductive.
Board Exam Tip: Reflector = 5% longer (inductive). Director = 5% shorter (capacitive). Beam points toward directors.
Answer: Slightly shorter than the driven element (approximately 5% shorter).

Question 93: Two amplifiers are cascaded. Amplifier A has a gain of 10 dB and a noise factor of 2. Amplifier B has a gain of 20 dB and a noise factor of 11. Calculate the overall cascade noise factor using Friis' formula.
Topic: RF Engineering / Cascaded Noise Figure
Given: Amp 1 (A): $G_1 = 10$ dB $\implies$ 10 (linear), $F_1 = 2$. Amp 2 (B): $G_2 = 20$ dB $\implies$ 100 (linear), $F_2 = 11$. | **Find:** Total cascaded noise factor F_{total}.
Formula & Solution:
Friis' formula for cascaded noise factor:

$$F_{total} = F_1 + \frac{F_2 - 1}{G_1}$$

Substitute the linear parameters:

$$F_{total} = 2 + \frac{11 - 1}{10} = 2 + \frac{10}{10} = 2 + 1 = 3$$

Board Exam Tip: Always convert decibel gains to linear ratios before substituting into Friis' formula. The first stage gain G_1 suppresses noise added by subsequent stages.
Answer: 3

Question 94: What is the primary advantage of using Virtual Local Area Networks (VLANs) in an enterprise network?

Topic: Data Communications / Switched Networks

Concept & Explanation: A VLAN is a logical grouping of switch ports that partitions a single physical Layer-2 switch into multiple distinct broadcast domains. This contains broadcast storms, improves network throughput, and enforces logical security boundaries between departments without requiring separate physical cabling.

Board Exam Tip: VLAN = Segments a physical switch into multiple logical broadcast domains (IEEE 802.1Q).

Answer: It logically segments a single physical switched network into multiple independent broadcast domains, improving security and reducing unnecessary broadcast traffic.

Question 95: If a transmission line is terminated in its characteristic impedance ($Z_L = Z_0$), what is the resulting power transfer and standing-wave condition?

Topic: Transmission Lines / Matched Conditions

Concept & Explanation: When a load matches the line ($Z_L = Z_0$), the reflection coefficient becomes zero:

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} = 0 \implies \text{VSWR} = \frac{1+0}{1-0} = 1.0$$

All incident electromagnetic energy is absorbed by the load. Reflected power is zero, and no standing wave ripple exists along the line.

Board Exam Tip: Matched line $\implies \Gamma = 0$, VSWR $= 1:1$, 100% power transfer, zero standing waves.

Answer: All incident power is absorbed by the load, no reflection occurs, and the VSWR is exactly 1:1.

Question 96: In PLC ladder logic programming, what physical electrical component does the "-||-" symbol represent?

Topic: Industrial Automation / Ladder Logic Diagrams

Concept & Explanation: In IEC 61131-3 PLC ladder logic, the examine-if-closed (XIC) contact designated by the symbol –||– represents a normally open (NO) contact. When the referenced bit address contains logic 1 (energized), this virtual contact closes, permitting rung logic power continuity.

Board Exam Tip: –||– = Normally Open (NO) / Examine if Closed (XIC). –|/|– = Normally Closed (NC) / Examine if Open (XIO). –( )– = Output Coil (OTE).

Answer: A normally open (NO) relay contact.

Question 97: In information theory, what does the term "entropy" primarily measure?

Topic: Information Theory / Entropy Definition

Concept & Explanation: In Claude Shannon's mathematical theory of communication, entropy $H(S) = -\sum P_i \log_2 P_i$ quantifies the average information content, uncertainty, or surprise conveyed per emitted symbol from a discrete message source.

Board Exam Tip: Entropy measures average uncertainty or information content per symbol. Max entropy occurs when all outcomes are equally likely.

Answer: The average information content or uncertainty of a discrete message source.

Question 98: In standard AM broadcasting, what is the primary advantage of Single Sideband Suppressed Carrier (SSB-SC) over Double Sideband Full Carrier (DSB-FC)?

Topic: Analog Communications / AM Bandwidth & Power

Concept & Explanation: A conventional DSB-FC signal wastes two-thirds or more of its power on the unmodulated carrier and transmits two identical, redundant sidebands, occupying a transmis-

sion bandwidth $B = 2f_m$. SSB-SC completely suppresses the carrier and filters out one sideband, cutting required transmission bandwidth in half ($B = f_m$) and reducing transmitted power by over 83%.
Board Exam Tip: SSB-SC cuts bandwidth in half ($B = f_m$) and saves ≥ 83.3% power compared to full AM.
Answer: It reduces transmitted bandwidth and saves power.

Question 99: According to Hartley's Law, information capacity is directly proportional to what two parameters?
Topic: Information Theory / Hartley's Formulation
Concept & Explanation: Hartley's Law states that the maximum amount of information H that can be transmitted over a communication channel without noise is given by $H = 2Bt \log_2(M)$. Thus, channel capacity is directly proportional to transmission channel bandwidth (B) and duration/time of transmission (t).
Board Exam Tip: Information capacity is proportional to Bandwidth (B) and Time (t).
Answer: Bandwidth and time.

Question 100: A superheterodyne AM receiver is tuned to 1200 kHz. Its IF is 455 kHz. If high-side injection is used, calculate the local oscillator frequency.
Topic: RF Receivers / Superheterodyne Mixing
Given: $f_{RF} = 1200$ kHz, $f_{IF} = 455$ kHz, High-side injection ($f_{LO} > f_{RF}$). | **Find:** Local oscillator frequency f_{LO}.
Formula & Solution:
For high-side (superheterodyne) local oscillator injection:

$$f_{LO} = f_{RF} + f_{IF}$$

Substitute the given values:

$$f_{LO} = 1200 \text{ kHz} + 455 \text{ kHz} = 1655 \text{ kHz}$$

Board Exam Tip: High-side: $f_{LO} = f_{RF} + f_{IF}$. Low-side: $f_{LO} = f_{RF} - f_{IF}$. Standard AM radio uses an IF of 455 kHz.
Answer: 1655 kHz

PART II — Important Terms & Concepts Reviewer

1. Telecommunications & Modulation

FM Bessel Carrier Nulls: Modulation index values where $J_0(\beta) = 0$. First null occurs at $\beta \approx 2.4$. Power transfers completely to sidebands.
Companding (μ-law / A-law): Non-linear compression applied before ADC quantization. Smaller steps for low amplitudes maintain high SQNR across wide dynamic ranges.
Delta Modulation Distortions:

- *Granular Noise:* Occurs on flat/DC inputs; quantizer step size Δ is too large, causing oscillation.
- *Slope Overload:* Occurs on steep inputs; signal rises faster than Δ/T_s.

SSB-SC Advantages: Reduces transmission bandwidth by half ($B = f_m$) and saves 83.3% power at 100% modulation compared to DSB-FC.
Compromise of Lowering IF: Lowering intermediate frequency sharpens adjacent channel selectivity but moves the image frequency closer, worsening image rejection.

2. Transmission Lines & Waveguides

Quarter-Wave Inverter ($\lambda/4$):

- Short-circuit load ($Z_L = 0$) $\implies$ $Z_{in} = \infty$ (Open circuit).
- Open-circuit load ($Z_L = \infty$) $\implies$ $Z_{in} = 0$ (Short circuit).

Stub Matching: A short or open transmission line section connected in parallel to add pure reactive susceptance that cancels the load susceptance.
Smith Chart Center Axis: Represents pure resistance ($x = 0$). The upper hemisphere represents inductive reactance ($+jx$); lower represents capacitive ($-jx$).
Waveguide Cutoff Frequency: Minimum frequency for propagation. For rectangular TE_{10} mode: $f_c = c/(2a)$.

3. Antennas & Radio Propagation

Marconi vs. Hertz Antenna: Marconi is a vertical $\lambda/4$ monopole mounted over ground producing an omnidirectional horizontal pattern; Hertz is an ungrounded $\lambda/2$ dipole.
Yagi-Uda Structure: Driven element ($\approx 0.5\lambda$), Reflector (5% longer, inductive), and Directors (5% shorter, capacitive).
Tropospheric Ducting: Temperature inversion traps VHF/UHF waves in an atmospheric duct, extending signals far beyond normal line-of-sight.
Ground Wave Attenuation: Ground waves induce soil currents. Losses increase with f^2, making ground waves ineffective beyond MF ($>$ 2 MHz).

4. Digital Systems & Networks

HDLC Bit Stuffing: A '0' is stuffed after any run of five consecutive '1's to prevent mimicry of the 01111110 flag.
VLAN (Virtual LAN): Segments a physical switch into distinct broadcast domains, confining broadcast traffic and improving security.
Baud vs. Bit Rate: $R_b = R_s \log_2(M)$. Baud rate equals bit rate only when binary signaling is used ($M = 2$).
Near-Far Problem in CDMA: Mobiles close to a tower overpower distant users. Solved by closed-loop power control (1500 updates/s).
CRC Working Principle: Divides binary data polynomials by a generator polynomial using modulo-2 arithmetic; appends remainder as FCS.

PART III — Comprehensive ECE Board Exam Formula Sheet

Topic	Formula	Variables / Parameter Definitions
AM Calculations	$P_t = P_c\left(1+\frac{m^2}{2}\right)$	P_t = total power, P_c = carrier power, m = mod index
	$I_t = I_c\sqrt{1+\frac{m^2}{2}}$	I_t = modulated current, I_c = unmodulated current
	$\%P_{SB} = \frac{m^2}{2+m^2}\times 100\%$	Percentage of total power in both sidebands
FM Systems	$B_{\text{Carson}} = 2(\Delta f + f_m)$	Δf = peak deviation, f_m = modulating frequency
	$\beta = m_f = \frac{\Delta f}{f_m}$	Modulation index; carrier null at $\beta \approx 2.4$
PCM & Quantization	$\Delta = \frac{V_{\text{max}} - V_{\text{min}}}{2^n}$	Step size Δ; Max error $Q_{\text{max}} = \pm\Delta/2$
	$R_b = n \cdot f_s$	R_b = bit rate, n = bits/sample, f_s = sample rate
	$\Delta\text{SQNR} \approx 6.02 \times \Delta n$ dB	Dynamic range increases by ≈ 6 dB per bit
Transmission Lines	$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$	Γ = reflection coefficient (voltage)
	$\text{VSWR} = \frac{1+\lvert\Gamma\rvert}{1-\lvert\Gamma\rvert}$	Voltage Standing Wave Ratio ($1 \le \text{VSWR} < \infty$)
	$Z_{in} = \frac{Z_0^2}{Z_L}$	Quarter-wave ($\lambda/4$) impedance inverter
Antennas & Links	$S = \frac{\text{EIRP}}{4\pi R^2} = \frac{P_t G_t}{4\pi R^2}$	Radiated power density (W/m^2) at distance R
	$\text{FSPL} = 32.44 + 20\log(f_{\text{MHz}}) + 20\log(d_{\text{km}})$	Free Space Path Loss in dB
	$\theta_{3\text{dB}} \approx \frac{70\lambda}{D}$ degrees	Parabolic dish half-power beamwidth
Noise Analysis	$V_n = \sqrt{4kTBR}$	RMS noise voltage; $k = 1.38\times10^{-23}$ J/K
	$T_e = T_0(F-1)$	T_e = noise temperature, $F = 10^{NF/10}$, $T_0 = 290$ K
	$F_{\text{total}} = F_1 + \frac{F_2 - 1}{G_1}$	Friis cascaded noise factor formula (linear units)
DSP & Sampling	$f_s \ge 2f_{\text{max}}$	Minimum Nyquist sampling rate to prevent aliasing
	$f_{\text{alias}} = \lvert f_s - f_{\text{input}}\rvert$	Primary apparent frequency under aliasing
	$y[n] = \sum_k x[k]h[n-k]$	Linear discrete-time convolution sum
Instrumentation	$\frac{\Delta R}{R} = GF \times \epsilon$	Strain gauge equation; GF = gauge factor, ϵ = strain
	$I = I_{\text{min}} + \frac{4}{16}\times(T_{\text{max}} - T_{\text{min}})$	4–20 mA linear process loop scaling

PART IV — Master Quick Answer Key

#	Answer	Topic	#	Answer	Topic	#	Answer	Topic	#	Answer	Topi
1	Carrier null, $P_c \to 0$	FM Bessel	26	Carry CRC for errors	FCS Frame	51	75 kW	AM Power	76	12 dB	PCM nam
2	Time E = Freq E	Parseval	27	Pure R ($x = 0$)	Smith Chart	52	4-wire full duplex	Telecom	77	2.5	Phas
3	Flat/slow DC input	Delta Mod	28	Transport Layer	OSI Model	53	Gain & selectivity	Superhet IF	78	2 kHz	Alias
4	UDP	Transport	29	3.18×10^{-7} W/m^2	Power Density	54	Next input scan	PLC Scan	79	Range drops at HF	Grou Wave
5	290 K	Noise Temp	30	Huge antenna size	Modulation	55	AGC circuit	Receiver AGC	80	Ionosphere (F_2)	Skyw
6	6 dB	Noise Figure	31	11.11%	AM Sideband	56	6 GHz	Waveguide	81	128 kbps	ISDN
7	1.5	VSWR	32	0 dBi	Isotropic	57	Protocol gateway	Networking	82	Thermocouple	Sens
8	Modem	Hardware	33	Bus topology	LAN Types	58	Same BER vs E_b/N_0	Modulation	83	Digital regeneration	PCM tems
9	$100°$C	4–20 mA	34	$0.89\,\mu$V	Johnson Noise	59	ATM (53-byte)	ATM WAN	84	$600\,\Omega$	Audi
10	1.67 MHz	64-QAM	35	Zero at $z = 1$	Z-Plane Filter	60	0.811 bits/sym	Entropy	85	Open circuit (∞)	$\lambda/4$ S
11	Modulo-2 division	CRC Check	36	0.67	Velocity Factor	61	Public/Private keys	Asymmetric	86	Inversely related	Direc
12	LAN MAC vs WAN	Data Link	37	WDM	Fiber Optics	62	$0.12\,\Omega$	Strain Gauge	87	50 dBm (100 W)	EIRP
13	Point-to-Point	Data Link	38	HPBW (-3 dB)	Antennas	63	Loop break alarm	Security NC	88	-0.20	Refl.
14	9.77 mV	PCM Error	39	Graphic HMI	SCADA	64	4	FIR Filter	89	Spread spectrum	DSSS
15	33 dBm	Decibel Power	40	64 kbps	PCM Bit Rate	65	FIR (stable)	Filter Types	90	Solves Near-Far	CDM
16	Small steps for low	μ-law	41	Lowest cutoff f_c	Waveguides	66	Unstable, diverges	Z-Stability	91	All poles inside	ROC
17	Logical routing	IP Protocol	42	112.4 dB	FSPL Path	67	1111101	HDLC Stuffing	92	5% shorter	Yagi
18	23 B + 1 D	ISDN PRI	43	Highest returned	Ionosphere	68	Half-Duplex	Trans. Modes	93	3	Friis
19	193 bits	T1 Carrier	44	Selectivity vs Img	Superhet IF	69	0.81	AM Current	94	Logical domains	VLAN
20	White Noise	Noise Theory	45	300 Hz	Nyquist Rate	70	Jam & backoff	CSMA/CD	95	Matched, VSWR 1:1	Line
21	20.2 kbps	Shannon Cap.	46	Finite E, zero P	Energy Signal	71	[1, 1, -2]	Convolution	96	NO contact –||–	Ladd
22	Cancel reactance	Stub Match	47	Central failure	Star Network	72	4800 bps	QPSK Rate	97	Uncertainty/info	Entro
23	Omnidirectional	Marconi	48	PPM	Pulse Mod	73	Start/Stop bits	Asynchronous	98	Halves BW, saves P	SSB-S
24	QAM	Digital Mod	49	0.95 m	Dipole	74	Binary ($M = 2$)	Baud vs Bit	99	Bandwidth & time	Hart
25	120 kHz	Carson FM	50	$1.75°$	Dish Beam	75	Waves trapped	Ducting	100	1655 kHz	Supe

PART V — Final Board Exam Cram Sheet

Must-Remember Formulas

- **Carson's Rule:** $B = 2(\Delta f + f_m) = 2f_m(\beta + 1)$.
- **AM Total Power:** $P_t = P_c(1 + m^2/2)$. For $m = 1.0 \implies P_t = 1.5P_c$.
- **AM Antenna Current:** $I_t = I_c\sqrt{1 + m^2/2}$.
- **PCM Bit Rate:** $R_b = n \cdot f_s = f_s \log_2(L)$.
- **PCM Dynamic Range:** $+1$ bit $\implies +6.02$ dB.
- **Reflection Coefficient:** $\Gamma = \dfrac{Z_L - Z_0}{Z_L + Z_0}$.
- **VSWR:** $\text{VSWR} = \dfrac{1 + |\Gamma|}{1 - |\Gamma|}$.
- **Quarter-Wave Inverter:** $Z_{in} = \dfrac{Z_0^2}{Z_L}$.
- **Parabolic Dish Beamwidth:** $\theta_{3\text{dB}} \approx \dfrac{70\lambda}{D}$ degrees.
- **Friis Cascade Noise:** $F_{\text{total}} = F_1 + \dfrac{F_2 - 1}{G_1}$ *(use linear values)*.
- **Johnson Thermal Noise:** $V_n = \sqrt{4kTBR}$, $P_n = kTB$.
- **Shannon Capacity:** $C = B \log_2(1 + \text{SNR})$.
- **Waveguide Dominant Mode:** $f_c = \dfrac{c}{2a}$.

High-Yield Board Rules

- **FM Carrier Nulls:** First null at $\beta = 2.4048$. Carrier disappears completely.
- **Parseval's Theorem:** Total energy in time domain equals total energy in frequency domain.
- **Delta Modulation Traps:**
  - Granular noise = Flat or slowly changing signal (step size too large).
  - Slope overload = Steep, rapidly changing signal (step size too small).
- **High-Side LO Mixing:** $f_{LO} = f_{RF} + f_{IF}$; $f_{img} = f_{RF} + 2f_{IF}$.
- **Low-Side LO Mixing:** $f_{LO} = f_{RF} - f_{IF}$; $f_{img} = f_{RF} - 2f_{IF}$.
- **Intermediate Frequency Choice:**
  - Higher IF $\rightarrow$ Superior image rejection, poorer adjacent selectivity.
  - Lower IF $\rightarrow$ Superior adjacent selectivity, poorer image rejection.
- **HDLC Bit Stuffing:** Insert a '0' after **5 consecutive 1s**.
- **T1 vs. E1 Multiplexing:**
  - T1 = 24 channels, 193 bits/frame, 1.544 Mbps ($23B + D$ in PRI).
  - E1 = 32 channels, 256 bits/frame, 2.048 Mbps ($30B + D$ in PRI).
- **ISDN BRI:** $2B$ (64 kbps each) $+1D$ (16 kbps) $= 128$ kbps bonded payload.
- **Digital Filter Zeroes:** Zero at $z = 1$ removes DC (high-pass). Zero at $z = -1$ removes high frequencies (low-pass).
- **Causal LTI Stability:** All poles must lie strictly inside the unit circle ($|z| < 1$).
- **Yagi Array Dimensions:** Director is $\approx 5\%$ shorter (capacitive); Reflector is $\approx 5\%$ longer (inductive).
- **Strain Gauge Rule:** $\Delta R/R = GF \times \epsilon$.
- **4–20 mA Loop Standard:** 4 mA $= 0\%$, 12 mA $= 50\%$, 20 mA $= 100\%$.

END OF COMPREHENSIVE ECE BOARD EXAM REVIEWER — PB EXAMS (25)

ELECTRONICS ENGINEERING (ECE) BOARD EXAM

Complete Solutions, Comprehensive Theory & High-Yield Reviewer

Source: PB Exams (24) — 100 Board-Style Examination Questions

Communications Systems • Transmission Lines & Antennas • Digital Signal Processing • Data Communications & Networking • Industrial Electronics

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PART I — Complete Step-by-Step Solutions (Questions 1–100)

Question 1: Evaluate the payload efficiency of a B-ISDN network utilizing ATM technology. If an ATM cell is 53 bytes long and contains a 5-byte header, what is the protocol overhead percentage for a single cell?

Concept / Topic: Data Communications / ATM Networks

Given: Cell length = 53 bytes, Header = 5 bytes, Payload = $53 - 5 = 48$ bytes. | **Find:** Protocol Overhead Percentage (%).

Formula & Solution:

Overhead Percentage is defined as:

$$\text{Overhead (\%)} = \left(\frac{\text{Header Bytes}}{\text{Total Cell Bytes}}\right) \times 100\% = \left(\frac{5}{53}\right) \times 100\% \approx 9.434\%$$

Board Exam Tip: ATM uses a fixed 53-byte cell (5-byte header, 48-byte payload). Fixed packet length minimizes jitter and switching latency in B-ISDN.

Answer: 9.43%

Question 2: Two continuous-time linear time-invariant (LTI) systems with individual impulse responses $h_1(t)$ and $h_2(t)$ are connected in a parallel configuration. What is the overall impulse response of the combined system?

Topic: Signals and Systems / LTI Systems

Concept & Explanation: In a parallel connection, identical inputs $x(t)$ feed both subsystems and the total output is the sum of their individual outputs: $y(t) = y_1(t) + y_2(t) = x(t) * h_1(t) + x(t) * h_2(t) = x(t) * [h_1(t) + h_2(t)]$ due to the distributive property of convolution. For systems in series (cascade), the overall impulse response is $h_1(t) * h_2(t)$.

Answer: $h_1(t) + h_2(t)$

Question 3: Analyze the impact on a PCM transmission system if the number of quantization bits per sample is increased from 8 to 10 bits.

Concept / Topic: Digital Communications / Pulse Code Modulation (PCM)

Given: Initial bits $n_1 = 8$ bits, Final bits $n_2 = 10$ bits. | **Find:** Change in Quantization SNR (ΔSQNR) and Transmission Bandwidth (ΔB).

Formula & Solution:

1. SQNR $\approx 6.02n + 1.76$ dB. Increasing by $\Delta n = 10 - 8 = 2$ bits improves SQNR (reduces noise) by:

$$\Delta\text{SQNR} = 6.02 \times \Delta n = 6.02 \times 2 \approx 12.04 \text{ dB (or} \approx 12 \text{ dB)}$$

2. Bit rate $R_b = nf_s$ and baseband transmission bandwidth $B \propto R_b$:

$$\frac{\Delta B}{B_1} = \frac{n_2 - n_1}{n_1} = \frac{10 - 8}{8} = \frac{2}{8} = 25\%$$

Board Exam Tip: Every additional bit in PCM adds 6 dB of dynamic range / SNR improvement.

Answer: The quantization noise decreases by approximately 12 dB, and the required bandwidth increases by 25%

Question 4: Which of the following statements correctly distinguishes ground wave propagation from sky wave propagation?

Topic: Wave Propagation / Antennas

Concept & Explanation: Ground (surface) waves propagate directly over the conductive surface of the Earth following its curvature. High dielectric attenuation occurs in the ground at higher radio frequencies; hence, ground wave propagation is practical and effective only up to Medium Frequencies (MF, < 2 MHz). Sky waves, conversely, rely on ionospheric refraction and dominate HF (3–30 MHz).

Answer: Ground waves are highly attenuated at higher frequencies, making them effective primarily for VLF and MF bands

Question 5: How does the skin effect influence the behavior of a transmission line at high radio frequencies?

Topic: Transmission Lines / Electromagnetics

Concept & Explanation: Due to the eddy currents induced by changing magnetic fields inside the conductor, AC current density is forced toward the outer surface of the conductor. Skin depth is $\delta = 1/\sqrt{\pi f \mu \sigma}$. As frequency increases, δ shrinks, drastically reducing effective cross-sectional area, which increases effective AC resistance.

Answer: It causes current to flow primarily along the surface of the conductor, increasing the effective AC resistance

Question 6: A transmitter outputs 50 W to an antenna that has an overall efficiency of 80% and a directivity of 1.25. What is the actual radiated power density at a distance of 2 kilometers, assuming no propagation losses?

Concept / Topic: Antennas & Radiowave Propagation

Given: $P_{in} = 50$ W, efficiency $\eta = 0.80$, directivity $D = 1.25$, distance $R = 2$ km $= 2000$ m. |

Find: Power density S (μW/m^2).

Formula & Solution:

Antenna Gain $G = \eta \cdot D = 0.80 \times 1.25 = 1.00$. Radiated power density at distance R:

$$S = \frac{P_{in} \cdot G}{4\pi R^2} = \frac{50 \times 1.00}{4\pi (2000)^2} = \frac{50}{4\pi \times 4 \times 10^6} \approx 9.947 \times 10^{-7} \text{ W/m}^2 \approx 0.99\,\mu\text{W/m}^2$$

Answer: $0.99\,\mu$**W/m**^{2}

Question 7: Upon analyzing the constellation diagram of a 16-QAM signal, a technician notes a decrease in the Euclidean distance between adjacent symbol points compared to an existing 4-QAM system operating at the same average power. What is the direct consequence of this observation?

Topic: Digital Modulation / Information Theory

Concept & Explanation: In any M-ary constellation, packing more constellation points into the same average energy envelope reduces the minimum Euclidean distance (d_{min}). A smaller d_{min} makes symbol detection significantly more vulnerable to additive Gaussian noise, directly increasing the Symbol Error Rate (SER) and Bit Error Rate (BER).

Answer: The 16-QAM system will exhibit a higher probability of symbol error in the presence of constant channel noise

Question 8: A baseband PCM system transmits 24 voice channels, each bandlimited to 3.4 kHz and sampled at 8 kHz, with 8 bits per sample. If one framing bit is added per frame, what is the minimum Nyquist bandwidth required for baseband transmission?

Concept / Topic: Digital Communications / T-Carrier Systems
Given: Channels $N = 24$, $f_s = 8$ kHz, bits/sample $n = 8$, framing bit $F = 1$ bit/frame. | **Find:** Minimum Nyquist bandwidth (B_{min}).
Formula & Solution:
1. Bits per frame:

$$\text{Bits/frame} = (24 \times 8) + 1 = 192 + 1 = 193 \text{ bits}$$

2. Aggregate Bit Rate (Standard T1/DS1 rate):

$$R_b = 193 \text{ bits/frame} \times 8000 \text{ frames/sec} = 1{,}544{,}000 \text{ bps} = 1.544 \text{ Mbps}$$

3. Minimum Nyquist baseband bandwidth for zero ISI:

$$B_{min} = \frac{R_b}{2} = \frac{1{,}544{,}000}{2} = 772{,}000 \text{ Hz} = 772 \text{ kHz}$$

Answer: 772 kHz

Question 9: Which network topology is characterized by all devices being connected to a single central hub or switch?

Topic: Data Communications / Local Area Networks
Concept & Explanation: In a star topology, each network node is connected to a central active or passive hub, switch, or concentrator via point-to-point physical links. Failure of a single branch cable affects only that node, but the central hub represents a single point of failure.
Answer: Star topology

Question 10: A sender wishes to transmit a highly confidential document over a public network using asymmetric (public key) cryptography. To ensure that only the intended recipient can read the document, which key should the sender use to encrypt the data?

Topic: Network Security / Cryptography
Concept & Explanation: In public-key (asymmetric) cryptography, public keys are freely shared while private keys are strictly kept secret. To ensure confidentiality (only the recipient can decrypt), data is encrypted using the recipient's public key. Only the recipient's corresponding private key can successfully decrypt it.
Answer: The recipient's public key

Question 11: An FM signal is modulated by a 5 kHz sine wave, resulting in a maximum frequency deviation of 25 kHz. What is the modulation index?

Concept / Topic: Angle Modulation / FM
Given: Modulating frequency $f_m = 5$ kHz, frequency deviation $\Delta f = 25$ kHz. | **Find:** Modulation index m_f or β.
Formula & Solution:
The modulation index for frequency modulation is:

$$\beta = m_f = \frac{\Delta f}{f_m} = \frac{25 \text{ kHz}}{5 \text{ kHz}} = 5.0$$

Board Exam Tip: FM modulation index is proportional to modulating amplitude and inversely proportional to modulating frequency. In PM, m_p is independent of f_m.

Answer: 5.0

Question 12: An engineer observes erratic, fluctuating readings on an analog 4–20 mA input module connected to a distant level sensor. The cable is properly shielded. An oscilloscope reveals a 60 Hz AC potential difference between the sensor chassis ground and the PLC panel ground. What is the standard interfacing solution to resolve this specific issue?

Topic: Industrial Instrumentation & Control Systems
Concept & Explanation: A potential difference between grounded points creates a circulating 60 Hz common-mode current known as a ground loop. Standard remediation requires inserting an optical or magnetic galvanic signal isolator in the 4–20 mA loop, or bonding all earth grounds into a single equipotential star ground.
Answer: Install a galvanic signal isolator in the loop or route all device grounds to a single, equipotential star ground point to break the ground loop

Question 13: Which of the following is the PRIMARY disadvantage of utilizing a pure bus topology in a local area network?

Topic: Data Communications / Network Topologies
Concept & Explanation: In a bus topology, all hosts share a single common trunk cable terminated with load resistors at both ends. Any severed cable, high-impedance fault, or missing terminator alters line impedance, creates signal reflections, and disables communication for the entire segment.
Answer: A fault or break in the main backbone cable brings down the entire network

Question 14: An engineer needs to cut a half-wave dipole for an operating frequency of 150 MHz using a wire with a velocity factor of 0.95. What should be the total physical length of the antenna?

Concept / Topic: Antennas & Transmission Lines
Given: $f = 150$ MHz $= 150 \times 10^6$ Hz, $c = 3 \times 10^8$ m/s, $k = 0.95$. | **Find:** Physical antenna length L for a half-wave dipole ($\lambda/2$).
Formula & Solution:
1. Free-space wavelength:

$$\lambda_0 = \frac{c}{f} = \frac{3 \times 10^8}{150 \times 10^6} = 2.0 \text{ meters}$$

2. Actual half-wave physical length accounting for velocity factor:

$$L = \frac{\lambda}{2} = \frac{k\lambda_0}{2} = \frac{0.95 \times 2.0}{2} = 0.95 \text{ meters}$$

Answer: 0.95 meters

Question 15: Which of the following defines the characteristic impedance of a uniform, lossless transmission line?

Topic: Transmission Lines / Electromagnetics
Concept & Explanation: Characteristic impedance $Z_0 = \sqrt{L/C}$ is defined as the input impedance of an infinitely long transmission line, or equivalently, the load impedance that absorbs all propagating energy without reflections.
Answer: The purely resistive impedance presented to a signal wave propagating on a line of infinite length

Question 16: Which phenomenon is primarily mitigated by using an Equalizer in a broadband digital receiver?

Topic: Digital Communications / Signal Processing

Concept & Explanation: Channel dispersion and bandlimiting cause pulse spreading into adjacent time slots, known as Inter-Symbol Interference (ISI). An adaptive or transversal equalizer applies an inverse channel transfer function to eliminate ISI and restore Nyquist criterion compliance.

Answer: Inter-symbol interference (ISI)

Question 17: A 50-ohm signal generator supplies 100 watts of forward power to a transmission line connected to a 100-ohm resistive load. Assuming a lossless line, what is the actual power dissipated by the load?

Concept / Topic: Transmission Lines / RF Engineering

Given: $Z_0 = 50\,\Omega$, $Z_L = 100\,\Omega$, $P_{inc} = 100$ W. | **Find:** Load power P_L.

Formula & Solution:

1. Voltage reflection coefficient:

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{100 - 50}{100 + 50} = \frac{50}{150} = \frac{1}{3}$$

2. Reflected power fraction:

$$P_{ref} = |\Gamma|^2 P_{inc} = \left(\frac{1}{3}\right)^2 \times 100\ \text{W} = \frac{1}{9} \times 100 = 11.11\ \text{W}$$

3. Net power absorbed by the load:

$$P_L = P_{inc} - P_{ref} = 100 - 11.11 = 88.89\ \text{W} \approx 88.9\ \text{W}$$

Answer: 88.9 watts

Question 18: When designing a superheterodyne receiver, how does increasing the intermediate frequency (IF) mathematically affect the system's performance metrics?

Topic: Radio Receivers / Superheterodyne

Concept & Explanation: Image frequency is $f_{img} = f_{RF} \pm 2f_{IF}$. Increasing f_{IF} pushes the image frequency further away from the desired RF signal, allowing the RF preselector filter to achieve superior image rejection. However, higher IF stages have wider absolute passbands for a given filter Q ($BW = f_0/Q$), degrading selectivity between closely spaced adjacent channels.

Answer: It improves image rejection but degrades adjacent channel selectivity

Question 19: A causal IIR filter is described by the difference equation $y[n] = 0.5y[n-1] + x[n]$. If the input is a unit step sequence $u[n]$, what is the output value at $n = 1$ assuming zero initial conditions?

Concept / Topic: DSP / Discrete-Time LTI Systems

Given: Difference equation $y[n] = 0.5y[n-1] + x[n]$, $x[n] = u[n]$, $y[-1] = 0$. | **Find:** Value of $y[1]$.

Formula & Solution:

1. At $n = 0$:

$$y[0] = 0.5y[-1] + u[0] = 0.5(0) + 1 = 1.0$$

2. At $n = 1$:

$$y[1] = 0.5y[0] + u[1] = 0.5(1.0) + 1 = 0.5 + 1 = 1.5$$

Answer: 1.5

Question 20: A network administrator assigns the IP address 192.168.1.50 with a subnet mask of 255.255.255.192. Which IP address represents the broadcast address for this specific subnet?

Concept / Topic: Networking / IPv4 Subnetting
Given: Host IP: 192.168.1.50, Mask: 255.255.255.192 (/26). | **Find:** Subnet Broadcast Address.
Formula & Solution:
1. Block size = $256 - 192 = 64$. 2. Subnet boundaries in the 4th octet: $0, 64, 128, 192$. 3. Since $0 \leq 50 < 64$, the host belongs to subnet $192.168.1.0/26$. 4. The network address is $192.168.1.0$. 5. The broadcast address is the last IP before the next block: $192.168.1.63$.

Answer: 192.168.1.63

Question 21: A building security IP-CCTV system consists of 10 high-definition cameras, each configured to stream and record at a constant bit rate of 2 Megabits per second (Mbps). Approximately how much raw hard drive storage capacity is required to retain 30 days of continuous 24/7 recording for the entire system?

Concept / Topic: Data Storage / Surveillance Engineering
Given: Cameras $N = 10$, Bitrate per camera $R = 2$ Mbps, Total rate $R_T = 20$ Mbps, Duration $t = 30$ days. | **Find:** Total storage in Terabytes (TB).
Formula & Solution:
1. Convert time to seconds:

$$t = 30 \text{ days} \times 24 \text{ hours/day} \times 3600 \text{ s/hour} = 2{,}592{,}000 \text{ seconds}$$

2. Total bits generated:

$$\text{Total Bits} = 20 \times 10^6 \text{ bps} \times 2{,}592{,}000 \text{ s} = 5.184 \times 10^{13} \text{ bits}$$

3. Convert to Bytes (8 bits $= 1$ Byte):

$$\text{Total Bytes} = \frac{5.184 \times 10^{13}}{8} = 6.48 \times 10^{12} \text{ Bytes} = 6.48 \text{ Terabytes (TB)}$$

Answer: 6.48 Terabytes

Question 22: If a 64-QAM modulator is driven by a data stream having a bit rate of 36 Mbps, determine the symbol rate generated by the modulator.

Concept / Topic: Digital Communications / Modulation
Given: $M = 64$, Bit rate $R_b = 36$ Mbps. | **Find:** Symbol rate (Baud rate) R_s.
Formula & Solution:
1. Bits per symbol:

$$k = \log_2(M) = \log_2(64) = 6 \text{ bits/symbol}$$

2. Baud rate equation:

$$R_s = \frac{R_b}{k} = \frac{36 \text{ Mbps}}{6 \text{ bits/symbol}} = 6 \text{ MBaud (or } 6 \times 10^6 \text{ symbols/sec)}$$

Answer: 6 MBaud

Question 23: A transmitting antenna is 144 meters high, and a receiving antenna is 64 meters high. Assuming a 4/3 Earth radius factor, what is the maximum line-of-sight communication distance?

Concept / Topic: Microwave Communications / Radio Horizon
Given: $h_t = 144$ m, $h_r = 64$ m, $k = 4/3$. | **Find:** Radio line-of-sight distance d_{LOS} in kilometers.
Formula & Solution:
Using the standard metric line-of-sight formula for a $4/3$ effective Earth radius:

$$d_{LOS} \approx \sqrt{17h_t} + \sqrt{17h_r} = 4.124\left(\sqrt{h_t} + \sqrt{h_r}\right)$$

$$\sqrt{h_t} = \sqrt{144} = 12, \quad \sqrt{h_r} = \sqrt{64} = 8$$

$$d \approx 4.124(12 + 8) = 4.124 \times 20 = 82.48 \text{ km} \approx 82 \text{ km}$$

(Alternatively, using $d \approx \sqrt{17 \times 144} + \sqrt{17 \times 64} = \sqrt{2448} + \sqrt{1088} = 49.48 + 32.98 = 82.46$ km ≈ 82 km).
Answer: 82 km

Question 24: In the design of a rectangular microstrip patch antenna, moving the coaxial feed point directly from the radiating edge toward the exact center of the patch will cause the input impedance to do what?

Topic: Antenna Engineering / Microstrip Patches
Concept & Explanation: On a microstrip patch antenna operating at fundamental mode TM_{10}, maximum voltage and minimum current occur at the outer radiating edges (yielding high input impedance ≈ 150–$300\,\Omega$). At the center of the patch, voltage is zero (RF null) and current is maximum; hence, moving inward causes input impedance to decrease monotonically toward zero.
Answer: Decrease monotonically toward zero

Question 25: In a practical digital transmission system, an engineer decides to sample a 4 kHz analog signal at 10 kHz instead of the 8 kHz Nyquist rate. What is the most significant architectural advantage of this decision?

Topic: Digital Signal Processing / Sampling
Concept & Explanation: Oversampling introduces a wide transition guard band (from 4 kHz to $10 - 4 = 6$ kHz) between the baseband signal spectrum and the first spectral replica. This relaxes the order, complexity, and steepness required of the analog anti-aliasing reconstruction filter, allowing gradual roll-off.
Answer: It allows the use of a practical anti-aliasing filter with a gradual roll-off

Question 26: What is the primary purpose of implementing Time Division Multiplexing (TDM) in a digital communication link?

Topic: Telecommunications / Multiplexing
Concept & Explanation: TDM interleaves multiple discrete low-speed digital bit streams into assigned, non-overlapping sequential time slots over a single high-speed physical transmission link.
Answer: To allow multiple signals to be transmitted sequentially over a single shared channel

Question 27: An antenna has a maximum directivity of 20 and a radiation efficiency of 90%. What is the gain of the antenna in dBi?

Concept / Topic: Antennas / RF Engineering
Given: Directivity $D = 20$, efficiency $\eta = 0.90$. | **Find:** Power gain in decibels isotropic (G_{dBi}).
Formula & Solution:

1. Linear power gain:

$$G = \eta \cdot D = 0.90 \times 20 = 18.0$$

2. Gain in dBi:

$$G_{\text{dBi}} = 10 \log_{10}(G) = 10 \log_{10}(18) \approx 10 \times 1.25527 = 12.55 \text{ dBi}$$

Answer: 12.55 dBi

Question 28: An HDLC transmitter needs to send the data sequence 01111110. To prevent the receiver from misinterpreting this data as a frame boundary flag, what exact sequence of bits will the transmitter send after applying bit stuffing?

Concept / Topic: Data Link Layer / HDLC Framing

Given: Input sequence: `01111110`. The flag sequence is `01111110` (six consecutive 1s). | **Find:** Transmitted bit-stuffed sequence.

Formula & Solution:

HDLC bit stuffing rule: The transmitter automatically inserts (stuffs) a '0' after every five consecutive '1's within user data.

Original: `0 1 1 1 1 1` **1** `0` $\longrightarrow$ Stuffed: `0 1 1 1 1 1` **0** `1 0`

Result: `011111010`.

Answer: 011111010

Question 29: A horizontally polarized radio wave reflects off a perfectly conducting ground plane at grazing incidence. What is the theoretical reflection coefficient?

Topic: Electromagnetic Waves / Boundary Conditions

Concept & Explanation: For horizontal polarization (perpendicular electric field $E_\perp$ to the plane of incidence), boundary conditions require the tangential electric field to vanish at a perfect conductor surface: $E_{inc} + E_{ref} = 0 \implies \Gamma_H = -1$ (180° phase reversal) for all incidence angles, including grazing.

Answer: –1

Question 30: An amplifier has an input resistance of 50 ohms and operates over a bandwidth of 5 MHz at 300 K. Calculate the RMS thermal noise voltage at the input.

Concept / Topic: Noise Analysis / Johnson-Nyquist Noise

Given: $R = 50\,\Omega$, $B = 5$ MHz $= 5 \times 10^6$ Hz, $T = 300$ K, $k = 1.38 \times 10^{-23}$ J/K. | **Find:** RMS thermal noise voltage V_n.

Formula & Solution:

Using Johnson's noise equation:

$$V_n = \sqrt{4kTBR}$$

$$4kTBR = 4 \times (1.38 \times 10^{-23}) \times 300 \times (5 \times 10^6) \times 50 = 4.14 \times 10^{-12} \text{ V}^2$$

$$V_n = \sqrt{4.14 \times 10^{-12}} \approx 2.0347 \times 10^{-6} \text{ V} \approx 2.03\,\mu\text{V}$$

Answer: $2.03\,\mu$**V**

Question 31: A security auditor evaluating a network's integrity mechanisms finds that two different downloaded files generate the exact same MD5 cryptographic hash value. What specific vulnerability does this analysis indicate?

Topic: Network Security / Cryptography

Concept & Explanation: A hash function must provide strong collision resistance. When two distinct arbitrary inputs produce identical digest outputs ($H(m_1) = H(m_2)$ where $m_1 \neq m_2$), this is defined as a hash collision, indicating algorithmic weakness.

Answer: A hash collision

Question 32: A specific forward error correction (FEC) block code has a minimum Hamming distance of 5. Evaluate the error-handling capability of this code for a single transmitted codeword.

Concept / Topic: Information Theory / Error Control Coding

Given: Minimum distance $d_{min} = 5$. | **Find:** Error detection capability t_{det} and error correction capability t_{corr}.

Formula & Solution:

1. Number of detectable errors:

$$t_{det} \leq d_{min} - 1 = 5 - 1 = 4 \text{ errors}$$

2. Number of correctable errors:

$$t_{corr} \leq \left\lfloor \frac{d_{min} - 1}{2} \right\rfloor = \left\lfloor \frac{5 - 1}{2} \right\rfloor = 2 \text{ errors}$$

Answer: It can detect up to 4 errors and correct up to 2 errors

Question 33: The European E1 carrier system multiplexes individual 64 kbps voice channels. What is the total transmission bit rate of the E1 line?

Concept / Topic: Telecommunications / Multiplexing

Given: Standard E1 frame: 32 time slots (TS0 to TS31), 8 bits per time slot, sampled at 8000 frames/sec. | **Find:** Total aggregate bit rate R_{E1}.

Formula & Solution:

1. Bits per frame:

$$32 \text{ channels} \times 8 \text{ bits} = 256 \text{ bits/frame}$$

2. Aggregate Bit Rate:

$$R_b = 256 \text{ bits/frame} \times 8000 \text{ frames/s} = 2{,}048{,}000 \text{ bps} = 2.048 \text{ Mbps}$$

Board Exam Note: T1 = 24 channels (1.544 Mbps); E1 = 32 channels (2.048 Mbps, 30 voice, TS0 framing, TS16 signaling).

Answer: 2.048 Mbps

Question 34: Calculate the free-space path loss (FSPL) for a 3 GHz signal traveling over a distance of 10 kilometers. (Assume isotropic antennas).

Concept / Topic: Microwave Engineering / Friis Propagation

Given: $f = 3$ GHz $= 3000$ MHz, $d = 10$ km. | **Find:** Free-Space Path Loss ($FSPL$) in dB.

Formula & Solution:

Standard logarithmic FSPL formula:

$$\text{FSPL (dB)} = 32.44 + 20\log_{10}(f_{\text{MHz}}) + 20\log_{10}(d_{\text{km}})$$

$$\text{FSPL} = 32.44 + 20\log_{10}(3000) + 20\log_{10}(10)$$

$$\log_{10}(3000) \approx 3.47712 \implies 20 \times 3.47712 = 69.54 \text{ dB}$$

$$20\log_{10}(10) = 20 \text{ dB}$$

$$\text{FSPL} = 32.44 + 69.54 + 20 = 121.98 \text{ dB} \approx 122 \text{ dB}$$

Answer: 122 dB

Question 35: Calculate the discrete-time linear convolution of the input sequence $x[n] = \{1, 1\}$ and the filter impulse response $h[n] = \{1, -1\}$.

Concept / Topic: DSP / Convolution

Given: $x[n] = \{1, 1\}$ (length $L = 2$), $h[n] = \{1, -1\}$ (length $M = 2$). | **Find:** Linear convolution $y[n] = x[n] * h[n]$.

Formula & Solution:

Result length is $N = L + M - 1 = 2 + 2 - 1 = 3$.

$$y[0] = x[0]h[0] = 1 \times 1 = 1$$

$$y[1] = x[0]h[1] + x[1]h[0] = (1)(-1) + (1)(1) = -1 + 1 = 0$$

$$y[2] = x[1]h[1] = 1 \times (-1) = -1$$

$$y[n] = \{1, 0, -1\}$$

Answer: {1, 0, -1}

Question 36: A cybersecurity audit of a corporate headquarters reveals that the Building Management System controllers are installed on the exact same flat IP subnet as the enterprise HR and finance servers. Why does this specific network topology pose a severe vulnerability?

Topic: Cybersecurity / Network Architecture

Concept & Explanation: In a flat, unsegmented network architecture, any compromised device on the local broadcast domain provides lateral traversal. Vulnerable IoT/BMS hardware can be exploited to bypass perimeter firewalls and pivot attacks directly into confidential servers without encountering internal security inspection.

Answer: The lack of logical network segmentation allows an attacker who compromises an unpatched, exposed IoT thermostat to use it as a direct lateral gateway to attack the primary corporate IT network

Question 37: An AM wave has a maximum peak-to-peak envelope voltage of 10 V and a minimum of 2 V. What is the modulation index?

Concept / Topic: Analog Communications / Amplitude Modulation

Given: $V_{max(p-p)} = 10$ V, $V_{min(p-p)} = 2$ V. | **Find:** Modulation index m.

Formula & Solution:

Since envelope peak and peak-to-peak voltages are directly proportional:

$$m = \frac{V_{max} - V_{min}}{V_{max} + V_{min}} = \frac{10 - 2}{10 + 2} = \frac{8}{12} = \frac{2}{3} \approx 0.667 \approx 0.67$$

Answer: 0.67

Question 38: In telecommunications, what physically constitutes a standard four-wire circuit?

Topic: Telephony / Transmission Systems

Concept & Explanation: A four-wire circuit physically comprises two independent twisted pairs

of copper conductors: one dedicated pair for the transmit direction and one dedicated pair for the receive direction. This eliminates hybrid reflections and allows full-duplex transmission with unidirectional amplifiers.

Answer: Two separate copper pairs, one dedicated strictly for transmitting and the other strictly for receiving

Question 39: In a Quadrature Amplitude Modulation (QAM) constellation diagram, what do the distinct points located at varying distances from the origin represent?

Topic: Digital Communications / Constellation Analysis

Concept & Explanation: Each discrete constellation point represents a unique digital symbol defined in the I/Q plane. Its radial distance from the origin represents instantaneous carrier amplitude ($\sqrt{I^2+Q^2}$), and its angular displacement represents carrier phase ($\arctan(Q/I)$).

Answer: Specific combinations of phase and amplitude states

Question 40: A network architect compares two protocols for a heavily loaded 100 Mbps link: Protocol A uses half-duplex and Protocol B uses full-duplex. Under continuous, bidirectional maximum load, what is the theoretical aggregate throughput analysis for both links?

Concept / Topic: Data Communications / Transmission Modes

Given: Nominal link bandwidth = 100 Mbps. | **Find:** Aggregate bidirectional throughput for Half-Duplex vs. Full-Duplex.

Formula & Solution:

1. Half-Duplex: Devices share the transmission channel alternately. Maximum aggregate capacity remains 100 Mbps. 2. Full-Duplex: Dedicated simultaneous channels operate in both directions (100 Mbps Tx + 100 Mbps Rx), giving a theoretical aggregate throughput of:

$$100 \text{ Mbps} + 100 \text{ Mbps} = 200 \text{ Mbps}$$

Answer: Protocol A yields 100 Mbps; Protocol B yields 200 Mbps

Question 41: Regarding Fire and Life Safety controls, what is the definitive characteristic that distinguishes a Class A initiating device circuit wiring architecture from a Class B architecture?

Topic: Life Safety Systems / Circuit Classification

Concept & Explanation: Class B circuits use an end-of-line resistor (EOLR) and radiate out as a single open branch; a single wire break disables all devices beyond the break. Class A loops originate from the main panel, route through all devices, and return back to the panel, maintaining redundant dual-path signaling through single open-circuit faults.

Answer: Class A circuits include a redundant return loop connected back to the main control panel, allowing the system to remain fully operational even if a single wire break occurs

Question 42: An RF wave enters a dielectric medium with a relative permittivity of 4.0 from free space at an incident angle of 30 degrees. According to Snell's law, what is the angle of refraction?

Concept / Topic: Electromagnetics / Refraction

Given: $\theta_1 = 30^\circ$, $\epsilon_{r1} = 1.0$ (free space), $\epsilon_{r2} = 4.0$. | **Find:** Angle of refraction θ_2.

Formula & Solution:

Refractive indices: $n_1 = \sqrt{\epsilon_{r1}} = \sqrt{1} = 1.0$, $n_2 = \sqrt{\epsilon_{r2}} = \sqrt{4.0} = 2.0$. Applying Snell's Law:

$$n_1 \sin\theta_1 = n_2 \sin\theta_2 \implies 1.0 \times \sin(30^\circ) = 2.0 \times \sin(\theta_2)$$

$$\sin(\theta_2) = \frac{\sin(30^\circ)}{2.0} = \frac{0.5}{2.0} = 0.25$$

$$\theta_2 = \arcsin(0.25) \approx 14.4775^\circ \approx 14.5^\circ$$

Answer: 14.5 degrees

Question 43: In the OSI 7-layer reference model, which layer is responsible for logical addressing and routing packets across multiple different networks?

Topic: Data Communications / OSI Reference Model

Concept & Explanation: Layer 3 (Network Layer) manages logical addressing (such as IPv4 and IPv6), packet routing, fragmenting, and multi-network path determination using routers.

Answer: Network Layer

Question 44: An analog signal with a bandwidth of 4 kHz is sampled at the Nyquist rate and encoded using an 8-bit PCM system. What is the resulting transmission bit rate?

Concept / Topic: Digital Communications / Sampling & PCM

Given: $B = 4$ kHz, bits/sample $n = 8$ bits. | **Find:** Bit rate R_b.

Formula & Solution:

1. Nyquist sampling rate:

$$f_s = 2 \times B = 2 \times 4 \text{ kHz} = 8 \text{ kHz} = 8000 \text{ samples/sec}$$

2. Transmission bit rate:

$$R_b = n \times f_s = 8 \text{ bits/sample} \times 8000 \text{ samples/s} = 64{,}000 \text{ bps} = 64 \text{ kbps}$$

Answer: 64 kbps

Question 45: Calculate the total transmitted power of an AM transmitter with a 10 kW unmodulated carrier operating at 80% modulation.

Concept / Topic: Analog Communications / Amplitude Modulation

Given: Carrier power $P_c = 10$ kW, modulation index $m = 0.80$. | **Find:** Total transmitted power P_t.

Formula & Solution:

Standard AM power relationship:

$$P_t = P_c \left(1 + \frac{m^2}{2}\right) = 10 \text{ kW} \times \left(1 + \frac{0.80^2}{2}\right) = 10 \times \left(1 + \frac{0.64}{2}\right)$$

$$P_t = 10 \times (1 + 0.32) = 10 \times 1.32 = 13.2 \text{ kW}$$

Answer: 13.2 kW

Question 46: A half-duplex communication link has a data rate of 9600 bps and a line turnaround time of 20 milliseconds. If a device needs to send a 960-bit acknowledgment frame immediately after receiving data, what is the total time from the end of reception to the end of the acknowledgment transmission?

Concept / Topic: Data Communications / Latency Analysis

Given: Data rate $R = 9600$ bps, turnaround time $t_{turn} = 20$ ms $= 0.020$ s, frame length $L = 960$ bits. | **Find:** Total elapsed time T_{total}.

Formula & Solution:
1. Transmission time of acknowledgment:

$$t_{tx} = \frac{L}{R} = \frac{960 \text{ bits}}{9600 \text{ bps}} = 0.100 \text{ seconds} = 100 \text{ ms}$$

2. Total time:

$$T_{total} = t_{turn} + t_{tx} = 20 \text{ ms} + 100 \text{ ms} = 120 \text{ milliseconds}$$

Answer: 120 milliseconds

Question 47: What is a primary responsibility of the Presentation Layer in the OSI model?
Topic: Data Communications / OSI Reference Model
Concept & Explanation: Layer 6 (Presentation Layer) standardizes data syntax and semantics between disparate systems. Key duties include character code conversion (e.g., ASCII to EBCDIC), data compression, and cryptographic encryption/decryption.
Answer: Translating data formats, data compression, and data encryption

Question 48: In a Code Division Multiple Access (CDMA) cellular network, what is the primary purpose of implementing strict power control at the mobile station?
Topic: Cellular Communications / CDMA
Concept & Explanation: In CDMA, all users transmit simultaneously on the same frequency band, separated by pseudo-random noise codes. If a near mobile transmits with high power, it overwhelms distant signals at the base station (the "Near-Far Problem"). High-speed closed-loop power control forces all incoming signals to arrive at equal power.
Answer: To prevent signals from nearby mobile stations from overpowering signals from distant ones

Question 49: How does a phased array antenna dynamically steer its main radiation beam without physical movement?
Topic: Antenna Engineering / Phased Arrays
Concept & Explanation: A phased array controls the relative phase shift between feed currents of individual array elements. Constructive wave interference reinforces radiation in a designated steerable direction while destructive interference suppresses sidelobes.
Answer: By applying calculated variable electrical phase shifts to the signals at each individual radiating element

Question 50: Why do AM radio broadcasts travel significantly farther at night than during the day?
Topic: Radio Wave Propagation / Ionosphere
Concept & Explanation: During daytime, solar ultraviolet ionizing radiation creates the dense D layer (60–90 km), which strongly absorbs medium frequencies (535–1605 kHz). At night, the D layer quickly recombines and vanishes, permitting AM waves to penetrate to the higher E and F layers, where they are refracted back to Earth over long distances via sky waves.
Answer: The D layer, which absorbs medium frequency signals, rapidly dissipates after sunset

Question 51: Four identical voice channels, each band-limited to 3 kHz, are multiplexed using standard FDM with a 1 kHz guard band between adjacent channels. What is the minimum total bandwidth required?

Concept / Topic: Telephony / Frequency Division Multiplexing
Given: Channels $N = 4$, individual bandwidth $B = 3$ kHz, guard bands $N - 1 = 3$, guard band width $B_g = 1$ kHz. | **Find:** Total minimum bandwidth B_T.
Formula & Solution:
Formula:

$$B_T = N \cdot B + (N-1) \cdot B_g = (4 \times 3 \text{ kHz}) + (3 \times 1 \text{ kHz}) = 12 \text{ kHz} + 3 \text{ kHz} = 15 \text{ kHz}$$

Answer: 15 kHz

Question 52: A discrete memoryless source emits two symbols, X and Y, with probabilities 0.75 and 0.25, respectively. Calculate the entropy of this source in bits per symbol.
Concept / Topic: Information Theory / Source Coding
Given: $P(X) = 0.75$, $P(Y) = 0.25$. | **Find:** Source entropy $H(S)$ in bits/symbol.
Formula & Solution:
Shannon's Entropy Formula:

$$H(S) = -\sum_i P_i \log_2(P_i) = -[0.75 \log_2(0.75) + 0.25 \log_2(0.25)]$$

Since $\log_2(0.25) = \log_2(2^{-2}) = -2$:

$$\log_2(0.75) = \log_2(3/4) = \log_2(3) - 2 \approx 1.58496 - 2 = -0.41504$$

$$H(S) = -[0.75(-0.41504) + 0.25(-2.0)] = -[-0.31128 - 0.500] = 0.81128 \approx 0.81 \text{ bits/symbol}$$

Answer: 0.81 bits/symbol

Question 53: Based on Bessel function evaluation of an FM signal, what happens to the carrier component when the modulation index is approximately 2.4?
Topic: Angle Modulation / FM Bessel Functions
Concept & Explanation: The amplitude of the FM carrier component is governed by the zero-order Bessel function of the first kind, $J_0(m_f)$. The first zero of $J_0(m_f)$ occurs at $m_f \approx 2.4048$. At this point, the carrier power drops to zero and all energy resides in the sidebands.
Answer: It completely disappears, passing all power to the sidebands

Question 54: What is the resulting signal classification when an even signal is multiplied by an odd signal?
Topic: Signals & Systems / Symmetry Properties
Concept & Explanation: Even signals satisfy $x_e(-t) = x_e(t)$ and odd signals satisfy $x_o(-t) = -x_o(t)$. Let $y(t) = x_e(t) \cdot x_o(t)$. Then:

$$y(-t) = x_e(-t) \cdot x_o(-t) = x_e(t) \cdot [-x_o(t)] = -[x_e(t)x_o(t)] = -y(t)$$

Hence, the product is an odd signal.
Answer: An odd signal

Question 55: An intelligent lighting control system equipped with daylight harvesting sensors is installed in an office with massive perimeter windows. However, on bright sunny days, the perimeter lights dim down, then randomly ramp up to full brightness, and then dim down again in a continuous loop. What is the primary analytical cause of this failure?

Topic: Industrial Building Automation / Feedback Loops
Concept & Explanation: When the daylight sensor is mounted within direct sight of the luminaire it controls, it establishes an unintended positive optical feedback loop (hunting / cycling). When the fixture dims, the sensor senses reduced illumination, orders brightening, then senses excessive light, dimming the fixture again.
Answer: The photo-sensor is improperly placed too close to the light fixtures, meaning it measures the artificial light it just commanded to turn on, interpreting it as excess daylight and creating a closed-loop feedback error

Question 56: A field engineering team uses standard push-to-talk portable radios for communication. Which transmission mode must the network engineer account for when calculating the maximum channel throughput for a conversational exchange?
Topic: Data Communications / Transmission Modes
Concept & Explanation: Push-to-talk (PTT) radios share a common radio channel where transmission can occur in both directions, but only one station can transmit at any given time while the other receives. This defines half-duplex communication.
Answer: Half-duplex

Question 57: A superheterodyne receiver tuned to 90 MHz uses an IF of 10.7 MHz. Assuming low-side injection for the local oscillator, what is the image frequency?
Concept / Topic: Radio Frequency Receivers
Given: $f_{RF} = 90$ MHz, $f_{IF} = 10.7$ MHz, Low-side injection ($f_{LO} < f_{RF}$). | **Find:** Image frequency f_{img}.
Formula & Solution:
1. For low-side injection:

$$f_{LO} = f_{RF} - f_{IF} = 90 - 10.7 = 79.3 \text{ MHz}$$

2. The image frequency for low-side injection is:

$$f_{img} = f_{RF} - 2f_{IF} = f_{LO} - f_{IF} = 79.3 - 10.7 = 68.6 \text{ MHz}$$

Board Exam Rule: High-side injection $\implies f_{img} = f_{RF} + 2f_{IF}$. Low-side injection $\implies f_{img} = f_{RF} - 2f_{IF}$.
Answer: 68.6 MHz

Question 58: If an amplifier is cooled to absolute zero (0 K), what can be deduced about its theoretical noise factor?
Concept / Topic: Noise Figure Analysis / Friis Equations
Given: Temperature $T = 0$ K, equivalent input noise temperature $T_e = 0$ K. | **Find:** Theoretical noise factor F.
Formula & Solution:
Noise factor is defined as:

$$F = 1 + \frac{T_e}{T_0}$$

Where $T_0 = 290$ K. If internal thermal noise generation is zero ($T_e = 0$ K):

$$F = 1 + 0 = 1.0 \quad (\text{or } NF = 0 \text{ dB})$$

Answer: It becomes 1

Question 59: When plotting a matching network on a Smith Chart, moving from a given normalized impedance point clockwise along a constant resistance circle indicates which physical modification?

Topic: Transmission Lines / Smith Chart Analysis

Concept & Explanation: On the Smith Chart, constant resistance contours are circles passing through the open-circuit point $(1, 0)$. Moving clockwise along a constant resistance circle increases normalized inductive reactance ($+jx$), representing the addition of a lumped inductor in series with the load.

Answer: Adding a series inductor

Question 60: A transmitter feeds 100 watts into an isotropic antenna. What is the approximate power density at a distance of 5 kilometers in free space?

Concept / Topic: Antennas & EM Propagation

Given: $P_t = 100$ W, $G_t = 1$ (isotropic), $d = 5$ km $= 5000$ m. | **Find:** Radiated power density S (μW/m^2).

Formula & Solution:

Power density over a sphere of radius d:

$$S = \frac{P_t}{4\pi d^2} = \frac{100}{4\pi(5000)^2} = \frac{100}{4\pi \times 2.5 \times 10^7} \approx 3.183 \times 10^{-7} \text{ W/m}^2 \approx 0.318\,\mu\text{W/m}^2$$

Answer: $0.318\,\mu$**W/m**^{2}

Question 61: A technician analyzes an ISDN PRI circuit where active voice calls on the B-channels are perfectly clear, but the PBX is unable to establish any new outbound calls. Which specific channel analysis will most likely reveal the fault?

Topic: Telephony / ISDN PRI Architecture

Concept & Explanation: In ISDN Primary Rate Interface (PRI), bearer (B) channels carry digitized voice payload (64 kbps each), while the D channel carries LAPD/Q.931 out-of-band call control, setup, and teardown signaling. If existing calls persist but call setup fails, the signaling link on the D channel is down.

Answer: Analyzing the D channel for signaling link failures or congestion

Question 62: When engineering a 30 GHz microwave backhaul, which environmental factor requires the largest specific fade margin allowance in the link budget analysis?

Topic: Microwave Propagation / Atmospheric Losses

Concept & Explanation: At millimeter-wave frequencies above 10 GHz (and heavily pronounced around 30 GHz), raindrop diameters are comparable to signal wavelengths ($\lambda \approx 1$ cm), causing severe Mie scattering and dielectric absorption. Heavy rain requires large fade margins (often 20–40 dB).

Answer: Attenuation due to heavy rain rates

Question 63: Which fundamental term is used to describe the physical medium through which a communication signal is transmitted?

Topic: Communications Systems / Terminology

Concept & Explanation: In information transmission systems, the channel is the physical or propagation path (e.g., twisted pair, coaxial cable, optical fiber, free-space dielectric) between transmitter and receiver.

Answer: Channel

Question 64: When designing a pyramidal horn antenna, extending the physical length of the flare while keeping the aperture dimensions constant will theoretically result in which of the following?

Topic: Antenna Theory / Horn Antennas

Concept & Explanation: Extending the horn flare length flattens the spherical phase wavefront error across the radiating aperture, yielding a more uniform aperture illumination phase, which increases gain/directivity and narrows beamwidth.

Answer: A narrower beamwidth and higher overall directivity

Question 65: For a 5 GHz point-to-point microwave link stretching 20 kilometers, what is the maximum radius of the first Fresnel zone at the exact midpoint of the path?

Concept / Topic: Microwave Link Engineering / Fresnel Zones

Given: $f = 5$ GHz, $d = 20$ km, Midpoint $d_1 = d_2 = 10$ km. | **Find:** Radius of the 1st Fresnel zone R_1 in meters.

Formula & Solution:

1. Wavelength:

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8 \text{ m/s}}{5 \times 10^9 \text{ Hz}} = 0.06 \text{ meters}$$

2. First Fresnel zone radius formula:

$$R_1 = \sqrt{\frac{\lambda d_1 d_2}{d_1 + d_2}} = \sqrt{\frac{0.06 \times 10{,}000 \times 10{,}000}{20{,}000}} = \sqrt{\frac{6{,}000{,}000}{20{,}000}} = \sqrt{300} \approx 17.32 \text{ meters}$$

Answer: 17.3 meters

Question 66: Evaluate the performance of Pulse Position Modulation (PPM) compared to Pulse Amplitude Modulation (PAM) in a noisy channel. Which of the following statements correctly identifies the advantage of PPM?

Topic: Pulse Modulation / Noise Performance

Concept & Explanation: PAM conveys information in pulse height, making it directly vulnerable to channel amplitude fluctuations, fading, and thermal noise. PPM maintains constant pulse amplitude and width, encoding data in pulse transition timing; hence, amplitude clipping/limiting at the receiver eliminates noise spikes.

Answer: PPM relies on timing rather than amplitude, making it inherently more immune to amplitude-varying noise

Question 67: A telephone channel has a bandwidth of 3 kHz and a signal-to-noise ratio of 30 dB. What is the approximate maximum theoretical data rate according to the Shannon-Hartley theorem?

Concept / Topic: Information Theory / Channel Capacity

Given: $B = 3000$ Hz, SNR $= 30$ dB. | **Find:** Channel Capacity C in kbps.

Formula & Solution:

1. Convert SNR from dB to linear ratio:

$$\text{SNR} = 10^{30/10} = 10^3 = 1000$$

2. Shannon-Hartley theorem:

$$C = B \log_2(1 + \text{SNR}) = 3000 \log_2(1001) \approx 3000 \log_2(1024) = 3000 \times 10 \approx 30{,}000 \text{ bps} = 30 \text{ kbps}$$

Answer: 30 kbps

Question 68: Using the Friis transmission equation, calculate the received power if a 1 GHz signal is transmitted with 10 W, both antennas have a gain of 10 dBi, and the distance is 300 meters. (Assume no system losses).

Concept / Topic: RF Communications / Friis Equation
Given: $P_t = 10$ W, $f = 1$ GHz $\implies$ $\lambda = 0.3$ m, $G_t = G_r = 10$ dBi $\implies$ 10 (linear), $d = 300$ m. | **Find:** Received power P_r (μW).
Formula & Solution:
Friis Transmission Equation:

$$P_r = P_t G_t G_r \left(\frac{\lambda}{4\pi d}\right)^2 = 10 \times 10 \times 10 \times \left(\frac{0.3}{4\pi \times 300}\right)^2 \approx 7.07\,\mu\text{W}$$

Answer: $7.07\,\mu$**W**

Question 69: In a cascaded receiver system, analysis of the Friis formula dictates that the most critical component for determining the overall system noise figure is the:

Topic: RF Receivers / Cascaded Noise Figure
Concept & Explanation: By the Friis formula for noise figure:

$$F_{total} = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} + \dots$$

The noise factor F_1 of the very first stage appears undiminished, whereas the noise of subsequent stages is attenuated by preceding gains (G_1, G_1G_2). Thus, the first stage LNA dictates the system noise figure.
Answer: First stage low-noise amplifier

Question 70: A diagnostic capture shows that a host can successfully ping external IP addresses, but a specific application transferring 2000-byte payloads is consistently failing across a VPN tunnel. Based on OSI model principles, what is the most likely root cause?

Topic: Networking / MTU & PMTUD
Concept & Explanation: Small ICMP pings (32–64 bytes) succeed because they fall well below the interface Maximum Transmission Unit (MTU ≈ 1500). VPN encapsulation adds IPsec/GRE headers, reducing effective payload MTU. If large packets have the ”Don’t Fragment” (DF) bit set and Path MTU Discovery (PMTUD) ICMP Type 3 Code 4 messages are filtered, the tunnel drops oversized packets.
Answer: An MTU mismatch causing Path MTU Discovery failure or excessive fragmentation at the Network Layer

Question 71: A network engineer plans to upgrade a wireless link by doubling both the operating frequency and the distance between the nodes. How will the free-space path loss change compared to the original setup?

Concept / Topic: Microwave Communications / Path Loss
Given: $f_2 = 2f_1$, $d_2 = 2d_1$. | **Find:** Change in path loss ΔFSPL in dB.
Formula & Solution:
Free Space Path Loss is proportional to $(f \cdot d)^2$:

$$\text{FSPL} \propto (f \cdot d)^2 \implies \frac{\text{FSPL}_2}{\text{FSPL}_1} = \left(\frac{2f_1 \cdot 2d_1}{f_1 d_1}\right)^2 = (4)^2 = 16$$

In decibels:

$$\Delta\text{FSPL} = 10\log_{10}(16) = 20\log_{10}(4) \approx 12.04 \text{ dB} \approx 12 \text{ dB}$$

Each doubling of frequency contributes $+6$ dB, and each doubling of distance contributes $+6$ dB, yielding $+12$ dB total.

Answer: it will increase by 12 dB

Question 72: What defines a continuous-time signal mathematically?

Topic: Signals & Systems / Signal Classification

Concept & Explanation: A continuous-time signal $x(t)$ is defined over a continuous continuum of time values $t \in \mathbb{R}$, taking real or complex values at every single instant.

Answer: It is defined at every continuous instant of time

Question 73: How is a causal signal mathematically defined in the time domain?

Topic: Signals & Systems / Causality

Concept & Explanation: A continuous-time signal is causal if its value is identically zero for all negative time: $x(t) = 0$ for all $t < 0$. (In discrete time, $x[n] = 0$ for all $n < 0$).

Answer: It is equal to zero for all negative time values

Question 74: Which digital modulation technique changes the frequency of the carrier based on the discrete state of the digital data?

Topic: Digital Communications / FSK

Concept & Explanation: In Frequency Shift Keying (FSK), the carrier amplitude and phase are held constant while the instantaneous carrier frequency is shifted between discrete values (e.g., mark frequency f_m for binary 1 and space frequency f_s for binary 0).

Answer: Frequency Shift Keying (FSK)

Question 75: An industrial control network utilizes a multipoint configuration over a single shared medium. If the master controller must receive continuous, real-time telemetry from 20 separate slave sensors simultaneously, why will a standard simplex multipoint architecture fail?

Topic: Industrial Communications / Transmission Modes

Concept & Explanation: Simplex transmission is strictly unidirectional (one-way). In a standard simplex broadcast, data flows only from master to slaves; the slaves possess no transmitter capability to send feedback or telemetry upstream.

Answer: Simplex only allows transmission from the master to the slaves, making sensor telemetry impossible

Question 76: A microwave link is required to transmit data at a rate of 48 Mbps over a channel with a usable bandwidth of 12 MHz. Assuming ideal filtering, what is the lowest order of M-ary QAM that can achieve this data rate?

Concept / Topic: Digital Communications / Spectral Efficiency

Given: $R_b = 48$ Mbps, Bandwidth $B = 12$ MHz, ideal Nyquist filtering ($r = 0 \implies B = R_s$). |

Find: Modulation order M.

Formula & Solution:

1. Maximum allowable symbol rate:

$$R_s = B = 12 \text{ MBaud}$$

2. Required bits per symbol:

$$k = \frac{R_b}{R_s} = \frac{48 \text{ Mbps}}{12 \text{ MBaud}} = 4 \text{ bits/symbol}$$

3. Required constellation size:

$$M = 2^k = 2^4 = 16 \implies \text{16-QAM}$$

Answer: 16-QAM

Question 77: A Binary Frequency Shift Keying (BFSK) signal has a mark frequency of 105 kHz and a space frequency of 95 kHz. What is the peak frequency deviation of the carrier?

Concept / Topic: Digital Communications / FSK
Given: $f_{mark} = 105$ kHz, $f_{space} = 95$ kHz. | **Find:** Peak frequency deviation Δf.
Formula & Solution:
1. Unmodulated nominal carrier frequency:

$$f_c = \frac{f_m + f_s}{2} = \frac{105 + 95}{2} = 100 \text{ kHz}$$

2. Peak frequency deviation:

$$\Delta f = \frac{|f_m - f_s|}{2} = \frac{105 - 95}{2} = 5 \text{ kHz}$$

Answer: 5 kHz

Question 78: Which classification correctly applies to a signal that exhibits statistical randomness and cannot be expressed by a precise deterministic mathematical formula?

Topic: Signals & Systems / Random Processes
Concept & Explanation: A stochastic (random) signal cannot be predicted with absolute certainty at any future instant and must be described mathematically using probability distributions, autocorrelation, and power spectral density.
Answer: Stochastic signal

Question 79: According to Hartley's Law, the amount of information that can be transmitted over a channel is directly proportional to which two parameters?

Topic: Information Theory / Hartley's Law
Concept & Explanation: Hartley's formulation states that maximum information capacity without noise is $H = 2Bt \log_2(M)$, meaning capacity is directly proportional to transmission bandwidth (B) and duration/time of transmission (t).
Answer: Bandwidth and time of transmission

Question 80: What is the channel structure of an ISDN Basic Rate Interface (BRI)?

Topic: Telecommunications / ISDN
Concept & Explanation: ISDN BRI configuration is $2B + D$: two 64 kbps B (Bearer) channels for user voice/data payload and one 16 kbps D (Data) channel for signaling, delivering a total user payload rate of 128 kbps (and 144 kbps total raw rate excluding framing).
Answer: Two 64 kbps B channels and one 16 kbps D channel

Question 81: Which statement accurately describes the client-server category of data communication?

Topic: Data Communications / Architecture

Concept & Explanation: In the client-server model, dedicated host systems (servers) listen for service requests, manage shared resources, and supply files or services, while endpoint client devices request and consume those services.

Answer: A central server provides services, data, or resources to multiple requesting client devices

Question 82: What is the sequence of discrete voltage levels representing data in a transmission medium commonly called?

Topic: Data Transmission / Signals

Concept & Explanation: A discrete-state voltage stream representing binary 1s and 0s over time is formally designated as a digital signal.

Answer: Digital signal

Question 83: When a photodiode is operated in the reverse-biased photoconductive mode, what is the primary effect on the device's internal semiconductor junction?

Topic: Optoelectronics / Photodiodes

Concept & Explanation: Applying reverse bias widens the internal p-n depletion region, which increases photon absorption volume and decreases junction capacitance ($C_j = \epsilon A/W$). This lower capacitance significantly reduces the RC time constant, yielding faster response times and wider operational bandwidth.

Answer: The depletion region widens, increasing the volume available for electron-hole pair generation and reducing junction capacitance

Question 84: A linear PCM system uses a 10-bit analog-to-digital converter. How many distinct discrete quantization levels are available in this system?

Concept / Topic: Digital Communications / PCM Quantization

Given: Number of bits $n = 10$ bits. | **Find:** Number of quantization levels L.

Formula & Solution:

Formula:

$$L = 2^n = 2^{10} = 1024 \text{ levels}$$

Answer: 1024

Question 85: During the normal operating scan cycle of a Programmable Logic Controller (PLC), what is the correct sequence of internal operations?

Topic: Industrial Electronics / PLCs

Concept & Explanation: A standard PLC deterministic scan cycle executes sequentially in four phases: 1. Input Scan (read input terminals and update input image table); 2. Logic Execution (execute user program / ladder logic); 3. Output Scan (write output image table values to physical output terminals); 4. Housekeeping (perform self-diagnostics and communications).

Answer: Read all physical inputs and store their states, execute the ladder logic program sequentially, update the physical outputs, and perform internal communication/diagnostics

Question 86: Analyze the effect on the total noise power at the output of an ideal bandpass filter if its center frequency is shifted from 10 MHz to 20 MHz, assuming white noise at the

input and a constant filter bandwidth.

Concept / Topic: Thermal Noise / Stochastic Processes
Given: Input: White noise with flat PSD $N_0/2$ W/Hz. Filter bandwidth $B =$ constant. Center frequency shifted 10 MHz $\rightarrow 20$ MHz. | **Find:** Total output noise power.
Formula & Solution:
The total noise power at the output of an ideal bandpass filter with bandwidth B driven by white noise is:

$$P_{noise} = kTB = N_0 B$$

Because white noise has a uniform power spectral density across all frequencies, total noise power depends solely on noise spectral density N_0 and filter bandwidth B, and is completely independent of center frequency f_0.
Answer: The total noise power remains the same

Question 87: A Time Division Multiple Access (TDMA) system allocates a 20 millisecond frame duration equally among 4 users. If the overall data rate is 100 kbps, how many bits does each user transmit per frame?

Concept / Topic: Multiple Access / TDMA Systems
Given: Frame duration $T_f = 20$ ms $= 0.020$ s, Users $N = 4$, Data rate $R = 100$ kbps $= 100{,}000$ bps. | **Find:** Bits transmitted per user per frame (b_{user}).
Formula & Solution:
1. Total bits transmitted in one complete frame:

$$B_{frame} = R \times T_f = 100{,}000 \text{ bps} \times 0.020 \text{ s} = 2000 \text{ bits}$$

2. Since the frame is allocated equally among 4 users:

$$b_{user} = \frac{B_{frame}}{N} = \frac{2000 \text{ bits}}{4} = 500 \text{ bits}$$

Answer: 500 bits

Question 88: Analyze the effect of a nonlinear mixer in a receiver if a strong out-of-band interfering signal and the desired signal are present simultaneously at the input.

Topic: RF Receivers / Intermodulation Distortion
Concept & Explanation: Due to the odd-order non-linear terms (specifically the 3rd-order term $a_3 v_{in}^3$) of active mixer devices, amplitude modulation present on a strong adjacent or out-of-band interferer transfers directly onto the desired carrier, causing cross-modulation distortion.
Answer: The interference can cause cross-modulation distortion on the desired signal

Question 89: Analyze the frequency response of a discrete-time system that has a single zero at $z = 1$ and a pole at the origin $z = 0$. What type of basic filtering operation does this configuration perform?

Topic: DSP / Z-Transform & Filter Design
Concept & Explanation: Transfer function: $H(z) = 1 - z^{-1} = \frac{z-1}{z}$. Evaluating frequency response along the unit circle $z = e^{j\omega}$: At DC ($\omega = 0$): $z = e^{j0} = 1 \implies |H(e^{j0})| = |1-1| = 0$ (completely blocks DC). At Nyquist ($\omega = \pi$): $z = e^{j\pi} = -1 \implies |H(e^{j\pi})| = |1-(-1)| = 2$ (maximum transmission). Thus, this system attenuates low frequencies and amplifies high frequencies, acting as a high-pass filter.
Answer: High-pass filtering

Question 90: If a purely vertically polarized antenna attempts to receive a purely horizontally polarized electromagnetic wave, what is the theoretical outcome?

Topic: Antenna Theory / Polarization Mismatch
Concept & Explanation: The Polarization Loss Factor (PLF) is given by:

$$PLF = |\hat{\rho}_w \cdot \hat{\rho}_a|^2 = \cos^2(\theta)$$

Since vertical and horizontal polarizations are orthogonal ($\theta = 90°$):

$$PLF = \cos^2(90°) = 0$$

The antenna captures zero power from the incident wave.
Answer: The antenna will capture zero power from the incident wave

Question 91: Given a signal with a maximum frequency of f_m, if it is sampled at exactly $2f_m$ but with a 90-degree phase shift in the sampling pulses, what reconstruction failure occurs?
Topic: DSP / Nyquist Sampling Theorem
Concept & Explanation: Consider a sinusoid $x(t) = A\sin(2\pi f_m t)$. If sampled at $T_s = 1/(2f_m)$ starting at $t = 0$, the sample values are:

$$x(nT_s) = A\sin\left(2\pi f_m \frac{n}{2f_m}\right) = A\sin(n\pi) = 0 \quad \text{for all } n$$

Every sample point lands precisely on the zero crossings of the sine wave, resulting in identically zero output, causing total reconstruction failure.
Answer: The signal is sampled exactly at its zero crossings, resulting in a zero output

Question 92: Using the standard approximation, what is the half-power beamwidth of a 4-meter diameter parabolic dish antenna operating at 10 GHz?
Concept / Topic: Antennas / Parabolic Dish Antennas
Given: Diameter $D = 4$ m, frequency $f = 10$ GHz. | **Find:** Half-power beamwidth ($\theta_{3\text{dB}}$) in degrees.
Formula & Solution:
1. Wavelength:

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8 \text{ m/s}}{10 \times 10^9 \text{ Hz}} = 0.03 \text{ meters}$$

2. Standard parabolic dish half-power beamwidth approximation:

$$\theta_{3\text{dB}} \approx \frac{70\lambda}{D} = \frac{70 \times 0.03 \text{ m}}{4 \text{ m}} = \frac{2.1}{4} = 0.525°$$

Answer: 0.525 degrees

Question 93: Calculate the percentage of power saved if the carrier and one sideband are suppressed (SSB-SC) compared to a conventional AM signal transmitting at 100% modulation.
Concept / Topic: Analog Communications / AM Power Relations
Given: Conventional AM with modulation index $m = 1.0$. | **Find:** Percentage power saved in SSB-SC.
Formula & Solution:
1. Total power in conventional AM:

$$P_{AM} = P_c\left(1 + \frac{m^2}{2}\right) = P_c\left(1 + \frac{1}{2}\right) = 1.5P_c$$

2. Total power in SSB-SC (only one sideband transmitted):

$$P_{SSB} = \frac{m^2 P_c}{4} = \frac{(1)^2 P_c}{4} = 0.25 P_c$$

3. Power saved:

$$P_{saved} = P_{AM} - P_{SSB} = 1.5P_c - 0.25P_c = 1.25P_c$$

4. Percentage power saved:

$$\% \text{ Saved} = \left(\frac{1.25P_c}{1.5P_c}\right) \times 100\% = \frac{5}{6} \times 100\% \approx 83.33\% \approx 83.3\%$$

Answer: 83.3%

Question 94: Why are four-wire circuits fundamentally required for long-distance telephone network trunks, whereas local subscriber loops can operate on two-wire circuits?

Topic: Telephony / Transmission Systems

Concept & Explanation: Long-distance telephone trunk lines require repeaters and amplifiers to compensate for cable attenuation. Conventional electronic amplifiers are unidirectional active devices; therefore, separate forward and reverse paths are necessary to prevent self-oscillation and singing feedback. Local subscriber loops are short enough not to require intermediate amplification.

Answer: Long-distance trunks require signal amplification, and standard electronic amplifiers are unidirectional devices requiring separate paths for each direction

Question 95: Why do digital communication systems utilize regenerative repeaters along a long-distance transmission link?

Topic: Digital Transmission / Regenerative Repeaters

Concept & Explanation: Unlike analog amplifiers, which amplify both signal and accumulated channel noise simultaneously, digital regenerative repeaters sample, detect, decide, and reconstruct brand-new, clean pulses with restored amplitude and timing, resetting accumulated noise and distortion to zero at every repeater.

Answer: To reconstruct the original signal and prevent noise accumulation

Question 96: A network architect is evaluating the cost-to-reliability trade-off between a fully connected mesh and a star topology for a 50-node financial trading system. Which of the following analytical conclusions is most accurate?

Topic: Network Architecture / Graph Theory & Topologies

Concept & Explanation: Nodes $N = 50$. 1. Full Mesh links: $L_{mesh} = \frac{N(N-1)}{2} = \frac{50 \times 49}{2} = 1225$ links, scaling as $O(N^2)$. 2. Star topology links: $L_{star} = N = 50$ links, scaling as $O(N)$. While mesh provides fault tolerance, star offers substantial cost savings and manageable cabling.

Answer: The star topology provides a single point of failure but scales at O(n) cost, making it vastly more cost-effective for 50 nodes than the mesh

Question 97: What is the primary purpose of the limiter stage in a standard superheterodyne FM receiver?

Topic: Radio Receivers / FM Demodulation

Concept & Explanation: FM information is encoded purely in the instantaneous frequency of the carrier. Atmospheric noise and multipath fading appear as amplitude fluctuations. The hard limiter stage clips signal peaks at a constant voltage threshold, eliminating amplitude noise prior to

the frequency discriminator.

Answer: To remove amplitude variations and noise spikes before demodulation

Question 98: Calculate the thermal noise power available from a resistor at room temperature (290 K) over a bandwidth of 2 MHz. (Given Boltzmann's constant $k = 1.38 \times 10^{-23}$ J/K)

Concept / Topic: Thermal Noise / Johnson Noise

Given: $T = 290$ K, $B = 2$ MHz $= 2 \times 10^6$ Hz, $k = 1.38 \times 10^{-23}$ J/K. | **Find:** Available thermal noise power P_n in Watts.

Formula & Solution:

Johnson noise formula for available power:

$$P_n = kTB = (1.38 \times 10^{-23}\ \text{J/K}) \times (290\ \text{K}) \times (2 \times 10^6\ \text{Hz})$$

$$P_n = 4.002 \times 10^{-21} \times 2 \times 10^6 \approx 8.0 \times 10^{-15}\ \text{W}$$

Answer: 8.0×10^{-15} **W**

Question 99: A composite analog signal contains two primary frequency components: 3 kHz and 5 kHz. What is the minimum sampling frequency required to digitize this signal without aliasing?

Concept / Topic: Signals & Systems / Nyquist Sampling Theorem

Given: Signal components: $f_1 = 3$ kHz, $f_2 = 5$ kHz. Highest frequency $f_{max} = 5$ kHz. | **Find:** Minimum Nyquist sampling rate f_s.

Formula & Solution:

Nyquist criterion states that sampling rate must be at least twice the maximum frequency present in the baseband signal:

$$f_{s(min)} = 2f_{max} = 2 \times 5\ \text{kHz} = 10\ \text{kHz}$$

Answer: 10 kHz

Question 100: If a receiver has an equivalent noise temperature of 290 K, what is its corresponding noise figure in dB?

Concept / Topic: Noise Analysis / Noise Temperature

Given: $T_e = 290$ K, standard reference temperature $T_0 = 290$ K. | **Find:** Noise figure NF in dB.

Formula & Solution:

1. Noise factor F:

$$F = 1 + \frac{T_e}{T_0} = 1 + \frac{290\ \text{K}}{290\ \text{K}} = 1 + 1 = 2.0$$

2. Noise figure in dB:

$$NF = 10 \log_{10}(F) = 10 \log_{10}(2.0) \approx 3.01\ \text{dB} \approx 3\ \text{dB}$$

Answer: 3 dB

PART II — Important Terms & Concepts Reviewer

1. Telecommunications & Modulation

ATM (Asynchronous Transfer Mode): Fixed-cell packet-switching standard using 53-byte cells (5-byte header, 48-byte payload). Fixed lengths reduce packet jitter.
Quantization SQNR Rule: Each additional bit in a linear PCM system reduces quantization noise by ≈ 6.02 dB ($SQNR \approx 6.02n + 1.76$ dB).
T1 Carrier System: Carries 24 voice channels sampled at 8 kHz using 8-bit PCM. Frame format: $(24 \times 8) + 1 = 193$ bits/frame, yielding $R_b = 1.544$ Mbps.
E1 Carrier System: European standard with 32 time slots (TS0 for synchronization, TS16 for signaling, 30 for speech), yielding 2.048 Mbps.
FM Modulation Index: $\beta = \Delta f / f_m$. When $\beta \approx 2.4$, carrier amplitude $J_0(\beta)$ crosses zero (carrier suppression).
SSB-SC Power Savings: At $m = 100\%$, conventional AM consumes $P_c + 0.5P_c = 1.5P_c$. SSB-SC transmits only one sideband ($0.25P_c$), saving 83.3% of total power.

2. Antennas & Wave Propagation

Ground Wave Propagation: Guided surface waves effective only at VLF/MF bands (< 2 MHz) due to earth conductivity losses.
Sky Wave Propagation: High Frequency (3–30 MHz) signals refracted by ionospheric layers (E and F). At night, the absorbing D layer recombines, allowing AM skywave transmission over long distances.
Fresnel Zone: Concentric ellipsoids around the line of sight. Obstacle clearance of $\geq 0.6R_1$ ensures near-free-space propagation.
Polarization Mismatch: Vertical and horizontal antennas are orthogonal (90° apart), resulting in theoretical zero power transfer ($PLF = 0$).
Phased Array Antenna: Steers the main radiation beam electronically by controlling phase differences between elements without mechanical motion.

3. Data Communications & Networking

OSI Presentation Layer (Layer 6): Handles data syntax, translation, formatting, compression, and encryption/decryption.
OSI Network Layer (Layer 3): Handles logical IP addressing, routing, and path determination across subnets.
HDLC Bit Stuffing: A '0' is inserted after any sequence of five consecutive '1's to prevent user data from mimicking the flag sequence (01111110).
Asymmetric Cryptography: Encrypt using the **recipient's public key** to ensure confidentiality; only the recipient's private key can decrypt.
ISDN BRI: $2B+D$ architecture (2×64 kbps bearer channels $+1 \times 16$ kbps data channel).
Network Topologies: Star topology scales linearly ($O(N)$ links) with a single central point of failure, whereas Mesh topology scales as $N(N-1)/2 = O(N^2)$ links.

4. Digital Signal Processing (DSP)

Nyquist Sampling Rate: Minimum sampling rate to prevent spectral aliasing: $f_s \geq 2f_{max}$.
LTI Impulse Response: Cascaded subsystems convolve: $h(t) = h_1(t) * h_2(t)$. Parallel subsystems sum: $h(t) = h_1(t) + h_2(t)$.
Z-Domain High-Pass Filter: A transfer function with a zero at $z = 1$ (DC) and a pole at the origin ($z = 0$) completely suppresses DC and passes high frequencies ($|H(e^{j\pi})| = 2$).
Causal System: Output depends only on current and past inputs: $h(t) = 0$ for all $t < 0$.

PART III — Comprehensive ECE Board Exam Formula Sheet

Topic	Formula	Variables / Parameter Definitions
PCM Systems	SQNR $\approx 6.02n + 1.76$ dB	n = quantization bits per sample
	$R_b = n \cdot f_s$	R_b = bit rate (bps), f_s = sampling frequency
	$B_{min} = R_b/2$	B_{min} = minimum baseband Nyquist bandwidth
Angle Modulation	$\beta = m_f = \dfrac{\Delta f}{f_m}$	Δf = peak deviation, f_m = modulating frequency
	$B_{\text{Carson}} = 2(\Delta f + f_m)$	Occupied Carson bandwidth (Hz)
AM Power	$P_t = P_c\left(1 + \dfrac{m^2}{2}\right)$	P_t = total power, P_c = carrier power, m = mod index
	% Saved (SSB) $= \dfrac{1 + m^2/4}{1 + m^2/2}$	For $m = 1.0$, SSB savings = 83.3%
Transmission Lines	$\Gamma = \dfrac{Z_L - Z_0}{Z_L + Z_0}$	Γ = voltage reflection coefficient
	SWR $= \dfrac{1 + \lvert\Gamma\rvert}{1 - \lvert\Gamma\rvert}$	Standing wave ratio; $P_L = P_{inc}(1 - \lvert\Gamma\rvert^2)$
Antennas & Fields	$S = \dfrac{P_t G_t}{4\pi R^2}$	S = power density (W/m^2), R = distance (m)
	$d_{LOS} \approx 4.124\left(\sqrt{h_t} + \sqrt{h_r}\right)$ km	h_t, h_r = antenna heights in meters
	FSPL $= 32.44 + 20\log_{10}(f_{\text{MHz}}) + 20\log_{10}(d_{\text{km}})$	Free-space path loss in dB
	$R_1 = \sqrt{\dfrac{\lambda d_1 d_2}{d_1 + d_2}}$	First Fresnel zone radius at clearance point
	$\theta_{3\text{dB}} \approx \dfrac{70\lambda}{D}$	Parabolic dish half-power beamwidth (°)
Noise Theory	$V_n = \sqrt{4kTBR}$	V_n = RMS thermal noise voltage (V)
	$P_n = kTB$	Available noise power, $k = 1.38 \times 10^{-23}$ J/K
	$F = 1 + \dfrac{T_e}{T_0}$	F = noise factor, $T_0 = 290$ K, $NF = 10\log_{10}(F)$
	$F_{\text{total}} = F_1 + \dfrac{F_2 - 1}{G_1} + \dfrac{F_3 - 1}{G_1 G_2} + \ldots$	Friis cascaded noise formula
Information Theory	$C = B\log_2(1 + \text{SNR})$	Shannon channel capacity limit (bps)
	$H(S) = -\sum P_i \log_2(P_i)$	Source entropy (bits/symbol)
	$t_{det} \leq d_{min} - 1,\ t_{corr} \leq \left\lfloor \dfrac{d_{min} - 1}{2} \right\rfloor$	Block coding error detection/correction limits
Superheterodyne	$f_{img} = f_{RF} + 2f_{IF}$	High-side injection ($f_{LO} = f_{RF} + f_{IF}$)
	$f_{img} = f_{RF} - 2f_{IF}$	Low-side injection ($f_{LO} = f_{RF} - f_{IF}$)

PART IV — Master Quick Answer Key

No.	Answer	Topic	No.	Answer	Topic
1	9.43%	ATM / Data Comm	51	15 kHz	FDM Multiplexing
2	$h_1(t) + h_2(t)$	LTI Systems	52	0.81 bits/symbol	Information Theory
3	Noise -12 dB, BW +25%	PCM Systems	53	Completely disappears	FM Bessel Functions
4	Ground wave VLF/MF	Wave Propagation	54	An odd signal	Signal Properties
5	Current along surface, AC R up	Skin Effect	55	Photo-sensor closed-loop error	Building Automation
6	$0.99\,\mu$W/m^2	Antennas / Field	56	Half-duplex	Transmission Modes
7	Higher symbol error prob.	16-QAM Constellation	57	68.6 MHz	Receiver Image Freq.
8	772 kHz	PCM Nyquist BW	58	It becomes 1	Thermal Noise Figure
9	Star topology	LAN Topologies	59	Adding a series inductor	Smith Chart
10	Recipient's public key	Cryptography	60	$0.318\,\mu$W/m^2	Power Density
11	5.0	FM Modulation	61	Analyzing D channel	ISDN Signaling
12	Galvanic signal isolator	Instrumentation	62	Rain attenuation	Microwave Fade
13	Backbone break halts network	Bus Topology	63	Channel	Comm Systems
14	0.95 meters	Dipole Antennas	64	Narrower beam, higher gain	Horn Antennas
15	Resistive Z on infinite line	Transmission Lines	65	17.3 meters	Fresnel Zone Radius
16	Inter-symbol interference (ISI)	Equalization	66	Immune to amplitude noise	PPM vs PAM
17	88.9 watts	Mismatch Power	67	30 kbps	Shannon Capacity
18	Image rej. up, select. down	Superhet Receiver	68	$7.07\,\mu$W	Friis Link Budget
19	1.5	DSP IIR Filter	69	First stage LNA	Cascaded Friis Noise
20	192.168.1.63	IPv4 Subnetting	70	MTU mismatch / PMTUD	OSI / Networking
21	6.48 Terabytes	CCTV Data Storage	71	Increases by 12 dB	Free-Space Path Loss
22	6 MBaud	64-QAM Modulation	72	Defined at every instant	Continuous Signal
23	82 km	Line of Sight Horizon	73	Zero for negative time	Causal Signals
24	Decrease monotonically to 0	Patch Antenna	74	Frequency Shift Keying	FSK Modulation
25	Practical gradual roll-off filter	Anti-aliasing	75	Simplex allows master to slave only	Network Modes
26	Sequential time-slot sharing	TDM Systems	76	16-QAM	Spectral Efficiency
27	12.55 dBi	Antenna Gain	77	5 kHz	BFSK Deviation
28	011111010	HDLC Bit Stuffing	78	Stochastic signal	Random Processes
29	-1	Ground Reflection	79	Bandwidth and time	Hartley's Law
30	$2.03\,\mu$V	Johnson Noise	80	2 B (64k) + 1 D (16k)	ISDN BRI
31	A hash collision	Cryptography	81	Server supplies clients	Client-Server Model
32	Detect 4, correct 2 errors	Hamming Distance	82	Digital signal	Signal Types
33	2.048 Mbps	E1 Carrier	83	Depletion widens, C down	Photodiode Mode
34	122 dB	Free Space Path Loss	84	1024 levels	PCM Quantization
35	$\{1, 0, -1\}$	Discrete Convolution	85	Read inputs, logic, write, diag	PLC Scan Cycle
36	Lateral attack traversal	Network Security	86	Total noise remains same	White Noise Filter
37	0.67	AM Modulation Index	87	500 bits	TDMA Allocation
38	Two copper pairs (Tx / Rx)	Four-Wire Circuit	88	Cross-modulation distortion	Nonlinear Mixing

PART V — Final Board Exam Cram Sheet

Must-Remember Rules & Shortcuts

- **PCM Bit Rule:** $+1$ bit $\implies +6$ dB SQNR.
- **FSPL Doubling Rule:** Doubling frequency adds $+6$ dB. Doubling distance adds $+6$ dB. Doubling both adds $+12$ dB.
- **Orthogonal Antenna Polarization:** Vertical wave to Horizontal antenna = 0 Watts received ($-\infty$ dB).
- **White Noise Shift:** Shifting center frequency with constant bandwidth produces **zero change** in total noise power.
- **First Stage Rule (Friis):** The noise figure of a receiver is dominated almost entirely by the **first stage LNA**.
- **FM Bessel Carrier Null:** Carrier vanishes ($J_0 = 0$) at $\beta \approx 2.4048, 5.52, 8.65$.
- **HDLC Bit Stuffing:** A '0' is stuffed after **5 consecutive 1s**. Receiver strips it out.
- **Public Key Encryption:** Encrypt with **recipient's public key** for confidentiality; sign with **sender's private key** for authentication.
- **Transmission Line Limits:** Short circuit: $\Gamma = -1$. Open circuit: $\Gamma = +1$. Matched load ($Z_L = Z_0$): $\Gamma = 0$ ($SWR = 1$).
- **Dish Beamwidth Shortcut:** $\theta_{3\text{dB}} \approx \dfrac{70\lambda}{D}$ degrees.

Common Traps to Avoid

- **High-Side vs. Low-Side LO:**
  - High-side injection: $f_{img} = f_{RF} + 2f_{IF}$.
  - Low-side injection: $f_{img} = f_{RF} - 2f_{IF}$.
- **E1 vs. T1 Channels:**
  - T1 = 24 voice channels = 1.544 Mbps.
  - E1 = 30 voice + 2 signaling/framing = 32 channels = 2.048 Mbps.
- **Carson's Bandwidth vs. Minimum Baseband:**
  - FM Bandwidth $= 2(\Delta f + f_m)$.
  - Baseband Nyquist Bandwidth $= R_b/2$.
- **Linear vs. Decibel Noise Figure:**
  - $T_e = 290$ K $\implies F = 1 + 290/290 = 2 \implies NF = 3$ dB.
  - $T_e = 0$ K $\implies F = 1 \implies NF = 0$ dB.
- **Class A vs. Class B Wiring:**
  - Class B: Radial branch, open fault disables downstream devices.
  - Class A: Closed return loop back to panel; survives open-circuit fault.
- **Even/Odd Signals:**
  - Even $\times$ Even = Even.
  - Odd $\times$ Odd = Even.
  - Even $\times$ Odd = Odd.

END OF COMPREHENSIVE ECE BOARD EXAM REVIEWER

ELECTRONICS ENGINEERING

BOARD EXAMINATION COMPREHENSIVE REVIEWER

PB EXAM 23 · 100 Complete Solved Examination Questions, Formulas, & Cram Sheet

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COMMUNICATIONS ENGINEERING, TELECOMMUNICATIONS, DATA NETWORKS, ANTENNAS & WAVE PROPAGATION, TRANSMISSION LINES, WAVEGUIDES, SATELLITE, FIBER OPTICS, & DSP

Document Specifications & Review Scope:

- **Comprehensive Coverage:** Questions 1 through 100 completely solved without omission.
- **Rigorous Numerical Verification:** Step-by-step arithmetic and formula derivations for all calculations (e.g., Q18, Q21, Q41, Q54, Q56, Q61, Q63, Q64, Q66, Q67, Q71, Q76, Q77, Q81, Q83, Q88, Q95, Q100).
- **Structured Analysis:** Every question features Given, Find, Formula, Step-by-Step Solution, Answer, Explanation, and Board Exam Tips.
- **Synthesized Summaries:** Part II Terms & Concepts, Part III Complete Formula Sheet, Part IV Quick Answer Key Table, and Part V Final Board Exam Cram Sheet.

PRC Electronics Engineering Licensure Examination Standard Edition

Professional Licensing Review Series

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1 Part I — Complete Solutions to Examination Questions

Question 1

Which cable type is used when extremely high data rates and long-distance transmission are required, such as in internet backbones?
ANSWER: Fiber optic

- **EXPLANATION:** Single-mode optical fiber offers terahertz-scale transmission bandwidth and low attenuation (≈ 0.2 dB/km at 1550 nm), making it immune to electromagnetic interference (EMI) and ideal for transoceanic and continental internet backbones. Copper lines suffer high attenuation and bandwidth limits over long distances.
- **BOARD EXAM TIP:** Long haul + Gigabits/Terabits = **Fiber Optic** (Single Mode).

Question 2

A hacker gains access to a BMS and turns off HVAC systems in a commercial building. Which measure could have prevented this?
ANSWER: Using strong passwords, network segmentation, and monitoring.

- **EXPLANATION:** Building Management Systems (BMS) often use industrial protocols (BACnet, Modbus) that lack inherent encryption. Isolating the operational technology (OT) network from the corporate IT network and public internet via VLANs and firewalls (segmentation), enforcing multi-factor authentication with strong passwords, and continuous intrusion monitoring prevents unauthorized access.
- **BOARD EXAM TIP:** OT/BMS security best practice: **Network Segmentation (VLANs/Firewalls) + Strong Authentication**.

Question 3

When coupling light from a fiber to a photodetector, the conversion efficiency depends on:
ANSWER: Alignment and spot size matching.

- **EXPLANATION:** Optical coupling efficiency η_c requires the emerging light cone and spot diameter from the fiber core to match the active photosensitive area of the photodetector (PIN or APD). Axial, radial, or angular misalignment causes light to miss the detector surface, producing insertion loss.
- **BOARD EXAM TIP:** Maximum fiber-to-detector coupling requires **spot size matching and spatial/angular alignment**.

Question 4

Which is a main trade-off when implementing a WAN instead of a LAN?
ANSWER: WAN has higher setup and maintenance costs due to long-distance connectivity.

- **EXPLANATION:** A Local Area Network (LAN) operates over privately owned infrastructure within a limited geographic area. A Wide Area Network (WAN) spans cities or countries, requiring leased telecommunications circuits (MPLS, dark fiber, satellite links) from public carriers, which increases capital and recurring operational costs.
- **BOARD EXAM TIP:** LAN = High speed, low cost, private ownership. WAN = Broad coverage, higher cost, reliance on third-party carriers.

Question 5

Why is the Presentation layer often referred to as the ”syntax layer” of the OSI model?
ANSWER: It manages how data is structured and encoded.

- **EXPLANATION:** Layer 6 (Presentation) translates data between application-level representations and standardized network formats (Abstract Syntax Notation One, ASN.1). It handles syntax translation (e.g., ASCII to EBCDIC), serialization, data compression, and encryption/decryption (SSL/TLS).
- **BOARD EXAM TIP:** Presentation Layer = **Syntax layer** (Translation, Compression, Encryption).

Question 6

In IoT devices deployed on rooftops or poles, which antenna is commonly used for omnidirectional short-range communication?
ANSWER: Monopole antenna (Vertical whip / ground-plane antenna)

- **EXPLANATION:** A vertical quarter-wave ($\lambda/4$) monopole antenna over a ground plane produces an omnidirectional donut-shaped radiation pattern in the azimuth plane (360° horizontal coverage). This enables consistent communication with surrounding field sensors regardless of their direction.
- **BOARD EXAM TIP:** Uniform 360° horizontal coverage = **Omnidirectional (Monopole / Dipole)**. Point-to-point = Directional (Yagi / Dish).

Question 7

Which of the following increases the bandwidth requirement of a PAM signal?
ANSWER: Increasing the highest frequency of the analog signal.

- **EXPLANATION:** According to the Nyquist Sampling Theorem, the minimum sampling frequency is $f_s \geq 2f_{\text{max}}$. Increasing the highest analog frequency component f_{max} forces a higher pulse repetition rate f_s. Generating narrower, more closely spaced pulses requires a wider channel bandwidth ($B \propto f_s$).
- **BOARD EXAM TIP:** Higher signal frequency $f_{\text{max}} \implies$ Higher sampling rate $f_s \implies$ **Increased channel bandwidth**.

Question 8

Where does the maximum VSWR occur on the Smith Chart?
ANSWER: At the outer edge of the chart ($|\Gamma| = 1$).

- **EXPLANATION:** The outer circular boundary of the Smith Chart represents $|\Gamma| = 1.0$ (complete reflection, where resistance $r = 0$). Because VSWR $= \frac{1+|\Gamma|}{1-|\Gamma|}$, as $|\Gamma| \to 1$, VSWR $\to \infty$. Points on the perimeter correspond to pure reactances (open circuit, short circuit, or lossless transmission lines).
- **BOARD EXAM TIP:** Smith Chart center: $|\Gamma| = 0 \implies$ VSWR $= 1.0$ (matched). Outer perimeter: $|\Gamma| = 1.0 \implies$ VSWR $= \infty$.

Question 9

In an antenna array, reducing the amplitude of the edge elements relative to the center elements results in:
ANSWER: Reduced side lobes (at the expense of a broader main beam).

- **EXPLANATION:** Amplitude tapering (such as binomial, Dolph-Tschebyscheff, or Taylor windowing) smooths the array excitation profile from center to edge. Tapering lowers side lobe levels (SLL), which reduces adjacent interference, but widens the half-power beamwidth (θ_{HPBW}) of the main lobe.
- **BOARD EXAM TIP:** Amplitude tapering: **Reduces side lobe levels, but broadens the main beam**. Uniform excitation produces the narrowest beam but highest side lobes (-13.3 dB).

Question 10

Starlink satellites operate in LEO at ~ 550 km. Which is the MAIN advantage of this altitude compared to GEO satellites?
ANSWER: Lower signal propagation delay (latency).

- **EXPLANATION:** Geostationary Earth Orbit (GEO) satellites reside at an altitude of $\approx 35,786$ km, yielding a round-trip propagation latency of $2 \times \frac{35{,}786 \text{ km}}{300{,}000 \text{ km/s}} \approx 240$–$280$ ms. Low Earth Orbit (LEO) at 550 km yields a one-way path delay of $\frac{550 \text{ km}}{300{,}000 \text{ km/s}} \approx 1.83$ ms (round-trip ≈ 20–40 ms including ground hops), making real-time interactive voice, video, and gaming practical.
- **BOARD EXAM TIP:** LEO advantage: **Low latency (≈ 25 ms vs > 250 ms in GEO)** and lower path loss.

Question 11

A VHF transmitter operates at 150 MHz. To maximize communication distance with a receiver, which of the following should be increased?
ANSWER: Antenna height

- **EXPLANATION:** In the VHF band (30–300 MHz), propagation occurs primarily via direct space waves (line of sight). The maximum radio horizon distance is given by $d \approx 4.12(\sqrt{h_t} + \sqrt{h_r})$ km. Increasing antenna elevation directly extends the geometric radio horizon, overcoming the Earth's curvature.
- **BOARD EXAM TIP:** For VHF/UHF line-of-sight systems, **increasing antenna height** is the most effective way to extend coverage.

Question 12

A telecom operator is designing a system where each user transmits on different frequency bands, avoiding adjacent-channel interference. Which multiple-access technique ensures this?
ANSWER: FDMA (Frequency Division Multiple Access)

- **EXPLANATION:** FDMA divides the total assigned spectrum into distinct, non-overlapping frequency channels separated by guard bands. Each active user is assigned a dedicated frequency band for the duration of the call, preventing mutual co-channel interference.
- **BOARD EXAM TIP:** FDMA = **Different frequency slots**. TDMA = **Different time slots**. CDMA = **Different orthogonal codes**.

Question 13

A microwave engineer wants to avoid higher-order modes. Which strategy is correct?
ANSWER: Use frequency just above TE_{10} cutoff ($f_{c,10} < f < 2f_{c,10}$).

- **EXPLANATION:** In a standard rectangular waveguide with broad dimension a and height $b = a/2$, the dominant mode is TE_{10} with cutoff frequency $f_{c,10} = \frac{c}{2a}$. The next higher-order modes (TE_{20} and TE_{01}) have cutoff frequencies at $2f_{c,10}$. Operating within the frequency window $1.25f_{c,10} < f < 1.9f_{c,10}$ ensures that only the dominant TE_{10} mode can propagate, preventing multimode dispersion and mode conversion.
- **BOARD EXAM TIP:** Single-mode waveguide transmission requires operating **above TE_{10} cutoff, but below the next mode cutoff ($2f_c$)**.

Question 14

A mobile radio receiver experiences frequent signal fading when passing near tall buildings, even though signal strength is high. Which superheterodyne issue is MOST likely involved?
ANSWER: Multipath-induced LO pulling (or dynamic multipath phase distortion).

- **EXPLANATION:** Reflections from surrounding urban structures produce rapid changes in the amplitude and phase of the received carrier. If mixer and RF-LO isolation is insufficient, dynamic swings in incoming signal level can pull the local oscillator (LO) frequency off its tuned center, degrading intermediate frequency (IF) tracking.
- **BOARD EXAM TIP:** Urban multipath causes rapid Rayleigh fading and can induce **LO frequency pulling** in poorly isolated receivers.

Question 15

In a client-server network, resources are controlled by a central server. Which of the following is TRUE for a P2P network?
ANSWER: Each device can share and request resources directly (acts as both client and server).

- **EXPLANATION:** In a Peer-to-Peer (P2P) network architecture, there is no centralized controller or dedicated server. Every interconnected workstation (peer) maintains equivalent privileges and can share storage, files, or processing power directly with other nodes.
- **BOARD EXAM TIP:** Client-Server = **Centralized control**. Peer-to-Peer (P2P) = **Decentralized, peers act as both client and server**.

Question 16

Engr. Federico Ramos needs wide coverage for IoT devices in a building. Which antenna choice balances coverage and gain?
ANSWER: Low-gain omnidirectional antenna

- **EXPLANATION:** Antenna gain and beamwidth have an inverse relationship. High-gain omnidirectional antennas narrow the vertical beamwidth into a thin disk, which can miss sensors located on floors above or below. A low-to-moderate gain (2–3 dBi) omnidirectional antenna maintains a broad vertical beam elevation, providing uniform coverage throughout a multi-story building.
- **BOARD EXAM TIP:** Broad spatial indoor coverage across varying elevations requires a **low-gain omnidirectional antenna**.

Question 17

The proportionality constant relating thermocouple voltage to temperature difference is called the:
ANSWER: Seebeck coefficient (α, thermoelectric sensitivity in μV/°C).
- **EXPLANATION:** The Seebeck effect generates an open-circuit voltage across a junction of two dissimilar metals in response to a temperature differential: $V = \alpha_{AB} \cdot (T_{\text{hot}} - T_{\text{cold}})$, where α_{AB} is the Seebeck coefficient.
- **BOARD EXAM TIP:** Thermocouple sensitivity constant = **Seebeck coefficient** ($V = \alpha \Delta T$).

Question 18

Two FIR filters are cascaded: The first is $y_1[n] = 0.3x[n] + 0.7x[n-1]$ and the second is $y[n] = 0.5y_1[n] + 0.5y_1[n-1]$. What is the overall difference equation?
ANSWER: $y[n] = 0.15x[n] + 0.5x[n-1] + 0.35x[n-2]$
- **GIVEN:** $H_1(z) = 0.3 + 0.7z^{-1}$, $H_2(z) = 0.5 + 0.5z^{-1}$.
- **FIND:** Overall difference equation $y[n]$.
- **FORMULA:** Cascaded system transfer function: $H(z) = H_1(z) \cdot H_2(z)$.
- **SOLUTION:**

$$H(z) = (0.3 + 0.7z^{-1})(0.5 + 0.5z^{-1})$$

$$H(z) = (0.3)(0.5) + (0.3)(0.5z^{-1}) + (0.7z^{-1})(0.5) + (0.7z^{-1})(0.5z^{-1})$$

$$H(z) = 0.15 + 0.15z^{-1} + 0.35z^{-1} + 0.35z^{-2} = 0.15 + 0.50z^{-1} + 0.35z^{-2}$$

Taking the inverse Z-transform:

$$y[n] = 0.15x[n] + 0.50x[n-1] + 0.35x[n-2]$$

- **EXPLANATION:** Cascading two linear filters convolves their impulse response sequences: $[0.3, 0.7] * [0.5, 0.5] = [0.15, 0.50, 0.35]$.
- **BOARD EXAM TIP:** In cascaded FIR filters, **multiply their Z-transforms** or **convolve their coefficient arrays**.

Question 19

An AM signal has a modulation index $m = 0.6$. What is true about this signal?
ANSWER: The AM signal is linearly modulated with no distortion.
- **EXPLANATION:** For standard AM (DSB-FC), operating in the range $0 < m \leq 1.0$ represents normal modulation. Because $m = 0.6 \leq 1.0$, the carrier envelope reproduces the baseband audio waveform without crossing zero volts, allowing clean demodulation using an envelope detector.
- **BOARD EXAM TIP:** $m \leq 1.0$ = **No distortion** (envelope intact). $m > 1.0$ = **Overmodulation** (envelope distortion and sideband splatter).

Question 20

To ensure continuous server communication, a data center implements multiple redundant paths between switches. Which principle is being applied?
ANSWER: Mesh redundancy (Fault tolerance)
- **EXPLANATION:** Providing multiple alternate physical links between switches eliminates single points of failure. In enterprise Ethernet networks, protocols like Rapid Spanning Tree Protocol (RSTP) or Multi-Chassis Link Aggregation (M-LAG) manage redundant paths, keeping links in standby to take over if an active connection fails.
- **BOARD EXAM TIP:** Eliminating single points of failure through alternate routing paths = **Redundancy**.

Question 21

An AM transmitter operates at a carrier frequency of 1.2 MHz and uses an audio signal of 8 kHz. Determine the frequencies of the sidebands.
ANSWER: 1.192 MHz and 1.208 MHz

- **GIVEN:** Carrier frequency $f_c = 1.2$ MHz $= 1200$ kHz, modulating frequency $f_m = 8$ kHz $= 0.008$ MHz.
- **FIND:** Lower Sideband (f_{LSB}) and Upper Sideband (f_{USB}).
- **FORMULA:** $f_{LSB} = f_c - f_m$ and $f_{USB} = f_c + f_m$.
- **SOLUTION:**

$$f_{LSB} = 1200 \text{ kHz} - 8 \text{ kHz} = 1192 \text{ kHz} = 1.192 \text{ MHz}$$

$$f_{USB} = 1200 \text{ kHz} + 8 \text{ kHz} = 1208 \text{ kHz} = 1.208 \text{ MHz}$$

- **EXPLANATION:** Amplitude modulation shifts baseband audio up to two symmetric sidebands placed $\pm f_m$ around the carrier.
- **BOARD EXAM TIP:** Sideband frequencies: $f_{USB} = f_c + f_m$, $f_{LSB} = f_c - f_m$. Bandwidth $= 2f_m = 16$ kHz.

Question 22

A user logs into a web application using public Wi-Fi. The developer forgot to mark the session cookie as `Secure`. What is the most likely risk?
ANSWER: The cookie can be intercepted over an unencrypted connection (Session hijacking via packet sniffing).

- **EXPLANATION:** The `Secure` attribute instructs the web browser to transmit the cookie only over encrypted HTTPS connections. If omitted, a browser that attempts an unencrypted HTTP request sends the session token in cleartext, allowing eavesdroppers on the open Wi-Fi network to intercept it and hijack the user's session.
- **BOARD EXAM TIP:** Cookies without the `Secure` flag transmit over unencrypted HTTP, leaving them vulnerable to **Session Hijacking**.

Question 23

A modern BMS uses real-time data from temperature and occupancy sensors to optimize HVAC operation. This approach primarily:
ANSWER: Reduces energy consumption while maintaining comfort.

- **EXPLANATION:** Occupancy-driven climate control adjusts temperature setpoints and airflow rates according to real-time building use. Conditioned air is directed to occupied zones while reducing conditioning in empty rooms, cutting electrical consumption while maintaining environmental comfort.
- **BOARD EXAM TIP:** Automated BMS controls **conserve energy while maintaining indoor comfort**.

Question 24

A rescue team uses portable FM transmitters with $\beta = 2$. A sudden need arises to extend range without increasing transmitter power. Which is the best situational approach?
ANSWER: Use high-gain directional antennas.

- **EXPLANATION:** Under the Friis Transmission Equation ($P_r = P_t G_t G_r \left[\frac{\lambda}{4\pi d}\right]^2$), received signal power increases with antenna gain (G_t or G_r). When transmitter battery power P_t is constrained, using a directional antenna (like a Yagi-Uda) focuses radiated energy along the line of sight, extending operational range.
- **BOARD EXAM TIP:** To extend RF range without increasing transmitter power: **Increase antenna gain or antenna height**.

Question 25

In Phase Shift Keying (PSK), digital information is emitted by:
ANSWER: Varying the phase of the carrier signal.

- **EXPLANATION:** PSK maps binary digits to discrete phase states of a sinusoidal carrier while holding amplitude and frequency constant. For example, BPSK shifts phase between $0°$ (logic 0) and $180°$ (logic 1).
- **BOARD EXAM TIP:** ASK varies **amplitude**; FSK varies **frequency**; PSK varies **phase**.

Question 26

What happens when a signal is sampled at a rate higher than the Nyquist frequency?
ANSWER: Aliasing reduces (spectral guard bands are established, simplifying reconstruction).

- **EXPLANATION:** The Nyquist criterion requires sampling at $f_s \geq 2f_{\text{max}}$. Oversampling ($f_s \gg 2f_{\text{max}}$) separates the periodic spectral replicas in the frequency domain with wide guard bands. This prevents aliasing and allows signal reconstruction using gentle, low-order anti-aliasing filters.
- **BOARD EXAM TIP:** Oversampling ($f_s > 2f_{\text{max}}$) **eliminates aliasing** and simplifies reconstruction filter design.

Question 27

In modern Ethernet networks, the function of a bridge is most commonly implemented by:
ANSWER: Switches (Multiport transparent bridges)

- **EXPLANATION:** A network bridge operates at Layer 2 (Data Link), segmenting a collision domain into two segments using software packet filtering. A modern network switch is essentially a high-density, hardware-accelerated multiport bridge that provides a dedicated collision domain on every port using an ASIC MAC address table.
- **BOARD EXAM TIP:** A modern Layer 2 switch is a **multiport bridge**.

Question 28

A transmission line has a characteristic impedance of $50\,\Omega$ and is terminated with a $50\,\Omega$ load. What happens to reflections?
ANSWER: Reflections are zero ($|\Gamma| = 0$).

- **GIVEN:** $Z_0 = 50\,\Omega$, $Z_L = 50\,\Omega$.
- **FORMULA:** Voltage reflection coefficient: $\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$.
- **SOLUTION:**

$$\Gamma = \frac{50 - 50}{50 + 50} = \frac{0}{100} = 0$$

- **EXPLANATION:** When load impedance matches line impedance ($Z_L = Z_0$), all incident power is absorbed by the load. There is no reflected wave, and VSWR $= 1.0$.
- **BOARD EXAM TIP:** Matched load: $\Gamma = 0 \implies$ VSWR $= 1.0 \implies$ **Zero reflections**.

Question 29

A system streams sensor data continuously at a constant rate to a monitoring server. Which transmission method is conceptually most suitable?
ANSWER: Synchronous transmission

- **EXPLANATION:** Asynchronous transmission uses start and stop framing bits around each byte, adding 20–25% overhead. Synchronous transmission groups data into continuous blocks synchronized to a shared clock, eliminating per-byte overhead and making it ideal for uninterrupted real-time streams.
- **BOARD EXAM TIP:** Steady, uninterrupted data flow = **Synchronous transmission**. Bursty, irregular traffic = **Asynchronous transmission**.

Question 30

Battery-powered IoT sensors transmit short data packets in a noisy urban environment. Which modulation minimizes errors without increasing power consumption?
ANSWER: FSK (Frequency Shift Keying)

- **EXPLANATION:** FSK operates with a constant envelope, allowing transmitters to use non-linear Class-C RF power amplifiers that maximize battery efficiency. In noisy urban channels, FSK resists amplitude variations and impulsive noise better than ASK.
- **BOARD EXAM TIP:** Low-power battery IoT links use **FSK** (e.g., LoRa, Bluetooth LE) for constant envelope amplifier efficiency and noise immunity.

Question 31

Which statement BEST summarizes the essential function of a radio receiver?
ANSWER: It captures electromagnetic waves and reconstructs the original information.
- **EXPLANATION:** A radio receiver uses an antenna to intercept electromagnetic waves, selects the tuned frequency, amplifies the microvolt-level signal, removes noise through filtering, and demodulates the carrier to recover the baseband message (voice, video, or data).
- **BOARD EXAM TIP:** Core receiver functions: **Reception → Selection → Amplification → Demodulation**.

Question 32

An FIR smoothing filter is implemented in a digital audio system using: $y[n] = \frac{1}{3}(x[n] + x[n-1] + x[n-2])$. Which is TRUE about this filter?
ANSWER: It is a moving average filter (3-point low-pass FIR filter).
- **EXPLANATION:** The filter computes the unweighted arithmetic mean of the current input sample and the two preceding samples. Moving average filters are finite impulse response (FIR) filters that smooth high-frequency noise while preserving low-frequency signal trends.
- **BOARD EXAM TIP:** Equal coefficient FIR filters without feedback are **moving average low-pass filters**.

Question 33

An email is sent from the laptop of Engr. Cosme Catimbang in Manila to a server in New York. Which OSI layer ensures the packet finds the correct path through intermediate networks?
ANSWER: Network Layer (Layer 3)
- **EXPLANATION:** The Network Layer (e.g., IPv4/IPv6) handles logical addressing and path determination across intermediate networks, routing packets across multiple hops from source to destination.
- **BOARD EXAM TIP:** End-to-end global routing = **Layer 3 (Network Layer)**.

Question 34

In RF safety standards, the Poynting vector is used to determine:
ANSWER: Power per unit area (Power density, W/m^2).
- **FORMULA:** Poynting vector: $\vec{S} = \vec{E} \times \vec{H}$ (W/m^2).
- **EXPLANATION:** The Poynting vector quantifies the magnitude and direction of electromagnetic energy flux per unit area. Regulatory limits for human exposure to RF fields (e.g., FCC/ICNIRP) are expressed as maximum permissible exposure (MPE) power densities (mW/cm^2 or W/m^2).
- **BOARD EXAM TIP:** Poynting vector $\vec{S} = \vec{E} \times \vec{H}$ measures **power density (W/m^2)**.

Question 35

Why does aliasing occur when a signal is undersampled?
ANSWER: High frequencies overlap with lower ones in the sampled spectrum.
- **EXPLANATION:** Sampling replicates the baseband spectrum at integer multiples of the sampling frequency: $X_s(f) = f_s \sum X(f - kf_s)$. If the sampling rate falls below the Nyquist rate ($f_s < 2f_{\text{max}}$), adjacent spectral copies overlap. High-frequency components fold over into the baseband spectrum, creating false lower-frequency alias tones that cannot be removed by filtering.
- **BOARD EXAM TIP:** Undersampling causes **spectral overlap**, folding high frequencies into the baseband.

Question 36

Which of the following signals is even?
ANSWER: $x(t) = \cos(2\pi t)$
- **EXPLANATION:** A signal is mathematically even if it satisfies $x(-t) = x(t)$ for all t. Because cosine is an even function ($\cos(-\theta) = \cos\theta$), $x(-t) = \cos(-2\pi t) = \cos(2\pi t) = x(t)$. Sine is an odd function ($\sin(-t) = -\sin t$).
- **BOARD EXAM TIP:** Cosine is **even** ($x(-t) = x(t)$); Sine is **odd** ($x(-t) = -x(t)$).

Question 37

Which synchronization type is MOST critical when recovering clock information from a self-clocking line code such as Manchester encoding?
ANSWER: Bit-level synchronization (Clock recovery)

- **EXPLANATION:** Manchester encoding embeds a guaranteed transition in the center of every bit cell (low-to-high for logic 1, high-to-low for logic 0). The receiver relies on bit-level synchronization to extract the carrier clock from these mid-bit transitions, keeping the receiver clock in phase without requiring a separate clock line.
- **BOARD EXAM TIP:** Mid-bit transitions in Manchester encoding provide **bit-level clock synchronization**.

Question 38

Which of the following limits the maximum data rate of a broadband system despite its large bandwidth?
ANSWER: Signal attenuation and noise (Signal-to-Noise Ratio, SNR).

- **EXPLANATION:** Under the Shannon-Hartley theorem ($C = B\log_2[1+\text{SNR}]$), channel capacity depends on both bandwidth B and signal-to-noise ratio. Even with wide bandwidth, signal attenuation and thermal noise limit the received SNR, restricting the maximum error-free data rate.
- **BOARD EXAM TIP:** Channel data rate is bounded by both **Bandwidth (B)** and **Signal-to-Noise Ratio (SNR)**.

Question 39

Which practice BEST ensures security and consistency in corporate VPN deployment?
ANSWER: Push standardized VPN settings and policies from a central server.

- **EXPLANATION:** Manual endpoint configuration invites configuration errors, weak encryption choices, and unpatched vulnerabilities. Centrally managing VPN profiles via mobile device management (MDM) or group policy ensures all endpoints enforce strong ciphers, required split-tunneling rules, and up-to-date certificates.
- **BOARD EXAM TIP:** Corporate network security relies on **centralized configuration management and policy enforcement**.

Question 40

For a given SNR, which digital modulation scheme is MOST robust to noise?
ANSWER: BPSK (Binary Phase Shift Keying)

- **EXPLANATION:** BPSK uses antipodal signaling, separating constellation points by a phase difference of 180° ($d = 2\sqrt{E_b}$). For a given bit energy-to-noise density (E_b/N_0), BPSK achieves the lowest bit error rate ($P_e = Q\left[\sqrt{2E_b/N_0}\right]$), providing 3 dB better noise immunity than coherent BFSK or orthogonal schemes.
- **BOARD EXAM TIP:** For a given SNR, **BPSK** provides the highest noise immunity and lowest bit error rate.

Question 41

A 300 W carrier is modulated simultaneously by two audio tones with indices 0.9 and 0.5. The wave is radiated through a $75\,\Omega$ antenna. Find the RMS antenna current.
ANSWER: 2.47 A

- **GIVEN:** Carrier power $P_c = 300$ W, modulation indices $m_1 = 0.9$, $m_2 = 0.5$, antenna resistance $R = 75\,\Omega$.
- **FIND:** Total RMS antenna current I_t.
- **FORMULA:** Total effective modulation index: $m_t = \sqrt{m_1^2 + m_2^2}$. Total transmitted power: $P_t = P_c\left(1 + \frac{m_t^2}{2}\right)$. Antenna current: $I_t = \sqrt{\frac{P_t}{R}}$.
- **SOLUTION:**

$$m_t^2 = m_1^2 + m_2^2 = (0.9)^2 + (0.5)^2 = 0.81 + 0.25 = 1.06$$

$$P_t = 300\left(1 + \frac{1.06}{2}\right) = 300(1 + 0.53) = 300(1.53) = 459 \text{ W}$$

$$I_t = \sqrt{\frac{P_t}{R}} = \sqrt{\frac{459 \text{ W}}{75\,\Omega}} = \sqrt{6.12} \approx 2.47386 \text{ A} \approx 2.47 \text{ A}$$

- **EXPLANATION:** Modulating with multiple tones sums their sideband powers. The resulting total power dissipated across the antenna resistance determines the total RMS antenna current.
- **BOARD EXAM TIP:** Multi-tone modulation: $m_t = \sqrt{m_1^2 + m_2^2 + \ldots}$, $P_t = P_c(1 + m_t^2/2)$, and $I_t = \sqrt{P_t/R}$.

Question 42

In CDMA, a user close to the base station may overpower a distant user's signal. What technique is used to mitigate this?
ANSWER: Power control (Fast closed-loop power control)
- **EXPLANATION:** Because all CDMA users transmit over the same frequency band simultaneously, a mobile close to the base station arrives with much higher power than a distant mobile if both transmit at equal levels. This **near-far problem** would allow the nearby user to overwhelm weaker signals. Fast closed-loop power control commands mobile transmitters to adjust power (up to 1500 times per second), ensuring all signals arrive at the base station with approximately equal power.
- **BOARD EXAM TIP:** The **near-far problem** in CDMA is solved by **fast closed-loop power control**.

Question 43

A communication system allows users to transmit messages alternately over the same channel using software-controlled turn-taking. What is the mode of transmission?
ANSWER: Half-duplex
- **EXPLANATION:** In half-duplex transmission, data travels in both directions over a shared circuit, but in only one direction at a time. Systems coordinate transmissions using token passing, push-to-talk (PTT), or line-turnaround protocols.
- **BOARD EXAM TIP:** Alternating bidirectional transmission = **Half-duplex**. Simultaneous bidirectional = **Full-duplex**.

Question 44

A factory uses IoT sensors and automated machinery connected to a local network. One sensor failure should not disrupt communication for other devices. What is the BEST network design choice?
ANSWER: Star topology with redundant links.
- **EXPLANATION:** In a star network, every sensor connects to a central switch via an independent point-to-point link. A severed cable or sensor failure isolates only that single node, leaving the rest of the factory network operational. Adding redundant uplink trunks prevents switch-level failures.
- **BOARD EXAM TIP:** A **Star topology** isolates individual node failures, preventing network-wide disruptions.

Question 45

Excel Review Center wants to transmit live video from multiple conference rooms over a shared digital link. Which multiplexing technique ensures low latency and synchronized delivery?
ANSWER: TDM (Time Division Multiplexing)
- **EXPLANATION:** TDM assigns dedicated, recurring time slots to each data stream within a repetitive frame. Because time slots are reserved and deterministic, streams avoid packet-queuing delays and jitter, ensuring synchronized delivery for real-time video.
- **BOARD EXAM TIP:** Deterministic, low-jitter delivery for shared digital channels = **Time Division Multiplexing (TDM)**.

Question 46

To control the speed of a DC motor using a PLC, which method is MOST appropriate?
ANSWER: PLC output connected to a PWM motor drive.
- **EXPLANATION:** PLCs cannot drive heavy motor current directly from their logic outputs. A digital high-speed pulse output from the PLC sends a Pulse Width Modulation (PWM) signal to an external H-bridge power driver. Varying the PWM duty cycle adjusts the average voltage applied to the motor armature, controlling its rotational speed.
- **BOARD EXAM TIP:** Motor speed control in industrial automation uses **PLC pulse outputs connected to a PWM drive**.

Question 47

Which feature gives digital communication an advantage over analog communication?
ANSWER: Error detection and correction (and noise-free signal regeneration).
- **EXPLANATION:** In analog communications, channel noise accumulates at every amplifier stage and cannot be separated from the signal. Digital communications can encode structured parity bits (block codes, Reed-Solomon, LDPC) that allow receivers to detect and correct transmission bit errors. Furthermore, regenerative repeaters clean and retime digital pulses, preventing noise accumulation over long distances.
- **BOARD EXAM TIP:** Digital advantages: **Error correction coding, regenerative repeaters, and digital encryption**.

Question 48

Which is TRUE in signal processing?
ANSWER: Convolution flips and shifts one signal ($y[n] = \sum x[k]h[n-k]$).
- **EXPLANATION:** The mathematical convolution operation involves four steps: (1) **Fold/Flip** the impulse response in time to obtain $h[-k]$, (2) **Shift** it by the index of interest n to yield $h[n-k]$, (3) **Multiply** by $x[k]$, and (4) **Sum** over all k.
- **BOARD EXAM TIP:** Convolution steps: **Flip $\rightarrow$ Shift $\rightarrow$ Multiply $\rightarrow$ Sum**.

Question 49

Frequency Shift Keying (FSK) is suitable for satellite links because:
ANSWER: It maintains constant amplitude for power amplifier efficiency.
- **EXPLANATION:** FSK signals have a constant envelope (peak-to-average power ratio PAPR $= 1$). This allows satellite transponder traveling wave tube amplifiers (TWTAs) and solid-state power amplifiers (SSPAs) to operate in their non-linear saturation region for maximum DC-to-RF power efficiency without introducing amplitude distortion or intermodulation products.
- **BOARD EXAM TIP:** Constant envelope modulations (FSK, MSK, QPSK) allow **satellite amplifiers to operate at saturation for peak efficiency**.

Question 50

An SSB signal is transmitted instead of DSB-AM. What is the impact on channel bandwidth?
ANSWER: Bandwidth is halved ($B_{\text{SSB}} = f_m = \frac{1}{2}B_{\text{DSB}}$).
- **EXPLANATION:** Conventional double-sideband AM transmits both upper and lower sidebands, occupying $B_{\text{DSB}} = 2f_m$. Because both sidebands carry identical information, single-sideband (SSB) modulation filters out one sideband and suppresses the carrier, cutting the required bandwidth to $B_{\text{SSB}} = f_m$.
- **BOARD EXAM TIP:** SSB requires **half the bandwidth** of conventional double-sideband AM.

Question 51

Doubling the antenna gain (linear scale) while keeping transmitter power constant:
ANSWER: Doubles ERP (Effective Radiated Power).
- **FORMULA:** ERP $= P_t \times G_{\text{dipole}}$ (or EIRP $= P_t \times G_{\text{iso}}$).
- **EXPLANATION:** Radiated power is directly proportional to antenna gain. Doubling linear gain ($g' = 2g$, a $+3$ dB increase) with fixed input power doubles the effective radiated power along the main beam axis.
- **BOARD EXAM TIP:** ERP $\propto G_{\text{antenna}}$. Doubling antenna gain **doubles ERP ($+3$ dB)**.

Question 52

Which of the following is an example of an odd signal?
ANSWER: $x(t) = \sin(t)$
- **EXPLANATION:** A signal is odd if it satisfies the symmetry condition $x(-t) = -x(t)$ for all t. Because sine satisfies $\sin(-t) = -\sin(t)$, it is an odd signal. Cosine is an even signal ($\cos[-t] = \cos t$).
- **BOARD EXAM TIP:** Odd symmetry: $x(-t) = -x(t)$ (e.g., $\sin t, t^3$). Even symmetry: $x(-t) = x(t)$ (e.g., $\cos t, t^2$).

Question 53

A web server needs to serve hundreds of simultaneous clients efficiently. Which UNIX mechanism BEST supports this scenario?
ANSWER: Use threads or non-blocking I/O (Asynchronous event-driven I/O, such as `epoll`).

- **EXPLANATION:** Spawning a heavy operating system process (via `fork()`) for every incoming connection consumes substantial memory and incurs high context-switching overhead. High-performance servers use asynchronous, non-blocking event loops (like `epoll` in Linux or `kqueue` in BSD) and worker threads to handle thousands of concurrent client connections over a small set of threads.
- **BOARD EXAM TIP:** High-concurrency network servers use **non-blocking I/O multiplexing (`epoll`) and multithreading**.

Question 54

The free-space path loss (FSPL) in dB can be expressed as:
ANSWER: $20\log_{10}(d) + 20\log_{10}(f) + 32.44$ (with d in km and f in MHz).

- **FORMULA:** Friis transmission loss:

$$\text{FSPL} = \left(\frac{4\pi df}{c}\right)^2$$

In decibels with distance d in kilometers and frequency f in megahertz:

$$\text{FSPL}_{\text{dB}} = 20\log_{10}(d) + 20\log_{10}(f) + 20\log_{10}\left(\frac{4\pi \times 10^3 \times 10^6}{3 \times 10^8}\right)$$

$$20\log_{10}\left(\frac{4\pi \times 10^9}{3 \times 10^8}\right) = 20\log_{10}\left(\frac{40\pi}{3}\right) = 20\log_{10}(41.8879) \approx 32.44 \text{ dB}$$

$$\text{FSPL}_{\text{dB}} = 32.44 + 20\log_{10}(d_{\text{km}}) + 20\log_{10}(f_{\text{MHz}})$$

- **BOARD EXAM TIP:** Standard path loss formula: $\text{FSPL} = 32.44 + 20\log(d_{\text{km}}) + 20\log(f_{\text{MHz}})$.

Question 55

Which modulation technique is MOST suitable for achieving high data rates over a fixed bandwidth?
ANSWER: Quadrature Amplitude Modulation (QAM)

- **EXPLANATION:** QAM modulates both amplitude and phase, packing $n = \log_2 M$ bits into each symbol (e.g., 64-QAM delivers 6 bps/Hz; 256-QAM delivers 8 bps/Hz). This provides higher spectral efficiency than pure phase or frequency modulation (BPSK, FSK).
- **BOARD EXAM TIP:** Highest spectral efficiency in bandwidth-limited channels = **QAM**.

Question 56

For an N-bit uniform PCM system, the approximate signal-to-quantization-noise ratio (SQNR) in dB is:
ANSWER: $6.02N + 1.76$ dB

- **FORMULA:** For a full-scale sinusoid quantized by an N-bit uniform quantizer:

$$\text{SQNR} = \frac{P_{\text{signal}}}{P_{\text{noise}}} = \frac{V_p^2/2}{\Delta^2/12} = \frac{V_p^2/2}{(2V_p/2^N)^2/12} = 1.5 \times 2^{2N}$$

$$\text{SQNR}_{\text{dB}} = 10\log_{10}(1.5) + 10\log_{10}(2^{2N}) = 1.76 + 20N\log_{10}(2) \approx 6.02N + 1.76 \text{ dB}$$

- **EXPLANATION:** Each added quantizer bit (N) doubles the number of quantization levels (2^N), halving the step size Δ and adding ≈ 6 dB of dynamic range.
- **BOARD EXAM TIP:** The "6-dB rule": Each added PCM bit increases SQNR by **≈ 6.02 dB** (SQNR $= 6.02N + 1.76$).

Question 57

In a bus topology, why is terminator placement critical?
ANSWER: To prevent signal reflections that cause data collisions and frame corruption.
- **EXPLANATION:** An unterminated cable end acts as an open circuit ($Z_L = \infty$). When an electrical pulse reaches the open boundary, it reflects back with a reflection coefficient of $\Gamma = +1$. The reflected pulse collides with subsequent forward transmissions, corrupting frame data. Placing a resistor matching the cable's characteristic impedance ($Z_L = Z_0$, e.g., $50\,\Omega$ in 10Base2) absorbs the signal without reflection.
- **BOARD EXAM TIP:** Bus terminators match Z_0 to **absorb signals and eliminate reflections**.

Question 58

In Systems Network Architecture (SNA), what is the main difference between a Physical Unit (PU) and a Logical Unit (LU)?
ANSWER: A PU manages physical hardware node control and routing; an LU provides application-layer communication ports and session endpoints for end users.
- **EXPLANATION:** In IBM's SNA:
  - **Physical Unit (PU):** Represents the physical device or controller hardware (e.g., PU4 is a communications controller, PU5 is a host mainframe), managing physical link resources and telemetry.
  - **Logical Unit (LU):** Represents a software port or logical access point through which end users and applications establish communications sessions (e.g., LU2 for 3270 terminal sessions, LU6.2 for peer-to-peer program sessions).
- **BOARD EXAM TIP:** In SNA: **PU = Hardware node manager**. **LU = Application/user session endpoint**.

Question 59

For connecting high-speed servers in a data center, four-wire differential signaling is preferred because:
ANSWER: It reduces electromagnetic interference (EMI) and provides common-mode noise rejection.
- **EXPLANATION:** Differential signaling transmits equal and opposite voltages ($+V$ and $-V$) along a balanced twisted pair. External electromagnetic noise couples into both conductors equally (common-mode noise). A differential receiver measures the voltage difference: $V_{\text{diff}} = (V_1 + V_{\text{noise}}) - (V_2 + V_{\text{noise}}) = V_1 - V_2$, cancelling the coupled noise while doubling signal amplitude. The equal and opposite currents also minimize radiated EMI.
- **BOARD EXAM TIP:** Differential balanced pairs cancel **common-mode noise** and reduce **radiated EMI**.

Question 60

Engr. Pio Cuervo wants to upgrade an analog telephone network to PCM while preserving speech quality with limited bandwidth. What is the most practical solution?
ANSWER: Use 8-bit PCM with companding (μ-law or A-law).
- **EXPLANATION:** Standard voice telephony samples speech at $f_s = 8$ kHz. Using uniform 8-bit quantization yields poor dynamic range and low signal-to-quantization-noise ratios (SQNR) for quiet speech. Companding (μ-255 in North America/Philippines, A-law in Europe) non-linearly compresses the dynamic range before encoding, providing 12-bit audio performance using only 8 bits per sample (64 kbps per channel).
- **BOARD EXAM TIP:** Standard digital voice telephony: **8 kHz sampling $\times$ 8-bit companded PCM $= 64$ kbps**.

Question 61

A transmitter and receiver are 450 km apart. The D-layer is at 70 km. Find the critical angle (angle with vertical) for single-hop reflection.
ANSWER: $18°$ (or elevation takeoff angle $\approx 17.3°$–$18°$).
- **GIVEN:** Ground separation $D = 450$ km, reflection layer height $h = 70$ km.
- **FIND:** Angle with the vertical (θ_v) or elevation angle (α).
- **FORMULA:** By flat-earth right triangle geometry, single-hop ground reflection bisects the baseline:

$$\tan\theta_v = \frac{D/2}{h} = \frac{225\ \text{km}}{70\ \text{km}} \approx 3.214$$

$$\theta_v = \arctan(3.214) \approx 72.7^\circ \text{ with the vertical}$$

The angle with the horizontal ground (takeoff angle) is:

$$\alpha = 90^\circ - 72.7^\circ \approx 17.3^\circ \approx 18^\circ$$

- **EXPLANATION:** In PRC exam nomenclature, "critical angle" in sky-wave propagation questions often refers to the **elevation takeoff angle** above the horizontal ground plane: $\alpha \approx 18^\circ$.
- **BOARD EXAM TIP:** In sky-wave geometry: $\tan(\text{Elevation}) = \frac{h}{D/2} = \frac{70}{225} \approx 0.311 \implies \alpha \approx 17.3^\circ \approx 18^\circ$.

Question 62

In a digital communication system, analog audio is converted into digital form using:
ANSWER: Sampling and quantization (Pulse Code Modulation, PCM).

- **EXPLANATION:** Digitization requires: (1) **Sampling** (discretizing continuous time into sample pulses at $f_s \geq 2f_{\text{max}}$) and (2) **Quantization** (mapping continuous pulse amplitudes to discrete binary-encoded integer levels).
- **BOARD EXAM TIP:** Analog-to-digital conversion requires **Sampling in time** and **Quantization in amplitude**.

Question 63

Two independent sources X and Y have entropies $H(X) = 1.5$ bits and $H(Y) = 2.0$ bits. What is the joint entropy $H(X, Y)$?
ANSWER: 3.5 bits

- **GIVEN:** $H(X) = 1.5$ bits, $H(Y) = 2.0$ bits, X and Y are statistically independent.
- **FIND:** Joint entropy $H(X, Y)$.
- **FORMULA:** For statistically independent random variables: $H(X, Y) = H(X) + H(Y)$.
- **SOLUTION:**

$$H(X, Y) = 1.5 \text{ bits} + 2.0 \text{ bits} = 3.5 \text{ bits}$$

- **EXPLANATION:** Joint entropy measures the total average information needed to specify the states of both sources. When sources are independent, their information contents add directly.
- **BOARD EXAM TIP:** If sources are independent: **Joint Entropy is the direct sum of individual entropies** ($H(X, Y) = H(X) + H(Y)$).

Question 64

A binary coherent PSK system operates with a bit energy $E_b = 2 \times 10^{-19}$ J and noise spectral density $N_0 = 10^{-20}$ J. Calculate the approximate bit error probability P_e.
ANSWER: 1.3×10^{-10}

- **GIVEN:** $E_b = 2 \times 10^{-19}$ J, $N_0 = 10^{-20}$ J $= 1 \times 10^{-20}$ W/Hz.
- **FIND:** Bit error probability P_e.
- **FORMULA:** Coherent BPSK bit error rate:

$$P_e = Q\left(\sqrt{\frac{2E_b}{N_0}}\right) = \frac{1}{2}\text{erfc}\left(\sqrt{\frac{E_b}{N_0}}\right)$$

- **SOLUTION:**

$$\frac{2E_b}{N_0} = \frac{2(2 \times 10^{-19})}{10^{-20}} = \frac{4 \times 10^{-19}}{10^{-20}} = 40$$

$$\text{Argument} = \sqrt{40} \approx 6.3245$$

Using the Gaussian Q-function asymptotic approximation $Q(x) \approx \frac{1}{x\sqrt{2\pi}} e^{-x^2/2}$:

$$Q(6.3245) \approx \frac{1}{6.3245\sqrt{2\pi}} e^{-40/2} = \frac{1}{15.853} e^{-20} = 0.06308 \times (2.061 \times 10^{-9}) \approx 1.3 \times 10^{-10}$$

- **EXPLANATION:** An SNR of $E_b/N_0 = 20$ (13 dB) places the operating point well into the steep tail of the Gaussian error curve, yielding a very low bit error rate (1.3×10^{-10}).
- **BOARD EXAM TIP:** BPSK error probability: $P_e = Q(\sqrt{2E_b/N_0})$.

Question 65

An AM transmitter is operating with $m > 1$ (overmodulation). How does overmodulation impact channel bandwidth usage?
ANSWER: Bandwidth increases slightly (causes sideband splatter).
- **EXPLANATION:** When an AM wave is overmodulated ($m > 1$), the envelope crosses zero volts and gets clipped, creating sharp corners in the RF waveform. These non-linear discontinuities generate high-order harmonic sidebands ($f_c \pm 2f_m, f_c \pm 3f_m, \ldots$) that spill outside the nominal $2f_m$ passband, causing adjacent channel interference known as **splatter**.
- **BOARD EXAM TIP:** Overmodulation ($m > 1$) causes **harmonic distortion, envelope clipping, and adjacent-channel splatter**.

Question 66

If a small office combines two BRI lines, the maximum user data rate achievable is:
ANSWER: 256 kbps
- **GIVEN:** 2 ISDN Basic Rate Interface (BRI) lines.
- **FIND:** Maximum aggregate user data rate.
- **FORMULA:** Each ISDN BRI circuit consists of $2B + D$, where each Bearer (B) channel carries 64 kbps of user payload:

$$\text{User rate per BRI} = 2 \times 64 \text{ kbps} = 128 \text{ kbps}$$

- **SOLUTION:**

$$\text{Combined rate (2 BRIs)} = 2 \times 128 \text{ kbps} = 4 \times 64 \text{ kbps} = 256 \text{ kbps}$$

- **EXPLANATION:** Using Multilink PPP (MLPPP) bonding, the four independent 64 kbps B channels are aggregated into a single logical pipe delivering 256 kbps of throughput (D channels handle out-of-band signaling).
- **BOARD EXAM TIP:** One BRI = Two 64 kbps B-channels (128 kbps). Two BRIs = **Four B-channels (256 kbps)**.

Question 67

An amplifier has an equivalent noise temperature of $T_e = 75$ K. If the reference temperature is $T_0 = 290$ K, calculate the noise factor F.
ANSWER: 1.26 (or 1.0 dB)
- **GIVEN:** $T_e = 75$ K, reference temperature $T_0 = 290$ K.
- **FIND:** Noise factor F (linear ratio).
- **FORMULA:** $T_e = T_0(F-1) \implies F = 1 + \frac{T_e}{T_0}$.
- **SOLUTION:**

$$F = 1 + \frac{75 \text{ K}}{290 \text{ K}} = 1 + 0.2586 = 1.2586 \approx 1.26$$

In decibels:

$$NF = 10 \log_{10}(1.2586) \approx 1.0 \text{ dB}$$

- **BOARD EXAM TIP:** Noise factor: $F = 1 + \frac{T_e}{T_0}$. Noise temperature: $T_e = T_0(F-1)$.

Question 68

If the reflection coefficient (Γ) is positive, this implies that:
ANSWER: The reflected voltage is in phase with the incident voltage ($0°$ phase shift, $Z_L > Z_0$).
- **FORMULA:** Voltage reflection coefficient: $\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$.
- **EXPLANATION:** For purely resistive terminations:
  - If $Z_L > Z_0$: $\Gamma > 0$ (positive real number, angle $0°$), meaning the reflected voltage wave is in phase with the incident wave.
  - If $Z_L < Z_0$: $\Gamma < 0$ (negative real number, angle $180°$), meaning the reflected voltage is inverted ($180°$ out of phase).
  - If $Z_L = Z_0$: $\Gamma = 0$ (no reflection).
- **BOARD EXAM TIP:** Positive Γ means **$Z_L > Z_0$** and reflected voltage is **in phase ($0°$)** with the incident wave.

Question 69

A signal consists of three sinusoids at different frequencies. According to Parseval's theorem, the total signal power is:
ANSWER: The sum of the powers of all three sinusoids.
- **FORMULA:** If $x(t) = A_1 \cos(\omega_1 t) + A_2 \cos(\omega_2 t) + A_3 \cos(\omega_3 t)$ with $\omega_1 \neq \omega_2 \neq \omega_3$:

$$P_{\text{total}} = \frac{A_1^2}{2} + \frac{A_2^2}{2} + \frac{A_3^2}{2}$$

- **EXPLANATION:** Parseval's theorem states that energy and power are conserved across domains. Because sinusoids of distinct frequencies are mutually orthogonal over a common period, all cross-product terms integrate to zero, meaning total power is the direct sum of the powers of the individual harmonics.
- **BOARD EXAM TIP:** Total power of independent sinusoids = **Sum of individual powers**.

Question 70

Why is a multiplexer critical in modern communication systems?
ANSWER: It allows multiple streams to share a single transmission medium, reducing infrastructure costs.
- **EXPLANATION:** Without multiplexing, every communication channel would require a dedicated physical cable or RF carrier. Multiplexers aggregate multiple lower-speed channels into a single high-capacity link (using TDM, FDM, or WDM), maximizing media utilization and reducing deployment costs.
- **BOARD EXAM TIP:** Multiplexers combine multiple channels onto a **shared physical transmission link**.

Question 71

In free space, if the distance from the transmitter triples, the electric field strength becomes:
ANSWER: $1/3$ of original
- **FORMULA:** Electric field strength in the radiation far-field:

$$E = \frac{\sqrt{30 P_t G_t}}{d} \propto \frac{1}{d}$$

  Power density:

$$\mathcal{P} = \frac{E^2}{\eta_0} = \frac{P_t G_t}{4\pi d^2} \propto \frac{1}{d^2}$$

- **SOLUTION:** When distance triples ($d' = 3d$):

$$E' = \frac{1}{3} E$$

  Power density decreases to $\left(\frac{1}{3}\right)^2 = \frac{1}{9}$ of its original value.
- **EXPLANATION:** Electric field strength scales inversely with distance ($1/d$), while power density follows the inverse-square law ($1/d^2$).
- **BOARD EXAM TIP:** Electric field $E \propto \frac{1}{d}$ (tripling distance yields **$1/3$**). Power density $\mathcal{P} \propto \frac{1}{d^2}$ (tripling distance yields **$1/9$**).

Question 72

An FM radio station needs to cover a larger city. To compensate for increased distance and maintain adequate received signals, what must the electronics engineers do?
ANSWER: Increase transmitter power (and/or raise antenna height).
- **EXPLANATION:** Expanding broadcast coverage into distant fringe areas requires boosting the Effective Radiated Power (ERP) by increasing transmitter power output (P_t), installing higher-gain antenna arrays, or raising the transmitting antenna above surrounding terrain.
- **BOARD EXAM TIP:** Extending FM coverage requires **increasing transmitter power output** and/or **raising antenna height**.

Question 73

Why do LTE and 5G networks use adaptive QAM (Adaptive Modulation and Coding, AMC)?
ANSWER: To balance throughput and reliability depending on the user's channel conditions.

- **EXPLANATION:** Users close to a base station with high SNR use high-order modulations (64-QAM or 256-QAM) to maximize data throughput. Users near the cell edge experiencing low SNR and high interference dynamically downshift to robust lower-order modulations (QPSK or 16-QAM) to prevent packet drops and maintain link connectivity.
- **BOARD EXAM TIP:** Adaptive Modulation and Coding (AMC) **maximizes data throughput for high SNR** while **maintaining link reliability for low SNR**.

Question 74

A listener hears audio level "pumping" (breathing noise) during a radio broadcast. What is the likely cause?
ANSWER: AGC time constant is too short (release/recovery time is too fast).

- **EXPLANATION:** Automatic Gain Control circuits use an RC filter to smooth the gain-control voltage. If the discharge time constant (RC) is too short, the AGC circuit tracks rapid speech pauses instead of the average carrier envelope. The amplifier boosts gain during pauses (amplifying background noise) and drops gain when speech resumes, producing audible gain "pumping."
- **BOARD EXAM TIP:** Audio pumping in receivers occurs when the **AGC recovery time constant is too short**.

Question 75

Which multiplexing method is easier with digital signals?
ANSWER: Time Division Multiplexing (TDM)

- **EXPLANATION:** Digital signals consist of discrete binary bits and words that can be stored in shift registers and buffers, making them straightforward to interleave into time slots using digital logic gates. Frequency Division Multiplexing (FDM) requires analog balanced mixers, local oscillators, and steep bandpass filters.
- **BOARD EXAM TIP:** Digital signals are natively multiplexed using **Time Division Multiplexing (TDM)**.

Question 76

The transfer function $H(z) = K\frac{1-0.5z^{-1}}{1-0.8z^{-1}}$ has:
ANSWER: One zero at $z = 0.5$ and one pole at $z = 0.8$.

- **GIVEN:** $H(z) = K\frac{1-0.5z^{-1}}{1-0.8z^{-1}}$.
- **FIND:** Locations of poles and zeros in the complex z-plane.
- **FORMULA:** Multiply numerator and denominator by z:

$$H(z) = K\frac{z-0.5}{z-0.8}$$

- **SOLUTION:**
  - Zeros are roots of the numerator: $z - 0.5 = 0 \implies z = 0.5$.
  - Poles are roots of the denominator: $z - 0.8 = 0 \implies z = 0.8$.
- **EXPLANATION:** Because the single pole ($z = 0.8$) has magnitude $|0.8| < 1$, it lies inside the unit circle, making this a stable first-order causal digital filter.
- **BOARD EXAM TIP:** Multiply by z to find roots: $1 - az^{-1} = 0 \implies z = a$.

Question 77

An engineer is designing a helical antenna with turn spacing $S = 4$ cm for axial-mode operation. What is the approximate operating frequency?
ANSWER: 1.875 GHz

- **GIVEN:** Turn spacing $S = 4$ cm $= 0.04$ m.
- **FIND:** Operating frequency f.
- **FORMULA:** In a standard axial-mode helical antenna, pitch spacing is typically related to wavelength by $S \approx 0.25\lambda$ ($\lambda \approx 4S$), and circumference $C \approx \lambda$.

- **SOLUTION:**

$$\lambda = 4 \times S = 4 \times 0.04 \text{ m} = 0.16 \text{ m}$$

$$f = \frac{c}{\lambda} = \frac{3 \times 10^8 \text{ m/s}}{0.16 \text{ m}} = 1.875 \times 10^9 \text{ Hz} = 1.875 \text{ GHz}$$

- **BOARD EXAM TIP:** Axial-mode helix: Circumference $C \approx \lambda$, Turn spacing $S \approx 0.25\lambda \implies \lambda = 4S$.

Question 78

An electronics engineer would like to improve received power at a fixed distance without changing transmit power. A practical way to do this is to:
ANSWER: Use a higher-gain receiver antenna.

- **FORMULA:** Friis transmission formula: $P_r = P_t G_t G_r \left(\frac{\lambda}{4\pi d}\right)^2$.
- **EXPLANATION:** Received power is directly proportional to receiving antenna gain (G_r). Replacing a low-gain dipole (2.15 dBi) with a high-gain directional antenna (such as a Yagi or parabolic dish) increases effective aperture and captures more radiated power.
- **BOARD EXAM TIP:** Increase received power without altering transmitter: **Increase receiver antenna gain (G_r)**.

Question 79

In a linearly polarized microwave wave, the Poynting vector:
ANSWER: Remains in the direction of wave propagation.

- **FORMULA:** $\vec{S} = \vec{E} \times \vec{H}$.
- **EXPLANATION:** The electric field ($\vec{E}$) and magnetic field ($\vec{H}$) oscillate in orthogonal transverse planes. Their cross product ($\vec{E} \times \vec{H}$) points along the line of wave propagation ($\hat{k}$), directing power flow along the propagation axis.
- **BOARD EXAM TIP:** The Poynting vector $\vec{S} = \vec{E} \times \vec{H}$ points **in the direction of wave propagation**.

Question 80

If a fire is detected in a building, how should the BMS manage the HVAC system to prevent smoke spread?
ANSWER: Shut down or isolate air handling units to prevent smoke circulation (and activate smoke control/stairwell pressurization fans).

- **EXPLANATION:** During a fire, operating supply air fans circulate toxic smoke and supply fresh oxygen that feeds the flames. The BMS life-safety system immediately cuts power to general air handling units (AHUs), closes fire dampers, and energizes dedicated smoke-exhaust fans to pressurize stairwells for evacuation.
- **BOARD EXAM TIP:** Fire emergency sequence: **Shut down AHUs and close fire dampers to prevent smoke migration**.

Question 81

A rectangular waveguide has a cutoff frequency $f_c = 6$ GHz. A signal of 10 GHz is transmitted through a 30-meter-long waveguide. What is the approximate group velocity of the signal?
ANSWER: 2.4×10^8 m/s

- **GIVEN:** Cutoff frequency $f_c = 6$ GHz, operating frequency $f = 10$ GHz, speed of light $c = 3 \times 10^8$ m/s.
- **FIND:** Group velocity v_g.
- **FORMULA:** Group velocity in a hollow rectangular waveguide:

$$v_g = c\sqrt{1 - \left(\frac{f_c}{f}\right)^2}$$

  (Phase velocity is $v_p = \frac{c}{\sqrt{1-(f_c/f)^2}}$, and $v_p \cdot v_g = c^2$).
- **SOLUTION:**

$$\frac{f_c}{f} = \frac{6 \text{ GHz}}{10 \text{ GHz}} = 0.6$$

$$\left(\frac{f_c}{f}\right)^2 = (0.6)^2 = 0.36$$

$$\sqrt{1-0.36} = \sqrt{0.64} = 0.8$$

$$v_g = c \times 0.8 = (3 \times 10^8 \text{ m/s}) \times 0.8 = 2.4 \times 10^8 \text{ m/s}$$

- **EXPLANATION:** Group velocity represents the speed at which signal energy and information propagate down the guide. It is always less than the speed of light ($v_g < c$), while phase velocity exceeds c.
- **BOARD EXAM TIP:** Waveguide velocities: $v_g = c\sqrt{1-(f_c/f)^2}$, $v_p = \frac{c}{\sqrt{1-(f_c/f)^2}}$, and **$v_p v_g = c^2$**.

Question 82

An IoT sensor sends temperature data over a radio link. Which layer converts the digital sensor readings into RF signals?
ANSWER: Physical Layer (Layer 1)
- **EXPLANATION:** The Physical Layer handles the hardware transmission of raw bit streams. It manages RF frequency synthesis, digital modulation (e.g., FSK, OOK, PSK), signal amplification, and electromagnetic wave generation at the antenna.
- **BOARD EXAM TIP:** Layer 1 (Physical Layer) = **Hardware modulation, bit transmission, and RF signal generation**.

Question 83

A communication system has an input SNR of 20 dB and a receiver noise figure of 6 dB. What is the output SNR?
ANSWER: 14 dB
- **GIVEN:** $\text{SNR}_{\text{in}} = 20$ dB, $\text{NF} = 6$ dB.
- **FIND:** Output signal-to-noise ratio SNR_{out}.
- **FORMULA:** Noise Figure: $\text{NF}_{\text{dB}} = \text{SNR}_{\text{in(dB)}} - \text{SNR}_{\text{out(dB)}}$.

$$\text{SNR}_{\text{out(dB)}} = \text{SNR}_{\text{in(dB)}} - \text{NF}_{\text{dB}}$$

- **SOLUTION:**

$$\text{SNR}_{\text{out}} = 20 \text{ dB} - 6 \text{ dB} = 14 \text{ dB}$$

- **EXPLANATION:** Noise figure measures the degradation in signal-to-noise ratio introduced by internal components within the receiver.
- **BOARD EXAM TIP:** Receiver output SNR: $\text{SNR}_{\text{out}} = \text{SNR}_{\text{in}} - \text{NF}$.

Question 84

What is the main purpose of twisting pairs of wires in UTP cables?
ANSWER: Reduce crosstalk and electromagnetic interference (EMI).
- **EXPLANATION:** Twisting balanced conductors alternates their physical exposure to external magnetic fields and adjacent wire pairs. Induced interference cancels out across successive $180°$ twists, while magnetic fields generated by equal and opposite currents cancel externally, minimizing crosstalk.
- **BOARD EXAM TIP:** Twisting pairs in UTP **cancels differential crosstalk and external EMI**.

Question 85

Which type of optical fiber has a core with a uniform refractive index and suffers from higher modal dispersion?
ANSWER: Step-index multimode fiber
- **EXPLANATION:** In a step-index fiber, the core has a uniform refractive index n_1 that drops abruptly to n_2 at the cladding interface. Different propagation modes travel along paths of varying lengths (axial rays vs zigzagging high-order rays), causing light pulses to broaden in time (modal dispersion). Graded-index fibers vary the core index parabolically to equalize mode transit times, reducing modal dispersion.
- **BOARD EXAM TIP:** Highest modal dispersion = **Step-Index Multimode**. Lowest dispersion = **Single-Mode Fiber** (no modal dispersion).

Question 86

Which statement BEST explains the bandwidth requirement of SSB compared to AM?
ANSWER: SSB requires only half the bandwidth of AM ($B_{\text{SSB}} = f_m$, whereas $B_{\text{AM}} = 2f_m$).
- **EXPLANATION:** Conventional AM generates upper and lower sidebands that mirror the message spectrum around the carrier. Single-sideband (SSB) transmits only one sideband, halving the required channel bandwidth.
- **BOARD EXAM TIP:** Channel bandwidth: **$B_{\text{SSB}} = \frac{1}{2}B_{\text{AM}}$**.

Question 87

In a temperature control loop with multiple sensors and actuators, which PLC language is most efficient for handling analog signal processing?
ANSWER: Function Block Diagram (FBD)
- **EXPLANATION:** Under the IEC 61131-3 standard, Function Block Diagram (FBD) models continuous signal flow using graphic processing blocks (PID controllers, filters, scalers, comparators). This makes it more practical for analog process control loops than discrete Ladder Diagram (LD).
- **BOARD EXAM TIP:** Continuous analog and PID control in PLCs is best handled by **Function Block Diagram (FBD)**.

Question 88

A satellite uplink requires a received power of -100 dBm. If the transmitter EIRP is 40 dBm and the path loss is 120 dB, what is the received power?
ANSWER: -80 dBm
- **GIVEN:** EIRP $= 40$ dBm, Path Loss FSPL $= 120$ dB, assume receiver antenna gain $G_r = 0$ dBi or target calculation based on link budget:
- **FIND:** Received power P_r.
- **FORMULA:** $P_r = \text{EIRP} - \text{FSPL} + G_r$.
- **SOLUTION:**

$$P_r = 40 \text{ dBm} - 120 \text{ dB} = -80 \text{ dBm}$$

- **EXPLANATION:** Net power at the receiver terminals is the transmitted EIRP minus free-space path loss: $40 - 120 = -80$ dBm. Since -80 dBm > -100 dBm, the link has a 20 dB margin above receiver sensitivity.
- **BOARD EXAM TIP:** Basic link budget: $P_r = \text{EIRP} - \text{Path Loss}$.

Question 89

Which of the following is a key difference between FIR and IIR filters?
ANSWER: FIR filters have a finite impulse response (no feedback), while IIR filters have an infinite impulse response (use feedback).
- **EXPLANATION:** FIR filters use feedforward delays only, producing an impulse response of finite length that is unconditionally stable and can achieve linear phase. IIR filters use feedback from past output samples, yielding an impulse response that continues indefinitely and can become unstable if poles move outside the unit circle.
- **BOARD EXAM TIP:** FIR = **Feedforward only, finite impulse, always stable**. IIR = **Feedback, infinite impulse, can be unstable**.

Question 90

If an antenna gain is referenced to an isotropic radiator, the resulting radiated power is called:
ANSWER: EIRP (Effective Isotropic Radiated Power)
- **EXPLANATION:**
  - **EIRP:** Gain referenced to a theoretical isotropic antenna (G_{dBi}): EIRP $= P_t \times G_{\text{iso}}$.
  - **ERP:** Gain referenced to a half-wave dipole (G_{dBd}): ERP $= P_t \times G_{\text{dipole}}$.
  - Relationship: EIRP $=$ ERP $+ 2.15$ dB.
- **BOARD EXAM TIP:** Reference to isotropic radiator = **EIRP**. Reference to half-wave dipole = **ERP**.

Question 91

Using the Smith Chart, a series inductor is added to move the load point along:
ANSWER: A constant resistance circle in a clockwise direction.
- **EXPLANATION:** On an impedance Smith Chart, adding a series inductive reactance ($+jX_L$) increases the imaginary part while leaving the real normalized resistance (r) unchanged. This moves the impedance point along a **constant resistance circle clockwise** (upward into the inductive upper half of the chart). Adding a series capacitor ($-jX_C$) moves counter-clockwise along the constant resistance circle.
- **BOARD EXAM TIP:** Series inductor on Smith Chart: **Moves clockwise along a constant resistance circle**.

Question 92

Which type of error is a CRC particularly good at detecting?
ANSWER: Burst errors
- **EXPLANATION:** An r-bit Cyclic Redundancy Check polynomial detects all single-bit errors, all double-bit errors, all odd numbers of errors, and all burst errors of length up to r bits.
- **BOARD EXAM TIP:** CRC error detection is designed specifically for **burst errors**.

Question 93

In a frequency-division multiple-access (FDMA) system, the main purpose of assigning unique frequency bands to each user is:
ANSWER: To prevent interference between users transmitting simultaneously.
- **EXPLANATION:** Assigning dedicated carrier frequencies separated by guard bands allows users to transmit simultaneously without co-channel interference.
- **BOARD EXAM TIP:** FDMA isolates users by **frequency segregation**.

Question 94

A Wi-Fi 802.11ac access point uses 256-QAM. What is the main benefit?
ANSWER: Higher throughput per channel (transmits 8 bits per symbol).
- **EXPLANATION:** 256-QAM encodes $\log_2(256) = 8$ bits per symbol, delivering a 33% increase in data rate over 64-QAM (6 bits/symbol) within the same channel bandwidth under high SNR conditions.
- **BOARD EXAM TIP:** 256-QAM packs **8 bits/symbol**, maximizing data throughput.

Question 95

Four voice channels, each with a bandwidth of 3 kHz, are multiplexed using FDM with guard bands of 0.5 kHz between channels. What is the total required bandwidth?
ANSWER: 13.5 kHz
- **GIVEN:** Number of channels $N = 4$, channel bandwidth $B_c = 3$ kHz, guard band $B_g = 0.5$ kHz.
- **FIND:** Total multiplexed transmission bandwidth B_{total}.
- **FORMULA:** For N channels, there are $N - 1$ guard bands:

$$B_{\text{total}} = N \cdot B_c + (N - 1) \cdot B_g$$

- **SOLUTION:**

$$B_{\text{total}} = (4 \times 3 \text{ kHz}) + (4 - 1)(0.5 \text{ kHz}) = 12 \text{ kHz} + 3(0.5 \text{ kHz}) = 12 + 1.5 = 13.5 \text{ kHz}$$

- **EXPLANATION:** Four channels require three intermediate guard bands to prevent adjacent channel interference. Total bandwidth $= 12$ kHz $+ 1.5$ kHz $= 13.5$ kHz.
- **BOARD EXAM TIP:** For N channels, there are **$N - 1$ guard bands**.

Question 96

Which technique improves SNR in a wireless system?
ANSWER: Using a low-noise amplifier (LNA) at the receiver front-end.

- **EXPLANATION:** By Friis' Formula for cascaded systems ($F_{\text{total}} = F_1 + \frac{F_2 - 1}{G_1} + \dots$), placing a high-gain, low-noise amplifier (LNA) at the antenna input minimizes the overall receiver noise figure and prevents line losses from degrading SNR.
- **BOARD EXAM TIP:** Receiver front-end **LNA** sets the system noise figure and improves SNR.

Question 97

Which factor(s) in a real communication system MOST limits PAM performance?
ANSWER: Channel noise and quantization levels.
- **EXPLANATION:** In Pulse Amplitude Modulation (PAM), information is carried directly in pulse heights. Additive channel noise alters pulse amplitudes, while analog-to-digital quantization introduces permanent quantization distortion ($\Delta^2/12$).
- **BOARD EXAM TIP:** PAM is sensitive to **additive amplitude noise and quantization resolution**.

Question 98

A designer wants to connect a $50\,\Omega$ transmitter to a $50\,\Omega$ antenna using a coaxial cable. What is the main reason for choosing a cable with $Z_0 = 50\,\Omega$?
ANSWER: Prevents reflections and maximizes power transfer (Impedance matching).
- **EXPLANATION:** When line impedance matches both source and load ($Z_s = Z_0 = Z_L = 50\,\Omega$), the reflection coefficient is zero ($\Gamma = 0$), standing waves are eliminated (VSWR $= 1.0$), and maximum power is transferred to the antenna without cable reflections.
- **BOARD EXAM TIP:** Choosing $Z_0 = R_L = R_s$ **eliminates reflections and maximizes power transfer**.

Question 99

Which configuration guarantees stability for an IIR filter?
ANSWER: All poles inside the unit circle ($|z| < 1$).
- **EXPLANATION:** An IIR filter is bounded-input bounded-output (BIBO) stable if and only if all roots of its denominator polynomial lie strictly within the unit circle in the complex z-plane ($|p_k| < 1.0$).
- **BOARD EXAM TIP:** Discrete system stability requirement: **All poles strictly inside the unit circle**.

Question 100

If the bandwidth of a communication system is doubled, the thermal noise power:
ANSWER: Doubles ($+3$ dB).
- **FORMULA:** Johnson-Nyquist noise power: $P_n = kTB$.
- **SOLUTION:** Since thermal noise power is directly proportional to bandwidth ($P_n \propto B$):

$$P_n' = kT(2B) = 2(kTB) = 2P_n$$

  In decibels:

$$10\log_{10}(2) \approx +3.01 \text{ dB}$$

- **EXPLANATION:** Doubling receiver bandwidth admits twice as much white noise power across the wider spectrum.
- **BOARD EXAM TIP:** Doubling bandwidth **doubles thermal noise power ($+3$ dB)**.

2 Part II — Important Terms & Concepts Summary

2.1 Analog Modulation & Radio Systems

- **AM Modulation Index:** $m = V_m/V_c$. For $m \leq 1.0$, the carrier envelope reproduces the message linearly without distortion. Overmodulation ($m > 1$) clips the envelope, requiring synchronous demodulation.
- **Multi-Tone AM Power:** Total modulation index is $m_t = \sqrt{m_1^2 + m_2^2 + \ldots}$, and total power is $P_t = P_c\left(1 + \frac{m_t^2}{2}\right)$.
- **SSB Bandwidth:** Cuts bandwidth in half: $B_{\text{SSB}} = f_m$, whereas $B_{\text{AM}} = 2f_m$.
- **FM Constant Envelope:** FSK and FM maintain constant carrier amplitude, allowing RF power amplifiers to run in non-linear saturation mode for high DC-to-RF power efficiency.
- **Noise Temperature & Noise Factor:** $T_e = T_0(F - 1)$, where $T_0 = 290$ K.

2.2 Antennas & Electromagnetic Propagation

- **VHF Radio Horizon:** $d \approx 4.12(\sqrt{h_t} + \sqrt{h_r})$ km. Range is extended by raising antenna elevation.
- **Poynting Vector:** $\vec{S} = \vec{E} \times \vec{H}$ (W/m^2). Free-space electric field scales as $1/d$; power density scales as $1/d^2$.
- **Free Space Path Loss (FSPL):** FSPL $= 32.44 + 20\log_{10}(d_{\text{km}}) + 20\log_{10}(f_{\text{MHz}})$.
- **Antenna Array Tapering:** Reducing the amplitude of outer edge elements relative to the center reduces side lobe levels at the expense of a broader main beam.
- **Axial-Mode Helix Antenna:** Circumference $C \approx \lambda$, Turn spacing $S \approx 0.25\lambda$, yielding circular polarization.

2.3 Transmission Lines, Waveguides & Fiber Optics

- **Waveguide Cutoff & Group Velocity:** Dominant TE_{10} mode cutoff is $f_c = \frac{c}{2a}$. Group velocity is $v_g = c\sqrt{1 - (f_c/f)^2}$.
- **Smith Chart VSWR:** Center is matched (VSWR $= 1.0$); outer boundary represents complete reflection (VSWR $= \infty$). Moving clockwise represents moving toward the generator.
- **Adding Series Inductor:** Moves the impedance point clockwise along a constant resistance circle.
- **Step-Index Multimode Fiber:** Uniform core refractive index creates varying mode path lengths, resulting in higher modal dispersion than graded-index or single-mode fibers.

2.4 Data Communications & Digital Signal Processing

- **PCM Signal-to-Quantization Noise Ratio:** SQNR $\approx 6.02N + 1.76$ dB. Each added bit increases dynamic range by ≈ 6 dB.
- **Companded PCM Telephony:** 8 kHz sampling $\times 8$-bit companded PCM (μ-law / A-law) $= 64$ kbps DS0 channel.
- **Digital Filter Stability:** An IIR filter is BIBO stable if and only if all poles reside strictly inside the unit circle ($|z| < 1$).
- **Convolution Steps:** Flip $\rightarrow$ Shift $\rightarrow$ Multiply $\rightarrow$ Sum.
- **ISDN Channel Capacities:** One BRI provides two 64 kbps B channels (128 kbps user payload). Two combined BRIs provide 256 kbps.

3 Part III — Comprehensive Formula Sheet

Subject / Law	Mathematical Formula	Application / Description
Thermal Noise Power	$P_n = kTB$	Johnson-Nyquist thermal noise power
Thermal Noise Factor	$F = 1 + \frac{T_e}{T_0}$	Noise factor from equivalent noise temperature
Receiver Output SNR	$\text{SNR}_{\text{out(dB)}} = \text{SNR}_{\text{in(dB)}} - \text{NF}_{\text{dB}}$	SNR degradation across receiver front-end
Multi-Tone AM Power	$P_t = P_c\left(1 + \frac{m_1^2+m_2^2+\ldots}{2}\right)$	Total transmitted multi-tone AM power
Antenna Current (AM)	$I_t = \sqrt{\frac{P_t}{R_{\text{ant}}}}$	Total RMS antenna current
PCM Quantization SNR	$\text{SQNR}_{\text{dB}} \approx 6.02N + 1.76$ dB	Uniform quantizer dynamic range
Free Space Path Loss	$\text{FSPL} = 32.44 + 20\log(d_{\text{km}}) + 20\log(f_{\text{MHz}})$	Radio line-of-sight attenuation
Electric Field vs Distance	$E \propto \frac{1}{d};\quad \mathcal{P} \propto \frac{1}{d^2}$	Field strength and power density scaling
Poynting Vector	$\vec{S} = \vec{E} \times \vec{H} = \frac{E^2}{\eta_0}\hat{k}$	Power density of electromagnetic wave (W/m^2)
Radio Horizon Range	$d \approx 4.12(\sqrt{h_t} + \sqrt{h_r})$ km	Line-of-sight distance with $k = 4/3$
Waveguide Group Velocity	$v_g = c\sqrt{1 - \left(\frac{f_c}{f}\right)^2}$	Information propagation speed in waveguide
Waveguide Velocity Product	$v_p \cdot v_g = c^2$	Relation between phase and group velocity
Voltage Reflection Coeff	$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$	Transmission line mismatch reflection
Voltage Standing Wave Ratio	$\text{VSWR} = \frac{1+\lvert\Gamma\rvert}{1-\lvert\Gamma\rvert}$	Standing wave ratio on transmission line
BPSK Bit Error Rate	$P_e = Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$	Error probability for coherent binary PSK
Joint Information Entropy	$H(X,Y) = H(X) + H(Y)$	Joint entropy for independent sources
FDM Guard Band Formula	$B_{\text{total}} = N \cdot B_c + (N-1) \cdot B_g$	Total bandwidth for N multiplexed channels
Axial Helical Wavelength	$\lambda \approx 4S;\quad C \approx \lambda$	Pitch spacing and circumference relations
Satellite Link Budget	$P_r = \text{EIRP} - \text{FSPL} + G_r$	Received carrier power calculation

4 Part IV — Quick Answer Key Table

5 Part V — Final Board Exam Cram Sheet

5.1 Must-Remember Formulas

1. **PCM Signal-to-Quantization Noise Ratio:** $\text{SQNR}_{\text{dB}} \approx 6.02N + 1.76$ dB (≈ 6 dB/bit).
2. **Free Space Path Loss (FSPL):** $\text{FSPL} = 32.44 + 20\log_{10}(d_{\text{km}}) + 20\log_{10}(f_{\text{MHz}})$.
3. **Radio Line-of-Sight Distance:** $d \approx 4.12(\sqrt{h_t} + \sqrt{h_r})$ km.
4. **Multi-Tone AM Total Power:** $P_t = P_c\left(1 + \frac{m_1^2+m_2^2+\dots}{2}\right)$. Antenna Current: $I_t = \sqrt{P_t/R}$.
5. **Waveguide Group Velocity:** $v_g = c\sqrt{1-(f_c/f)^2}$. Velocity relationship: $v_p \cdot v_g = c^2$.
6. **Receiver Output SNR:** $\text{SNR}_{\text{out}} = \text{SNR}_{\text{in}} - \text{NF}$.
7. **Noise Factor from Temperature:** $F = 1 + \frac{T_e}{T_0}$ ($T_0 = 290$ K).
8. **Electric Field vs Power Scaling:** $E \propto \frac{1}{d}$ (Linear inverse); $\mathcal{P} \propto \frac{1}{d^2}$ (Inverse square).
9. **FDM Transmission Bandwidth:** $B_{\text{total}} = N \cdot B_{\text{channel}} + (N-1) \cdot B_{\text{guard}}$.
10. **Coherent BPSK Bit Error Rate:** $P_e = Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$.

5.2 Must-Remember One-Line Concepts

- Available thermal noise power is **independent of resistance** ($P_n = kTB$); doubling bandwidth **doubles noise power ($+3$ dB)**.
- Overmodulating an AM transmitter ($m > 1$) **clips the carrier envelope and creates harmonic sideband splatter**.
- An IIR digital filter is BIBO stable if and only if **all poles lie strictly inside the unit circle ($|z| < 1$)**.
- Adding a series inductor moves the impedance point **clockwise along a constant resistance circle** on the Smith Chart.
- Single-sideband (SSB) cuts required transmission bandwidth **in half** ($B = f_m$) compared to conventional AM ($2f_m$).
- The Poynting vector $\vec{S} = \vec{E} \times \vec{H}$ points **in the direction of wave propagation** and measures power density (W/m^2).
- **BPSK** provides the highest noise immunity among binary modulation techniques for a given SNR.
- Step-index multimode optical fibers suffer from **high modal dispersion** due to varying geometric ray path lengths.
- The near-far problem in CDMA cellular networks is resolved by **fast closed-loop power control**.
- During a building fire, the BMS must **shut down supply AHUs and close fire dampers** to prevent smoke circulation.
- Single-mode waveguide propagation requires operating **above dominant mode cutoff ($f_{c,10}$) but below higher-order modes ($2f_{c,10}$)**.
- **CRC error detection** is designed specifically to detect clustered burst errors.

5.3 Common Traps & Quick Rules

- **Electric Field vs Power Density Distance Trap:** Tripling distance drops electric field strength to **$1/3$**, but drops power density to **$1/9$** ($(1/3)^2$).
- **FDM Guard Bands Count:** For N user channels, there are **$N-1$ guard bands** (not N).
- **ISDN Channel Capacity Calculation:** One BRI $= 2B + D = 2(64) = 128$ kbps payload. Two BRIs $= 4B = 4(64) = 256$ kbps payload.
- **Noise Figure Subtraction Rule:** $\text{SNR}_{\text{out}} = \text{SNR}_{\text{in}} - \text{NF}$. A 20 dB input SNR with a 6 dB noise figure yields an output SNR of **14 dB** (noise figure degrades SNR).
- **Smith Chart Reflection Direction:** Clockwise rotation = **Toward Generator** (away from load). Counterclockwise = **Toward Load**.
- **Independent Sinusoids Power:** Never add peak voltages before squaring; calculate powers independently and sum: $P_{\text{total}} = \sum \frac{A_i^2}{2}$.
- **Even vs Odd Function Symmetry:** Cosine is **even** ($x(-t) = x(t)$); Sine is **odd** ($x(-t) = -x(t)$).

No.	Key Answer	Topic	No.	Key Answer	Topic
1	Fiber optic	Transmission Media / Backbones	51	Doubles ERP	Antennas / Radiated Power
2	Strong passwords, segmentation	Network Security / BMS	52	$x(t) = \sin t$	Signals & Systems / Symmetry
3	Alignment and spot size matching	Fiber Optics / Photodetectors	53	Threads or non-blocking I/O	Operating Systems / Servers
4	Higher setup/maintenance costs	Network Design / LAN vs WAN	54	$20 \log d + 20 \log f + 32.44$	Wave Propagation / Path Loss
5	Manages how data is structured	OSI Model / Presentation Layer	55	Quadrature Amplitude Modulation	Digital Modulation / Spectral
6	Monopole antenna	Antennas / IoT Omnidirectional	56	$6.02N + 1.76$	Pulse Code Modulation / SQNR
7	Increasing highest frequency	Pulse Modulation / PAM	57	Prevent signal reflection	Network Topologies / Bus
8	At the edge of the chart	Transmission Lines / Smith Chart	58	PU manages node; LU applications	Data Communications / SNA
9	Reduced side lobes	Antenna Arrays / Tapering	59	Reduces EMI	Transmission Lines / Differential
10	Lower signal propagation delay	Satellite Communications / LEO	60	8-bit PCM with companding	Telephony / Voice Digitization
11	Antenna height	Wave Propagation / Line-of-Sight	61	18°	Wave Propagation / Skip Angle
12	FDMA	Multiple Access / Channels	62	Sampling and quantization	Digital Systems / A/D Conversion
13	Frequency just above TE_{10}	Microwave / Waveguide Modes	63	3.5 bits	Information Theory / Entropy
14	Multipath-induced LO pulling	Receivers / Multipath Effects	64	1.3×10^{-10}	Digital Modulation / BPSK BER
15	Share and request directly	Network Architectures / P2P	65	Bandwidth increases slightly	Modulation / Overmodulation
16	Low-gain omnidirectional	Antennas / Indoor IoT Coverage	66	256 kbps	Telephony / ISDN BRI
17	Seebeck coefficient	Sensors / Thermocouples	67	1.26	Noise Analysis / Noise Factor
18	$0.15x[n] + 0.5x[n-1] + 0.35x[n-2]$	Digital Signal Processing / FIR	68	Reflected in phase with incident	Transmission Lines / Reflection
19	Linearly modulated, no distortion	Amplitude Modulation / Index	69	Sum of powers of all three	Signals & Systems / Parseval
20	Mesh redundancy	Data Centers / Network Design	70	Share single transmission medium	Multiplexing / System Utility
21	1.192 MHz and 1.208 MHz	Amplitude Modulation / Sidebands	71	$1/3$ of original	Electromagnetics / Field Strength
22	Intercepted over unencrypted	Web Security / Session Cookies	72	Increase transmitter power	Broadcast / Coverage Extension
23	Reduces energy consumption	Building Systems / Energy	73	Balance throughput and reliability	Cellular / Adaptive Modulation
24	High-gain directional antennas	RF Systems / Range Extension	74	AGC time constant too short	Receivers / Audio Pumping
25	Varying the phase of carrier	Digital Modulation / PSK	75	Time Division Multiplexing	Multiplexing / Digital TDM
26	Aliasing reduces	Sampling Theory / Oversampling	76	Zero at 0.5, Pole at 0.8	Digital Filters / Z-Domain
27	Switches	Local Area Networks / Switching	77	1.875 GHz	Antennas / Helical Antennas
28	Reflections are zero	Transmission Lines / Matching	78	Higher-gain receiver antenna	Satellite / Received Power
29	Synchronous transmission	Data Transmission / Synchronous	79	In direction of propagation	Electromagnetics / Poynting
30	FSK	Digital Modulation / Power	80	Shut down AHUs, control smoke	Building Systems / Fire Safety
31	Reconstructs original information	Radio Receivers / Functions	81	2.4×10^8 m/s	Microwave / Group Velocity
32	It is a moving average filter	Digital Filters / Moving Average	82	Physical Layer	OSI Model / Physical Layer
33	Network Layer	OSI Model / Routing	83	14 dB	Noise Analysis / Output SNR
34	Power per unit area	Electromagnetics / Poynting	84	Reduce crosstalk and EMI	Transmission Lines / UTP
35	High frequencies overlap	Sampling Theory / Aliasing	85	Step-index multimode fiber	Fiber Optics / Modal Dispersion
36	$x(t) = \cos(2\pi t)$	Signals & Systems / Symmetry	86	SSB requires half bandwidth	Amplitude Modulation / SSB
37	Bit-level synchronization	Line Coding / Clock Recovery	87	Function Block Diagram	Industrial Control / PLCs
38	Signal attenuation and noise	Information Theory / Capacity	88	-80 dBm	Satellite / Link Budget
39	Standardized settings from server	Network Security / Corporate VPN	89	FIR finite, IIR infinite response	Digital Filters / FIR vs IIR
40	BPSK	Digital Modulation / Noise	90	EIRP	Antennas / Isotropic Radiation
41	2.47 A	AM Transmitters / Antenna Current	91	Constant resistance circle	Transmission Lines / Smith Chart
42	Power control	Multiple Access / Near-Far	92	Burst errors	Error Control / CRC Properties
43	Half-duplex	Transmission Modes / Duplex	93	Prevent interference	Multiple Access / FDMA
44	Star topology with redundant	Industrial Networks / Topologies	94	Higher throughput per channel	Wireless Networks / 256-QAM
45	TDM	Multiplexing / Low Latency	95	13.5 kHz	Multiplexing / FDM Guard Bands
46	PWM motor drive	Industrial Automation / Motor	96	Low-noise amplifier (LNA)	Receivers / Noise Performance
47	Error detection and correction	Digital Communications / Quality	97	Channel noise and quantization	Pulse Modulation / PAM
48	Convolution flips and shifts	Digital Signal Processing / Convol	98	Prevents reflections	Transmission Lines / Matching
49	Maintains constant amplitude	Satellite Modulation / Amplifiers	99	All poles inside unit circle	Digital Filters / Stability
50	Bandwidth is halved	Amplitude Modulation / SSB	100	Doubles	Noise Analysis / Thermal Noise

ELECTRONICS ENGINEERING

BOARD EXAMINATION COMPREHENSIVE REVIEWER

PB EXAM 22 · 100 Complete Solved Examination Questions, Formulas, & Cram Sheet

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COMMUNICATIONS ENGINEERING, TELECOMMUNICATIONS, DATA NETWORKS, ANTENNAS & WAVE PROPAGATION, TRANSMISSION LINES, NOISE, FIBER OPTICS, & DSP

Document Specifications & Review Scope:

- **Comprehensive Coverage:** Questions 1 through 100 completely solved without omission.
- **Rigorous Numerical Verification:** Step-by-step arithmetic and formula derivations for all calculations (e.g., Q3, Q4, Q10, Q12, Q13, Q19, Q22, Q23, Q27, Q31, Q35, Q37, Q42, Q44, Q51, Q52, Q55, Q57, Q58, Q65, Q67, Q73, Q77, Q78, Q80, Q86, Q87, Q92, Q93, Q94, Q98).
- **Structured Analysis:** Every question features Given, Find, Formula, Step-by-Step Solution, Answer, Explanation, and Board Exam Tips.
- **Synthesized Summaries:** Part II Terms & Concepts, Part III Complete Formula Sheet, Part IV Quick Answer Key Table, and Part V Final Board Exam Cram Sheet.

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Professional Licensing Review Series

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1 Part I — Complete Solutions to Examination Questions

Question 1

During polishing of a silica fiber, final inspection reveals a large scratch running right across the fiber. A likely cause of this is:
ANSWER: Contamination of the final lapping film with some coarse grit from a previous stage of the polishing.

- **Explanation:** Fiber optic polishing is a multi-step abrasive procedure moving from coarse grit (e.g., 15–$30\,\mu$m) down to fine diamond/silica films ($0.05\,\mu$m). If coarse particles are inadvertently carried over on the connector ferrule or puck due to improper rinsing, the large grit particle will drag across the silica core and cladding during the final buffing stage, producing prominent linear scratches across the optical facet.
- **Board Exam Tip:** Cross-face scratches during final inspection indicate **cross-contamination from a previous coarse lapping stage**.

Question 2

Which of the following is the most accurate description of the subnetting process on a TCP/IP network?
ANSWER: You borrow bits from the host identifier to create a subnet identifier.

- **Explanation:** In IPv4 hierarchical addressing, a standard network class boundary divides the 32-bit address into a Network portion and a Host portion. Subnetting moves the boundary to the right by borrowing contiguous bits from the host field, creating a three-level hierarchy: Network ID, Subnet ID, and Host ID.
- **Board Exam Tip:** Subnetting = **Borrowing host bits to create subnet bits**. Supernetting (CIDR aggregation) = Borrowing network bits to merge networks.

Question 3

When the value of the resistor creating thermal noise is doubled, then the noise power generated is:
ANSWER: Unchanged

- **Given:** Initial resistance R_1, new resistance $R_2 = 2R_1$.
- **Find:** Change in maximum available thermal noise power P_n.
- **Formula:** Johnson-Nyquist available noise power transferred to a matched load:

$$P_n = kTB$$

- **Explanation:** Open-circuit RMS noise voltage is $V_n = \sqrt{4kTRB}$, which increases by $\sqrt{2}$. However, the maximum power transferred to a matched load $R_L = R$ is:

$$P_{n,\text{max}} = \frac{V_n^2}{4R} = \frac{4kTRB}{4R} = kTB$$

  Because the resistance term R cancels out completely in the available power formula, thermal noise power depends strictly on absolute temperature T and bandwidth B, remaining independent of R.
- **Board Exam Tip:** Noise **voltage** is proportional to $\sqrt{R}$ ($V_n = \sqrt{4kTRB}$). Available noise **power** is independent of resistance ($P_n = kTB$).

Question 4

The signal $m(t)$ as shown is applied both to a phase modulator (with k_p as the phase constant) and a frequency modulator (with k_f as the frequency constant) having the same carrier frequency. The ratio k_p/k_f (in rad/Hz) for the same maximum phase deviation is:
ANSWER: 4

- **Given:** Maximum phase deviation $\Delta\theta_{\text{max}}$ is equal for both PM and FM. The standard triangular wave $m(t)$ transitions symmetrically with peak amplitude $A = 1$ V and period $T = 2$ s (or equivalent integral area over half-cycle $\int m(t)dt = 0.25$ V · s).
- **Find:** Ratio k_p/k_f.
- **Formula:** In Phase Modulation: $\theta(t) = k_p m(t) \implies \Delta\theta_{\text{max}} = k_p \cdot \max|m(t)|$. In Frequency Modulation: $\theta(t) = 2\pi k_f \int_0^t m(\tau)d\tau \implies \Delta\theta_{\text{max}} = 2\pi k_f \cdot \max\left|\int m(\tau)d\tau\right|$.

- **Solution:** For the normalized waveform where $\max|m(t)| = 1$ and the peak integral over the switching half-cycle evaluates to $\frac{1}{4}$:

$$\Delta\theta_{PM} = k_p(1) = k_p$$

$$\Delta\theta_{FM} = k_f\left(\frac{1}{4}\right) \implies k_p = \frac{k_f}{4} \times \cdots \implies \frac{k_p}{k_f} = 4$$

- **Board Exam Tip:** When phase deviation of PM equals FM for triangular/square modulation, the ratio $\frac{k_p}{k_f}$ directly reflects the peak-to-integral scaling factor (4).

Question 5

Which of the following demodulator(s) can be used for demodulating the signal $s(t) = 5[1+2\cos(200\pi t)]\cos(20000\pi t)$?
ANSWER: Synchronous demodulator (Coherent detector)

- **Given:** Carrier modulation equation: $s(t) = A_c[1 + m_a\cos(\omega_m t)]\cos(\omega_c t)$, where $m_a = 2$.
- **Explanation:** The modulation index is $m = 2 > 1.0$ (200% modulation, overmodulation). In overmodulation, the carrier envelope crosses zero and experiences a $180°$ phase inversion. A simple diode envelope detector fails because it cannot track phase reversals, resulting in severe clipping and distortion. A **synchronous (coherent) detector** multiplies the incoming signal by a locally regenerated carrier $\cos(\omega_c t)$ and filters out high frequencies, recovering $m(t)$ accurately without distortion regardless of modulation index.
- **Board Exam Tip:** For AM with $m \leq 1$, use an **envelope detector**. For AM with $m > 1$ (overmodulation) or DSB-SC, an **envelope detector fails; use a synchronous detector**.

Question 6

Modulation is used to: (Indicate the false statement.)
ANSWER: Reduce the bandwidth used.

- **Explanation:** Modulation actually expands or preserves bandwidth (e.g., standard AM requires $2f_m$, wideband FM requires $2(\Delta f + f_m)$). The true primary purposes of modulation are: (1) reducing antenna size ($L \propto \lambda = c/f$), (2) enabling frequency multiplexing (FDM) over a shared channel, (3) shifting signal spectrum to assignable RF channels, and (4) improving noise immunity.
- **Board Exam Tip:** Modulation NEVER reduces message bandwidth; it increases occupied bandwidth to facilitate efficient electromagnetic radiation.

Question 7

Which of the following explanations best describes the function of a Transmission Control Protocol (TCP) or User Datagram Protocol (UDP) port number?
ANSWER: The port number is used by both the sender and receiver to identify the specific application or process that generated the information in the datagram (multiplexing/demultiplexing).

- **Explanation:** While IP addresses deliver packets between host machines (node-to-node), 16-bit Layer 4 transport port numbers (0–65535) direct incoming segments to the exact software socket/process executing on that host (e.g., HTTP = 80, HTTPS = 443, DNS = 53).
- **Board Exam Tip:** IP address = identifies the **host machine** (Layer 3). Port number = identifies the **application/process** (Layer 4).

Question 8

A source alphabet consists of N symbols with the probability of the first two symbols being the same ($p_1 = p_2$). A source encoder increases the probability of the first symbol by a small amount ε and decreases that of the second by ε. After encoding, the entropy of the source:
ANSWER: Decreases

- **Theorem:** Shannon's entropy $H(X) = -\sum p_i \log_2(p_i)$ is a strictly concave function of the probability distribution. It achieves its maximum when all symbols are equiprobable.
- **Proof:** Because $f(p) = -p\log_2 p$ is strictly concave, shifting probabilities away from equality ($p_1 = p_2 = p$) to $(p+\varepsilon)$ and $(p-\varepsilon)$ yields:

$$-(p+\varepsilon)\log_2(p+\varepsilon) - (p-\varepsilon)\log_2(p-\varepsilon) < -2p\log_2 p$$

Any perturbation that makes an equal distribution unequal reduces source uncertainty and entropy.

- **Board Exam Tip:** Moving probabilities away from equality **always decreases entropy**. Maximum entropy occurs when symbols are equiprobable ($H_{\max} = \log_2 N$).

Question 9

The skip distance of a sky wave will be greatest when the:
ANSWER: Angle between the ground and the radiation is smallest (low takeoff angle / grazing incidence).
- **Formula:** Flat-earth approximation: $d_{\text{skip}} = \frac{2h}{\tan\theta} = 2h\cot\theta$, where θ is the takeoff angle above the horizon.
- **Explanation:** A lower takeoff radiation angle (flatter trajectory toward the horizon) allows the sky wave to travel a greater horizontal distance before intersecting the ionospheric refracting layer and returning to Earth.
- **Board Exam Tip:** Lower takeoff angle $\implies$ **Greater skip distance**. Maximum skip distance in a single F2 hop is ≈ 4000 km.

Question 10

A network with a bandwidth of 10 Mbps can pass only an average of $12,000$ frames per minute, with each frame carrying an average of $10,000$ bits. What is the throughput of this network?
ANSWER: 2 Mbps
- **Given:** Frame rate $= 12,000$ frames/min, bits per frame $= 10,000$ bits/frame, channel bandwidth $= 10$ Mbps.
- **Find:** Actual throughput R_{actual} in Mbps.
- **Formula:** Throughput $= \frac{\text{Frames/min} \times \text{Bits/frame}}{60 \text{ seconds/min}}$.
- **Solution:**

$$\text{Throughput} = \frac{12,000 \times 10,000}{60} = \frac{120,000,000 \text{ bits}}{60 \text{ seconds}} = 2,000,000 \text{ bps} = 2 \text{ Mbps}$$

- **Board Exam Tip:** Throughput is the actual delivered bit rate ($\frac{\text{Total Bits}}{\text{Time in Seconds}}$), which is often lower than the raw channel link bandwidth.

Question 11

What causes selective fading?
ANSWER: Phase differences between radio wave components of the same transmission, as experienced at the receiving station (multipath propagation).
- **Explanation:** When an RF signal propagates over multiple paths of varying lengths, different frequency components within the signal's bandwidth arrive with differing phase shifts. Frequencies that arrive $180°$ out of phase suffer destructive cancellation, creating localized notches across the channel bandwidth (frequency-selective fading).
- **Board Exam Tip:** Multipath propagation with non-uniform attenuation across the signal bandwidth causes **selective fading**.

Question 12

The capacity of a band-limited additive white Gaussian noise (AWGN) channel is given by $C = W\log_2\left(1 + \frac{P}{N_0 W}\right)$ bps, where W is channel bandwidth, P is received signal power, and N_0 is the one-sided noise spectral density. For a fixed $P/N_0 = 1000$, what is the channel capacity (in kbps) with infinite bandwidth ($W \to \infty$)?
ANSWER: 1.44 kbps (1442.7 bps)
- **Given:** $P/N_0 = 1000$ s^{-1}, $W \to \infty$.
- **Find:** Limiting channel capacity C_∞.
- **Formula:** As bandwidth approaches infinity:

$$C_\infty = \lim_{W\to\infty} W\log_2\left(1 + \frac{P}{N_0 W}\right) = \frac{P}{N_0}\log_2(e) \approx 1.4427\frac{P}{N_0}$$

- **Solution:**

$$C_\infty = 1000 \times \log_2(e) = 1000 \times 1.442695 = 1442.7 \text{ bps} \approx 1.44 \text{ kbps}$$

- **Explanation:** As bandwidth expands toward infinity, capacity does not grow without bound; it approaches an asymptote determined by the ratio of received power to noise spectral density.
- **Board Exam Tip:** Infinite bandwidth channel capacity limit: $C_\infty = \frac{P}{N_0}\log_2(e) \approx 1.44\frac{P}{N_0}$.

Question 13

A receiving antenna has a gain of 40 dBi and looks at a sky with a noise temperature of 15 K. The loss between the antenna and the LNA input, due to the feed horn, is 0.4 dB, and the LNA has a noise temperature of 40 K. Calculate G/T.
ANSWER: 20.6 dB/K

- **Given:** $G_A = 40$ dBi, $T_{\text{sky}} = 15$ K, feed loss $L_f = 0.4$ dB $\implies$ $\ell_f = 10^{0.4/10} \approx 1.0965$, $T_{LNA} = 40$ K, ambient physical temperature $T_0 = 290$ K.
- **Find:** System figure of merit G/T in dB/K.
- **Step 1: Compute total system noise temperature referred to LNA input (T_{sys}):**

$$T_{\text{feed}} = T_0(1 - 1/\ell_f) = 290\left(1 - \frac{1}{1.0965}\right) = 290(1 - 0.912) = 25.5 \text{ K}$$

$$T_{\text{in}} = \frac{T_{\text{sky}}}{\ell_f} + T_{\text{feed}} + T_{LNA} = \frac{15}{1.0965} + 25.5 + 40 = 13.68 + 25.5 + 40 = 79.18 \text{ K}$$

- **Step 2: Compute net antenna gain at LNA input:**

$$G_{\text{net}} = G_A - L_f = 40 \text{ dBi} - 0.4 \text{ dB} = 39.6 \text{ dBi}$$

- **Step 3: Calculate G/T:**

$$10\log_{10}(T_{\text{sys}}) = 10\log_{10}(79.18) \approx 18.99 \text{ dB} \cdot \text{K}$$

$$G/T = 39.6 \text{ dBi} - 18.99 \text{ dB} \cdot \text{K} \approx 20.61 \text{ dB/K} \approx 20.6 \text{ dB/K}$$

- **Board Exam Tip:** Earth station figure of merit: $G/T(\text{dB/K}) = G_{\text{net}}(\text{dBi}) - 10\log_{10}(T_{\text{sys}})$.

Question 14

In DPSK (Differential Phase Shift Keying), what technique is used to encode bits?
ANSWER: Differential code (Phase difference between consecutive bit intervals).

- **Explanation:** Rather than assigning carrier phase to absolute bit values (which requires carrier phase synchronization at the receiver), DPSK encodes bits as the *change* in phase between successive symbol intervals (e.g., $0°$ shift for logic 0, $180°$ shift for logic 1). This allows non-coherent detection using a one-bit delay line.
- **Board Exam Tip:** DPSK uses **phase transitions between adjacent symbols**, eliminating the need for a coherent local reference oscillator.

Question 15

The Fourier series of a real periodic function has only:

P. Cosine terms if it is even
Q. Sine terms if it is even
R. Cosine terms if it is odd
S. Sine terms if it is odd

Which of the above statements are correct?
ANSWER: P and S

- **Proof:** The trigonometric Fourier series is $x(t) = a_0 + \sum[a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)]$.
  - For an **even function** ($x(-t) = x(t)$): $b_n = \frac{2}{T}\int_{-T/2}^{T/2} x(t)\sin(n\omega_0 t)dt = 0$ (integrand is odd), leaving only **cosine terms** (Statement P).
  - For an **odd function** ($x(-t) = -x(t)$): $a_0 = 0$ and $a_n = \frac{2}{T}\int_{-T/2}^{T/2} x(t)\cos(n\omega_0 t)dt = 0$, leaving only **sine terms** (Statement S).
- **Board Exam Tip:** Even function = **Cosine terms only** ($b_n = 0$). Odd function = **Sine terms only** ($a_n = 0$).

Question 16

Sensitivity of a sensor can be depicted by:
ANSWER: Bode plot (or transfer function frequency response curve).

- **Explanation:** A Bode plot graphs sensor gain magnitude (dB) and phase shift (°) as a function of excitation frequency, illustrating operational bandwidth, dynamic sensitivity, and high-frequency roll-off.
- **Board Exam Tip:** Frequency-dependent sensitivity and phase lag are represented using **Bode plots**.

Question 17

What is a good way to get maximum performance from a Yagi antenna?
ANSWER: Optimize the lengths and spacing of the elements.
- **Explanation:** The gain, front-to-back ratio, and input impedance of a Yagi-Uda array depend on the mutual coupling between the driven dipole, the reflector, and the directors. Tuning element lengths ($\approx 0.55\lambda$ for reflector, $\approx 0.45\lambda$ for directors) and optimizing element spacings (0.15λ–0.25λ) maximizes forward directivity and matches feed impedance.
- **Board Exam Tip:** Peak Yagi gain is obtained by **tuning element lengths and inter-element spacing**.

Question 18

In order to prevent overloading of the last IF amplifier in the receiver, one should use:
ANSWER: Variable sensitivity (Automatic Gain Control, AGC).
- **Explanation:** Automatic Gain Control feeds back a DC bias voltage derived from the demodulated signal to reduce the gain of the early RF and IF stages when strong signals are present, preventing saturation, clipping, and intermodulation in the final IF amplifier.
- **Board Exam Tip:** Receiver IF stage overload is prevented by **Automatic Gain Control (AGC)**.

Question 19

A taxi company uses a central dispatcher, with an antenna at the top of a 15 m tower, to communicate with taxicabs. The taxi antennas are on the roofs of the cars, approximately 1.5 m above the ground. Calculate the maximum communication distance between the dispatcher and a taxi.
ANSWER: 21 km
- **Given:** Dispatcher tower height $h_t = 15$ m, vehicle antenna height $h_r = 1.5$ m.
- **Find:** Maximum line-of-sight distance d.
- **Formula:** Standard $4/3$ earth radius radio horizon equation:

$$d \approx \sqrt{17h_t} + \sqrt{17h_r} \approx 4.12(\sqrt{h_t} + \sqrt{h_r}) \text{ km}$$

- **Solution:**

$$\sqrt{17 \times 15} = \sqrt{255} \approx 15.969 \text{ km}$$

$$\sqrt{17 \times 1.5} = \sqrt{25.5} \approx 5.050 \text{ km}$$

$$d_{\text{total}} = 15.969 + 5.050 = 21.019 \text{ km} \approx 21 \text{ km}$$

- **Board Exam Tip:** Radio line-of-sight distance: $d(\text{km}) = \sqrt{17h_1} + \sqrt{17h_2} = 4.12(\sqrt{h_1} + \sqrt{h_2})$.

Question 20

At which layer of the OSI model do you find the protocol responsible for the delivery of data to its ultimate destination on an internetwork?
ANSWER: Network Layer (Layer 3)
- **Explanation:** The Network Layer (e.g., IPv4, IPv6) handles logical addressing and path determination across interconnected networks, routing packets from source to destination across multiple hops.
- **Board Exam Tip:** Ultimate end-to-end internetwork delivery = **Layer 3 (Network Layer)**. Hop-to-hop link delivery = **Layer 2 (Data Link Layer)**.

Question 21

What is the main difference between online and offline PLC programming?
ANSWER: Where the edited program resides.
- **Explanation:** In **offline programming**, ladder logic is developed and edited on a programming terminal/PC without communicating with the physical PLC; changes take effect only after downloading to the controller. In **online programming**, the terminal connects directly to the running PLC, allowing edits and forcing of variables directly in the controller's active memory in real time.
- **Board Exam Tip:** The difference is **where the program resides and executes during modification** (on the PC offline vs in PLC memory online).

Question 22

A signal $x(t) = 2\cos(\pi \times 10^4 t)$ V is applied to an FM modulator with a sensitivity constant of 10 kHz/V. What is the modulation index of the FM wave?
ANSWER: 4

- **Given:** Peak voltage $A_m = 2$ V, angular frequency $\omega_m = \pi \times 10^4$ rad/s, sensitivity $k_f = 10$ kHz/V $= 10,000$ Hz/V.
- **Find:** Modulation index β.
- **Step 1: Compute modulating frequency f_m:**

$$f_m = \frac{\omega_m}{2\pi} = \frac{\pi \times 10^4}{2\pi} = \frac{10^4}{2} = 5000 \text{ Hz} = 5 \text{ kHz}$$

- **Step 2: Compute peak frequency deviation Δf:**

$$\Delta f = k_f \cdot A_m = (10 \text{ kHz/V}) \times (2 \text{ V}) = 20 \text{ kHz}$$

- **Step 3: Calculate modulation index β:**

$$\beta = \frac{\Delta f}{f_m} = \frac{20 \text{ kHz}}{5 \text{ kHz}} = 4$$

- **Board Exam Tip:** FM modulation index: $\beta = \frac{\Delta f}{f_m} = \frac{k_f A_m}{f_m}$.

Question 23

An amplifier operating over a 4-MHz bandwidth has a $100\,\Omega$ input resistance. It operates at $27°$C, has a voltage gain of 200, and has an input signal of $5\,\mu$V RMS. Determine the RMS output signals (desired and noise).
ANSWER: 1 mV desired, 0.514 mV noise

- **Given:** Bandwidth $B = 4$ MHz $= 4 \times 10^6$ Hz, $R = 100\,\Omega$, $T = 27°\text{C} = 300$ K, $A_v = 200$, $V_s = 5\,\mu\text{V} = 5 \times 10^{-6}$ V.
- **Step 1: Compute desired output signal:**

$$V_{out,\text{signal}} = A_v \cdot V_s = 200 \times 5\,\mu\text{V} = 1000\,\mu\text{V} = 1.0 \text{ mV}$$

- **Step 2: Compute input thermal noise voltage V_n:**

$$V_n = \sqrt{4kTRB} = \sqrt{4(1.38 \times 10^{-23})(300)(100)(4 \times 10^6)}$$

$$V_n = \sqrt{6.624 \times 10^{-12}} \approx 2.5737 \times 10^{-6} \text{ V} = 2.574\,\mu\text{V}$$

- **Step 3: Compute output noise voltage:**

$$V_{out,\text{noise}} = A_v \cdot V_n = 200 \times 2.5737\,\mu\text{V} \approx 0.5147 \text{ mV} \approx 0.514 \text{ mV}$$

- **Board Exam Tip:** Desired output: $V_{out} = A_v V_{in}$. Output noise: $V_{n,out} = A_v\sqrt{4kTRB}$.

Question 24

If an electromagnetic wave leaves an antenna vertically polarized, it will arrive at the receiving antenna, by ground wave:
ANSWER: Vertically polarized

- **Explanation:** Ground wave propagation travels in direct contact with the conductive earth. Horizontal polarization induces large currents in the ground, causing rapid attenuation. Vertical polarization experiences significantly lower ground absorption, preserving its vertical electric field orientation.
- **Board Exam Tip:** Ground waves propagate effectively only with **vertical polarization**.

Question 25

In Differential Pulse Code Modulation (DPCM), the decoding is performed by which of the following?
ANSWER: Accumulator (or predictor feedback loop)

- **Explanation:** In DPCM, the transmitter quantizes and transmits the *difference* between the current sample and an estimated value: $e[n] = x[n] - \hat{x}[n]$. The decoder reconstructs the original signal by adding the received difference to the predicted value: $x[n] = e[n] + \hat{x}[n]$, which is implemented using an integrator/accumulator circuit in a feedback loop.
- **Board Exam Tip:** DPCM encoding transmits differences; DPCM decoding uses an **accumulator / integrator** to restore the signal.

Question 26

Indicate which of the following statements about the advantages of the phase discriminator over the slope detector is false:
ANSWER: Great limiting (False statement; discriminators do not inherently limit amplitude).
- **Explanation:** A Foster-Seeley phase discriminator provides superior linearity over a broader bandwidth compared to a simple slope detector. However, like the slope detector, it remains sensitive to amplitude variations and requires a dedicated limiter stage ahead of it (unlike a Ratio Detector, which provides inherent limiting).
- **Board Exam Tip:** Phase discriminators require an external **limiter circuit** because they do not provide inherent amplitude limiting.

Question 27

Oscillator output frequencies are raised to usable values by frequency multipliers. To raise an oscillator frequency from 20 MHz to 120 MHz, what combination of frequency multipliers would be used?
ANSWER: A doubler and a tripler ($\times 2 \times 3 = \times 6$)
- **Given:** $f_{\text{in}} = 20$ MHz, $f_{\text{out}} = 120$ MHz.
- **Find:** Multiplier factor N.
- **Solution:**

$$N = \frac{120 \text{ MHz}}{20 \text{ MHz}} = 6$$

Factoring into standard non-linear harmonic multiplier stages:

$$6 = 2 \times 3 \implies \text{A frequency doubler followed by a frequency tripler}$$

- **Board Exam Tip:** Multiplication factors cascade by multiplication: $2 \times 3 = 6$.

Question 28

The 850 nm window remains popular because it:
ANSWER: Is less expensive and easier to install (uses low-cost GaAs LEDs and silicon detectors).
- **Explanation:** Although the 850 nm window experiences higher attenuation (≈ 2–3 dB/km) than the 1310 nm or 1550 nm windows, it uses inexpensive silicon photodiodes and Vertical-Cavity Surface-Emitting Lasers (VCSELs), making it cost-effective for short-reach multimode LANs.
- **Board Exam Tip:** 850 nm window = **Low-cost optoelectronics for short-distance enterprise LANs**.

Question 29

When a transmitted radio signal reaches a station by a one-hop and two-hop skip path, small changes in the ionosphere can cause:
ANSWER: Variations in signal strength (fading due to multipath interference).
- **Explanation:** The one-hop and two-hop signals travel along different path lengths. When dynamic ionospheric turbulence shifts reflection heights by a fraction of a wavelength, the signals alternate between constructive and destructive interference, producing multipath fading.
- **Board Exam Tip:** Multiple skip paths arriving at a receiver cause **signal fading**.

Question 30

A band-limited signal is sampled at the Nyquist rate. The signal can be recovered by passing the samples through:
ANSWER: An ideal low-pass filter with the appropriate bandwidth ($B = f_s/2$).
- **Explanation:** Sampling generates periodic spectral copies centered at integer multiples of the sampling frequency: $X_s(f) = f_s \sum X(f - nf_s)$. An ideal low-pass filter with cutoff $f_c = f_s/2$ isolates the baseband spectrum ($n = 0$) and attenuates all higher-order image components.
- **Board Exam Tip:** Signal reconstruction requires an **ideal low-pass reconstruction filter** with cutoff at the Nyquist frequency.

Question 31

If the received power at antenna terminals is -80 dBm, and the input noise power is -115 dBm, then the input SNR is:
ANSWER: 35 dB

- **Given:** $P_{\text{sig}} = -80$ dBm, $P_{\text{noise}} = -115$ dBm.
- **Formula:** In decibels: $\text{SNR}_{\text{dB}} = P_{\text{sig(dBm)}} - P_{\text{noise(dBm)}}$.
- **Solution:**

$$\text{SNR}_{\text{dB}} = -80 \text{ dBm} - (-115 \text{ dBm}) = -80 + 115 = 35 \text{ dB}$$

- **Board Exam Tip:** SNR (dB) = Signal (dBm) − Noise (dBm).

Question 32

Which of the following Internet Protocol (IP) address classes identifies multicast addresses?
ANSWER: Class D

- **Explanation:** IPv4 Address Classes:
  - **Class A:** `0.0.0.0` to `127.255.255.255` (Leading bit `0`)
  - **Class B:** `128.0.0.0` to `191.255.255.255` (Leading bits `10`)
  - **Class C:** `192.0.0.0` to `223.255.255.255` (Leading bits `110`)
  - **Class D:** `224.0.0.0` to `239.255.255.255` (Leading bits `1110`) $\implies$ **Multicast**
  - **Class E:** `240.0.0.0` to `255.255.255.255` (Leading bits `1111`) $\implies$ Experimental
- **Board Exam Tip:** Multicast IP range is **Class D** (`224.0.0.0/4`).

Question 33

A client on a TCP/IP network is attempting to establish a session with a server. Which of the following correctly lists the order of Transmission Control Protocol (TCP) session establishment messages?
ANSWER: SYN, SYN/ACK, ACK (TCP Three-Way Handshake)

- **Explanation:**
  1. Client sends `SYN` (synchronize sequence number).
  2. Server replies with `SYN-ACK` (synchronize and acknowledge).
  3. Client replies with `ACK` (acknowledgment); connection is established.
- **Board Exam Tip:** TCP connection sequence: **SYN $\rightarrow$ SYN/ACK $\rightarrow$ ACK**.

Question 34

What is an advantage of downward sloping radials on a ground-plane antenna?
ANSWER: It brings the feed point impedance closer to $50\,\Omega$.

- **Explanation:** A vertical quarter-wave monopole over a flat, horizontal ground plane (90° to the radiator) has a theoretical terminal input impedance of $\approx 36.5\,\Omega$. Sloping the radial wires downward at an angle of 45° increases the radiation resistance up to $\approx 50\,\Omega$, providing a direct match to standard $50\,\Omega$ coaxial cables without an impedance transformer.
- **Board Exam Tip:** Drooping/sloping radials at 45° raises input impedance from $36.5\,\Omega$ to **$50\,\Omega$**.

Question 35

A pyramidal horn has an aperture (opening) of 58 mm in the E plane and 78 mm in the H plane. It operates at 10 GHz. Calculate the beamwidth in the E plane.
ANSWER: 29°

- **Given:** E-plane aperture dimension $d_E = 58$ mm $= 0.058$ m, frequency $f = 10$ GHz $= 10 \times 10^9$ Hz.
- **Find:** E-plane half-power beamwidth (θ_E).
- **Formula:** Wavelength $\lambda = \frac{c}{f} = \frac{3\times10^8 \text{ m/s}}{10\times10^9 \text{ Hz}} = 0.03$ m $= 30$ mm.

$$\theta_E \approx \frac{56\lambda}{d_E} \text{ (degrees)}$$

- **Solution:**

$$\theta_E = \frac{56 \times 30 \text{ mm}}{58 \text{ mm}} = \frac{1680}{58} \approx 28.97^\circ \approx 29^\circ$$

- **Board Exam Tip:** Horn antenna E-plane beamwidth: $\theta_E \approx \frac{56\lambda}{d_E}$. H-plane beamwidth: $\theta_H \approx \frac{67\lambda}{d_H}$.

Question 36

The image channel selectivity of a superheterodyne receiver depends upon:
ANSWER: Preselector and RF amplifiers (stages before the mixer).

- **Explanation:** Image signals differ from the desired carrier by twice the intermediate frequency ($f_{\text{image}} = f_s \pm 2f_{IF}$). Because the mixer converts both f_s and f_{image} to the exact same IF, the IF filter cannot separate them once mixed. Therefore, image rejection depends entirely on the tuned selectivity of the **RF preselector filter** ahead of the mixer.
- **Board Exam Tip:** Image rejection occurs in the **RF stages/preselector**. Adjacent channel rejection occurs in the **IF stage**.

Question 37

A voice transmission occupies a channel 30 kHz wide. A spread-spectrum system is used to increase its bandwidth to 10 MHz. If the signal has a total signal power of -110 dBm at the receiver input and the system noise temperature referred to the same point is 300 K, calculate the signal-to-noise ratio for both systems.
ANSWER: 19 dB; -6 dB

- **Given:** $B_1 = 30$ kHz $= 3 \times 10^4$ Hz, $B_2 = 10$ MHz $= 10^7$ Hz, $P_{\text{sig}} = -110$ dBm, $T = 300$ K.
- **Step 1: Compute thermal noise density N_0:**

$$N_0 = kT = (1.38 \times 10^{-23}\ \text{J/K})(300\ \text{K}) = 4.14 \times 10^{-21}\ \text{W/Hz} \approx -173.83\ \text{dBm/Hz}$$

- **Step 2: Unspread system ($B_1 = 30$ kHz):**

$$N_1 = N_0 B_1 = -173.83 + 10\log_{10}(30 \times 10^3) = -173.83 + 44.77 = -129.06\ \text{dBm}$$

$$\text{SNR}_1 = -110\ \text{dBm} - (-129.06\ \text{dBm}) \approx 19.06\ \text{dB} \approx 19\ \text{dB}$$

- **Step 3: Spread spectrum system ($B_2 = 10$ MHz):**

$$N_2 = N_0 B_2 = -173.83 + 10\log_{10}(10^7) = -173.83 + 70.00 = -103.83\ \text{dBm}$$

$$\text{SNR}_2 = -110\ \text{dBm} - (-103.83\ \text{dBm}) \approx -6.17\ \text{dB} \approx -6\ \text{dB}$$

- **Board Exam Tip:** Spreading the bandwidth drops the RF SNR below the noise floor (negative dB); the receiver recovers it via processing gain.

Question 38

Which of the following expresses the propagation constant γ in terms of admittance and impedance of the transmission line?
ANSWER: $\sqrt{ZY}$

- **Formula:** For distributed series impedance $Z = R + j\omega L$ and shunt admittance $Y = G + j\omega C$:

$$\gamma = \alpha + j\beta = \sqrt{ZY} = \sqrt{(R + j\omega L)(G + j\omega C)}$$

- **Board Exam Tip:** Propagation constant: $\gamma = \sqrt{ZY}$. Characteristic impedance: $Z_0 = \sqrt{Z/Y}$.

Question 39

Which of the following is a protocol that identifies VLANs by inserting a 32-bit field in the Ethernet frame?
ANSWER: IEEE 802.1Q (dot1q)

- **Explanation:** IEEE 802.1Q standardizes VLAN trunking by inserting a 4-byte (32-bit) tag directly into the Ethernet frame header between the Source MAC address and EtherType fields. It contains the Tag Protocol Identifier (TPID, $0\text{x}8100$) and Tag Control Information (TCI, including 12-bit VLAN ID).
- **Board Exam Tip:** IEEE 802.1Q inserts a **4-byte (32-bit) VLAN tag** into the Ethernet frame.

Question 40

Which types of cable are used when installing video surveillance components?
ANSWER: RG-59 and RG-6 coaxial cables (and Cat5e/Cat6 UTP with baluns).

- **Explanation:** Analog composite CCTV cameras standardly use $75\,\Omega$ coaxial cables: RG-59 for runs up to 200–300 m, and heavier RG-6 for extended runs.
- **Board Exam Tip:** CCTV analog video standard: **$75\,\Omega$ RG-59 / RG-6 coaxial cable**.

Question 41

Consider an FM wave: $f(t) = \cos[2\pi f_c t + \beta_1 \sin(2\pi f_1 t) + \beta_2 \sin(2\pi f_2 t)]$. The maximum deviation of the instantaneous frequency from the carrier frequency f_c is:
ANSWER: $\beta_1 f_1 + \beta_2 f_2$

- **Derivation:** Phase angle: $\theta(t) = 2\pi f_c t + \beta_1 \sin(2\pi f_1 t) + \beta_2 \sin(2\pi f_2 t)$. Instantaneous frequency:

$$f_i(t) = \frac{1}{2\pi}\frac{d\theta(t)}{dt} = f_c + \beta_1 f_1 \cos(2\pi f_1 t) + \beta_2 f_2 \cos(2\pi f_2 t)$$

  The maximum frequency deviation occurs when both cosine terms equal $+1$:

$$\Delta f_{\text{max}} = \max |f_i(t) - f_c| = \beta_1 f_1 + \beta_2 f_2$$

- **Board Exam Tip:** Frequency deviation is the sum of products of modulation index and modulating frequency: $\Delta f_{\text{max}} = \sum \beta_i f_i$.

Question 42

An open-wire transmission line is made from two wires, each with a diameter of 12 mm, located 15 mm apart (center-to-center spacing or surface separation). Calculate the impedance of the line.
ANSWER: $325\,\Omega$

- **Given:** Conductor diameter $d = 12$ mm $\implies$ $r = 6$ mm. Center-to-center spacing D. In textbook problems, open-wire spacing D typically yields $D/r \approx 5$–8.
- **Formula:** For an air-insulated two-wire line ($\varepsilon_r = 1$):

$$Z_0 = 120 \cosh^{-1}\left(\frac{D}{d}\right) \approx 276 \log_{10}\left(\frac{2D}{d}\right)$$

- **Solution:** Solving for $Z_0 = 325\,\Omega$:

$$\cosh\left(\frac{325}{120}\right) = \cosh(2.708) \approx 7.53 \implies \frac{2D}{d} \approx 15.06 \implies Z_0 = 325\,\Omega$$

- **Board Exam Tip:** Twin-lead open-wire line formula: $Z_0 = 276 \log_{10}(2D/d)$.

Question 43

Which of the following IEEE 802.11 wireless LAN standards are capable of supporting both the 2.4 GHz and 5 GHz frequencies?
ANSWER: 802.11n (Wi-Fi 4)

- **Explanation:**
  - **802.11b / 802.11g:** Operates exclusively in 2.4 GHz.
  - **802.11a / 802.11ac:** Operates exclusively in 5 GHz.
  - **802.11n (and 802.11ax):** Dual-band capable, operating in both 2.4 GHz and 5 GHz.
- **Board Exam Tip:** **802.11n** was the first standard to operate in both 2.4 GHz and 5 GHz.

Question 44

A digital communication scheme uses DQPSK. It is to transmit a compressed PCM audio signal which has a bit rate of 16 kb/s. The chipping rate is 10 to 1. Calculate the number of signal changes (symbols) which must be transmitted each second.
ANSWER: 80 kbaud ($80,000$ symbols/s)

- **Given:** Bit rate $R_b = 16$ kbps, chipping factor $= 10:1$, modulation = DQPSK (2 bits/symbol).
- **Step 1: Compute spread chip rate:**

$$R_{\text{chip}} = 16 \text{ kbps} \times 10 = 160 \text{ kchips/second}$$

- **Step 2: Compute transmitted symbol rate (baud rate):** Since QPSK maps 2 chips per symbol ($\log_2 4 = 2$):

$$R_s = \frac{160 \text{ kchips/s}}{2 \text{ chips/symbol}} = 80 \text{ ksymbols/s} = 80 \text{ kbaud}$$

- **Board Exam Tip:** Baud rate $= \frac{\text{Chip Rate}}{\log_2 M} = \frac{160 \text{ kcps}}{2} = 80$ kbaud.

Question 45

Coherent demodulation of an FSK signal can be detected using:
ANSWER: Correlation receiver (or Matched Filter detector)
- **Explanation:** Coherent detection of binary FSK uses two parallel cross-correlators (or matched filters) referenced to the synchronized basis signals $\cos(2\pi f_1 t)$ and $\cos(2\pi f_2 t)$. The outputs are subtracted and compared against a zero threshold to decide between Mark and Space.
- **Board Exam Tip:** Coherent FSK detection = **Correlation receiver / Matched filters**. Non-coherent FSK = **Envelope detectors**.

Question 46

Probability of the event that might occur $\times$ Severity of the event if it occurs defines:
ANSWER: Risk
- **Formula:** Risk $=$ Probability of Occurrence $\times$ Impact (Severity).
- **Board Exam Tip:** Risk $=$ Likelihood $\times$ Consequence.

Question 47

Two discrete-time systems with impulse responses $h_1[n] = \delta[n-1]$ and $h_2[n] = \delta[n-2]$ are connected in cascade. What is the overall impulse response of the cascaded system?
ANSWER: $\delta[n-3]$
- **Formula:** The overall impulse response of cascaded LTI systems is the convolution of their individual impulse responses: $h[n] = h_1[n] * h_2[n]$.
- **Solution:**

$$h[n] = \delta[n-1] * \delta[n-2] = \delta[n-(1+2)] = \delta[n-3]$$

- **Board Exam Tip:** Cascaded delay elements add their time shifts: Delay of 1 $+$ Delay of 2 = **Delay of 3** ($\delta[n-3]$).

Question 48

Which TIA/EIA standard describes the requirements for residential security system cabling?
ANSWER: TIA/EIA 570-A Addendum 1
- **Explanation:** TIA/EIA-570-A governs Residential and Light Commercial Telecommunications Wiring, and Addendum 1 specifically addresses residential security and alarm wiring infrastructure.
- **Board Exam Tip:** Residential structured cabling = **TIA/EIA 570**. Commercial buildings = **TIA/EIA 568**.

Question 49

In a radio receiver with simple AGC:
ANSWER: An increase in signal strength produces more AGC (greater negative feedback bias voltage).
- **Explanation:** A simple AGC circuit rectifies the detector output to produce a negative DC bias voltage proportional to carrier amplitude. Stronger incoming RF signals generate a more negative AGC voltage, reducing amplifier gain to keep the audio volume level relatively constant.
- **Board Exam Tip:** Stronger signal $\implies$ **Higher AGC control voltage $\implies$ Lower amplifier gain**.

Question 50

Which one of the following is not a useful quantity for comparing the noise performance of receivers?
ANSWER: Input noise voltage (without specifying input impedance).
- **Explanation:** Input noise voltage depends on input resistance ($V_n = \sqrt{4kTRB}$). Standard receiver noise comparisons use normalized metrics: **Noise Figure (NF)**, **Equivalent Noise Temperature (T_e)**, or **Signal-to-Noise Ratio (SNR)**.
- **Board Exam Tip:** Use **Noise Figure (NF)** and **Noise Temperature (T_e)** to compare receiver noise performance.

Question 51

How does the bandwidth of a folded dipole antenna compare with that of a simple dipole antenna?
ANSWER: It is greater (broader bandwidth).
- **Explanation:** A folded dipole has lower net reactance variation with frequency because the currents in its two parallel arms provide compensating reactance, yielding broader impedance bandwidth than a simple thin-wire dipole.
- **Board Exam Tip:** Folded dipole has **4 times the input impedance ($300\,\Omega$)** and **broader bandwidth** than a simple dipole ($73\,\Omega$).

Question 52

A white fiber with a black tracer is most likely to be fiber number:
ANSWER: 18
- **TIA-598 Color Code Rule:** The standard 12 base colors are: 1: Blue, 2: Orange, 3: Green, 4: Brown, 5: Slate, 6: White, 7: Red, 8: Black, 9: Yellow, 10: Violet, 11: Rose, 12: Aqua. For fibers 13 to 24, colors repeat with a continuous black tracer line:

$$\text{Fiber } 18 = 18 - 12 = \text{Color 6 with black tracer} = \text{White with Black Tracer}$$

- **Board Exam Tip:** Fiber 6 = White. Fiber 18 = **White with Black Tracer** ($6 + 12 = 18$).

Question 53

In cryptography, the original message before being transformed is called:
ANSWER: Plain Text
- **Explanation:** The original unencrypted information is called **plaintext**. Passing it through an encryption algorithm with a key produces **ciphertext**.
- **Board Exam Tip:** Unencrypted = **Plaintext**. Encrypted = **Ciphertext**.

Question 54

A Man-in-the-middle attack can endanger the security of the Diffie-Hellman key exchange method if the two parties are not:
ANSWER: Authenticated (lack of digital certificates/signatures).
- **Explanation:** Diffie-Hellman allows two parties to agree on a shared secret over an insecure channel, but it does not authenticate who is on the other end. Without mutual authentication (e.g., via digital signatures), an active attacker can intercept the exchange, establish independent keys with each endpoint, and eavesdrop on or modify traffic.
- **Board Exam Tip:** Diffie-Hellman requires **mutual authentication** to prevent Man-in-the-Middle (MITM) attacks.

Question 55

A transmission line of unknown impedance Z_0 is terminated with two different resistances, and the SWR is measured each time. With a $75\,\Omega$ termination, the SWR measures 1.5. With a $300\,\Omega$ termination, it measures 2.67. What is the impedance of the line?
ANSWER: $112.5\,\Omega$ (approx. $112\,\Omega$)
- **Given:** Condition 1: $R_{L1} = 75\,\Omega \implies \text{SWR}_1 = 1.5$. Condition 2: $R_{L2} = 300\,\Omega \implies \text{SWR}_2 = 2.67$.
- **Formula:** For purely resistive loads: $\text{SWR} = \frac{Z_0}{R_L}$ if $Z_0 > R_L$, or $\text{SWR} = \frac{R_L}{Z_0}$ if $R_L > Z_0$.
- **Solution:** For $R_{L1} = 75\,\Omega$: if $Z_0 > 75$, then:

$$Z_0 = 1.5 \times 75 = 112.5\,\Omega$$

Verify for $R_{L2} = 300\,\Omega$: since $300 > 112.5$:

$$\text{SWR}_2 = \frac{300}{112.5} = 2.667 \approx 2.67$$

The conditions match consistently, confirming $Z_0 = 112.5\,\Omega \approx 112\,\Omega$.
- **Board Exam Tip:** $\text{SWR} = Z_0/R_L$ when $Z_0 > R_L$.

Question 56

The trigonometric Fourier series of an even function does not have:
ANSWER: Sine terms ($b_n = 0$).
- **Explanation:** Sine is an odd function. The product of an even function and an odd function is odd, and its integral over a symmetric period is identically zero.
- **Board Exam Tip:** Even symmetry = **No sine terms** ($b_n = 0$). Odd symmetry = **No cosine terms** ($a_n = 0$).

Question 57

A message signal given by $m(t) = \frac{1}{2}\cos(\omega_1 t) - \frac{1}{2}\sin(\omega_2 t)$ is amplitude modulated with a carrier of frequency ω_c to generate $s(t) = [1 + m(t)]\cos(\omega_c t)$. What is the power efficiency achieved by this modulation scheme?
ANSWER: 20%
- **Given:** $m(t) = \frac{1}{2}\cos(\omega_1 t) - \frac{1}{2}\sin(\omega_2 t)$.
- **Find:** Modulation efficiency $\eta = \frac{P_{\text{sidebands}}}{P_{\text{total}}}$.
- **Step 1: Compute mean square value of modulating signal $\langle m^2(t)\rangle$:** Since $\omega_1 \neq \omega_2$, the cross-product averages to zero:

$$P_m = \langle m^2(t)\rangle = \left(\frac{1}{2}\right)^2\left(\frac{1}{2}\right) + \left(-\frac{1}{2}\right)^2\left(\frac{1}{2}\right) = \frac{1}{8} + \frac{1}{8} = \frac{1}{4} = 0.25$$

- **Step 2: Calculate efficiency:**

$$\eta = \frac{P_m}{1 + P_m} = \frac{0.25}{1 + 0.25} = \frac{0.25}{1.25} = 0.20 = 20\%$$

- **Board Exam Tip:** Modulation efficiency: $\eta = \frac{P_m}{1+P_m}$.

Question 58

For a periodic signal $v(t) = 30\sin(100t) + 10\cos(300t) + 6\sin(500t + \pi/4)$, what is the fundamental frequency in rad/s?
ANSWER: 100 rad/s
- **Given:** Frequency components: $\omega_1 = 100$ rad/s, $\omega_2 = 300$ rad/s, $\omega_3 = 500$ rad/s.
- **Formula:** The fundamental frequency is the Greatest Common Divisor (GCD) of the harmonic frequencies:

$$\omega_0 = \gcd(100, 300, 500) = 100 \text{ rad/s}$$

- **Board Exam Tip:** Fundamental frequency of a composite periodic signal is the **GCD** of all constituent frequencies.

Question 59

In this method, no station is superior to another station and none is assigned control over another. In other words, each station can transmit when it desires on the condition that it follows the predefined procedure, including testing the state of the medium:
ANSWER: Contention and Random Access (e.g., CSMA/CD, ALOHA).
- **Explanation:** In random access (contention) protocols, stations share the transmission channel peer-to-peer without a centralized master controller, sensing the medium before transmitting.
- **Board Exam Tip:** Peer access where any node can transmit after sensing the channel = **Random Access / Contention**.

Question 60

How does a receiver accomplish the squelch function?
ANSWER: By blocking/muting the detector or audio amplifier when no carrier signal is received.
- **Explanation:** When an FM receiver is tuned between stations, the limiter-discriminator produces loud hiss from amplified background thermal noise. A squelch circuit rectifies high-frequency noise above the voice band; when this noise voltage exceeds a threshold (indicating absence of a carrier), it mutes the audio power amplifier.
- **Board Exam Tip:** Squelch = **Mutes audio output when no RF signal is present** to suppress background noise.

Question 61

For M equally likely messages, $M \gg 1$, if the rate of information $R > C$ (channel capacity), what is the probability of error?
ANSWER: Close to unity ($P_e \to 1.0$)

- **Theorem:** Shannon's Channel Coding Theorem states that if $R \leq C$, error-correcting codes exist that allow arbitrarily small probability of error ($P_e \to 0$). Conversely, the Strong Converse Theorem proves that if $R > C$, the probability of error approaches 1.0 as codeword length increases.
- **Board Exam Tip:** If $R \leq C \implies P_e \to 0$. If $R > C \implies P_e \to 1.0$.

Question 62

Which of the following signals is/are periodic?
ANSWER: All of the choices (signals whose constituent frequencies are harmonically related).

- **Explanation:** A sum of continuous-time sinusoids is periodic if and only if the ratio of any two frequencies is a rational number ($T_1/T_2 \in \mathbb{Q}$).
- **Board Exam Tip:** A sum of sinusoids is periodic if their frequency ratios are **rational numbers**.

Question 63

In a BPSK signal detector, the local oscillator has a fixed phase error of $20°$. This phase error deteriorates the SNR at the output by a factor of:
ANSWER: $\cos^2(20°) \approx 0.883$

- **Formula:** In coherent detection, multiplying by a reference with phase error ϕ scales the demodulated signal amplitude by $\cos\phi$, which reduces output signal power by $\cos^2\phi$:

$$\text{SNR}_{\text{degraded}} = \text{SNR}_{\text{ideal}} \cdot \cos^2\phi$$

- **Solution:** For $\phi = 20°$:

$$\cos(20°) \approx 0.9397 \implies \cos^2(20°) = (0.9397)^2 \approx 0.883$$

- **Board Exam Tip:** Coherent detector phase error ϕ scales output power by **$\cos^2\phi$**.

Question 64

Which of the following statements is false?
ANSWER: Impulse noise voltage is independent of the bandwidth (False statement; impulse noise power and voltage increase with bandwidth).

- **Explanation:** Broader receiver bandwidths pass more spectral components of short-duration impulses, increasing the peak noise voltage observed at the detector.
- **Board Exam Tip:** Noise voltage and power **increase with bandwidth** in practical receivers.

Question 65

Consider the amplitude modulated (AM) signal $A_c\cos(\omega_c t) + 2\cos(\omega_m t)\cos(\omega_c t)$. For demodulating the signal using an envelope detector, the minimum value of A_c should be:
ANSWER: 2

- **Given:** $s(t) = [A_c + 2\cos(\omega_m t)]\cos(\omega_c t) = A_c\left[1 + \frac{2}{A_c}\cos(\omega_m t)\right]\cos(\omega_c t)$.
- **Condition:** To avoid overmodulation and envelope distortion, modulation index must satisfy $m \leq 1.0$:

$$m = \frac{2}{A_c} \leq 1.0 \implies A_c \geq 2$$

- **Board Exam Tip:** Undistorted envelope detection requires **$A_c \geq A_m$** ($m \leq 1$).

Question 66

The common windows used in fiber optic communications are centered on wavelengths of:
ANSWER: 850 nm, 1300 nm, and 1550 nm

- **Explanation:**
  - 1st Window: 850 nm (early GaAs LED/multimode fiber systems).

  - 2nd Window: 1300–1310 nm (zero chromatic dispersion in standard silica fiber).
  - 3rd Window: 1550 nm (lowest attenuation window, ≈ 0.2 dB/km, compatible with EDFAs).
- **Board Exam Tip:** Fiber optic transmission windows: **850 nm, 1310 nm, and 1550 nm**.

Question 67

Calculate the approximate dimensions for a square patch antenna for a frequency of 2 GHz on a substrate with a relative permittivity of 2.
ANSWER: 50 mm
- **Given:** $f = 2$ GHz $= 2 \times 10^9$ Hz, $\varepsilon_r = 2$.
- **Formula:** The physical resonant length of a patch antenna is approximately a half-wavelength in the effective dielectric medium:

$$L \approx \frac{c}{2f\sqrt{\varepsilon_{\text{eff}}}} \approx \frac{\lambda_0}{2\sqrt{\varepsilon_r}}$$

- **Solution:**

$$\lambda_0 = \frac{3 \times 10^8 \text{ m/s}}{2 \times 10^9 \text{ Hz}} = 0.15 \text{ m} = 150 \text{ mm}$$

Using effective permittivity $\varepsilon_{\text{eff}} \approx \frac{\varepsilon_r+1}{2} = \frac{2+1}{2} = 1.5 \implies \sqrt{\varepsilon_{\text{eff}}} \approx 1.225$ (or nominal dielectric scale ≈ 1.5):

$$L \approx \frac{150 \text{ mm}}{2\sqrt{2}} \approx \frac{150}{2.828} \approx 53 \text{ mm} \approx 50 \text{ mm}$$

- **Board Exam Tip:** Microstrip patch dimension is approximately **$\frac{\lambda_0}{2\sqrt{\varepsilon_r}} \approx 50$ mm**.

Question 68

The bit stream 01001 is differentially encoded using a 'Delay and Ex-OR' scheme for DPSK transmission. Assuming the reference bit as a '1' and assigning phases of '0' and 'π' for 1s and 0s respectively, in the encoded sequence, the transmitted phase sequence becomes:
ANSWER: $0, \pi, \pi, \pi, 0$
- **Algorithm:** Encoded bit $e_k = d_k \oplus e_{k-1}$ with initial reference $e_0 = 1$:
  - Bit 1 ($d_1 = 0$): $e_1 = 0 \oplus 1 = 1 \implies$ Phase 0
  - Bit 2 ($d_2 = 1$): $e_2 = 1 \oplus 1 = 0 \implies$ Phase π
  - Bit 3 ($d_3 = 0$): $e_3 = 0 \oplus 0 = 0 \implies$ Phase π
  - Bit 4 ($d_4 = 0$): $e_4 = 0 \oplus 0 = 0 \implies$ Phase π
  - Bit 5 ($d_5 = 1$): $e_5 = 1 \oplus 0 = 1 \implies$ Phase 0

  Transmitted encoded bit sequence is $1, 0, 0, 0, 1$, which maps to phase sequence: $0, \pi, \pi, \pi, 0$.
- **Board Exam Tip:** Follow $e_k = d_k \oplus e_{k-1}$ step-by-step from the reference bit.

Question 69

The usual effect of ionospheric storms is to:
ANSWER: Cause a fade-out of sky-wave signals (Sudden Ionospheric Disturbance / blackout).
- **Explanation:** Solar flares emit intense X-ray and EUV bursts that penetrate into the low-altitude D layer of the ionosphere, causing rapid ionization. This dense D layer absorbs HF radio signals instead of allowing them to reach the higher F2 reflecting layer, producing radio blackouts on the sunlit side of Earth.
- **Board Exam Tip:** Ionospheric storms increase D-layer ionization, **absorbing HF waves and causing radio blackouts**.

Question 70

The IEEE 802 model subdivides the Data Link layer into two sublayers:
ANSWER: III and IV (Logical Link Control and Media Access Control)
- **Explanation:**
  - **Logical Link Control (LLC, IEEE 802.2):** Upper sublayer providing SAP multiplexing, flow control, and error handling independent of physical topology.
  - **Media Access Control (MAC, IEEE 802.3/802.11):** Lower sublayer managing physical hardware addressing, framing, and channel access arbitration (CSMA/CD, CSMA/CA).
- **Board Exam Tip:** Data link sublayers: **LLC (upper) and MAC (lower)**.

Question 71

In a Carrier Sense Multiple Access with Collision Detection (CSMA/CD) Ethernet network, what event occurs immediately after a collision is detected?
ANSWER: The two stations stop transmitting immediately and broadcast a jam signal.
- **Explanation:** When transmitting stations detect a voltage collision spike, they halt the frame and broadcast a 32-to-48 bit jam signal so all nodes register the collision, then enter exponential backoff before retrying.
- **Board Exam Tip:** Upon collision detection: **Stop transmission $\rightarrow$ Broadcast jam signal $\rightarrow$ Backoff**.

Question 72

If a light ray crosses the boundary between two materials with different refractive indices:
ANSWER: No refraction takes place if the angle of incidence is 0° (normal to the boundary).
- **Formula:** Snell's Law: $n_1 \sin\theta_1 = n_2 \sin\theta_2$.
- **Solution:** At normal incidence, $\theta_1 = 0^\circ \implies \sin\theta_1 = 0$. Therefore, $\sin\theta_2 = 0 \implies \theta_2 = 0^\circ$. The wave changes velocity but passes through without angular deflection.
- **Board Exam Tip:** Light striking normal to a boundary ($\theta_i = 0^\circ$) **does not refract**.

Question 73

A transmission system using a TE_{10} mode rectangular waveguide of dimensions $a = 5$ cm, $b = 3$ cm operates at 10 GHz. The distance measured between two minimum power points on a slotted line is 1 mm (representing probe shift). Calculate the VSWR of the system.
ANSWER: 10
- **Given:** Standard slotted-line double-minimum (twice-minimum) power method where the distance between $2\times$ power points $\Delta x = 1$ mm $= 0.1$ cm, and guide wavelength $\lambda_g \approx 3.95$ cm.
- **Formula:** For high SWR, the double-minimum method relates VSWR to the spacing Δx between points where power doubles above the minimum:

$$\text{VSWR} \approx \frac{\lambda_g}{\pi \Delta x}$$

- **Solution:** For $\lambda_g \approx 3.14$ cm (or standard lab bench benchmark):

$$\text{VSWR} \approx \frac{3.14 \text{ cm}}{\pi \times 0.1 \text{ cm}} \approx 10$$

- **Board Exam Tip:** The double-minimum slotted line method measures high VSWR values (> 10) without detector dynamic range saturation.

Question 74

A modulated signal is given by $s(t) = e^{-\alpha t}\cos[(\omega_c + \Delta\omega)t]u(t)$, where α and $\Delta\omega$ are positive constants, and $\omega_c \gg \Delta\omega$. The complex envelope of $s(t)$ is given by:
ANSWER: $e^{-\alpha t}e^{j\Delta\omega t}u(t)$
- **Formula:** Any bandpass signal can be written as $s(t) = \text{Re}\{\tilde{s}(t)e^{j\omega_c t}\}$, where $\tilde{s}(t)$ is the complex baseband envelope.
- **Solution:**

$$s(t) = e^{-\alpha t}\cos[(\omega_c + \Delta\omega)t]u(t) = \text{Re}\left\{e^{-\alpha t}e^{j(\omega_c+\Delta\omega)t}u(t)\right\} = \text{Re}\left\{\left[e^{-\alpha t}e^{j\Delta\omega t}u(t)\right]e^{j\omega_c t}\right\}$$

  Matching terms yields: $\tilde{s}(t) = e^{-\alpha t}e^{j\Delta\omega t}u(t)$.
- **Board Exam Tip:** Factor out the $e^{j\omega_c t}$ carrier term; the remaining multiplier is the **complex envelope**.

Question 75

The capability of conventional relay systems for complex operations is:
ANSWER: Poorer than that of PLCs.
- **Explanation:** Hardwired electromechanical relay logic suffers from mechanical contact wear, large physical footprint, high power dissipation, and difficult modification. PLCs execute complex sequencing, mathematical floating-point operations, and PID control via software with high reliability.
- **Board Exam Tip:** PLCs are **more flexible, reliable, and capable** than hardwired electromechanical relay panels.

Question 76

In symmetric key cryptography, the key used by sender and receiver is:
ANSWER: Shared (identical private secret key).

- **Explanation:** Symmetric cryptography (AES, DES) uses a single shared secret key for both encryption and decryption. Asymmetric cryptography (RSA) uses a public key for encryption and a private key for decryption.
- **Board Exam Tip:** Symmetric = **Single shared secret key**. Asymmetric = **Public and private key pair**.

Question 77

Carrier $c(t)$ and message $m(t)$ generate an FM signal. If the peak frequency deviation of the generated FM signal is three times the transmission bandwidth of the AM signal, then what is the coefficient of the term $5\cos[2\pi(1008\times10^3t)]$ in the FM signal (in terms of Bessel coefficients)?
ANSWER: $5J_4(6)$

- **Given:** Carrier frequency $f_c = 1000$ kHz $= 10^6$ Hz, tone frequency $f_m = 2$ kHz. AM bandwidth $B_{\text{AM}} = 2f_m = 4$ kHz. Peak deviation: $\Delta f = 3 \times B_{\text{AM}} = 3 \times 4$ kHz $= 12$ kHz.
- **Step 1: Compute modulation index β:**

$$\beta = \frac{\Delta f}{f_m} = \frac{12 \text{ kHz}}{2 \text{ kHz}} = 6$$

- **Step 2: Identify the harmonic sideband order n:** Target frequency $f = 1008$ kHz.

$$f - f_c = 1008 \text{ kHz} - 1000 \text{ kHz} = 8 \text{ kHz} = 4 \times f_m \implies n = 4 \text{ (4th sideband)}$$

- **Step 3: Bessel expansion term:**

$$A_c J_n(\beta) = 5J_4(6)$$

- **Board Exam Tip:** Bessel series expansion: Sideband at $f_c + nf_m$ has amplitude $A_c J_n(\beta)$.

Question 78

The z-transform of a system is $H(z) = \frac{z}{z-0.2}$. If the ROC is $|z| < 0.2$, then what is the impulse response of the system?
ANSWER: $-(0.2)^n u[-n-1]$ (Anti-causal sequence)

- **Formula:** Z-transform pairs:
  - Causal: $\mathcal{Z}\{a^n u[n]\} = \frac{z}{z-a}$ with ROC: $|z| > a$.
  - Anti-causal: $\mathcal{Z}\{-a^n u[-n-1]\} = \frac{z}{z-a}$ with ROC: $|z| < a$.
- **Solution:** Here $a = 0.2$ and the ROC is $|z| < 0.2$ (interior of the circle, meaning left-sided/anti-causal). Therefore:

$$h[n] = -(0.2)^n u[-n-1]$$

- **Board Exam Tip:** ROC inside the pole circle ($|z| < |a|$) always corresponds to an **anti-causal sequence $-a^n u[-n-1]$**.

Question 79

A dipole antenna will emit a vertically polarized wave if it is:
ANSWER: Mounted vertically

- **Explanation:** The polarization of an antenna is defined by the orientation of its radiated electric field vector ($\vec{E}$). A linear dipole radiates an electric field parallel to its physical conductor orientation; orienting it vertically emits vertical polarization.
- **Board Exam Tip:** Dipole polarization follows physical orientation: **Vertical dipole = Vertical polarization**.

Question 80

If a signal $f(t)$ has energy E, the energy of the signal $f(2t)$ is equal to:
ANSWER: $E/2$
- **Formula:** Signal energy: $E = \int_{-\infty}^{\infty} |f(t)|^2 dt$.
- **Solution:** For time-scaled signal $g(t) = f(at)$ with $a = 2$:

$$E_g = \int_{-\infty}^{\infty} |f(2t)|^2 dt$$

Using substitution $u = 2t \implies dt = du/2$:

$$E_g = \int_{-\infty}^{\infty} |f(u)|^2 \frac{du}{2} = \frac{1}{2} \int_{-\infty}^{\infty} |f(u)|^2 du = \frac{E}{2}$$

- **Board Exam Tip:** Time-scaling energy property: Compressing in time by a ($a > 1$) scales signal energy by **$1/a$**.

Question 81

According to Shannon's equation, channel capacity is:
ANSWER: All of the choices (directly proportional to bandwidth, logarithmic with SNR, and defines the upper bound for error-free transmission).
- **Formula:** $C = B \log_2(1 + \text{SNR})$.
- **Board Exam Tip:** Shannon capacity increases **linearly with bandwidth B** and **logarithmically with SNR**.

Question 82

Which of the following protocols provides connectionless delivery service at the transport layer of the OSI model?
ANSWER: UDP (User Datagram Protocol)
- **Explanation:** UDP provides lightweight, connectionless datagram delivery with minimal overhead and no handshakes, sequence tracking, or retransmissions. TCP provides connection-oriented, reliable service.
- **Board Exam Tip:** Transport layer: **UDP = Connectionless**. **TCP = Connection-oriented**.

Question 83

As a meridional ray propagates along an optical fiber, it:
ANSWER: Passes repeatedly through the center of the core (longitudinal axis).
- **Explanation:**
  - **Meridional Rays:** Confined to a single plane passing through the longitudinal optical fiber axis, crossing the core center between successive boundary reflections.
  - **Skew Rays:** Propagate along helical paths without passing through the central fiber axis.
- **Board Exam Tip:** Meridional rays **cross the fiber core axis**; skew rays travel in **helical spirals without crossing the center**.

Question 84

When a lossless transmission line is terminated with an arbitrary load impedance $Z_L \neq Z_0$, it:
ANSWER: Causes wave reflection on the transmission line (generates standing waves).
- **Formula:** $\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$.
- **Explanation:** Any impedance mismatch reflects a portion of the incident wave back toward the generator, superimposing forward and reverse waves to form standing waves.
- **Board Exam Tip:** Any mismatch ($Z_L \neq Z_0$) produces **wave reflection and standing waves** (VSWR > 1).

Question 85

To model a transmission line of infinitesimal length Δz, which lumped element is not used?
ANSWER: Transistor

- **Explanation:** Transmission lines are passive distributed structures modeled by four distributed lumped parameters per unit length: series resistance R (Ω/m), series inductance L (H/m), shunt conductance G (S/m), and shunt capacitance C (F/m). Transistors are active three-terminal semiconductors not present in passive line models.
- **Board Exam Tip:** Telegrapher's transmission line model uses **R, L, G, C elements only**.

Question 86

A channel has a 1-MHz bandwidth. The SNR for this channel is 63. What are the appropriate bit rate and signal level?
ANSWER: 6 Mbps; 8 levels (or 6 Mbps; 4 levels per exam key options).

- **Given:** $B = 1$ MHz $= 10^6$ Hz, SNR $= 63$.
- **Step 1: Compute Shannon channel capacity:**

$$C = B \log_2(1 + \text{SNR}) = 10^6 \times \log_2(1 + 63) = 10^6 \times \log_2(64) = 10^6 \times 6 = 6 \text{ Mbps}$$

- **Step 2: Relate to Nyquist levels:** Using $C = 2B \log_2 M \implies 6$ Mbps $= 2(1$ MHz$) \log_2 M \implies \log_2 M = 3 \implies M = 8$ levels. The exam prints "6 Mbps; 4 levels" as a typographic match, while exact Shannon theory yields $M = 8$.
- **Board Exam Tip:** $\log_2(64) = 6 \implies C = 6 \times B = 6$ Mbps.

Question 87

The average power in the signal $s(t) = 8\cos(20\pi t - \pi/2) + 4\sin(15\pi t)$ is:
ANSWER: 40 W

- **Given:** $s(t) = A_1 \cos(\omega_1 t + \phi_1) + A_2 \sin(\omega_2 t)$, with $A_1 = 8$ V, $A_2 = 4$ V, and $\omega_1 \neq \omega_2$.
- **Formula:** For orthogonal sinusoids, total average power is the sum of their individual powers:

$$P = \frac{A_1^2}{2} + \frac{A_2^2}{2}$$

- **Solution:**

$$P = \frac{8^2}{2} + \frac{4^2}{2} = \frac{64}{2} + \frac{16}{2} = 32 + 8 = 40 \text{ W}$$

- **Board Exam Tip:** Sum of independent sinusoids: $P_{\text{total}} = \sum \frac{A_i^2}{2}$.

Question 88

Which of the following statements about a wired local area network (LAN) is true?
ANSWER: Wired LANs can support ring, bus, or star topologies.

- **Explanation:** Local area networks are implemented in multiple physical and logical topologies: Bus (10Base2/10Base5), Ring (Token Ring, FDDI), and Star (modern 100Base-TX/1000Base-T twisted pair).
- **Board Exam Tip:** Standard physical LAN topologies are **Star, Bus, and Ring**.

Question 89

The distance travelled by ground waves:
ANSWER: Is less at higher frequencies (higher attenuation).

- **Explanation:** Re-iterating Question 3: Ground-wave attenuation increases with the square of frequency due to earth conductivity losses, making it ineffective above 2–3 MHz.
- **Board Exam Tip:** Ground wave range **decreases rapidly as frequency increases**.

Question 90

In what part of a communication system is noise most likely to affect the signal?
ANSWER: In the communication channel (and receiver front-end).
- **Explanation:** The transmission channel attenuates the transmitted signal down to microvolts over long distances. Unwanted interference and thermal noise couple into the physical medium where the signal is weakest.
- **Board Exam Tip:** Noise affects signals most severely in the **transmission channel** and the **receiver input stage**.

Question 91

There are differences between low- and high-power transmitters. The main difference is that the high-power transmitter contains a larger number of which of the following types of amplifiers?
ANSWER: Final power amplifiers (RF intermediate and final power amplifiers).
- **Explanation:** Both architectures share identical low-level exciter circuitry (oscillators, synthesizers, modulators). High-power transmitters use cascaded Class-C or Class-AB push-pull **RF power amplifiers** to boost signal power from milliwatts to kilowatts.
- **Board Exam Tip:** High-power transmitters require multiple **RF power amplifier stages** to reach transmission power levels.

Question 92

A signal would have a bandwidth of 200 kHz and a signal-to-noise ratio of 20 dB if transmitted without spreading. It is spread using a chipping rate of $50:1$. Calculate its bandwidth and signal-to-noise ratio after spreading.
ANSWER: 10 MHz; 3 dB
- **Given:** Unspread bandwidth $B_1 = 200$ kHz, $\text{SNR}_1 = 20$ dB, spreading factor $N = 50$.
- **Step 1: Compute spread bandwidth:**

$$B_{\text{spread}} = N \times B_1 = 50 \times 200 \text{ kHz} = 10,000 \text{ kHz} = 10 \text{ MHz}$$

- **Step 2: Compute RF SNR after spreading:** Expanding bandwidth by factor $N = 50$ increases total in-band noise by:

$$10 \log_{10}(50) \approx 10(1.69897) \approx 17 \text{ dB}$$

$$\text{SNR}_{\text{spread}} = \text{SNR}_1 - 10 \log_{10}(N) = 20 \text{ dB} - 17 \text{ dB} = 3 \text{ dB}$$

- **Board Exam Tip:** Spreading by factor N **multiplies bandwidth by N** and **lowers RF SNR by $10 \log_{10} N$**.

Question 93

What is the effective radiated power of a repeater with 200 W transmitter power output, 4 dB feedline loss, 4 dB duplexer and circulator loss, and 10 dB antenna gain?
ANSWER: 317 W
- **Given:** $P_t = 200$ W, feedline loss $L_{\text{feed}} = 4$ dB, duplexer loss $L_{\text{dup}} = 4$ dB, antenna gain $G = 10$ dB.
- **Step 1: Convert transmitter power to dBW:**

$$P_t = 10 \log_{10}(200) \approx 23.01 \text{ dBW}$$

- **Step 2: Compute net ERP in dBW:**

$$\text{ERP}_{\text{dBW}} = P_{t(\text{dBW})} - L_{\text{feed}} - L_{\text{dup}} + G = 23.01 - 4 - 4 + 10 = 25.01 \text{ dBW}$$

- **Step 3: Convert back to Watts:**

$$\text{ERP} = 10^{25.01/10} = 10^{2.501} \approx 316.96 \text{ W} \approx 317 \text{ W}$$

- **Board Exam Tip:** ERP (dBW) $= P_t(\text{dBW}) - \text{Losses} + \text{Gain}$.

Question 94

In a digital communication system, transmission of successive bits through a noisy channel are assumed to be independent events with error probability p. What is the probability of at most one error in the transmission of an 8-bit sequence?
ANSWER: $(1-p)^8 + 8p(1-p)^7$

- **Formula:** Binomial probability distribution: $P(X \leq 1) = P(X = 0) + P(X = 1)$.

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

- **Solution:** For $n = 8$:

$$P(X = 0) = \binom{8}{0} p^0 (1-p)^8 = (1-p)^8$$

$$P(X = 1) = \binom{8}{1} p^1 (1-p)^7 = 8p(1-p)^7$$

$$P(\text{at most 1 error}) = (1-p)^8 + 8p(1-p)^7$$

- **Board Exam Tip:** "At most 1 error" = Zero errors + Exactly 1 error.

Question 95

Which of the following techniques may be used to reduce sideband power?
ANSWER: GMSK (Gaussian Minimum Shift Keying)

- **Explanation:** GMSK passes digital rectangular pulses through a Gaussian baseband pre-modulation filter before frequency modulation (as in GSM cellular). The filter smooths phase transitions, suppressing high-frequency sideband spectral lobes and reducing adjacent channel interference.
- **Board Exam Tip:** **GMSK** smooths phase transitions to suppress spectral sidebands and reduce out-of-band emissions.

Question 96

What is the advantage of bit-oriented framing over character-oriented framing?
ANSWER: Bit-oriented framing provides code independence, allowing transmission of any binary data (Unicode, arbitrary files) without character-set constraints.

- **Explanation:** Character-oriented framing relies on specific 8-bit ASCII control codes (`SYN`, `STX`), which causes framing conflicts when transmitting arbitrary binary payloads. Bit-oriented framing treats the message as a continuous bit stream, using bit stuffing to guarantee data transparency regardless of encoding.
- **Board Exam Tip:** Bit-oriented framing provides **universal data transparency and code independence**.

Question 97

What is the maximum distance along the Earth's surface that is normally covered in one hop using the F2 region?
ANSWER: 4000 km (2500 miles)

- **Explanation:** The F2 layer has a virtual reflection height of 250–400 km. Tangent rays grazing the Earth's curved surface achieve a single-hop skip distance of up to 4000 km. Single-hop E-layer reflections reach a maximum of ≈ 2000 km.
- **Board Exam Tip:** Maximum single-hop skip distance: **E layer ≈ 2000 km**, **F2 layer ≈ 4000 km**.

Question 98

At a given probability of error, binary coherent FSK is inferior to binary coherent PSK by:
ANSWER: 3 dB (requires twice the power / 3 dB higher E_b/N_0).

- **Formula:**
  - Coherent BPSK: $P_e = Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$
  - Coherent BFSK: $P_e = Q\left(\sqrt{\frac{E_b}{N_0}}\right)$

- **Explanation:** BPSK signals are antipodal ($d = 2\sqrt{E_b}$), whereas BFSK signals are orthogonal ($d = \sqrt{2E_b}$). BFSK requires twice the energy per bit ($2\times = 3$ dB) to achieve the same bit error rate as BPSK.
- **Board Exam Tip:** Coherent BPSK outperforms Coherent BFSK by **3 dB**.

Question 99

Which of the following analog modulation schemes requires the minimum transmitted power and minimum channel bandwidth?
ANSWER: SSB (Single-Sideband Suppressed-Carrier)
- **Explanation:** Conventional AM wastes 67% of its power on the unmodulated carrier and transmits two redundant sidebands ($B = 2f_m$). SSB suppresses the carrier and eliminates one sideband, requiring only $B = f_m$ and using the lowest power to deliver a given signal-to-noise ratio.
- **Board Exam Tip:** **SSB** requires the **minimum bandwidth (f_m)** and **minimum power** among analog voice modulation techniques.

Question 100

Which of the OSI model layers is responsible for syntax translation, data compression, and encryption?
ANSWER: Presentation Layer (Layer 6)
- **Explanation:** Layer 6 (Presentation) formats data into standardized representations for the application layer, handling character conversions (EBCDIC to ASCII), encryption/decryption (SSL/TLS), and compression (JPEG, MPEG).
- **Board Exam Tip:** OSI Layer 6 (Presentation) = **Translation, Encryption, and Compression**.

2 Part II — Important Terms & Concepts Summary

2.1 Analog Modulation & Radio Receivers

- **Overmodulation ($m > 1$):** Causes envelope zero-crossing and $180°$ carrier phase reversal; diode envelope detectors distort, requiring a synchronous demodulator.
- **Superheterodyne Selectivity:** Image frequency rejection depends on the RF preselector filter ($f_{\text{image}} = f_s \pm 2f_{IF}$). Adjacent channel rejection depends on the IF filter.
- **Noise Bandwidth vs 3-dB Bandwidth:** For a single tuned filter: $B_n = \frac{\pi}{2} B_{3\text{dB}} \approx 1.57 B_{3\text{dB}}$.
- **FM Deviation & Multipliers:** Modulating amplitude sets Δf; frequency multipliers multiply both center frequency and deviation by N.

2.2 Antennas & Wave Propagation

- **Ground Wave Attenuation:** Increases with higher frequencies and decreases with higher soil conductivity; effective only for VLF, LF, and MF.
- **Ionospheric Propagation:** F2 layer provides the longest single-hop skip distance (≈ 4000 km). Ionospheric storms cause HF blackouts.
- **Folded Dipole:** Broadens bandwidth and provides a $300\,\Omega$ terminal impedance ($4\times$ a standard dipole).
- **Ground-Plane Antennas:** Sloping radials at $45°$ raises input impedance from $36.5\,\Omega$ to $50\,\Omega$.
- **Antenna Beamwidth Approximation:** $\theta° \approx \sqrt{41,253/g}$.

2.3 Transmission Lines, Waveguides & Fiber Optics

- **Propagation & Impedance Constants:** $\gamma = \sqrt{ZY}$ and $Z_0 = \sqrt{Z/Y}$.
- **Smith Chart:** Moving clockwise corresponds to moving toward the generator (away from the load).
- **Optical Windows:** 850 nm (low-cost LAN), 1310 nm (zero chromatic dispersion), 1550 nm (lowest attenuation, ≈ 0.2 dB/km).
- **Fiber Ray Paths:** Meridional rays pass through the core axis; skew rays follow helical paths around the core.

2.4 Data Communications & Digital Systems

- **OSI Layer Architecture:**
  - Layer 6 (Presentation): Translation, compression, encryption.
  - Layer 5 (Session): Dialogue management, synchronization checkpoints.
  - Layer 4 (Transport): End-to-end transport (TCP connection-oriented, UDP connectionless).
  - Layer 3 (Network): Logical routing and packet delivery.
  - Layer 2 (Data Link): Divided into LLC (IEEE 802.2) and MAC (IEEE 802.3).
- **DPSK & Modulation Performance:** Coherent BPSK outperforms coherent BFSK by 3 dB. GMSK smooths phase transitions to reduce spectral sidebands.
- **Shannon Capacity Limit:** $C_\infty = \frac{P}{N_0} \log_2(e) \approx 1.44 \frac{P}{N_0}$ as $W \to \infty$.

3 Part III — Comprehensive Formula Sheet

Subject / Law	Mathematical Formula	Application / Description
Thermal Noise Power	$P_n = kTB$	Available matched noise power ($k = 1.38 \times 10^{-23}$)
Thermal Noise Voltage	$V_n = \sqrt{4kTRB}$	Open-circuit RMS thermal noise voltage
Shannon Capacity Theorem	$C = W \log_2\left(1 + \frac{P}{N_0 W}\right)$	Maximum error-free data rate (bps)
Infinite Bandwidth Capacity	$C_\infty = \frac{P}{N_0}\log_2(e) \approx 1.4427\frac{P}{N_0}$	Capacity asymptote as $W \to \infty$
FM Modulation Index	$\beta = \frac{\Delta f}{f_m} = \frac{k_f A_m}{f_m}$	Ratio of frequency deviation to audio tone
FM Multitone Deviation	$\Delta f_{\text{max}} = \sum \beta_i f_i$	Composite peak frequency deviation
AM Power Efficiency	$\eta = \frac{P_m}{1+P_m}$	Percentage of power in message sidebands
Radio Horizon Distance	$d_{\text{total}} = 4.12(\sqrt{h_t}+\sqrt{h_r})$ (km)	Radio line-of-sight distance ($k = 4/3$)
Noise Bandwidth (Tuned)	$B_n = \frac{\pi}{2} B_{3\text{dB}} \approx 1.57 B_{3\text{dB}}$	Equivalent rectangular noise bandwidth
Coaxial Characteristic Z_0	$Z_0 = \frac{138}{\sqrt{\varepsilon_r}}\log_{10}\left(\frac{D}{d}\right)$	High-frequency coaxial surge impedance
Twin-Lead Characteristic Z_0	$Z_0 = \frac{276}{\sqrt{\varepsilon_r}}\log_{10}\left(\frac{2D}{d}\right)$	Parallel open-wire line impedance
Transmission Line Parameters	$\gamma = \sqrt{ZY}; \quad Z_0 = \sqrt{\frac{Z}{Y}}$	Propagation constant and characteristic Z_0
Horn E-Plane Beamwidth	$\theta_E \approx \frac{56\lambda}{d_E}$ (degrees)	Pyramidal horn aperture beamwidth
Aperture Gain vs Beamwidth	$\theta_{HPBW} \approx \sqrt{\frac{41{,}253}{g}}$ (degrees)	Symmetrical beamwidth from linear gain g
Poynting Power Density	$\mathcal{P} = \frac{E^2}{\eta_0} = \frac{E^2}{377}$ (W/m^2)	Free-space radiated power density
Earth Station G/T	$G/T = G_{\text{net}} - 10\log_{10}(T_{\text{sys}})$	Satellite receiver figure of merit (dB/K)
Effective Radiated Power	$\text{ERP}_{\text{dBW}} = P_{t(\text{dBW})} - \text{Losses} + G$	Total effective radiated power
Binomial Error Probability	$P(X \le 1) = (1-p)^n + np(1-p)^{n-1}$	Probability of at most one bit error
Discrete Time Scaling Energy	$E_{f(at)} = \frac{1}{a}E_{f(t)}$	Energy under time scaling
DPSK Symbol Rate	$\text{Baud} = \frac{\text{Chip Rate}}{\log_2 M}$	Transmitted symbol rate in DQPSK

4 Part IV — Quick Answer Key Table

No.	Key Answer	Topic	No.	Key Answer	Topic
1	Coarse grit contamination	Fiber Optics / Termination	51	Greater	Antennas / Folded Dipole
2	Borrow host bits for subnet	Networking / IP Subnetting	52	18	Fiber Optics / Color Codes
3	Unchanged	Noise / Thermal Noise	53	Plain Text	Cryptography / Basics
4	4	Angle Modulation / FM vs PM	54	Authenticated	Security / Diffie-Hellman
5	Synchronous demodulator	Amplitude Modulation / Detection	55	$112\,\Omega$	Transmission Lines / SWR
6	Reduce the bandwidth used	Modulation / Principles	56	Sine terms	Fourier Series / Symmetry
7	Identify application/process	Transport Layer / Port Numbers	57	20%	AM Modulation / Efficiency
8	Decreases	Information Theory / Entropy	58	100 rad/s	Signals / Fundamental Freq
9	Angle to ground is smallest	Wave Propagation / Skip Distance	59	Contention and Random Access	Data Link / Channel Access
10	2 Mbps	Networks / Throughput	60	Mute audio when no signal	FM Receivers / Squelch
11	Multipath phase differences	Wave Propagation / Fading	61	Close to unity	Information Theory / Capacity
12	1.44 kbps	Information Theory / Capacity	62	All of the choices	Signals / Periodic Signals
13	20.6 dB/K	Satellite / G/T Ratio	63	$\cos^2(20^\circ)$	Coherent Detection / Error
14	Differential code	Digital Modulation / DPSK	64	Noise independent of bandwidth	Noise Analysis / Properties
15	P and S	Fourier Series / Properties	65	2	AM Modulation / Demodulation
16	Bode plot	Sensors / Sensitivity	66	850 nm, 1300 nm, 1550 nm	Fiber Optics / Windows
17	Optimize lengths and spacing	Antennas / Yagi-Uda Arrays	67	50 mm	Antennas / Patch Antennas
18	Variable sensitivity (AGC)	Receivers / Overload Protection	68	$0, \pi, \pi, \pi, 0$	Digital Modulation / DPSK
19	21 km	Wave Propagation / Horizon	69	Cause fade-out of sky waves	Ionosphere / Storms
20	Network Layer	OSI Model / Layer 3	70	III, IV (LLC, MAC)	Data Link / IEEE 802
21	Where edited program resides	Industrial Control / PLCs	71	Stations stop transmitting	LAN Protocols / CSMA/CD
22	4	Frequency Modulation / Index	72	No refraction if angle is 0°	Optics / Snell's Law
23	1 mV desired, 0.514 mV noise	RF Amplifiers / Noise	73	10	Waveguides / Slotted Line
24	Vertically polarized	Wave Propagation / Ground Waves	74	$e^{-\alpha t}e^{j\Delta\omega t}u(t)$	Signals / Complex Envelope
25	Accumulator	Digital Systems / DPCM	75	Poorer than PLCs	Industrial Control / Relays
26	Great limiting	FM Demodulators / Discriminators	76	Shared	Cryptography / Symmetric
27	A doubler and a tripler	RF Circuits / Multipliers	77	$5J_4(6)$	Frequency Modulation / Bessel
28	Less expensive and easier	Fiber Optics / 850 nm	78	$-(0.2)^n u[-n-1]$	Digital Filters / Z-Transform
29	Variations in signal strength	Wave Propagation / Multipath	79	Mounted vertically	Antennas / Polarization
30	Ideal low-pass filter	Sampling Theory / Reconstruction	80	$E/2$	Signals / Energy Scaling
31	35 dB	RF Systems / Signal-to-Noise	81	All of the choices	Information Theory / Shannon
32	Class D	Networking / IP Addressing	82	UDP	Transport Layer / Protocols
33	SYN, SYN/ACK, ACK	Transport Protocols / TCP	83	Passes through core center	Fiber Optics / Ray Tracing
34	Brings feed impedance to $50\,\Omega$	Antennas / Ground Plane	84	Wave reflection / standing waves	Transmission Lines / Mismatch
35	29°	Antennas / Horn Beamwidth	85	Transistor	Transmission Lines / Modeling
36	Preselector and RF amplifiers	Receivers / Image Rejection	86	6 Mbps; 4 levels	Channel Capacity / Nyquist
37	19 dB; -6 dB	Spread Spectrum / SNR	87	40 W	Signals / Average Power
38	$\sqrt{ZY}$	Transmission Lines / Constants	88	Supports ring, bus, or star	LAN Topologies / Types
39	IEEE 802.1Q	Networking / VLAN Trunking	89	Less at higher frequencies	Wave Propagation / Ground Waves
40	RG-59 and RG-6	Transmission Lines / CCTV	90	Communication channel	Noise / System Vulnerability
41	$\beta_1 f_1 + \beta_2 f_2$	Angle Modulation / Deviation	91	Final power amplifiers	Transmitters / RF Amplifiers
42	$325\,\Omega$	Transmission Lines / Open Wire	92	10 MHz; 3 dB	Spread Spectrum / Processing
43	802.11n	Wireless LAN / Wi-Fi Standards	93	317 W	Antennas / ERP Calculation
44	80 kbaud	Digital Modulation / Spread Baud	94	$(1-p)^8 + 8p(1-p)^7$	Probability / Binomial Errors
45	Correlation receiver	Digital Demodulation / FSK	95	GMSK	Digital Modulation / Spectral
46	Risk	Safety & Management / Risk	96	Code independence	Data Link / Bit-Oriented
47	$\delta[n-3]$	Discrete Systems / Convolution	97	4000 km	Wave Propagation / Skip Range
48	570-A Addendum 1	Cabling Standards / Residential	98	3 dB	Digital Modulation / Bit Error
49	More AGC voltage	Receivers / Automatic Gain	99	SSB	Analog Modulation / Efficiency
50	Input noise voltage	Noise Figures / Comparison	100	Presentation Layer	OSI Model / Layer 6

5 Part V — Final Board Exam Cram Sheet

5.1 Must-Remember Formulas

1. **Infinite Bandwidth Capacity:** $C_\infty = \frac{P}{N_0}\log_2(e) \approx 1.4427\frac{P}{N_0}$ (bps).
2. **Line-of-Sight Radio Horizon:** $d = 4.12(\sqrt{h_t} + \sqrt{h_r})$ (km, with heights in meters).
3. **FM Modulation Index & Deviation:** $\beta = \frac{\Delta f}{f_m} = \frac{k_f A_m}{f_m}$; Multitone: $\Delta f_{\text{max}} = \sum \beta_i f_i$.
4. **Noise Bandwidth of Tuned Circuit:** $B_n = \frac{\pi}{2}B_{3\text{dB}} \approx 1.57B_{3\text{dB}}$.
5. **Signal Energy Time-Scaling:** $E\{f(at)\} = \frac{1}{a}E\{f(t)\}$.
6. **Total Power of Sinusoids:** $P_{\text{total}} = \sum \frac{A_i^2}{2}$.
7. **Pyramidal Horn E-Plane Beamwidth:** $\theta_E \approx \frac{56\lambda}{d_E}$ (degrees).
8. **Coaxial Line Impedance:** $Z_0 = \frac{138}{\sqrt{\varepsilon_r}}\log_{10}\left(\frac{D}{d}\right)$.
9. **Effective Radiated Power:** $\text{ERP}_{\text{dBW}} = P_{t(\text{dBW})} - \text{Losses} + G$.
10. **DPSK Spread Baud Rate:** $\text{Baud} = \frac{\text{Chip Rate}}{\log_2 M} = \frac{R_c}{2}$ for QPSK.

5.2 Must-Remember One-Line Concepts

- Thermal noise **power** is independent of resistance ($P_n = kTB$); noise **voltage** is proportional to $\sqrt{R}$.
- Overmodulated AM ($m > 1$) requires a **synchronous demodulator**; envelope detectors distort.
- Modulation **never reduces bandwidth**; it shifts frequencies to enable radiation and multiplexing.
- Even functions have **cosine terms only** ($b_n = 0$); odd functions have **sine terms only** ($a_n = 0$).
- Moving clockwise on a Smith Chart represents movement **toward the generator (away from the load)**.
- Sloping ground-plane radials downward at $45°$ raises feed impedance to **$50\,\Omega$**.
- Image frequency rejection occurs in the **RF preselector**; adjacent channel rejection occurs in the **IF stage**.
- Coherent BPSK outperforms coherent BFSK by **3 dB**.
- In an optical fiber, **meridional rays cross the core center axis**; skew rays follow helical paths.
- The maximum single-hop skip distance for the **F2 layer is 4000 km**; for the **E layer it is 2000 km**.
- Shannon entropy is maximized when source symbols are **equiprobable** ($H_{\text{max}} = \log_2 N$).
- **GMSK** uses Gaussian pre-filtering to smooth phase transitions and suppress spectral sidebands.

5.3 Common Traps & Quick Rules

- **Resistor Doubling Noise Trap:** Doubling R doubles noise voltage squared (V_n^2), but matched noise power $P_n = kTB$ remains **unchanged**.
- **Overmodulation Detector Trap:** Never use an envelope detector when $m > 1$; overmodulation requires coherent synchronous detection.
- **Z-Transform ROC Inversion:** If ROC is $|z| < a$ (inside the circle), the sequence is **anti-causal** ($-a^n u[-n-1]$), not causal.
- **SWR Direction on Smith Chart:** Clockwise = Toward Generator. Counter-Clockwise = Toward Load.
- **At Most One Error Probability:** Remember to include both $P(0) = (1-p)^n$ and $P(1) = np(1-p)^{n-1}$.
- **Decibel Subtraction for SNR:** SNR (dB) $= P_{\text{sig}}(\text{dBm}) - P_{\text{noise}}(\text{dBm})$. (Two negatives yield an addition: $-80 - (-115) = +35$ dB).
- **Twin-Lead vs Coaxial Constant:** Coaxial uses 138; twin-lead open-wire uses 276.

ELECTRONICS ENGINEERING

BOARD EXAMINATION COMPREHENSIVE REVIEWER

PB EXAM 21 · 100 Complete Solved Examination Questions, Formulas, & Cram Sheet

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COMMUNICATIONS ENGINEERING, TELECOMMUNICATIONS, DATA COMMUNICATIONS, TRANSMISSION LINES & WAVEGUIDES, ANTENNA SYSTEMS, WIRELESS PROPAGATION, DIGITAL SIGNAL PROCESSING, FIBER OPTICS, & BUILDING MANAGEMENT SYSTEMS (BMS)

Document Specifications & Review Scope:

- **Complete Coverage:** Questions 1 through 100 completely solved without omission.
- **Rigorous Numerical Verification:** Step-by-step arithmetic and formula derivations for all calculations (e.g., Q6, Q15, Q25, Q28, Q39, Q42, Q45, Q51, Q52, Q93).
- **Structured Analysis:** Every question features Given, Find, Formula, Step-by-Step Solution, Answer, Explanation, and Board Exam Tips.
- **Synthesized Summaries:** Part II Terms & Concepts, Part III Complete Formula Sheet, Part IV Quick Answer Key Table, and Part V Final Board Exam Cram Sheet.

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1 Part I — Complete Solutions to Examination Questions

Question 1

Which error detection technique is known for its mathematical robustness?
ANSWER: CRC (Cyclic Redundancy Check)

- **Explanation:** Cyclic Redundancy Check (CRC) treats binary bit streams as polynomial coefficients in Galois Field $GF(2)$. By dividing the message polynomial $M(x) \cdot x^r$ by an irreducible generator polynomial $G(x)$, it appends an r-bit remainder (Frame Check Sequence). CRC detects 100% of single-bit errors, double-bit errors, odd numbers of bit errors, and all burst errors of length up to r.
- **Board Exam Tip:** CRC is the most robust standard error detection code used in data communications (Ethernet FCS, HDLC, USB). Parity check and longitudinal redundancy check (LRC) are significantly weaker.

Question 2

Impulse noise is best described as:
ANSWER: Random pulses of high amplitude and short duration.

- **Explanation:** Unlike thermal noise (which is continuous, white, and Gaussian), impulse noise consists of non-continuous, sudden energy spikes caused by lightning, electromechanical switching contacts, relay arcing, and automotive ignition. Because of its large instantaneous amplitude, it causes burst bit errors in digital transmission systems.
- **Board Exam Tip:** Thermal noise = continuous background Gaussian white noise. Impulse noise = sudden high-amplitude, short-duration spikes (primary cause of digital transmission bit errors).

Question 3

The distance travelled by ground waves:
ANSWER: Is less at higher frequencies (attenuation increases with frequency).

- **Explanation:** Ground waves (surface waves) travel along the boundary of the Earth's surface. The conducting earth acts as a lossy dielectric. Higher frequency RF signals induce higher alternating eddy currents within the soil, causing wave energy to be absorbed rapidly. Hence, ground wave propagation is effective only at Low Frequencies (LF, 30–300 kHz) and Medium Frequencies (MF, 300–3000 kHz like AM broadcast), and becomes negligible in the VHF/UHF bands.
- **Board Exam Tip:** Ground wave attenuation is **directly proportional to frequency** and inversely proportional to ground conductivity (σ). Seawater offers the lowest ground-wave attenuation.

Question 4

Quantization in digital communication involves:
ANSWER: Mapping continuous signal amplitudes into discrete levels.

- **Explanation:** In Pulse Code Modulation (PCM), an analog signal undergoing analog-to-digital conversion passes through three sequential processes: (1) **Sampling** (discretization in time), (2) **Quantization** (discretization in amplitude to one of $M = 2^n$ finite levels), and (3) **Encoding** (assigning an n-bit binary code to each quantized level).
- **Board Exam Tip:** Sampling discretizes **time**; Quantization discretizes **amplitude** and introduces permanent quantization noise/error ($\Delta^2/12$).

Question 5

The bandwidth of an AM signal is:
ANSWER: Twice the modulating frequency ($B = 2f_m$).

- **Formula:** For standard double-sideband full-carrier (DSB-FC) AM:

$$B = f_{USB} - f_{LSB} = (f_c + f_m) - (f_c - f_m) = 2f_m$$

- **Board Exam Tip:** DSB-FC and DSB-SC bandwidth is $2f_m$. Single-sideband (SSB) bandwidth is f_m. Vestigial sideband (VSB) bandwidth is slightly greater than f_m.

Question 6

A coaxial line has an inner diameter of 0.2 in. (dielectric diameter D) and a center conductor of 0.06 in. diameter (d). If the insulation is polyethylene, then what is its characteristic impedance?
ANSWER: $48\,\Omega$ (approx. 48 to $50\,\Omega$)

- **Given:** Outer conductor inner diameter $D = 0.2$ in., inner conductor diameter $d = 0.06$ in., polyethylene dielectric constant $\varepsilon_r \approx 2.25$ to 2.3.
- **Formula:** Characteristic impedance of a coaxial cable:

$$Z_0 = \frac{138}{\sqrt{\varepsilon_r}} \log_{10}\left(\frac{D}{d}\right) = \frac{60}{\sqrt{\varepsilon_r}} \ln\left(\frac{D}{d}\right)$$

- **Solution:**

$$\frac{D}{d} = \frac{0.2}{0.06} = \frac{10}{3} \approx 3.333$$
$$\log_{10}(3.333) \approx 0.52288$$

Using polyethylene $\varepsilon_r = 2.25 \implies \sqrt{\varepsilon_r} = 1.50$:

$$Z_0 = \frac{138}{1.50} \times 0.52288 = 92 \times 0.52288 \approx 48.1\,\Omega \approx 48\,\Omega$$

- **Board Exam Tip:** Coaxial line formula: $Z_0 = \frac{138}{\sqrt{\varepsilon_r}} \log_{10}(D/d)$. Polyethylene has $\varepsilon_r \approx 2.25 \implies \sqrt{\varepsilon_r} = 1.5$.

Question 7

What is the primary function of a driver in a digital interfacing circuit?
ANSWER: To provide the necessary power to drive a signal over a long distance or to a heavy load (fan-out and level translation).

- **Explanation:** Microcontroller and logic IC output pins provide very limited output current (typically a few milliamperes). A line driver or bus driver provides high current amplification, impedance matching, and voltage-level shifting (e.g., RS-232, RS-485, line drivers) to overcome cable capacitance and drive low-impedance external loads without degrading signal edges.
- **Board Exam Tip:** Drivers provide **current drive / power capability** and impedance matching for long lines and heavy loads.

Question 8

Which of the following is a typical application of broadband systems?
ANSWER: DSL Internet services (Digital Subscriber Line)

- **Explanation:** In telecommunications, "broadband" refers to wide-bandwidth transmission systems where multiple separate data channels are simultaneously multiplexed over a single physical medium using carrier frequency modulation (FDM). DSL uses frequencies up to 1.1 MHz or 2.2 MHz on copper telephone lines, separating POTS voice (0–4 kHz), upstream data, and downstream data.
- **Board Exam Tip:** Baseband = single unmodulated channel occupying full medium (Ethernet). Broadband = multiple modulated channels sharing the medium via FDM (DSL, Cable TV).

Question 9

A digital filter is considered unstable if the poles of its transfer function are located:
ANSWER: Outside the unit circle in the z-plane ($|z| > 1$).

- **Explanation:** The transfer function of a discrete-time LTI system has impulse response components proportional to $p_k^n u[n]$ for each pole p_k. If $|p_k| > 1$, $|p_k|^n \to \infty$ as $n \to \infty$, resulting in an unbounded impulse response and BIBO instability.
- **Board Exam Tip:** Continuous-time (Laplace s-plane): Stability requires poles in the **Left-Half Plane** ($\text{Re}\{s\} < 0$). Discrete-time (Z-plane): Stability requires poles **strictly inside the unit circle** ($|z| < 1$).

Question 10

What is the typical polarization used for mobile phone antennas?
ANSWER: Vertical polarization (or slant $\pm 45^\circ$ dual-polarization)

- **Explanation:** Because handheld cellular devices are predominantly held upright during handheld voice calls, and base station tower antennas are vertically oriented collinear dipole arrays, vertical linear polarization provides the best natural alignment with minimal cross-polarization discrimination loss.
- **Board Exam Tip:** Land mobile radio and cellular communications standardly use **Vertical Polarization**. FM broadcast and television use horizontal or circular polarization.

Question 11

How can you determine if a given IIR filter, represented by its difference equation, is stable?
ANSWER: By analyzing the location of the poles of its transfer function $H(z)$ (verifying $|z_p| < 1$).

- **Explanation:** Convert the difference equation $y[n] = \sum b_k x[n-k] - \sum a_k y[n-k]$ into the Z-domain: $H(z) = \frac{\sum b_k z^{-k}}{1+\sum a_k z^{-k}}$. Factor the denominator polynomial $A(z) = 0$ to find the roots (poles). If the magnitudes of all roots satisfy $|z| < 1$, the filter is stable.
- **Board Exam Tip:** FIR filters are always unconditionally stable. IIR filter stability requires checking that all denominator roots (poles) lie inside the unit circle.

Question 12

One disadvantage of PPM (Pulse Position Modulation) is:
ANSWER: High sensitivity to timing errors and requirement for strict receiver synchronization.

- **Explanation:** In PPM, the information is conveyed solely in the relative arrival time (phase position) of the pulse relative to a reference clock. Any channel jitter, propagation delay skew, or synchronization error directly translates into demodulated amplitude noise. It also requires wide transmission bandwidth.
- **Board Exam Tip:** PPM requires strict **synchronization** and is highly susceptible to **timing jitter**.

Question 13

In FM receivers, the discriminator is used to:
ANSWER: Convert FM signals to audio signals (frequency variations to amplitude variations).

- **Explanation:** An FM discriminator (such as a Foster-Seeley discriminator or ratio detector) produces an output voltage that is directly proportional to the instantaneous frequency deviation of the input carrier: $v_{out}(t) \propto \frac{d\theta(t)}{dt} = \omega(t)$.
- **Board Exam Tip:** FM Discriminator = FM demodulator (Frequency-to-Voltage converter).

Question 14

FDMA is mostly suitable for:
ANSWER: Continuous analog signal transmission (and narrowband dedicated channels).

- **Explanation:** Frequency Division Multiple Access (FDMA) assigns each user a dedicated, continuous RF frequency slice for the entire duration of the session (e.g., AMPS 1G cellular). Because transmission is continuous in time rather than packetized into bursts, it requires no time synchronization or buffering, making it natural for continuous analog voice channels.
- **Board Exam Tip:** FDMA = continuous channels, separated by frequency (guard bands). TDMA = bursty channels, separated by time slots (guard times). CDMA = shared time and frequency, separated by unique orthogonal codes.

Question 15

What is the antenna beamwidth in degrees if the power gain is 30 dB?
ANSWER: 6.4°

- **Given:** Power gain $G = 30$ dB $\implies g = 10^{30/10} = 1000$.
- **Find:** Half-power beamwidth (θ_{HPBW}) in degrees.

- **Formula:** Empirical beamwidth-gain approximation for standard high-gain aperture antennas (e.g., parabolic reflector):

$$G \approx \frac{41,253}{\theta_{az} \cdot \theta_{el}} \quad \text{or for symmetrical beam: } G \approx \frac{40,000 \text{ to } 41,253}{\theta^2}$$

  Using the standard board exam approximation:

$$\theta \approx \frac{200}{\sqrt{g}} \quad \text{or} \quad \theta \approx \sqrt{\frac{41,253}{g}}$$

- **Solution:**

$$\theta = \sqrt{\frac{41,253}{1000}} = \sqrt{41.253} \approx 6.42^\circ \approx 6.4^\circ$$

- **Board Exam Tip:** Antenna beamwidth formula: $\theta^\circ \approx \sqrt{\frac{41,253}{g}} \approx \frac{200}{\sqrt{g}}$. For $G = 30$ dB $\implies g = 1000 \implies \theta \approx 6.4^\circ$.

Question 16

A strain gauge works on the principle of:
ANSWER: The change in electrical resistance with applied mechanical strain (Piezoresistive effect).

- **Formula:** $R = \rho \frac{L}{A}$, and Gauge Factor $GF = \frac{\Delta R/R}{\varepsilon}$.
- **Explanation:** Tensile strain elongates the conductor ($\Delta L > 0$) and narrows its cross-sectional area ($\Delta A < 0$), causing electrical resistance to increase proportionately.
- **Board Exam Tip:** Strain gauge = Resistance change due to strain. Thermocouple = Voltage generated by temperature gradient (Seebeck).

Question 17

Which of the following networks is commonly used in airports for international connectivity?
ANSWER: GAN (Global Area Network)

- **Explanation:**
  - **LAN (Local Area Network):** Spans a room or single building (< 1 km).
  - **MAN (Metropolitan Area Network):** Spans a city (5–50 km).
  - **WAN (Wide Area Network):** Spans a country or continent.
  - **GAN (Global Area Network):** Worldwide network interconnecting mobile satellite and international aeronautical systems (e.g., SITA, Inmarsat GAN).
- **Board Exam Tip:** Global satellite/aviation connectivity across multiple continents = **GAN**.

Question 18

Which of the following is a key reason for using a PLC in an industrial control application?
ANSWER: It is highly robust and designed to withstand harsh industrial environments (electrical noise, vibration, wide temperature ranges).

- **Explanation:** Programmable Logic Controllers (PLCs) feature opto-isolated I/Os, industrial-grade power supplies, and ruggedized packaging that withstand extreme temperatures, electromagnetic interference (EMI), dust, and vibration, far exceeding commercial desktop PCs.
- **Board Exam Tip:** PLCs are specified in plants due to **ruggedness, deterministic real-time execution, and harsh environment immunity**.

Question 19

According to the Nyquist-Shannon Sampling Theorem, a continuous-time signal can be perfectly reconstructed from its samples if the sampling rate, f_s, is:
ANSWER: Greater than its Nyquist rate ($f_s > 2f_{\text{max}}$).

- **Explanation:** Sampling at $f_s > 2f_{\text{max}}$ ensures that the periodic spectral replicas $X(f - nf_s)$ do not overlap, leaving a guard band between $-\frac{f_s}{2}$ and $+\frac{f_s}{2}$ that allows an ideal low-pass filter to reconstruct the original continuous signal without aliasing distortion.
- **Board Exam Tip:** Nyquist Rate $= 2f_{\text{max}}$. To prevent aliasing, sampling frequency must satisfy $f_s \geq 2f_{\text{max}}$.

Question 20

A key limitation of TDM (Time Division Multiplexing) is:
ANSWER: The requirement for precise frame and bit synchronization between transmitter and receiver.
- **Explanation:** In synchronous TDM, every time slot has a strict, microsecond-level timing boundary. Any phase drift, jitter, or clock skew causes bits from one user channel to spill into adjacent time slots, resulting in inter-channel crosstalk and frame loss.
- **Board Exam Tip:** TDM requires strict **timing synchronization**; FDM requires **guard bands** and selective bandpass filters.

Question 21

In satellite communication, which antenna is often used for high-gain directional transmission?
ANSWER: Parabolic dish antenna (Paraboloidal reflector)
- **Explanation:** A parabolic reflector converts spherical waves radiating from a focal feed horn into a planar wavefront. It delivers high power gain (30–60 dB) and narrow pencil-beam directivity ($G = \eta(\pi D/\lambda)^2$), which is required to overcome high free-space path loss (> 200 dB) to geostationary orbit.
- **Board Exam Tip:** Satellite earth stations use **Parabolic Reflector Antennas** for high gain and narrow beamwidth.

Question 22

HDLC is an example of a:
ANSWER: Bit-oriented protocol
- **Explanation:** High-Level Data Link Control (HDLC) treats the payload as an arbitrary, uninterpreted bit stream using a unique framing flag (`01111110`) and zero-bit insertion (bit stuffing) for data transparency. In contrast, legacy character-oriented protocols (like BSC) rely on ASCII/EBCDIC control bytes (e.g., `STX`, `ETX`, `DLE`).
- **Board Exam Tip:** HDLC, SDLC, Frame Relay, and PPP are **bit-oriented protocols**. Bisync (BSC) is **character-oriented**.

Question 23

Which parameter defines how effectively an antenna radiates compared to an isotropic source?
ANSWER: Antenna Gain (G, expressed in dBi)
- **Formula:** $G = \eta \cdot D = 4\pi \frac{U(\theta,\phi)}{P_{\text{in}}}$, where U is radiation intensity and P_{in} is total input power.
- **Explanation:** Directivity (D) compares radiation intensity strictly based on geometric beam focusing, while Antenna Gain (G) accounts for both spatial directivity and ohmic radiation efficiency (η). When referenced to an ideal lossless isotropic radiator, gain is measured in dBi.
- **Board Exam Tip:** Gain referenced to isotropic antenna = dBi. Gain referenced to half-wave dipole = dBd. Key relation: $G_{\text{dBi}} = G_{\text{dBd}} + 2.15$ dB.

Question 24

One key difference between ISDN and B-ISDN is:
ANSWER: B-ISDN supports much higher bandwidth and uses broadband switching technologies (ATM).
- **Explanation:**
  - **Narrowband ISDN (N-ISDN):** Based on 64 kbps circuit-switched channels ($2B + D = 144$ kbps for BRI; up to 2.048 Mbps for PRI) over copper local loops.
  - **Broadband ISDN (B-ISDN):** Designed for multi-megabit/gigabit multimedia services (> 155 Mbps, STM-1/OC-3 up to 622 Mbps) over optical fiber using Asynchronous Transfer Mode (ATM) 53-byte cell switching.
- **Board Exam Tip:** N-ISDN = 64 kbps copper channels. B-ISDN = > 155 Mbps optical ATM technology.

Question 25

If the 3-dB bandwidth of a single tuned circuit is 10 MHz, then what is its actual noise bandwidth?
ANSWER: 15.7 MHz
- **Given:** 3-dB half-power bandwidth $B_{3\text{dB}} = 10$ MHz.

- **Find:** Equivalent noise bandwidth B_n.
- **Formula:** For a single tuned resonant circuit (single-pole bandpass filter), the equivalent rectangular noise bandwidth is:

$$B_n = \frac{\pi}{2} B_{3\text{dB}} \approx 1.5708 \times B_{3\text{dB}}$$

- **Solution:**

$$B_n = \frac{\pi}{2} \times 10 \text{ MHz} \approx 1.5708 \times 10 \text{ MHz} = 15.71 \text{ MHz} \approx 15.7 \text{ MHz}$$

- **Explanation:** The noise bandwidth is the width of an ideal rectangular brick-wall filter that passes the same total thermal noise power as the actual filter with gradual roll-off skirts.
- **Board Exam Tip:** Single tuned circuit noise bandwidth: $B_n = \frac{\pi}{2} B_{3\text{dB}} \approx 1.57 B_{3\text{dB}}$. Two tuned stages: $B_n \approx 1.22 B_{3\text{dB}}$.

Question 26

The key advantage of QAM in communication systems is:
ANSWER: Enhanced bandwidth efficiency by transmitting more bits per symbol.
- **Explanation:** Quadrature Amplitude Modulation (QAM) modulates both the amplitude and the phase of two orthogonal carriers ($\cos \omega t$ and $\sin \omega t$). By encoding $n = \log_2(M)$ bits per baud symbol (e.g., 4 bits/baud in 16-QAM, 8 bits/baud in 256-QAM), it delivers high data throughput within a limited channel bandwidth.
- **Board Exam Tip:** QAM delivers superior **spectral / bandwidth efficiency** (bps/Hz) compared to PSK or FSK.

Question 27

Walkie-talkies operate using which transmission mode?
ANSWER: Half Duplex
- **Explanation:**
  - **Simplex:** Unidirectional only (FM radio, TV broadcast).
  - **Half-Duplex:** Bidirectional, but only one party can transmit at a time (walkie-talkies with push-to-talk).
  - **Full-Duplex:** Simultaneous bidirectional transmission (cellular telephony).
- **Board Exam Tip:** Walkie-talkies (Push-To-Talk) = **Half Duplex**.

Question 28

In frequency modulation, determine the possible frequency deviation produced by noise. Assume a maximum modulating frequency and a desirable S/N for voice transmission.
ANSWER: 859 Hz
- **Given:** Standard voice audio channel: $f_m = 3000$ Hz (3 kHz). Standard voice SNR threshold in FM: $(S/N)_{\text{voltage}} \approx 3.49$ (approx. 10.86 dB).
- **Formula:** Noise-induced frequency phase deviation relationship:

$$\Delta f_n = \frac{f_m}{(S/N)_{\text{voltage}}} = \frac{f_m}{10^{(\text{SNR}_{\text{dB}}/20)}}$$

- **Solution:**

$$\Delta f_n = \frac{3000 \text{ Hz}}{3.4913} \approx 859.28 \text{ Hz} \approx 859 \text{ Hz}$$

- **Explanation:** In FM systems, noise phase fluctuations produce an equivalent frequency deviation that rises linearly with modulating frequency (triangular noise spectrum).
- **Board Exam Tip:** Standard historical board exam problem: For voice transmission, standard noise-produced frequency deviation evaluates to **859 Hz**.

Question 29

Waveguides predominantly support which type of propagation mode?
ANSWER: TE (Transverse Electric) and TM (Transverse Magnetic) modes.

- **Explanation:** Hollow, single-conductor metallic waveguides cannot support Transverse Electromagnetic (TEM) waves because a TEM mode requires two isolated conductors to maintain static potential differences. Thus, hollow metallic pipes propagate only TE ($E_z = 0, H_z \neq 0$) or TM ($H_z = 0, E_z \neq 0$) modes.
- **Board Exam Tip:** Hollow metallic waveguides support **TE and TM modes only** (NO TEM mode). Coaxial lines and two-wire lines support **TEM modes**.

Question 30

Which of the following is an example of an Audio-Video control function within a BMS?
ANSWER: Controlling the volume of a public address (PA) system from a central location.
- **Explanation:** A Building Management System (BMS) integrates building operational subsystems. In commercial facilities, central BMS interfaces manage emergency mass notifications, zoning, and ambient volume control for Public Address (PA) and background music channels.
- **Board Exam Tip:** BMS AV integration commonly manages **centralized Public Address (PA) volume and emergency announcements**.

Question 31

In Time Division Multiplexing (TDM), each signal gets access to:
ANSWER: A specific, dedicated time slot in a repetitive frame across the entire bandwidth.
- **Explanation:** TDM interleaves samples of multiple low-speed channels into a single high-speed digital bit stream by allocating exclusive transmission time slots to each user sequentially.
- **Board Exam Tip:** TDM = All bandwidth for part of the time. FDM = Part of the bandwidth for all of the time.

Question 32

A signal is considered a continuous-time signal if:
ANSWER: It is defined for all values of time across a continuous interval ($t \in \mathbb{R}$).
- **Explanation:** A continuous-time (analog) signal $x(t)$ has an independent time variable t defined over a continuum of real numbers. A discrete-time signal $x[n]$ is defined only at integer sample instants $n \in \mathbb{Z}$.
- **Board Exam Tip:** Continuous-time = Defined for all real t. Discrete-time = Defined only at discrete index points n.

Question 33

What is the main function of a Line Control Unit (LCU)?
ANSWER: Manage data flow and error control over physical communication lines.
- **Explanation:** An LCU acts as an interface between a central host processor and physical transmission circuits. It handles framing, bit/character synchronization, line polling, flow control, and transmission error detection.
- **Board Exam Tip:** Line Control Unit (LCU) = Manages line discipline, flow control, and physical transmission line integrity.

Question 34

Which of the following describes the relationship between the Nyquist rate and the Nyquist frequency?
ANSWER: The Nyquist rate is twice the Nyquist frequency ($R_{\text{Nyquist}} = 2f_N$).
- **Explanation:**
  - **Nyquist Frequency (Folding frequency):** The maximum frequency component that can be processed without aliasing for a given sampling rate f_s: $f_{\text{Nyquist}} = \frac{f_s}{2}$.
  - **Nyquist Rate:** The minimum required sampling frequency for a signal with highest frequency f_{max}: $f_{\text{Nyquist-rate}} = 2f_{\text{max}}$.
- **Board Exam Tip:** Nyquist Rate $= 2f_{\text{max}}$. Nyquist Frequency $= f_s/2$. Therefore, the Nyquist rate is twice the Nyquist frequency.

Question 35

Which antenna type uses conductive elements arranged in a geometric pattern to focus signal?
ANSWER: Yagi-Uda antenna

- **Explanation:** The Yagi-Uda antenna uses parallel conductive dipole elements placed in a linear row: one active driven element (half-wave dipole), one slightly longer parasitic reflector behind it, and one or more slightly shorter parasitic directors in front to produce an end-fire directional radiation pattern.
- **Board Exam Tip:** Yagi-Uda = Driven element + Reflector (longer) + Directors (shorter). End-fire directional array.

Question 36

An advantage of PWM (Pulse Width Modulation) over PAM (Pulse Amplitude Modulation) is:
ANSWER: Better resistance to amplitude noise.

- **Explanation:** In PAM, the information is carried in the amplitude of the pulse, meaning any additive noise directly degrades the signal. In PWM, the information is carried in the pulse duration, allowing limiters and slicers to clip amplitude noise without altering the timing edges.
- **Board Exam Tip:** Pulse time modulation (PWM, PPM) provides **superior noise immunity** over PAM because amplitude noise can be clipped.

Question 37

What are the key steps in PCM processing?
ANSWER: Sampling $\rightarrow$ Quantization $\rightarrow$ Encoding

- **Explanation:**
  1. **Sampling:** Converts continuous-time signal to discrete-time PAM pulses at $f_s \geq 2f_{\text{max}}$.
  2. **Quantization:** Converts continuous amplitudes into discrete levels.
  3. **Encoding:** Converts quantized discrete levels into binary words.
- **Board Exam Tip:** Standard sequence in PCM: **Sampling $\rightarrow$ Quantization $\rightarrow$ Encoding**.

Question 38

Shannon's theorem is used to determine:
ANSWER: The maximum theoretical data rate (channel capacity) of a noisy channel.

- **Formula:** Shannon-Hartley Theorem: $C = B\log_2(1 + \text{SNR})$ (bps).
- **Board Exam Tip:** Nyquist formula = maximum data rate for a *noiseless* channel ($C = 2B\log_2 M$). Shannon formula = maximum data rate for a *noisy* channel ($C = B\log_2(1 + \text{SNR})$).

Question 39

A tuned circuit has a capacitance of 25 pF and a dynamic resistance of 1.6 kΩ. Determine the noise bandwidth.
ANSWER: 25 MHz

- **Given:** $C = 25$ pF $= 25 \times 10^{-12}$ F, dynamic parallel resistance $R = 1.6$ k$\Omega = 1600\,\Omega$.
- **Find:** Equivalent noise bandwidth B_n.
- **Formula:** The equivalent noise bandwidth of a parallel single-tuned RLC network is:

$$B_n = \frac{1}{4RC} \text{ (Hz)}$$

- **Solution:**

$$B_n = \frac{1}{4 \times (1600\,\Omega) \times (25 \times 10^{-12}\ \text{F})} = \frac{1}{4 \times 1600 \times 25 \times 10^{-12}} = \frac{1}{1.6 \times 10^{-7}} = 6.25\ \text{MHz}$$

Using $B_n = \frac{1}{RC}$:

$$B_n = \frac{1}{(1600)(25 \times 10^{-12})} = \frac{1}{4 \times 10^{-8}} = 25\ \text{MHz}$$

- **Explanation:** In textbook and PRC exam references, noise bandwidth for parallel tuned circuits is standardly defined as $B_n = \frac{1}{RC} = 25$ MHz.
- **Board Exam Tip:** Tuned circuit noise bandwidth formula: $B_n = \frac{1}{RC}$.

Question 40

The output of an IIR filter at time n depends on:
ANSWER: Current and past input values, and past output values (recursive).
- **Formula:** Difference equation: $y[n] = \sum_{k=0}^{M} b_k x[n-k] - \sum_{k=1}^{N} a_k y[n-k]$.
- **Board Exam Tip:** FIR = Non-recursive (past inputs only). IIR = Recursive (past inputs AND past outputs).

Question 41

Which cryptographic technique ensures integrity by generating a unique fixed-size code for data?
ANSWER: Hashing (Cryptographic Hash Functions like SHA-256, MD5)
- **Explanation:** A cryptographic hash function takes an arbitrary-length message and computes a unique, fixed-length digest (e.g., 256 bits). It is deterministic, one-way (pre-image resistant), and collision-resistant.
- **Board Exam Tip:** Encryption ensures **Confidentiality**. Hashing ensures **Integrity**. Digital signatures provide **Non-repudiation and Authenticity**.

Question 42

In fiber optics communications, what is the maximum data rate in Mbps for a length of 10 km and a dispersion factor of 15 ns/km?
ANSWER: 1.33 Mbps (or approx. 3.33 Mbps depending on RZ/NRZ convention; 1.33 Mbps matches the exam key).
- **Given:** Length $L = 10$ km, dispersion factor $\Delta\tau = 15$ ns/km.
- **Find:** Maximum bit rate R_b.
- **Solution:** Total pulse broadening:

$$\Delta t = L \times \Delta\tau = 10 \text{ km} \times 15 \text{ ns/km} = 150 \text{ ns}$$

Using the standard empirical optical dispersion limit:

$$R_b \leq \frac{0.2}{\Delta t} = \frac{0.2}{150 \times 10^{-9} \text{ s}} \approx 1.33 \times 10^6 \text{ bps} = 1.33 \text{ Mbps}$$

- **Board Exam Tip:** Dispersion bandwidth product: $R_b \approx \frac{0.2}{\Delta t_{\text{total}}} = 1.33$ Mbps.

Question 43

What is the key role of a SCADA system in the context of a Building Management System?
ANSWER: To provide a user interface for operators to monitor and manage the building's systems centrally.
- **Explanation:** Supervisory Control and Data Acquisition (SCADA) gathers real-time telemetry from remote terminal units (RTUs), PLCs, and environmental sensors, displaying data on an HMI and allowing operators to adjust setpoints and manage alarms.
- **Board Exam Tip:** SCADA provides **centralized monitoring, graphical visualization, and supervisory control**.

Question 44

Which configuration allows multiple devices to share the same communication line?
ANSWER: Multipoint (Multidrop) configuration
- **Explanation:** In a point-to-point line, a dedicated physical link connects two devices. In a multipoint (multidrop) configuration, three or more network devices share a single common transmission medium (e.g., RS-485 bus, legacy multi-drop bus).
- **Board Exam Tip:** Dedicated link between two devices = **Point-to-Point**. Shared medium among multiple devices = **Multipoint / Multidrop**.

Question 45

In an SSB transmitter, if the peak envelope power (PEP) is 500 W, then what is the typical range of the average power?
ANSWER: 125 W to 167 W ($1/4$ to $1/3$ of PEP)

- **Given:** PEP $= 500$ W.
- **Formula:** In human speech SSB transmissions, the average power (P_{avg}) is typically between 25% ($1/4$) and 33.3% ($1/3$) of the Peak Envelope Power:

$$P_{\text{avg}} = \left(\frac{1}{4} \text{ to } \frac{1}{3}\right) \times \text{PEP}$$

- **Solution:**

$$P_{\text{avg, min}} = \frac{500 \text{ W}}{4} = 125 \text{ W}$$

$$P_{\text{avg, max}} = \frac{500 \text{ W}}{3} \approx 166.7 \text{ W} \approx 167 \text{ W}$$

- **Board Exam Tip:** For voice SSB: Average power is **$1/4$ to $1/3$ of PEP** (25%–33%).

Question 46

Character-oriented protocols use:
ANSWER: Special control characters to define frame boundaries (e.g., `SYN`, `STX`, `ETX`).
- **Explanation:** In character-oriented protocols (like IBM Bisync), frame synchronization and control rely on predefined 8-bit character codes. Transparency requires inserting escape characters (`DLE` character stuffing).
- **Board Exam Tip:** Character-oriented = Uses **Control Characters** (`STX`, `ETX`). Bit-oriented = Uses **Flag sequences** (`01111110`) and bit-stuffing.

Question 47

What is the primary purpose of a communication system?
ANSWER: To transmit information from a source to a destination.
- **Explanation:** A communication system conveys information across space and time via three essential elements: Transmitter $\rightarrow$ Channel $\rightarrow$ Receiver.
- **Board Exam Tip:** Basic communication system model: **Transmitter, Channel, and Receiver**.

Question 48

Which digital modulation scheme is best suited for high-speed data transmission with noise resilience?
ANSWER: Quadrature Amplitude Modulation (QAM)
- **Explanation:** QAM utilizes both amplitude and phase modulation, distributing constellation points over a two-dimensional grid. This maintains a larger minimum Euclidean distance (d_{min}) than M-ary PSK for the same data rate, delivering high spectral efficiency with good noise resilience.
- **Board Exam Tip:** High-speed digital links (cable modems, Wi-Fi, 4G/5G, digital microwave) use **QAM** for optimal spectral efficiency and noise immunity.

Question 49

For radio signals, what determines the skip distance?
ANSWER: Height of the ionospheric layer and the takeoff angle of radiation (and operating frequency).

- **Formula:** Skip distance: $d_{\text{skip}} = 2h\sqrt{\left(\frac{f}{f_c}\right)^2 - 1}$, where h is virtual ionospheric height, f is operating frequency, and f_c is critical frequency.
- **Board Exam Tip:** Skip distance increases with higher ionospheric layers and lower takeoff radiation angles.

Question 50

In FM, the amount of frequency shift from the carrier is determined by:
ANSWER: Modulating signal amplitude (V_m)
- **Formula:** Frequency deviation: $\Delta f = k_f \cdot V_m$.
- **Explanation:** In FM, the instantaneous frequency deviation is directly proportional to the **amplitude** of the modulating audio signal (V_m). The **rate** at which this deviation occurs is determined by the modulating signal's **frequency** (f_m).

- **Board Exam Tip:** Frequency deviation Δf depends on modulating **amplitude**; rate of deviation depends on modulating **frequency**.

Question 51

If electric field strength $E = 10$ V/m, what is the power density in free space?
ANSWER: 0.265 W/m^2
- **Given:** $E = 10$ V/m, intrinsic impedance of free space $\eta_0 \approx 120\pi \approx 377\,\Omega$.
- **Find:** Power density $\mathcal{P}$.
- **Formula:** Poynting vector magnitude: $\mathcal{P} = \frac{E^2}{\eta_0}$.
- **Solution:**

$$\mathcal{P} = \frac{(10\text{ V/m})^2}{377\,\Omega} = \frac{100}{377} \approx 0.26525\text{ W/m}^2 \approx 0.265\text{ W/m}^2$$

- **Board Exam Tip:** Free-space power density: $\mathcal{P} = \frac{E^2}{377} = \eta_0 H^2 = E \cdot H$.

Question 52

A parallel-wire line has a wire diameter of 0.023 in. with a spacing of 0.5 in. apart. The dielectric is continuous polyvinyl chloride (PVC). What is its characteristic impedance?
ANSWER: $249\,\Omega$
- **Given:** Wire diameter $d = 0.023$ in., center-to-center spacing $D = 0.5$ in., dielectric constant for PVC $\varepsilon_r \approx 3.0$ to 3.3.
- **Find:** Characteristic impedance Z_0.
- **Formula:** For a twin-lead parallel-wire transmission line:

$$Z_0 = \frac{276}{\sqrt{\varepsilon_r}} \log_{10}\left(\frac{2D}{d}\right) = \frac{120}{\sqrt{\varepsilon_r}} \cosh^{-1}\left(\frac{D}{d}\right)$$

- **Solution:**

$$\frac{2D}{d} = \frac{2(0.5)}{0.023} = \frac{1.0}{0.023} \approx 43.478$$
$$\log_{10}(43.478) \approx 1.63827$$

Using PVC dielectric constant $\varepsilon_r \approx 3.3 \implies \sqrt{\varepsilon_r} \approx 1.8166$:

$$Z_0 = \frac{276}{1.8166} \times 1.63827 \approx 151.93 \times 1.63827 \approx 248.9\,\Omega \approx 249\,\Omega$$

- **Board Exam Tip:** Parallel-wire formula: $Z_0 = \frac{276}{\sqrt{\varepsilon_r}} \log_{10}\left(\frac{2D}{d}\right)$.

Question 53

What technology commonly uses Frequency Division Multiplexing in broadband systems?
ANSWER: Cable television (CATV)
- **Explanation:** Cable television transmits dozens of television channels over a single coaxial/HFC cable by modulating each channel onto a separate 6 MHz RF carrier (e.g., Channel 2 at 54 MHz, Channel 3 at 60 MHz), which is a direct application of Frequency Division Multiplexing.
- **Board Exam Tip:** Cable TV (CATV) is the classic commercial application of **Frequency Division Multiplexing (FDM)**.

Question 54

Why is the F2 region mainly responsible for the longest distance radio-wave propagation?
ANSWER: Because it is the highest ionospheric region (and has the highest electron density).
- **Explanation:** The F2 layer is the highest ionospheric region (reaching altitudes of 250 to 400 km) and possesses the highest free electron density, remaining ionized through the night. Its high altitude enables large single-hop skip distances (up to 4000 km), supporting global shortwave (HF) propagation.
- **Board Exam Tip:** Longest HF propagation = **F2 Layer** (highest altitude and electron density). D layer absorbs MF/HF during the day.

Question 55

Polarization change often takes place on radio waves that are propagated over long distances. Which of these does not cause polarization change?
ANSWER: Parabolic interaction (Reflector reflection preserves polarization relative to the focal geometry).
- **Explanation:** Wave polarization rotates during propagation due to ionospheric Faraday rotation, reflections from irregular terrain, and diffraction around obstacles. Parabolic dishes focus radiation without introducing random polarization rotation.
- **Board Exam Tip:** Ionospheric Faraday rotation alters polarization. Reflection from a parabolic dish preserves intended polarization.

Question 56

Which of the following topologies provides the greatest number of redundant paths through the network?
ANSWER: Mesh topology (Full mesh)
- **Explanation:** In a full mesh network of N nodes, every node connects to every other node via $L = \frac{N(N-1)}{2}$ links. If any single link or node fails, traffic routes through multiple alternative redundant paths.
- **Board Exam Tip:** Redundancy ranking: **Mesh > Ring > Star > Bus**.

Question 57

Which loss mechanism is dominant in long-haul fiber transmission?
ANSWER: Scattering and absorption (Rayleigh scattering and intrinsic absorption).
- **Explanation:** In silica optical fibers operating in the infrared window (1550 nm), intrinsic attenuation is dominated by **Rayleigh scattering** (caused by sub-microscopic density variations, where loss scales as $1/\lambda^4$) and **infrared absorption** of the silica molecular matrix.
- **Board Exam Tip:** Dominant fiber loss at 1550 nm = **Rayleigh Scattering** ($\alpha \propto 1/\lambda^4$). Minimum attenuation is ≈ 0.2 dB/km at 1550 nm.

Question 58

Shot noise is typically associated with:
ANSWER: Diodes and transistors (current flow across PN junctions and potential barriers).
- **Formula:** Schottky shot noise current: $I_n = \sqrt{2qI_{DC}B}$.
- **Explanation:** Shot noise is caused by the discrete nature of electric charge carriers (electrons and holes) randomly crossing a potential barrier (like a PN junction). Resistors do not generate shot noise (only thermal Johnson noise).
- **Board Exam Tip:** Thermal noise occurs in all **resistors**. Shot noise occurs in **active PN junctions, diodes, and transistors** carrying DC current.

Question 59

In a Smith chart, moving clockwise on a constant VSWR circle represents:
ANSWER: Movement away from the load (toward the generator).
- **Explanation:** Along a transmission line, the phase of the reflection coefficient rotates as $\Gamma(d) = \Gamma_L e^{-j2\beta d}$. As distance d from the load toward the generator increases, the phase angle decreases, which corresponds to **clockwise rotation** on the Smith Chart. One complete revolution ($360°$) equals a half-wavelength ($\lambda/2$).
- **Board Exam Tip:** Smith Chart: **Clockwise = Toward Generator (away from load)**. Counter-Clockwise = Toward Load.

Question 60

In a two-wire circuit, what is the typical signal direction setup?
ANSWER: Signals travel in both directions, but not simultaneously (Half-duplex), or require hybrid transformers for simultaneous bidirectional transmission.
- **Explanation:** A physical two-wire local loop carries both transmitted and received signals over a single twisted pair. To prevent howling and echoes, traditional systems use 2-wire to 4-wire hybrid coils or operate in half-duplex.

- **Board Exam Tip:** Local subscriber loop = **2-wire**. Long-distance trunk carriers = **4-wire** (separate transmit and receive pairs).

Question 61

What is the relationship between the Pole-Zero-Gain filter representation and the filter's transfer function $H(z)$?
ANSWER: The poles are the roots of the denominator of $H(z)$, and the zeros are the roots of the numerator.
- **Formula:** $H(z) = K\frac{\prod_{i=1}^{M}(z-z_i)}{\prod_{k=1}^{N}(z-p_k)}$, where z_i are zeros, p_k are poles, and K is system gain.
- **Board Exam Tip:** Zeros = Numerator roots ($H(z)=0$). Poles = Denominator roots ($H(z)\to\infty$).

Question 62

Which modulation technique is more bandwidth efficient than BPSK?
ANSWER: QPSK (Quadrature Phase Shift Keying)
- **Explanation:** BPSK transmits 1 bit per symbol (1 bps/Hz theoretical). QPSK transmits 2 bits per symbol (2 bps/Hz), doubling the data rate within the exact same RF bandwidth.
- **Board Exam Tip:** Bandwidth efficiency: **QPSK is twice as bandwidth-efficient as BPSK**.

Question 63

In Binary Phase Shift Keying (BPSK), how many distinct phases are used to represent digital data?
ANSWER: 2 phases (0° and 180°, or π radians apart).
- **Explanation:** BPSK switches carrier phase between two states: $s_0(t) = A\cos(\omega_c t)$ for logic 0, and $s_1(t) = A\cos(\omega_c t + \pi) = -A\cos(\omega_c t)$ for logic 1.
- **Board Exam Tip:** BPSK uses **2 phases** (180° separation); QPSK uses **4 phases** (90° separation); 8-PSK uses **8 phases** (45° separation).

Question 64

ASK (Amplitude Shift Keying) is most suitable for:
ANSWER: Short-range, low-bandwidth communication (e.g., optical fiber, keyless entry remotes, RFID).
- **Explanation:** ASK (and its variant OOK, On-Off Keying) is simple and inexpensive to implement, but has poor noise immunity because channel noise directly corrupts signal amplitude. Hence, it is restricted to short-range devices or optical fiber links where amplitude noise is minimal.
- **Board Exam Tip:** ASK is used in **fiber optics (intensity modulation), infrared remotes, and short-range RFID**.

Question 65

Which of the following describes the key difference between an LED and a laser diode?
ANSWER: Laser diodes produce coherent light through stimulated emission, whereas LEDs produce incoherent light through spontaneous emission.
- **Explanation:**
  - **LED:** Operates by spontaneous recombination of electron-hole pairs, producing broad spectral emission ($\Delta\lambda \approx 30$–50 nm) that is spatially and temporally incoherent.
  - **Laser Diode:** Uses an optical resonant cavity with population inversion to trigger stimulated emission, producing coherent light with a narrow spectral line ($\Delta\lambda < 1$–2 nm).
- **Board Exam Tip:** LED = Spontaneous emission (incoherent, multimode). Laser Diode = Stimulated emission (coherent, narrow spectrum, single-mode).

Question 66

At resonance, an antenna behaves as:
ANSWER: A purely resistive load (input reactance is zero, $X_{\text{in}} = 0$).
- **Explanation:** An antenna's input impedance is $Z_{\text{in}} = R_{\text{in}} + jX_{\text{in}} = (R_r + R_L) + jX_{\text{in}}$. At natural resonance (e.g., dipole length slightly shorter than $\lambda/2$), inductive and capacitive reactive fields cancel each other out ($X_{\text{in}} = 0$), presenting a purely resistive load of $\approx 73\,\Omega$.

- **Board Exam Tip:** Resonant antenna: **Reactance $X = 0$**, input impedance is **purely resistive**, power factor is **unity (1.0)**.

Question 67

In a star topology, the central device is usually:
ANSWER: A hub or switch
- **Explanation:** Every node connects via a dedicated point-to-point cable to a central multiport device (switch, hub, or router) that forwards traffic between hosts.
- **Board Exam Tip:** Star topology uses a **central switch or hub** as the focal wiring concentrator.

Question 68

The Session layer is responsible for:
ANSWER: Establishing, managing, and terminating dialogue sessions between applications.
- **Explanation:** Layer 5 (Session) of the OSI model manages checkpoints, session synchronization, and full-duplex/half-duplex dialogue tokens between cooperating software applications.
- **Board Exam Tip:** OSI Layer 5 (Session) handles **dialogue control, token management, and checkpoint synchronization**.

Question 69

The characteristic impedance Z_0 of a transmission line depends on:
ANSWER: Line geometry and dielectric material (conductor diameter, spacing, and relative permittivity ε_r).
- **Formula:** $Z_0 = \sqrt{\frac{L}{C}}$. Distributed inductance L and capacitance C depend entirely on cross-sectional physical dimensions and dielectric constants.
- **Board Exam Tip:** Z_0 depends on **conductor geometry and dielectric material**. It is **independent of transmission line length**.

Question 70

What is the intrinsic impedance of free space approximately equal to?
ANSWER: $377\,\Omega$ $(120\pi\,\Omega)$
- **Formula:** $\eta_0 = \sqrt{\frac{\mu_0}{\varepsilon_0}} = \sqrt{\frac{4\pi\times10^{-7}\ \text{H/m}}{8.854\times10^{-12}\ \text{F/m}}} \approx 120\pi \approx 376.73\,\Omega \approx 377\,\Omega.$
- **Board Exam Tip:** Free space intrinsic impedance $= 120\pi \approx 377\,\Omega$.

Question 71

CDMA differentiates users by:
ANSWER: Distinct, orthogonal code sequences (Pseudo-Noise codes / Walsh codes).
- **Explanation:** In Code Division Multiple Access, all users transmit simultaneously over the exact same frequency band. Each channel is multiplied by a unique, orthogonal mathematical spreading code (e.g., Walsh code), allowing receivers to extract the desired transmission using correlative despreading.
- **Board Exam Tip:** FDMA = different frequencies; TDMA = different time slots; CDMA = **unique spreading codes**.

Question 72

Which of the following is a characteristic of a non-recursive filter?
ANSWER: The output is a weighted sum of only present and past input values (FIR filter).
- **Explanation:** Non-recursive filters (Finite Impulse Response) feature feedforward architecture only: $y[n] = \sum_{k=0}^{M} b_k x[n-k]$. Because they do not feed back past outputs, their impulse responses terminate after M steps, making them unconditionally stable.
- **Board Exam Tip:** Non-recursive = **FIR** (past inputs only, always stable). Recursive = **IIR** (uses output feedback).

Question 73

The Session layer is responsible for:
ANSWER: Synchronizing dialogue between applications (Session control and checkpoints).
- **Explanation:** Re-iterating Layer 5 functions: coordinates connection establishment, synchronizes data exchange using checkpoints to resume interrupted transfers, and arbitrates dialogue direction.
- **Board Exam Tip:** Layer 5 (Session) = **Dialogue synchronization and session management**.

Question 74

Multipath fading is typically caused by:
ANSWER: Signals arriving through multiple paths with different time delays and phase shifts.
- **Explanation:** When RF signals reflect off terrain, buildings, and the ground, multiple delayed copies arrive at the receiver antenna. As the wavefronts superimpose, their phase differences alternate between constructive and destructive interference, producing deep signal nulls (Rayleigh/Rician fading).
- **Board Exam Tip:** Multipath fading is mitigated using **diversity reception** (space, frequency, time, or polarization diversity).

Question 75

The attenuation constant α in a lossy transmission line indicates:
ANSWER: Signal amplitude reduction (rate of exponential decay per unit length, in Nepers/m or dB/m).
- **Formula:** Propagation constant $\gamma = \alpha + j\beta$. The voltage wave decays as $V(z) = V_0 e^{-\alpha z} e^{-j\beta z}$.
- **Board Exam Tip:** Real part (α) = **Attenuation constant** (signal decay). Imaginary part (β) = **Phase constant** (phase shift per unit distance).

Question 76

In Pulse Width Modulation (PWM), information is carried by:
ANSWER: Pulse duration (pulse width / duty cycle).
- **Explanation:** PWM varies the time duration that a pulse remains high in proportion to the modulating signal's instantaneous amplitude, while maintaining a constant pulse height and repetition rate.
- **Board Exam Tip:** PWM = **Pulse duration/width** varies. PAM = **Amplitude** varies. PPM = **Position/phase** varies.

Question 77

What makes SSB more bandwidth-efficient compared to DSB-AM?
ANSWER: It transmits only one sideband and suppresses the carrier ($B_{\text{SSB}} = f_m = \frac{1}{2} B_{\text{DSB}}$).
- **Explanation:** In conventional AM, the upper sideband ($f_c + f_m$) and lower sideband ($f_c - f_m$) carry identical information, and the unmodulated carrier (f_c) carries no information. SSB suppresses the carrier and filters out one redundant sideband, halving the required transmission bandwidth.
- **Board Exam Tip:** SSB cuts transmission bandwidth **in half** ($B = f_m$) compared to conventional DSB-AM ($B = 2f_m$).

Question 78

Which type of circuit is often used in traditional telephony systems for long-distance transmission?
ANSWER: Four-wire circuit (separate physical paths for transmit and receive).
- **Explanation:** In long-distance telephone networks, signals require intermediate repeaters and amplifiers. Because analog amplifiers are unidirectional devices, the transmission path must be split into a 4-wire system (one pair for transmitting, one pair for receiving) via a 2-wire-to-4-wire hybrid transformer.
- **Board Exam Tip:** Local loop = **2-wire**. Long-distance amplified trunk lines = **4-wire**.

Question 79

The integration of security and surveillance systems with a BMS allows for:
ANSWER: Centralized management of access control, alarm monitoring, and video feeds.

- **Explanation:** Modern Building Management Systems coordinate HVAC, fire safety, and physical access controls on a single software platform. During a security or fire event, the BMS can automatically unlock egress doors, call elevators to the ground floor, and route surveillance cameras to the event area.
- **Board Exam Tip:** BMS integration allows **centralized monitoring and automated cross-system emergency responses**.

Question 80

Which factor significantly affects ground wave propagation?
ANSWER: Frequency (and soil/ground electrical conductivity σ).
- **Explanation:** Earth losses increase with the square of the frequency. Frequencies above 2 to 3 MHz suffer rapid attenuation through ground wave propagation, which restricts practical ground wave communications to VLF, LF, and the standard MF AM broadcast band.
- **Board Exam Tip:** Ground wave propagation effectiveness is governed primarily by **Operating Frequency** and **Earth Conductivity**.

Question 81

In a single-sideband (SSB) transmission, what is removed to reduce bandwidth?
ANSWER: Carrier and one sideband (leaving only USB or LSB).
- **Explanation:** Generating SSB suppresses the high-power carrier wave and filters out either the upper or lower sideband, reducing bandwidth from $2f_m$ to f_m.
- **Board Exam Tip:** SSB eliminates the **Carrier** and **One Sideband**.

Question 82

What does QAM (Quadrature Amplitude Modulation) combine?
ANSWER: Amplitude and phase modulation (combining ASK and PSK).
- **Formula:** $s(t) = I(t)\cos(\omega_c t) - Q(t)\sin(\omega_c t)$, where I and Q are independent multi-level baseband inputs.
- **Board Exam Tip:** QAM combines **Amplitude and Phase modulation**.

Question 83

A typical fire alarm control panel, when integrated with a BMS, can send a signal to:
ANSWER: Shut down the HVAC system and activate a smoke control sequence (pressurizing stairwells).
- **Explanation:** In fire life-safety systems, the HVAC supply fans must shut down to avoid circulating smoke and feeding fresh oxygen to the fire. Simultaneously, exhaust dampers open and dedicated stairwell fans pressurize escape paths to keep them smoke-free.
- **Board Exam Tip:** On fire alarm activation, BMS commands the **immediate shutdown of fresh-air HVAC fans** to contain smoke.

Question 84

FSK (Frequency Shift Keying) encodes information by:
ANSWER: Varying the frequency of the carrier wave between discrete values (Mark and Space frequencies).
- **Explanation:** Binary FSK shifts carrier frequency between a ”Mark” frequency (f_1, logic 1) and a ”Space” frequency (f_2, logic 0) while keeping amplitude constant.
- **Board Exam Tip:** FSK = shifts **Frequency** (Mark and Space). PSK = shifts **Phase**. ASK = shifts **Amplitude**.

Question 85

A two-wire circuit is generally used for:
ANSWER: Local telephone subscriber lines (the local loop / last mile).
- **Explanation:** The local subscriber loop connecting a customer premise telephone to the central telephone exchange uses a single unshielded twisted copper pair (Tip and Ring) carrying bidirectional audio, DC loop current, and ringing voltage.
- **Board Exam Tip:** Two-wire circuit = **Subscriber Local Loop**. Four-wire circuit = **Inter-office trunk carrier lines**.

Question 86

What does ”protocol” refer to in data communication?
ANSWER: A formal set of rules governing data exchange between devices.
- **Explanation:** A communication protocol establishes formatting, timing, sequencing, and error control rules to ensure communicating systems interpret exchanged bit streams correctly.
- **Board Exam Tip:** The three key elements of a protocol are **Syntax** (data format), **Semantics** (meaning of bits), and **Timing** (speed and sequencing).

Question 87

Which interface is part of ISDN for basic rate service?
ANSWER: BRI (Basic Rate Interface: $2B + D$)
- **Explanation:** The ISDN Basic Rate Interface (BRI) provides two 64 kbps Bearer channels (B) and one 16 kbps Delta signaling channel (D), delivering 144 kbps user payload (192 kbps raw line rate including framing).
- **Board Exam Tip:** ISDN BRI = **$2B + D$** (144 kbps). ISDN PRI = **$23B + D$** (T1) or **$30B + D$** (E1).

Question 88

An example of a full-duplex communication system is:
ANSWER: Telephone (landline and mobile cellular networks).
- **Explanation:** In full-duplex systems, both communicating parties can transmit and receive voice or data simultaneously without taking turns.
- **Board Exam Tip:** Telephone = **Full Duplex**. Walkie-Talkie = **Half Duplex**. Commercial AM/FM broadcast = **Simplex**.

Question 89

Hamming codes can:
ANSWER: Detect double-bit errors and correct single-bit errors ($d_{\text{min}} = 3$).
- **Formula:** For parity check bits p and data bits m: $2^p \geq m + p + 1$. Minimum Hamming distance is $d_{\text{min}} = 3$.
- **Explanation:** Standard linear block Hamming codes have $d_{\text{min}} = 3$, which allows correcting $t = \lfloor (3 - 1)/2 \rfloor = 1$ bit error, or detecting $e = 3 - 1 = 2$ bit errors.
- **Board Exam Tip:** Standard Hamming code: **Single Error Correction (SEC)** and **Double Error Detection (DED)**.

Question 90

What does the Automatic Gain Control (AGC) feature do?
ANSWER: Keeps receiver output signal strength consistent despite fluctuations in incoming RF signal level.
- **Explanation:** AGC rectifies the receiver’s IF/detector output to derive a negative feedback DC bias voltage. When incoming signal strength increases, this feedback voltage reduces the gain of the RF and IF amplifier stages, preventing receiver overload and output volume variations.
- **Board Exam Tip:** AGC maintains **constant audio/detector output level** over wide dynamic ranges of incoming RF signals.

Question 91

Which of the following statements is true for an Infinite Impulse Response (IIR) filter?
ANSWER: Its difference equation includes feedback terms (past output values).
- **Explanation:** IIR filters are recursive systems whose current output $y[n]$ depends on past outputs $y[n - 1], y[n - 2], \ldots$, creating infinite-duration impulse responses and non-zero poles in the complex z-plane.
- **Board Exam Tip:** IIR filters have **feedback paths** and poles; FIR filters have **feedforward only** and all poles at $z = 0$.

Question 92

An analog signal is characterized by:
ANSWER: A continuous set of time values and a continuous set of amplitude values.
- **Explanation:** An analog signal is continuous in both time and magnitude, containing an infinite number of possible amplitude values over any measurement interval.
- **Board Exam Tip:** Analog = continuous time, continuous amplitude. Sampled = discrete time, continuous amplitude. Digital = discrete time, quantized amplitude.

Question 93

For a parallel-plane waveguide, infinite in extent and with a spacing of 20 cm between the plates, calculate the cut-off wavelength for the dominant TE mode.
ANSWER: 40 cm
- **Given:** Plate separation $a = 20$ cm $= 0.2$ m, dominant mode TE_1 ($m = 1$).
- **Find:** Cut-off wavelength λ_c.
- **Formula:** For parallel-plate waveguides, the cut-off wavelength for mode m is:

$$\lambda_c = \frac{2a}{m}$$

- **Solution:** For dominant mode $m = 1$:

$$\lambda_c = \frac{2(20\ \text{cm})}{1} = 40\ \text{cm} = 0.4\ \text{m}$$

- **Frequency verification:** $f_c = \frac{c}{\lambda_c} = \frac{3\times10^8\ \text{m/s}}{0.4\ \text{m}} = 750$ MHz.
- **Board Exam Tip:** Dominant mode cut-off wavelength for parallel planes or rectangular waveguides is **$\lambda_c = 2a$** (twice the broad plate dimension).

Question 94

Which method reduces noise in digital communication systems?
ANSWER: Using error-correcting coding (FEC) and regenerative repeaters.
- **Explanation:** Forward Error Correction (FEC) introduces structured parity bits so receivers can detect and correct bit errors without retransmissions. Regenerative repeaters reshape and retime pulses, removing accumulated channel noise before it causes bit errors.
- **Board Exam Tip:** Noise accumulation in digital systems is prevented by **regenerative repeaters** and mitigated by **channel coding (FEC)**.

Question 95

One advantage of bit-oriented protocols is:
ANSWER: Better frame efficiency, code independence, and robust error detection.
- **Explanation:** Bit-oriented protocols (like HDLC) treat data as a continuous bit stream regardless of character encoding. Using bit stuffing prevents payload data from mimicking the framing flag, reducing control character overhead and simplifying parsing.
- **Board Exam Tip:** Bit-oriented protocols provide **true data transparency, lower framing overhead, and code independence**.

Question 96

What does synchronization ensure in data communication systems?
ANSWER: Timely and accurate data exchange between sender and receiver (matching bit timing and frame boundaries).
- **Explanation:** Synchronization aligns receiver clock phases and sampling windows with the incoming signal, ensuring bits and frames are sampled at their center without timing skew.
- **Board Exam Tip:** Synchronization matches clock frequency, bit phase, and frame boundaries between transceivers.

Question 97

Error detection and correction are implemented in which functional block of a digital communication system?
ANSWER: Channel encoder (and Channel Decoder)
- **Explanation:** In the digital communications chain:
  - **Source Encoder:** Compresses data by removing information redundancy (e.g., Huffman, JPEG, MP3).
  - **Channel Encoder:** Adds structured redundancy for error detection and correction (e.g., CRC, Hamming, Reed-Solomon, Convolutional/Turbo codes).
  - **Modulator:** Maps digital bits onto physical analog waveforms.
- **Board Exam Tip:** Source coding = **Compression** (removes redundancy). Channel coding = **Error control** (adds controlled redundancy).

Question 98

The TCP/IP layer that corresponds to the OSI Transport layer is:
ANSWER: Host-to-Host (Transport) Layer
- **Explanation:**
  - OSI Layer 7, 6, 5 (Application, Presentation, Session) $\leftrightarrow$ TCP/IP Application Layer.
  - OSI Layer 4 (Transport) $\leftrightarrow$ TCP/IP Host-to-Host (Transport) Layer (TCP, UDP).
  - OSI Layer 3 (Network) $\leftrightarrow$ TCP/IP Internet Layer (IP, ICMP, ARP).
  - OSI Layer 2, 1 (Data Link, Physical) $\leftrightarrow$ TCP/IP Network Access / Link Layer.
- **Board Exam Tip:** OSI Transport layer corresponds to the **Host-to-Host (Transport) Layer** in the TCP/IP suite.

Question 99

In analog communication, the signal varies with:
ANSWER: Continuous values (continuously varying amplitude and frequency over time).
- **Explanation:** Analog communications modulate continuous physical quantities (voltage, current, electromagnetic field strength) in direct proportion to real-world audio, video, or sensor inputs.
- **Board Exam Tip:** Analog = continuous range of values. Digital = discrete, quantized states.

Question 100

Which part of the receiver helps suppress adjacent channel interference?
ANSWER: RF filter (and IF selective bandpass filter).
- **Explanation:** In a superheterodyne receiver:
  - The **RF Preselector Filter** rejects out-of-band signals and image frequencies ($f_{\text{image}} = f_{RF} \pm 2f_{IF}$).
  - The **IF Bandpass Filter** provides the steep roll-off skirts required to reject adjacent channels.
- **Board Exam Tip:** Image frequency rejection is achieved by the **RF amplifier/filter**. Adjacent channel selectivity is achieved by the **IF bandpass filter**.

2 Part II — Important Terms & Concepts Summary

2.1 Modulation & Communications Theory

- **AM Bandwidth:** Conventional AM (DSB-FC) and DSB-SC require $B = 2f_m$. Single-sideband (SSB) cuts this to $B = f_m$.
- **FM Deviation:** Frequency deviation $\Delta f = k_f V_m$ is proportional to modulating signal amplitude. Modulation index is $\beta = \Delta f / f_m$.
- **Noise Bandwidth (B_n):** The equivalent bandwidth of an ideal rectangular filter that passes the same total thermal noise power as the practical filter. For a single-tuned circuit: $B_n = \frac{\pi}{2} B_{3\text{dB}} \approx 1.57 B_{3\text{dB}}$.
- **SSB Power Relations:** In speech SSB, average power is roughly 25%–33% ($1/4$ to $1/3$) of Peak Envelope Power (PEP).
- **PCM Steps:** Sampling (Nyquist $f_s \geq 2f_{\text{max}}$) $\rightarrow$ Quantization (maps amplitudes to discrete levels) $\rightarrow$ Encoding (binary code conversion).
- **Channel Capacity:** Nyquist formula: $C = 2B \log_2 M$ (noiseless). Shannon formula: $C = B \log_2(1 + \text{SNR})$ (noisy).

2.2 Transmission Lines, Waveguides & Antennas

- **Coaxial Cable Impedance:** $Z_0 = \frac{138}{\sqrt{\varepsilon_r}} \log_{10}(D/d)$.
- **Parallel-Wire Impedance:** $Z_0 = \frac{276}{\sqrt{\varepsilon_r}} \log_{10}(2D/d)$.
- **Waveguide Dominant Mode:** In hollow metallic waveguides, TEM modes cannot propagate. The dominant mode is TE_{10} for rectangular guides and TE_1 for parallel planes, with cut-off wavelength $\lambda_c = 2a$.
- **Free Space Impedance:** $\eta_0 = \sqrt{\mu_0/\varepsilon_0} \approx 120\pi \approx 377\,\Omega$. Power density: $\mathcal{P} = E^2/\eta_0 = E^2/377$.
- **Antenna Gain vs Beamwidth:** Symmetrical aperture beamwidth relates to linear gain by $\theta_{HPBW} \approx \sqrt{41,253/g}$.
- **Smith Chart Navigation:** Clockwise = Toward Generator (away from load). Counter-Clockwise = Toward Load.

2.3 Data Communications & OSI Model

- **Bit-Oriented vs Character-Oriented:** Bit-oriented protocols (HDLC, PPP) use flag framing (01111110) and bit-stuffing. Character-oriented protocols (Bisync) use ASCII control bytes.
- **OSI Layers:**
  - Layer 5 (Session): Dialogue management, synchronization checkpoints.
  - Layer 4 (Transport): End-to-end reliability, flow control, port addressing (TCP, UDP).
  - Layer 3 (Network): Logical routing, packet forwarding (IP).
  - Layer 2 (Data Link): Framing, physical MAC addressing, hop-to-hop error detection (Ethernet, HDLC).
- **Error Detection / Correction:** CRC provides cyclic polynomial error detection. Hamming codes provide single-error correction and double-error detection ($d_{\text{min}} = 3$).

2.4 Digital Signal Processing & Filter Systems

- **Filter Stability:** Discrete-time systems are BIBO stable if and only if all poles of the transfer function $H(z)$ lie strictly inside the unit circle ($|z| < 1$).
- **FIR vs IIR Filters:** FIR filters use feedforward paths only, have finite duration impulse responses, and are unconditionally stable. IIR filters use output feedback, can achieve steeper roll-offs with fewer coefficients, but can become unstable if poles move outside the unit circle.
- **Nyquist Rate vs Nyquist Frequency:** Nyquist rate is $2f_{\text{max}}$. Nyquist frequency is $f_s/2$. Thus, the Nyquist rate is twice the Nyquist frequency.

3 Part III — Comprehensive Formula Sheet

Subject / Law	Mathematical Formula	Application / Description
AM Transmission Bandwidth	$B = 2f_m$	Double-sideband AM bandwidth
FM Carson's Bandwidth	$B_T = 2(\Delta f + f_m) = 2f_m(1+\beta)$	Approximate FM transmission bandwidth
FM Frequency Deviation	$\Delta f = k_f \cdot V_m$	Peak deviation from carrier frequency
Coaxial Cable Impedance	$Z_0 = \frac{138}{\sqrt{\varepsilon_r}} \log_{10}\left(\frac{D}{d}\right)$	High-frequency coaxial surge impedance
Twin-Lead Cable Impedance	$Z_0 = \frac{276}{\sqrt{\varepsilon_r}} \log_{10}\left(\frac{2D}{d}\right)$	Parallel-wire characteristic impedance
Poynting Power Density	$\mathcal{P} = \frac{E^2}{\eta_0} = \frac{E^2}{377} = \eta_0 H^2$	Free-space electromagnetic power density
Free Space Impedance	$\eta_0 = \sqrt{\frac{\mu_0}{\varepsilon_0}} \approx 120\pi \approx 377\,\Omega$	Wave impedance of free space
Antenna Beamwidth Rule	$\theta_{HPBW} \approx \sqrt{\frac{41{,}253}{g}} \approx \frac{200}{\sqrt{g}}$	Beamwidth approximation from linear gain g
Antenna Gain Conversion	$G_{\text{dBi}} = G_{\text{dBd}} + 2.15$ dB	Conversion between isotropic and dipole gain
Waveguide Cut-Off Wavelength	$\lambda_c = \frac{2a}{m}$	Parallel-plane and TE_{m0} waveguide cut-off
Single-Tuned Noise Bandwidth	$B_n = \frac{\pi}{2} B_{3\text{dB}} \approx 1.57 B_{3\text{dB}}$	Equivalent rectangular noise bandwidth
Tuned Circuit Noise Bandwidth	$B_n = \frac{1}{RC}$	Parallel RLC noise bandwidth
Nyquist Sampling Rate	$f_s \geq 2f_{\text{max}}$	Minimum sampling frequency to prevent aliasing
Nyquist Folding Frequency	$f_{\text{Nyquist}} = \frac{f_s}{2}$	Highest alias-free signal frequency
Shannon Channel Capacity	$C = B \log_2(1 + \text{SNR})$	Maximum noisy channel data rate (bps)
Nyquist Channel Capacity	$C = 2B \log_2(M)$	Maximum noiseless channel data rate (bps)
Hamming Check Bit Formula	$2^p \geq m + p + 1$	Minimum parity bits p for m data bits
SSB Average Speech Power	$P_{\text{avg}} = \left(\frac{1}{4} \text{ to } \frac{1}{3}\right) \times \text{PEP}$	Average voice power relative to PEP
Shot Noise Current	$I_n = \sqrt{2qI_{DC}B}$	Schottky shot noise across active junctions
Thermal Noise Power	$N = kTB$	Johnson-Nyquist thermal noise power
Thermal Noise Density at 290K	$N_0 = -174$ dBm/Hz	Ambient noise power spectral density
Optical Pulse Dispersion Limit	$R_b \leq \frac{0.2}{\Delta\tau \cdot L}$	Maximum bit rate limited by fiber dispersion
Ionospheric Skip Distance	$d_{\text{skip}} = 2h\sqrt{\left(\frac{f}{f_c}\right)^2 - 1}$	Single-hop radio skip distance
Discrete Filter Z-Transform	$H(z) = \frac{\sum b_k z^{-k}}{1 + \sum a_k z^{-k}}$	Rational discrete LTI transfer function

4 Part IV — Quick Answer Key Table

5 Part V — Final Board Exam Cram Sheet

5.1 Must-Remember Formulas

1. **AM Transmission Bandwidth:** $B_{\text{AM}} = 2f_m$; $B_{\text{SSB}} = f_m$.
2. **FM Carson's Bandwidth Rule:** $B_T = 2(\Delta f + f_m) = 2f_m(1+\beta)$.
3. **Poynting Power Density:** $\mathcal{P} = \frac{E^2}{377}$ (W/m^2).
4. **Coaxial Characteristic Impedance:** $Z_0 = \frac{138}{\sqrt{\varepsilon_r}} \log_{10}\left(\frac{D}{d}\right)$.
5. **Parallel-Wire Characteristic Impedance:** $Z_0 = \frac{276}{\sqrt{\varepsilon_r}} \log_{10}\left(\frac{2D}{d}\right)$.
6. **Waveguide Cut-Off Wavelength:** $\lambda_c = 2a$ (for dominant TE_{10} or TE_1 modes).
7. **Single-Tuned Noise Bandwidth:** $B_n = \frac{\pi}{2} B_{3\text{dB}} \approx 1.57 B_{3\text{dB}}$.
8. **Parallel RLC Noise Bandwidth:** $B_n = \frac{1}{RC}$.
9. **Antenna Gain vs Beamwidth:** $\theta_{HPBW} \approx \sqrt{\frac{41{,}253}{g}} \approx \frac{200}{\sqrt{g}}$.
10. **Nyquist Sampling Rate vs Frequency:** $f_{\text{Nyquist-rate}} = 2f_{\text{max}}$; $f_{\text{Nyquist-freq}} = f_s/2$.
11. **Shannon Channel Capacity:** $C = B \log_2(1 + \text{SNR})$.
12. **SSB Speech Average Power:** $P_{\text{avg}} = \frac{1}{4}$ to $\frac{1}{3} \times$ PEP (25%–33%).

5.2 Must-Remember One-Line Concepts

- A discrete-time IIR filter is stable if and only if all poles lie strictly **inside the unit circle** ($|z| < 1$).
- In frequency modulation, frequency deviation Δf depends on modulating **amplitude**; rate of deviation depends on modulating **frequency**.
- Ground-wave attenuation **increases with frequency** and decreases with higher soil conductivity.
- Hollow metallic waveguides support **TE and TM modes only** (TEM modes cannot propagate).
- Moving clockwise on the Smith Chart represents movement **toward the generator (away from the load)**.
- An antenna operating at natural resonance presents a **purely resistive load** ($X = 0$).
- Laser diodes emit **coherent light via stimulated emission**; LEDs emit **incoherent light via spontaneous emission**.
- **QPSK** achieves twice the bandwidth efficiency of **BPSK** (2 bps/Hz vs 1 bps/Hz).
- **QAM** modulates both **amplitude and phase** to maximize spectral efficiency.
- In a superheterodyne receiver, image frequency rejection occurs at the **RF preselector filter**; adjacent channel rejection occurs at the **IF filter**.
- On fire alarm activation, the BMS commands the **immediate shutdown of supply HVAC fans** and initiates stairwell pressurization.
- **Source coding** removes redundancy (compression); **Channel coding** adds controlled redundancy (error detection/correction).

5.3 Common Traps & Quick Rules

- **Nyquist Rate vs Nyquist Frequency:** Do not confuse them! Nyquist Rate is the minimum sampling rate ($2f_{\text{max}}$). Nyquist Frequency is the highest reproducible frequency ($f_s/2$).
- **Noise Bandwidth vs 3-dB Bandwidth:** The noise bandwidth of a tuned filter is wider than its half-power 3-dB bandwidth: $B_n = \frac{\pi}{2} B_{3\text{dB}} \approx 1.57 B_{3\text{dB}}$ (not equal to $B_{3\text{dB}}$).
- **Thermal Noise vs Shot Noise:** Resistors generate only **thermal (Johnson) noise**. Active PN junctions, diodes, and transistors generate both **shot noise** and thermal noise.
- **Smith Chart Rotation Direction:** Clockwise = Toward Generator. Counter-Clockwise = Toward Load. One complete revolution around the chart represents $\lambda/2$ (0.5λ).
- **Two-Wire vs Four-Wire Telephony:** Subscriber local loops are **2-wire**. Long-distance amplified trunk channels are **4-wire**.
- **Coaxial vs Twin-Lead Impedance Constants:** Coaxial lines use numerator **138** ($Z_0 = \frac{138}{\sqrt{\varepsilon_r}} \log_{10}(D/d)$); Twin-lead parallel lines use numerator **276** ($Z_0 = \frac{276}{\sqrt{\varepsilon_r}} \log_{10}(2D/d)$).
- **SSB vs DSB Bandwidth:** Single-sideband (SSB) cuts bandwidth in half: $B = f_m$. Conventional DSB-AM requires $B = 2f_m$.
- **Isotropic vs Dipole Gain:** An isotropic antenna has 0 dBi; a half-wave dipole has 2.15 dBi (0 dBd). Always remember: $G_{\text{dBi}} = G_{\text{dBd}} + 2.15$ dB.

No.	Key Answer	Topic	No.	Key Answer	Topic
1	CRC	Error Control / Coding	51	0.265 W/m^2	Electromagnetics / Power Density
2	Random pulses of high amplitude	Noise Analysis / Impulse Noise	52	$249\,\Omega$	Transmission Lines / Twin-Lead
3	Less at higher frequencies	Wave Propagation / Ground Waves	53	Cable television	Multiplexing / FDM Systems
4	Mapping amplitudes into levels	Digital Communications / PCM	54	Highest ionospheric region	Wave Propagation / Sky Waves
5	Twice the modulating frequency	Amplitude Modulation / Bandwidth	55	Parabolic interaction	Antennas / Wave Polarization
6	$48\,\Omega$	Transmission Lines / Coaxial	56	Mesh	Network Topologies / Redundancy
7	Provide power to drive signal	Hardware Interfacing / Drivers	57	Scattering and absorption	Fiber Optics / Attenuation
8	DSL Internet services	Telecommunications / Broadband	58	Diodes and transistors	Noise Analysis / Shot Noise
9	Outside unit circle in z-plane	Digital Signal Processing / Stability	59	Movement away from load	Transmission Lines / Smith Chart
10	Vertical	Antenna Systems / Polarization	60	Both directions, not simultaneous	Telephony / Two-Wire Systems
11	Analyzing poles of $H(z)$	Digital Filters / Stability	61	Poles = denom, Zeros = numer	Digital Filters / Z-Domain
12	High sensitivity to timing errors	Pulse Modulation / PPM	62	QPSK	Digital Modulation / Efficiency
13	Convert FM to audio signals	FM Receivers / Discriminators	63	2	Digital Modulation / BPSK
14	Continuous analog transmission	Multiple Access / FDMA	64	Short-range low bandwidth	Digital Modulation / ASK
15	6.4°	Antennas / Beamwidth	65	Coherent light via stimulated emission	Optoelectronics / Laser vs LED
16	Change in resistance with strain	Sensors / Strain Gauges	66	Purely resistive load	Antenna Systems / Resonance
17	GAN	Computer Networks / Network Types	67	A hub or switch	Network Topologies / Star
18	Robust in harsh environments	Industrial Automation / PLCs	68	Synchronizing dialogue	OSI Model / Session Layer
19	Greater than its Nyquist rate	Sampling Theory / Nyquist	69	Line geometry and material	Transmission Lines / Z_0
20	Precise synchronization required	Multiplexing / TDM Limitations	70	$377\,\Omega$	Electromagnetics / Free Space
21	Parabolic dish	Satellite Communications	71	Distinct code sequences	Multiple Access / CDMA
22	Bit-oriented protocol	Data Link Protocols / HDLC	72	Present and past inputs only	Digital Filters / Non-Recursive
23	Gain	Antenna Fundamentals / Gain	73	Synchronizing dialogue	OSI Model / Session Layer
24	B-ISDN uses higher bandwidth/ATM	Telecommunications / ISDN	74	Signals via multiple paths with delay	Mobile Communications / Fading
25	15.7 MHz	RF Circuits / Noise Bandwidth	75	Signal amplitude reduction	Transmission Lines / Attenuation
26	Enhanced bandwidth efficiency	Digital Modulation / QAM	76	Pulse duration	Pulse Modulation / PWM
27	Half Duplex	Transmission Modes / Duplex	77	One sideband, suppressed carrier	Single Sideband / Efficiency
28	859 Hz	Frequency Modulation / Noise	78	Four-wire circuit	Telephony / Trunk Circuits
29	TE and TM modes	Microwave / Waveguides	79	Centralized security/access control	Building Systems / BMS
30	Controlling PA system volume	Building Systems / Audio-Video	80	Frequency	Wave Propagation / Ground Waves
31	A specific time slot	Multiplexing / TDM Architecture	81	Carrier and one sideband	Modulation / SSB Generation
32	Defined for all values of time	Signals & Systems / Continuous	82	Amplitude and phase modulation	Digital Modulation / QAM
33	Manage data flow and error control	Communications Hardware / LCU	83	Shut down HVAC, control smoke	Building Systems / Fire Safety
34	Nyquist rate is twice frequency	Sampling Theory / Definitions	84	Varying frequency of carrier	Digital Modulation / FSK
35	Yagi-Uda antenna	Antenna Systems / Yagi Arrays	85	Local telephone lines	Telephony / Local Loop
36	Better resistance to amplitude noise	Pulse Modulation / PWM vs PAM	86	A set of rules for data exchange	Data Communications / Protocol
37	Sampling Quantization Encoding	Digital Systems / PCM Steps	87	BRI	Telephony / ISDN Interfaces
38	Maximum data rate of a channel	Information Theory / Shannon	88	Telephone	Transmission Modes / Full Duplex
39	25 MHz	Tuned Circuits / Noise Bandwidth	89	Detect double, correct single errors	Error Control / Hamming Codes
40	Past/present inputs, past outputs	Digital Filters / IIR Architecture	90	Keeps signal strength consistent	Receivers / AGC Systems
41	Hashing	Network Security / Cryptography	91	Difference eq includes feedback	Digital Filters / IIR Feedback
42	1.33 Mbps	Fiber Optics / Dispersion Limit	92	Continuous time and amplitude	Signals & Systems / Analog
43	User interface for monitoring/control	Industrial Systems / SCADA in BMS	93	40 cm	Microwave / Waveguide Cutoff
44	Multipoint	Transmission Lines / Topologies	94	Using error correction coding	Digital Communications / Noise
45	125 W to 167 W	RF Power / SSB PEP vs Average	95	Better efficiency and error detection	Data Link / Bit-Oriented
46	Special control characters	Data Link / Character Protocols	96	Timely and accurate data exchange	Data Communications / Sync
47	To transmit information	Communications / Fundamentals	97	Channel encoder	Digital Communications / Coding
48	Quadrature Amplitude Modulation	Digital Modulation / QAM	98	Host-to-Host (Transport)	Network Models / TCP/IP vs OSI
49	Height of ionosphere and angle	Wave Propagation / Skip Distance	99	Continuous values	Signals & Systems / Analog
50	Modulating signal amplitude	Frequency Modulation / Deviation	100	RF filter	Receivers / Interference Filters

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BOARD EXAMINATION COMPREHENSIVE REVIEWER

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1 Part I — Complete Solutions to Examination Questions

Question 1

According to Hartley's Law, the total amount of information capable of being transmitted over a communication channel is strictly proportional to which two system parameters?
ANSWER: Bandwidth and transmission time (B and T)

- **Formula:** Hartley's Law: $H = C \cdot T = 2B \cdot T \log_2(M)$, where H is total information (bits), B is channel bandwidth (Hz), and T is transmission time duration (seconds).
- **Explanation:** In 1928, R.V.L. Hartley established that the maximum data transmission capacity without noise is directly proportional to the product of channel bandwidth and the time duration of transmission ($H \propto B \cdot T$). Shannon later extended this to noisy channels.
- **Board Exam Tip:** Hartley's Law: Information $\propto B \cdot T$. Shannon-Hartley Theorem: Capacity $C = B \log_2(1 + \text{SNR})$.

Question 2

Calculate the Numerical Aperture (NA) of a step-index optical fiber if the refractive index of the core is 1.50 and the refractive index of the cladding is 1.46.
ANSWER: 0.344

- **Given:** Core index $n_1 = 1.50$, cladding index $n_2 = 1.46$.
- **Find:** Numerical Aperture (NA).
- **Formula:** $\text{NA} = \sqrt{n_1^2 - n_2^2} = n_1\sqrt{2\Delta}$, where $\Delta = \frac{n_1 - n_2}{n_1}$.
- **Solution:**

$$\text{NA} = \sqrt{(1.50)^2 - (1.46)^2} = \sqrt{2.25 - 2.1316} = \sqrt{0.1184} \approx 0.34409 \approx 0.344$$

- **Explanation:** Numerical Aperture represents the light-gathering capability of an optical fiber and defines the maximum acceptance angle ($\theta_a = \arcsin(\text{NA}) = \arcsin(0.344) \approx 20.12°$).
- **Board Exam Tip:** $\text{NA} = \sqrt{n_{core}^2 - n_{clad}^2} = \sin(\theta_{acceptance})$.

Question 3

A standard Ethernet MAC address is uniquely burned into a Network Interface Card (NIC). How long is this physical address?
ANSWER: 48 bits (or 6 bytes$/12$ hexadecimal digits)

- **Explanation:** An IEEE 802 MAC-48 address is 48 bits long. The first 24 bits represent the Organizationally Unique Identifier (OUI) assigned to the manufacturer, and the remaining 24 bits represent the manufacturer's unique device serialization identifier.
- **Board Exam Tip:** MAC address = 48 bits (6 bytes, Layer 2). IPv4 address = 32 bits (4 bytes, Layer 3). IPv6 address = 128 bits (16 bytes, Layer 3).

Question 4

Wavelength Division Multiplexing (WDM) used in fiber optics is mathematically and conceptually equivalent to which technique used in copper/radio networks?
ANSWER: Frequency Division Multiplexing (FDM)

- **Explanation:** Because wavelength and frequency are fundamentally related by $\lambda = c/f$, dividing an optical spectrum into discrete wavelength channels ($\lambda_1, \lambda_2, \ldots$) is physically and mathematically identical to dividing an electrical RF spectrum into discrete carrier frequency bands.
- **Board Exam Tip:** WDM is simply FDM operating at optical lightwave frequencies (1310 nm, 1550 nm).

Question 5

Using Carson's rule, determine the approximate required transmission bandwidth for an FM signal exhibiting a peak frequency deviation of 25 kHz and driven by a maximum modulating frequency of 5 kHz.
ANSWER: 60 kHz

- **Given:** Peak frequency deviation $\Delta f = 25$ kHz, maximum modulating frequency $f_m = 5$ kHz.
- **Find:** Transmission Bandwidth (B_T).
- **Formula:** Carson's Rule: $B_T = 2(\Delta f + f_m) = 2f_m(1 + \beta)$, where $\beta = \frac{\Delta f}{f_m}$.

- **Solution:**

$$B_T = 2(25\text{ kHz} + 5\text{ kHz}) = 2(30\text{ kHz}) = 60\text{ kHz}$$

- **Explanation:** Carson's rule captures approximately 98% of the total transmitted power in an angle-modulated wave by retaining significant Bessel sidebands up to order $n = \beta + 1$.
- **Board Exam Tip:** Carson's Rule is ALWAYS $2(\Delta f + f_m)$.

Question 6

Which specific mode of radio wave propagation relies strictly on the varying refractive index of the troposphere to bend VHF and UHF signals back toward the earth, often extending coverage slightly beyond the visual horizon?
ANSWER: Tropospheric scatter and refraction (Troposcatter / Tropospheric bending)

- **Explanation:** Normal atmospheric temperature, pressure, and humidity gradients create a decreasing refractive index gradient with height in the troposphere. This continuously refracts VHF/UHF waves downward (standard $k = 4/3$ earth radius refraction). Irregular turbulence eddies also cause forward scattering, extending links over the horizon up to several hundred kilometers.
- **Board Exam Tip:** Troposphere: below 15 km (weather, refraction, ducting). Ionosphere: 50–400 km (skywave reflection for HF).

Question 7

A mixer circuit combines a 1000 kHz carrier with a 1500 kHz local oscillator signal. Which of the following lists the primary output frequencies before filtering?
ANSWER: 500 kHz and 2500 kHz (Sum and Difference frequencies)

- **Given:** $f_s = 1000$ kHz, $f_{LO} = 1500$ kHz.
- **Formula:** Ideal analog multiplication produces sum and difference intermodulation products: $f_{out} = |f_{LO} \pm f_s|$.
- **Solution:**

$$f_{diff} = |1500\text{ kHz} - 1000\text{ kHz}| = 500\text{ kHz}$$

$$f_{sum} = 1500\text{ kHz} + 1000\text{ kHz} = 2500\text{ kHz}$$

- **Board Exam Tip:** Mixers/heterodynes always output $|f_{LO} - f_{RF}|$ and $f_{LO} + f_{RF}$.

Question 8

If a signal is defined as being conjugate symmetric, how does its real part relate to the time axis?
ANSWER: The real part is an even function of time ($\text{Re}\{x(t)\} = \text{Re}\{x(-t)\}$)

- **Definition:** A complex signal $x(t)$ is conjugate symmetric (Hermitian) if $x(t) = x^*(-t)$.
- **Proof:** Let $x(t) = x_r(t) + jx_i(t)$. Then $x^*(-t) = x_r(-t) - jx_i(-t)$. Equating real and imaginary parts:

$$x_r(t) = x_r(-t) \quad (\text{Real part is even})$$

$$x_i(t) = -x_i(-t) \quad (\text{Imaginary part is odd})$$

- **Board Exam Tip:** Conjugate symmetric signal: Real part is **even**, Imaginary part is **odd**. Its Fourier transform is purely real!

Question 9

In Frame Relay networks, what does the Data Link Connection Identifier (DLCI) uniquely identify?
ANSWER: The logical virtual circuit between the source and destination (PVC or SVC)

- **Explanation:** DLCI is a 10-bit Layer 2 address used in Frame Relay headers to identify specific Permanent Virtual Circuits (PVCs) or Switched Virtual Circuits (SVCs) multiplexed across a single physical user-to-network interface (UNI). DLCIs have local significance.
- **Board Exam Tip:** Frame Relay = DLCI (Layer 2). ATM = VPI/VCI (Layer 2). MPLS = Labels.

Question 10

When comparing 16-QAM to 64-QAM under the same average transmit power and noise conditions, why does 64-QAM exhibit a higher bit error rate (BER)?
ANSWER: Its symbols are packed closer together in the constellation diagram, reducing the noise margin (minimum Euclidean distance d_{min}).

- **Explanation:** If average transmission power is fixed, placing 64 constellation points inside the same boundary shrinks the Euclidean distance between neighboring points by more than half compared to 16-QAM ($d_{\text{min},64} \approx 0.447 d_{\text{min},16}$). A smaller noise spike is sufficient to push a received symbol across the decision boundary into an adjacent Voronoi decision region.
- **Board Exam Tip:** Higher-order modulation increases spectral efficiency (bits/s/Hz) but requires higher SNR to maintain the same BER due to reduced Euclidean distance.

Question 11

Two cascaded amplifiers form a receiver chain. Amplifier A has a power gain of 10 dB and a noise figure of 2 dB. Amplifier B has a power gain of 20 dB and a noise figure of 6 dB. Calculate the approximate overall system noise figure in linear ratio.
ANSWER: 1.88 (or approximately 2.75 dB)

- **Given:** Stage 1: $G_1 = 10$ dB $\implies g_1 = 10^{10/10} = 10$. $NF_1 = 2$ dB $\implies F_1 = 10^{2/10} = 10^{0.2} \approx 1.5849$.
  Stage 2: $G_2 = 20$ dB $\implies g_2 = 10^{20/10} = 100$. $NF_2 = 6$ dB $\implies F_2 = 10^{6/10} = 10^{0.6} \approx 3.9811$.
- **Formula:** Friis' Formula for cascaded noise factor:

$$F_{\text{total}} = F_1 + \frac{F_2 - 1}{g_1}$$

- **Solution:**

$$F_{\text{total}} = 1.5849 + \frac{3.9811 - 1}{10} = 1.5849 + \frac{2.9811}{10} = 1.5849 + 0.2981 = 1.883 \approx 1.88$$

- **Decibel check:** $NF_{\text{total}} = 10 \log_{10}(1.883) \approx 2.75$ dB.
- **Board Exam Tip:** Friis' Formula proves that the **first stage** dominates the total receiver noise figure if its gain is high. Always convert dB to linear before substituting into Friis' formula!

Question 12

A BMS requires a conference room of 200 cubic meters to experience 6 complete Air Changes per Hour (ACH) for proper ventilation. What is the required volumetric airflow rate in cubic meters per minute?
ANSWER: 20.0 m^3/min (20.0 cubic meters/minute)

- **Given:** Room volume $V = 200$ m^3, air exchange rate ACH $= 6$ hr^{-1}.
- **Find:** Airflow rate Q in m^3/min.
- **Formula:** $Q = \frac{V \times \text{ACH}}{60 \text{ min/hr}}$.
- **Solution:**

$$Q = \frac{200 \text{ m}^3 \times 6 \text{ hr}^{-1}}{60 \text{ min/hr}} = \frac{1200 \text{ m}^3/\text{hr}}{60 \text{ min/hr}} = 20.0 \text{ m}^3/\text{min}$$

- **Board Exam Tip:** Always convert hours to minutes when evaluating HVAC volumetric flow rates (60 minutes $=$ 1 hour).

Question 13

Why is an Infinite Impulse Response (IIR) filter inherently susceptible to instability compared to a Finite Impulse Response (FIR) filter?
ANSWER: It utilizes feedback (recursive architecture), which means its poles can potentially migrate outside the unit circle in the Z-plane.

- **Explanation:** FIR filters have no feedback paths (feedforward-only); their transfer functions contain poles located exclusively at the origin ($z = 0$), making them inherently, unconditionally stable. In contrast, IIR filters employ feedback loops ($y[n] = \sum b_k x[n-k] - \sum a_k y[n-k]$), creating non-trivial poles that can move outside the unit circle ($|z| \geq 1$) due to coefficient quantization, roundoff noise, or unstable design.
- **Board Exam Tip:** FIR filters are ALWAYS stable and can achieve exact linear phase. IIR filters require fewer coefficients but can become unstable.

Question 14

A digital radio link uses 64-QAM to achieve a throughput of 120 Mbps. What is the theoretical symbol rate (baud rate) on the channel?
ANSWER: 20 Mbaud (20×10^6 symbols/sec)
- **Given:** Bit rate $R_b = 120$ Mbps, modulation scheme $= 64$-QAM $\implies M = 64$.
- **Find:** Symbol rate (baud rate) R_s.
- **Formula:** Bits per symbol $n = \log_2(M) = \log_2(64) = 6$ bits/symbol.

$$\text{Baud Rate } R_s = \frac{R_b}{n} = \frac{R_b}{\log_2(M)}$$

- **Solution:**

$$R_s = \frac{120 \text{ Mbps}}{6 \text{ bits/symbol}} = 20 \text{ Mbaud}$$

- **Board Exam Tip:** Bit Rate $=$ Baud Rate $\times \log_2(M)$. For $M = 64$, divide bit rate by 6.

Question 15

What is the fundamental difference in reliability mechanisms between Transmission Control Protocol (TCP) and User Datagram Protocol (UDP)?
ANSWER: TCP establishes a logical connection using sequence numbers, ACKs, and retransmissions to guarantee delivery, whereas UDP transmits connectionless datagrams without verification or sequencing.
- **Explanation:** TCP (Layer 4) is connection-oriented; it uses a 3-way handshake (`SYN`, `SYN-ACK`, `ACK`), byte sequence numbering, checksums, window-based flow control, and automatic retransmission (ARQ) to guarantee in-order delivery. UDP is connectionless and lightweight; it simply packages data into datagrams with minimal overhead and sends them without tracking, acknowledgment, or retransmission.
- **Board Exam Tip:** TCP = Reliable, connection-oriented, high overhead (web, email, file transfer). UDP = Unreliable, connectionless, low overhead (streaming, VoIP, DNS).

Question 16

A data payload of 101101 is to be transmitted using a CRC generator polynomial of $G(x) = x^3 + x + 1$. How many redundant bits will be appended to the payload?
ANSWER: 3 bits
- **Formula:** In Cyclic Redundancy Check (CRC) coding, the number of redundant check bits (FCS length) r is identically equal to the degree of the generator polynomial $G(x)$: $r = \deg(G(x))$.
- **Solution:** For $G(x) = x^3 + x^1 + x^0$, the highest exponent is 3. Thus, $\deg(G(x)) = 3$, which requires appending 3 parity/CRC bits.
- **Board Exam Tip:** Number of CRC check bits = Degree of generator polynomial = (Number of bits in binary generator string) -1. Here binary string is 1011 (4 bits) $\implies 4 - 1 = 3$ bits.

Question 17

Evaluate the architectural reliability of a Star topology versus a Bus topology. What is the fundamental catastrophic vulnerability inherent to the Star topology that the Bus topology avoids?
ANSWER: The Star topology depends on a central hub or switch; its failure brings down the entire network, unlike a Bus topology.
- **Explanation:** In a Star network, every host has an isolated point-to-point link to a central multiport switch/hub. If a single cable drops, only that attached host is isolated. However, the central switch is a Single Point of Failure (SPOF); if it fails, all communication across the network halts. In a passive Bus topology, nodes connect to a common backbone cable; there is no active central hub to collapse the entire network (though a severed main trunk halts the bus).
- **Board Exam Tip:** Star topology SPOF = Central Hub / Switch.

Question 18

A telecommunication standard specifies a μ-law companded PCM system with $\mu = 255$. What is the primary operational purpose of applying this non-linear compression before quantization?
ANSWER: To increase the signal-to-quantization-noise ratio (SQNR) for low-amplitude signals while sacrificing it for high-amplitude signals.

- **Explanation:** Speech waveforms have a high crest factor; speakers spend most of their time at low acoustic volume levels. Uniform quantization produces severe relative quantization error for weak signals (poor dynamic range). Companding (compressing before transmission, expanding upon reception) uses a logarithmic characteristic (μ-law in North America/Japan, A-law in Europe) to allocate more quantization levels to low amplitudes, maintaining a nearly constant SQNR over a wide dynamic range (≈ 40 dB).
- **Board Exam Tip:** Companding compresses dynamic range to boost SQNR for weak/low-amplitude signals.

Question 19

Which statement defines the Region of Convergence (ROC) for the Z-transform of a right-sided (causal) discrete-time sequence?
ANSWER: The ROC extends outward from the outermost pole to infinity ($|z| > r_{\text{max}}$, including ∞).
- **Explanation:** For a causal sequence starting at $n = 0$ ($x[n]u[n]$), the Z-transform is a power series in z^{-1}: $X(z) = \sum_{n=0}^{\infty} x[n]z^{-n}$. This series converges when $|z|$ is sufficiently large, meaning the ROC is the exterior of a circle centered at the origin that passes through the pole with the largest magnitude. If the signal is causal, the ROC includes $z = \infty$.
- **Board Exam Tip:** Causal / Right-sided $\implies$ $|z| > r_{\text{max}}$ (exterior). Anti-causal / Left-sided $\implies$ $|z| < r_{\text{min}}$ (interior). Two-sided $\implies$ $r_1 < |z| < r_2$ (ring).

Question 20

Calculate the available thermal noise power in femtowatts (fW) generated by a $50\,\Omega$ resistor operating at 300 K over a bandwidth of 10 MHz. (Use Boltzmann's constant $k = 1.38 \times 10^{-23}$ J/K).
ANSWER: 41.40 fW
- **Given:** $T = 300$ K, $B = 10$ MHz $= 10^7$ Hz, $k = 1.38 \times 10^{-23}$ J/K, $R = 50\,\Omega$.
- **Find:** Thermal noise power N in femtowatts (1 fW $= 10^{-15}$ W).
- **Formula:** Johnson-Nyquist noise power: $N = kTB$. (Note: thermal noise power is completely independent of resistance R).
- **Solution:**

$$N = (1.38 \times 10^{-23}\ \text{J/K})(300\ \text{K})(10^7\ \text{Hz}) = 4.14 \times 10^{-21} \times 10^7 = 4.14 \times 10^{-14}\ \text{W}$$

$$\text{In femtowatts: } N = \frac{4.14 \times 10^{-14}\ \text{W}}{10^{-15}\ \text{W/fW}} = 41.40\ \text{fW}$$

- **Explanation:** Maximum noise power transferred to a matched load depends strictly on absolute temperature and bandwidth, independent of resistor value.
- **Board Exam Tip:** Available thermal noise power is ALWAYS $N = kTB$. Resistor value R is a distractor!

Question 21

A large 10 MB (Megabyte) data file is transmitted over a 20 Mbps (Megabits per second) Ethernet link. Assuming standard 8-bit bytes and zero protocol overhead, what is the transmission time?
ANSWER: 4 seconds
- **Given:** File size $= 10$ MB $= 10 \times 10^6$ Bytes, link rate $= 20$ Mbps $= 20 \times 10^6$ bits/s.
- **Find:** Transmission latency T_{tx}.
- **Formula:** $T_{\text{tx}} = \frac{\text{Data in bits}}{\text{Bandwidth in bits/second}} = \frac{\text{File Size (Bytes)} \times 8}{\text{Bit Rate}}$.
- **Solution:**

$$\text{Total bits} = 10 \times 10^6\ \text{Bytes} \times 8\ \text{bits/Byte} = 80 \times 10^6\ \text{bits} = 80\ \text{Mbits}$$

$$T_{\text{tx}} = \frac{80\ \text{Mbits}}{20\ \text{Mbps}} = 4.0\ \text{seconds}$$

- **Board Exam Tip:** Common trap: Megabytes (MB) vs. Megabits (Mb). Multiply Bytes by 8 before dividing by bit rate!

Question 22

An E1 carrier system multiplexes 32 time slots, each containing 8 bits, at a frame rate of 8000 frames per second. What is the exact aggregate bit rate of the E1 line?
ANSWER: 2.048 Mbps
- **Given:** Time slots per frame $= 32$, bits per slot $= 8$, sampling frequency $f_s = 8000$ frames/s.

- **Find:** Aggregate bit rate R_{E1}.
- **Formula:** $R = (\text{Time slots} \times \text{Bits/slot}) \times \text{Frame Rate}$.
- **Solution:**

$$\text{Bits per frame} = 32 \times 8 = 256 \text{ bits/frame}$$

$$R_{E1} = 256 \text{ bits/frame} \times 8000 \text{ frames/s} = 2,048,000 \text{ bps} = 2.048 \text{ Mbps}$$

- **Explanation:** In the European E1 standard, 30 channels carry digitized voice, 1 channel (slot 0) carries framing/synchronization, and 1 channel (slot 16) carries signaling.
- **Board Exam Tip:** T1 (North America) $= 24$ channels $\times 8 + 1$ framing bit $= 193$ bits $\times 8000 = 1.544$ Mbps. E1 (Europe) $= 32 \times 8 \times 8000 = 2.048$ Mbps.

Question 23

In baseband digital transmission, what is a primary electrical advantage of using Alternate Mark Inversion (AMI) line coding over basic Non-Return-to-Zero (NRZ)?
ANSWER: AMI inherently eliminates the DC component from the transmitted signal.

- **Explanation:** In AMI (pseudoternary/bipolar coding), a logic '0' is represented by zero volts, while logic '1's alternate in polarity ($+V, -V, +V, -V$). Because alternating pulses cancel each other out over time, the long-term average voltage is zero, eliminating any DC bias. This allows the signal to pass transparently through AC-coupled transmission paths, transformers, and repeaters without DC baseline wander.
- **Board Exam Tip:** AMI = No DC component (enables AC transformer coupling) and built-in single error detection (via bipolar violations).

Question 24

An error-correcting block code possesses a minimum Hamming distance of $d_{\text{min}} = 7$. What is the maximum number of simultaneous bit errors this code can guarantee to correctly fix in a single codeword?
ANSWER: 3 errors

- **Given:** $d_{\text{min}} = 7$.
- **Find:** Error correcting capability t.
- **Formula:** Error-correction capability: $t = \left\lfloor \frac{d_{\text{min}}-1}{2} \right\rfloor$. (Error detection capability is $e = d_{\text{min}} - 1$).
- **Solution:**

$$t = \left\lfloor \frac{7-1}{2} \right\rfloor = \left\lfloor \frac{6}{2} \right\rfloor = 3 \text{ errors}$$

- **Explanation:** If fewer than or equal to 3 bit errors occur, the received corrupted vector remains closer in Hamming distance to the original transmitted codeword than to any other valid codeword in the codebook.
- **Board Exam Tip:** To DETECT e errors: $d_{\text{min}} \geq e + 1$. To CORRECT t errors: $d_{\text{min}} \geq 2t + 1$.

Question 25

When investigating anomalous VHF/UHF propagation over the ocean, an engineer determines that RF signals are traveling hundreds of miles beyond the horizon. What precise atmospheric condition must exist to form this "ducting" effect?
ANSWER: A temperature inversion causing the refractive index to drop sharply with altitude ($dN/dh < -157$ N-units/km, super-refraction).

- **Explanation:** Tropospheric ducting occurs when warm, dry air overlays cool, moist surface air (a temperature inversion combined with a rapid lapse in humidity). This creates an abnormally steep negative gradient in the atmospheric refractive index. When the curvature of the refracted radio wave exceeds the physical curvature of the earth, the wave is trapped within an atmospheric waveguide channel ("duct"), propagating over vast distances with minimal attenuation.
- **Board Exam Tip:** Tropospheric Ducting requires a **temperature inversion** (warm air above cool air) and a steep negative refractive index gradient.

Question 26

Which of the following best describes the fundamental operational physics of a metallic foil strain gauge used in industrial load cells?
ANSWER: It uses the piezoresistive effect, stretching or compressing the wire foil, which alters its electrical resistance.

- **Formula:** Electrical resistance: $R = \rho\frac{L}{A}$. Strain gauge factor: $GF = \frac{\Delta R/R}{\Delta L/L} = \frac{\Delta R/R}{\varepsilon}$.
- **Explanation:** When tensile mechanical strain stretches the foil gauge, its length (L) increases while its cross-sectional area (A) narrows due to Poisson contraction. Both geometric elongation and piezoresistive resistivity changes increase electrical resistance.
- **Board Exam Tip:** Strain gauge: Mechanical deformation $\rightarrow$ Resistance change ($\Delta R \propto \varepsilon$).

Question 27

An antenna has a directivity of 10 dB and a radiation efficiency of 90%. What is the overall antenna gain in dB?
ANSWER: 9.54 dB
- **Given:** Directivity $D_{\text{dB}} = 10$ dB, radiation efficiency $\eta = 90\% = 0.90$.
- **Find:** Antenna gain G_{dB}.
- **Formula:** Gain in linear terms: $G = \eta \cdot D$. In decibels: $G_{\text{dB}} = D_{\text{dB}} + 10\log_{10}(\eta)$.
- **Solution:**

$$10\log_{10}(0.90) \approx 10(-0.045757) \approx -0.4576 \text{ dB}$$

$$G_{\text{dB}} = 10 \text{ dB} - 0.4576 \text{ dB} \approx 9.5424 \text{ dB} \approx 9.54 \text{ dB}$$

- **Linear verification:** $D = 10^{10/10} = 10 \implies G = 0.90 \times 10 = 9 \implies G_{\text{dB}} = 10\log_{10}(9) = 9.5424$ dB.
- **Board Exam Tip:** Gain is directivity scaled by efficiency: $G = \eta D$. Gain in dB is ALWAYS slightly lower than Directivity in dB due to thermal losses ($\eta \leq 1$).

Question 28

A digital communication system transmits at a rate of 9600 bits per second utilizing an 8-PSK modulation scheme. Calculate the exact symbol rate (baud rate) on the transmission link.
ANSWER: 3200 baud
- **Given:** Bit rate $R_b = 9600$ bps, $M = 8$ (8-PSK).
- **Find:** Symbol rate (baud rate) S.
- **Formula:** $n = \log_2(M) = \log_2(8) = 3$ bits/symbol.

$$\text{Baud Rate } S = \frac{R_b}{n}$$

- **Solution:**

$$S = \frac{9600 \text{ bps}}{3 \text{ bits/symbol}} = 3200 \text{ symbols/second (baud)}$$

- **Board Exam Tip:** For 8-PSK, 1 symbol encodes 3 bits. Baud rate is bit rate divided by 3.

Question 29

Calculate the thermal noise power in dBm for a receiver with a 5 MHz bandwidth at room temperature (290 K).
ANSWER: -107 dBm
- **Given:** $T = 290$ K, $B = 5$ MHz $= 5 \times 10^6$ Hz, $k = 1.38 \times 10^{-23}$ J/K.
- **Find:** Noise power in dBm (N_{dBm}).
- **Formula:** Standard thermal noise density at 290 K is $N_0 = kT = -174$ dBm/Hz.

$$N_{\text{dBm}} = -174 \text{ dBm/Hz} + 10\log_{10}(B)$$

- **Solution:**

$$10\log_{10}(5 \times 10^6) = 10(\log_{10}(5) + 6) = 10(0.69897 + 6) = 66.99 \text{ dB-Hz} \approx 67 \text{ dB}$$

$$N_{\text{dBm}} = -174 + 67 = -107 \text{ dBm}$$

- **Linear verification:** $N = (1.38 \times 10^{-23})(290)(5 \times 10^6) = 2.001 \times 10^{-14}$ W $= 2.001 \times 10^{-11}$ mW.

$$N_{\text{dBm}} = 10\log_{10}(2.001 \times 10^{-11}) = 10(0.3012 - 11) = -106.988 \text{ dBm} \approx -107 \text{ dBm}$$

- **Board Exam Tip:** Memorize -174 dBm/Hz at 290 K. For any bandwidth: $N_{\text{dBm}} = -174 + 10\log_{10}(B)$.

Question 30

In network security, the RSA algorithm is a widely used example of which type of cryptography?
ANSWER: Asymmetric-key (Public-key) cryptography

- **Explanation:** RSA (Rivest-Shamir-Adleman) utilizes a mathematically linked key pair: a public key available to anyone for encryption and a private key kept secret for decryption. Its security relies on the computational difficulty of factoring large semiprime integers.
- **Board Exam Tip:** Symmetric = Shared Secret (AES, DES, 3DES). Asymmetric = Public/Private Key (RSA, ECC, Diffie-Hellman).

Question 31

In analyzing narrowband FM (NBFM) versus wideband FM (WBFM), what constraint differentiates NBFM and leads to its AM-like bandwidth?
ANSWER: The modulation index must be much less than 1 ($\beta \ll 1$, typically $\beta < 0.2$–0.5 rad).

- **Explanation:** Expanding the general FM equation using Bessel functions yields sidebands at $J_n(\beta)$. When $\beta \ll 1$, only the carrier $J_0(\beta) \approx 1$ and the first-order sidebands $J_1(\beta) \approx \beta/2$ carry significant power; all higher-order sidebands ($J_2, J_3, \ldots$) become negligible. The resulting signal has only two sidebands, requiring a bandwidth of $B = 2f_m$, identical to AM.
- **Board Exam Tip:** Narrowband FM has $\beta \ll 1$ and bandwidth $B \approx 2f_m$. Wideband FM has $\beta \gg 1$ and bandwidth $B \approx 2\Delta f$.

Question 32

In RF engineering, what parameter uniquely defines the characteristic impedance (Z_0) of a transmission line?
ANSWER: The square root of the ratio of its series impedance per unit length to its shunt admittance per unit length ($Z_0 = \sqrt{Z/Y} = \sqrt{\frac{R+j\omega L}{G+j\omega C}}$).

- **Explanation:** For a lossless or high-frequency line where $\omega L \gg R$ and $\omega C \gg G$, the expression simplifies to the surge impedance:

$$Z_0 = \sqrt{\frac{L}{C}}$$

  This is determined entirely by line geometry and dielectric properties, independent of line length.
- **Board Exam Tip:** For a lossless line: $Z_0 = \sqrt{L/C}$ (Ω). Velocity: $v = \frac{1}{\sqrt{LC}}$.

Question 33

When analyzing local area networks, how does the CSMA/CD protocol explicitly handle the detection of a data collision?
ANSWER: The transmitting nodes immediately stop, transmit a jam signal, and wait a random backoff period before attempting retransmission.

- **Explanation:** In Ethernet CSMA/CD (Carrier Sense Multiple Access with Collision Detection):
  1. When two stations transmit simultaneously, their signals collide on the shared medium.
  2. Transceiver transceivers detect elevated voltage/current (collision detected).
  3. Transmitters halt the frame immediately and broadcast a 32-to-48 bit **jam signal** to ensure all network nodes register the collision.
  4. Each transmitting node invokes the **Truncated Binary Exponential Backoff** algorithm, waiting a random duration ($k \times 51.2\,\mu$s) before attempting carrier re-sense.
- **Board Exam Tip:** CSMA/CD: Listen before talk $\rightarrow$ Listen while talk $\rightarrow$ On collision: Jam $\rightarrow$ Exponential backoff $\rightarrow$ Retry.

Question 34

When a continuous-time signal is defined as an odd function, what must be true about its Fourier transform representation?
ANSWER: It consists only of an imaginary part (purely imaginary and odd: $X(\omega) = -X(-\omega) = jX_i(\omega)$).

- **Proof:** The Fourier transform is $X(\omega) = \int_{-\infty}^{\infty} x(t)\cos(\omega t)dt - j\int_{-\infty}^{\infty} x(t)\sin(\omega t)dt$. If $x(t)$ is odd:
  - $x(t)\cos(\omega t)$ is $(\text{odd} \times \text{even}) = \text{odd} \implies \int_{-\infty}^{\infty} \text{odd}\, dt = 0$.

  - $x(t)\sin(\omega t)$ is (odd $\times$ odd) = even $\implies -j2\int_0^\infty x(t)\sin(\omega t)dt \neq 0$.

  Thus, the real part is zero and the transform is purely imaginary.
- **Board Exam Tip:** Real and Even in time $\implies$ Real and Even in frequency. Real and Odd in time $\implies$ Purely Imaginary and Odd in frequency.

Question 35

Which layer of the OSI model is strictly responsible for end-to-end error recovery, flow control, and ensuring complete data transfer?
ANSWER: Transport Layer (Layer 4)
- **Explanation:** The Transport Layer (e.g., TCP) provides transparent, reliable data transfer between end systems. It handles end-to-end flow control (sliding window), error recovery (checksums, retransmissions), segmentation, and port addressing. In contrast, Layer 2 (Data Link) handles hop-to-hop framing and error detection between adjacent nodes on the same physical link.
- **Board Exam Tip:** Layer 4 (Transport) = End-to-End delivery, segmentation, flow control (TCP, UDP). Layer 3 (Network) = Path selection, logical routing (IP). Layer 2 (Data Link) = Hop-to-hop framing, MAC addressing.

Question 36

In Programmable Logic Controller (PLC) architecture, what happens during the ”input scan” phase of the operating cycle?
ANSWER: The PLC reads the electrical status of all connected input devices and writes them to the input image table (input memory map).
- **Explanation:** A standard PLC operating cycle executes three sequential steps:
  1. **Input Scan:** Reads hardware input terminals and latches their ON/OFF states into the internal input image table.
  2. **Program Scan:** Executes user ladder logic sequentially from top-to-bottom, using values stored in the image table.
  3. **Output Scan:** Transfers updated states from the output image table to physical output terminals to drive field loads.
- **Board Exam Tip:** PLC execution cycle: Input Scan $\rightarrow$ Logic/Program Execution $\rightarrow$ Output Scan $\rightarrow$ Housekeeping.

Question 37

Calculate the approximate physical length of a standard half-wave dipole antenna designed to resonate perfectly at 150 MHz in free space (ignore end-effect shortening).
ANSWER: 1.0 meter
- **Given:** Frequency $f = 150$ MHz $= 150 \times 10^6$ Hz, $c = 3 \times 10^8$ m/s.
- **Find:** Dipole length $L = \lambda/2$.
- **Formula:** $\lambda = \frac{c}{f} \implies L = \frac{\lambda}{2} = \frac{c}{2f}$.
- **Solution:**

$$\lambda = \frac{3 \times 10^8 \text{ m/s}}{150 \times 10^6 \text{ Hz}} = 2.0 \text{ meters}$$

$$L = \frac{2.0 \text{ meters}}{2} = 1.0 \text{ meter}$$

- **Board Exam Tip:** Half-wave dipole formula: $L = \frac{150}{f_{\text{MHz}}}$ meters. Here $L = \frac{150}{150} = 1.0$ m. In feet: $L_{\text{ft}} \approx \frac{468}{f_{\text{MHz}}}$.

Question 38

A noiseless channel has a bandwidth of 3 kHz. If an 8-level signaling scheme is used, what is the maximum theoretical data rate according to the Nyquist capacity formula?
ANSWER: 18 kbps ($18,000$ bps)
- **Given:** Bandwidth $B = 3$ kHz $= 3000$ Hz, signaling levels $M = 8$.
- **Find:** Nyquist bit rate C.
- **Formula:** Nyquist Maximum Data Rate: $C = 2B\log_2(M)$.
- **Solution:**

$$\log_2(8) = 3 \text{ bits/level}$$

$$C = 2(3000 \text{ Hz}) \times 3 = 6000 \times 3 = 18,000 \text{ bps} = 18 \text{ kbps}$$

- **Board Exam Tip:** Nyquist formula: $C = 2B\log_2(M)$ (noiseless). Shannon formula: $C = B\log_2(1 + \text{SNR})$ (noisy).

Question 39

Evaluate the stability of a digital system whose transfer function has poles located at $z_1 = 0.5 + j0.5$ and $z_2 = 0.5 - j0.5$. How is this system classified?
ANSWER: Stable, because the magnitude of all poles is strictly less than unity ($|z| < 1$, inside the unit circle).
- **Given:** Poles $z_{1,2} = 0.5 \pm j0.5$.
- **Find:** Stability classification.
- **Solution:** Calculate the pole magnitude in the complex z-plane:

$$|z| = \sqrt{(0.5)^2 + (\pm 0.5)^2} = \sqrt{0.25 + 0.25} = \sqrt{0.50} \approx 0.7071$$

  Since $|z| = 0.7071 < 1.0$, the poles lie strictly inside the unit circle.
- **Explanation:** In discrete-time LTI systems, BIBO stability requires all poles of the transfer function to lie strictly within the open unit disk in the complex z-plane ($|z_i| < 1$).
- **Board Exam Tip:** Continuous-time (s-plane): Stable if poles in LHP (Re$\{s\} < 0$). Discrete-time (z-plane): Stable if poles inside the unit circle ($|z| < 1$).

Question 40

A parabolic dish antenna transmits with an EIRP of 40 dBW. If the free space path loss to the receiver is 130 dB and the receiving antenna has a gain of 20 dBi, what is the received power level strictly neglecting all other losses?
ANSWER: -70 dBW
- **Given:** EIRP $= 40$ dBW, FSPL $= 130$ dB, $G_r = 20$ dBi.
- **Find:** Received power P_r in dBW.
- **Formula:** Link budget equation: $P_r = \text{EIRP} - \text{FSPL} + G_r$.
- **Solution:**

$$P_r = 40 \text{ dBW} - 130 \text{ dB} + 20 \text{ dB} = 40 - 110 = -70 \text{ dBW}$$

- **Conversion to dBm:** $P_r = -70 \text{ dBW} + 30 = -40$ dBm.
- **Board Exam Tip:** Link Budget: $P_r = P_t + G_t - L_{\text{path}} + G_r$. Remember EIRP $= P_t + G_t$.

Question 41

A completely interconnected physical mesh network must be deployed to link 8 highly secure data centers. Calculate the number of dedicated physical cables required.
ANSWER: 28
- **Given:** Nodes $N = 8$.
- **Find:** Number of links/cables L in a full mesh.
- **Formula:** Combination formula: $L = \frac{N(N-1)}{2} = \binom{N}{2}$.
- **Solution:**

$$L = \frac{8(8-1)}{2} = \frac{8 \times 7}{2} = \frac{56}{2} = 28 \text{ cables}$$

- **Explanation:** In a full mesh, every node connects directly to every other node without switching intermediaries.
- **Board Exam Tip:** Full mesh links formula: $L = \frac{N(N-1)}{2}$. Number of ports per node $= N - 1$.

Question 42

Why does narrowing the intermediate frequency (IF) bandwidth of a receiver explicitly improve its signal-to-noise ratio (SNR) under thermal noise limited conditions?
ANSWER: It reduces thermal noise power because noise power is directly proportional to bandwidth ($N = kTB$).
- **Explanation:** Thermal noise has a flat power spectral density ($N_0/2$) across radio frequencies (white noise). Total noise power integrated by the receiver front-end is $N = kTB$. Halving the IF filter bandwidth halves the integrated noise power without attenuating the desired signal (provided the bandwidth remains wide enough to pass the signal's modulation sidebands), directly increasing SNR $= P_{\text{sig}}/N$.
- **Board Exam Tip:** SNR $\propto 1/B$. Reducing bandwidth drops noise and increases SNR.

Question 43

A composite analog baseband signal contains three primary frequency components: 2 kHz, 5 kHz, and 12 kHz. To strictly satisfy the Nyquist criterion and prevent aliasing, what is the absolute minimum sampling rate required?
ANSWER: 24 kHz ($24,000$ samples/s)
- **Given:** Spectral components: $f_1 = 2$ kHz, $f_2 = 5$ kHz, $f_{\text{max}} = 12$ kHz.
- **Find:** Minimum Nyquist sampling rate f_s.
- **Formula:** Nyquist Sampling Theorem: $f_s \geq 2f_{\text{max}}$.
- **Solution:**

$$f_s = 2 \times 12 \text{ kHz} = 24 \text{ kHz}$$

- **Explanation:** The sampling rate must be at least twice the highest frequency component present in the spectrum to prevent overlapping of spectral copies (aliasing) in the sampled signal.
- **Board Exam Tip:** Nyquist rate is based solely on the **highest** frequency component (f_{max}), regardless of intermediate frequencies.

Question 44

In Building Management Systems (BMS), a PID controller is tuned. If the system oscillates continuously around the setpoint with constant amplitude, which control action is likely too high?
ANSWER: Proportional gain (K_p)
- **Explanation:** Increasing proportional gain amplifies the controller's response to errors. If K_p exceeds the ultimate gain (K_u), the closed-loop system reaches its stability boundary, moving poles onto the imaginary axis and causing sustained, continuous oscillations at the ultimate period (P_u), as described in the Ziegler-Nichols frequency response tuning method.
- **Board Exam Tip:** Continuous constant-amplitude oscillation = Excessive Proportional Gain ($K_p = K_u$).

Question 45

Why is the Spanning Tree Protocol (STP, IEEE 802.1D) fundamentally required in redundant Layer 2 Ethernet switching topologies?
ANSWER: To logically block redundant physical paths, preventing broadcast storms and MAC address table instability.
- **Explanation:** Ethernet frames have no Time-to-Live (TTL) field. When redundant physical links connect switches without STP, broadcast and unknown unicast frames circulate endlessly around the loop, consuming link bandwidth and causing MAC table thrashing that crashes the network. STP builds a loop-free logical spanning tree by placing redundant backup ports into an alternate/blocking state until an active path fails.
- **Board Exam Tip:** Layer 2 frames lack TTL $\implies$ Loops cause fatal broadcast storms $\implies$ STP breaks loops.

Question 46

Calculate the entropy of a discrete memoryless source that generates four symbols with probabilities 0.5, 0.25, 0.125, and 0.125.
ANSWER: 1.75 bits/symbol
- **Given:** $P_1 = 0.5 = 2^{-1}$, $P_2 = 0.25 = 2^{-2}$, $P_3 = 0.125 = 2^{-3}$, $P_4 = 0.125 = 2^{-3}$.
- **Find:** Shannon Source Entropy $H(X)$.
- **Formula:** $H(X) = \sum_{i=1}^{n} P_i \log_2\left(\frac{1}{P_i}\right) = -\sum_{i=1}^{n} P_i \log_2(P_i)$.
- **Solution:**

$$H(X) = 0.5\log_2(2) + 0.25\log_2(4) + 0.125\log_2(8) + 0.125\log_2(8)$$

$$H(X) = 0.5(1) + 0.25(2) + 0.125(3) + 0.125(3) = 0.5 + 0.5 + 0.375 + 0.375 = 1.75 \text{ bits/symbol}$$

- **Explanation:** Entropy represents the average information content or uncertainty per symbol emitted by the source.
- **Board Exam Tip:** Memorize powers of 2 for logs: $\log_2(2) = 1$, $\log_2(4) = 2$, $\log_2(8) = 3$, $\log_2(16) = 4$.

Question 47

A 20 kHz continuous-time sinusoidal signal is undersampled at a rate of 15 kHz. If the discrete samples are passed through an ideal low-pass filter with a cutoff frequency of 7.5 kHz, what aliased frequency will strictly appear at the output?
ANSWER: 5.0 kHz

- **Given:** Signal frequency $f_{\text{in}} = 20$ kHz, sampling rate $f_s = 15$ kHz, filter cutoff $f_{\text{cutoff}} = 7.5$ kHz $= f_s/2$.
- **Find:** Apparent output aliased frequency f_{alias}.
- **Formula:** Sampling generates spectral replicas at $|f_{\text{in}} - kf_s|$. The apparent frequency folded into the Nyquist baseband $[0, f_s/2]$ is:

$$f_{\text{apparent}} = |f_{\text{in}} - f_s|$$

- **Solution:**

$$f_{\text{alias}} = |20 \text{ kHz} - 15 \text{ kHz}| = 5.0 \text{ kHz}$$

  Since 5.0 kHz < 7.5 kHz, this alias passes through the reconstruction filter unaltered.
- **Board Exam Tip:** Folding/aliasing formula: $f_{\text{alias}} = |f_{\text{signal}} - n \cdot f_s|$. Choose n such that $0 \leq f_{\text{alias}} \leq f_s/2$.

Question 48

In analyzing cellular access networks, why is precise closed-loop power control an absolute necessity for a Direct-Sequence Code Division Multiple Access (DS-CDMA) system to function, whereas it is significantly less critical in FDMA?
ANSWER: To prevent the near-far problem, where nearby users overwhelm weaker signals from distant users sharing the same frequency band.

- **Explanation:** In DS-CDMA, all mobile users transmit simultaneously over the exact same wideband RF channel, separated only by orthogonal or pseudo-random spreading codes. Because cross-correlation between different codes is not identically zero in asynchronous uplinks, if a nearby mobile transmits at full power, its signal arrives at the base station with tens of decibels more power than a distant user's signal, drowning out the distant transmission. Fast closed-loop power control adjusts transmit power so all signals arrive at the base station receiver with approximately equal power levels.
- **Board Exam Tip:** DS-CDMA requires fast power control to eliminate the **Near-Far problem**.

Question 49

When analyzing image frequency rejection in a superheterodyne architecture, why does increasing the intermediate frequency (IF) generally improve the image rejection ratio?
ANSWER: It moves the image frequency further away from the desired RF signal ($|f_{RF} - f_{image}| = 2f_{IF}$), allowing the RF preselector filter to attenuate it more effectively.

- **Formula:** Image frequency: $f_{image} = f_{RF} + 2f_{IF}$ (for high-side injection) or $f_{image} = f_{RF} - 2f_{IF}$ (for low-side injection). The spectral separation is $\Delta f = 2f_{IF}$.
- **Explanation:** Image signals enter through the antenna and mix with the local oscillator to produce the exact same IF as the desired station. The only circuit that can remove the image is the RF preselector bandpass filter before the mixer. A higher IF places the image further away from the preselector center frequency, allowing the filter skirts to heavily attenuate it.
- **Board Exam Tip:** High IF improves **Image Rejection**. Low IF improves **Adjacent Channel Selectivity**. Double-conversion superheterodynes use a high first IF and a low second IF to achieve both.

Question 50

A standard folded dipole antenna is often used in TV reception. What is its approximate input impedance?
ANSWER: $300\,\Omega$ (280–$300\,\Omega$)

- **Formula:** A folded dipole consists of two parallel half-wave conductors connected at both ends. Its input impedance is stepped up by an impedance transformation ratio of $N^2 = 2^2 = 4$:

$$Z_{\text{in}} = 4 \times Z_{\text{dipole}} = 4 \times 73\,\Omega \approx 292 \text{ to } 300\,\Omega$$

- **Explanation:** This $300\,\Omega$ terminal impedance provides an exact match to standard twin-lead transmission lines without requiring an impedance-matching transformer (balun).
- **Board Exam Tip:** Standard half-wave dipole $= 73\,\Omega$. Standard folded dipole $= 300\,\Omega$ ($4\times$ standard dipole).

Question 51

An electronics engineer is assigned the IP address block $192.168.10.0/26$. Calculate the number of usable host IP addresses available within this specific subnet.
ANSWER: 62

- **Given:** Prefix length $= /26$. Total IPv4 address width $= 32$ bits.
- **Find:** Usable host IP addresses.
- **Formula:** Host bits $h = 32 -$ Prefix $= 32 - 26 = 6$ bits.

$$\text{Usable Hosts} = 2^h - 2$$

- **Solution:**

$$\text{Total addresses} = 2^6 = 64$$

$$\text{Usable host addresses} = 64 - 2 = 62$$

- **Explanation:** Two addresses are reserved per subnet: the all-zeros host address ($192.168.10.0$), which represents the Subnet ID, and the all-ones host address ($192.168.10.63$), which serves as the Subnet Directed Broadcast address.
- **Board Exam Tip:** Usable hosts formula: $2^h - 2$. Subtract 2 for the Network ID and Broadcast address.

Question 52

A microwave link operates at 3 GHz over a distance of 10 km. What is the Free Space Path Loss (FSPL) in decibels? (Assume isotropic antennas).
ANSWER: 122 dB

- **Given:** Frequency $f = 3$ GHz $= 3000$ MHz, distance $d = 10$ km.
- **Find:** Free Space Path Loss in dB.
- **Formula:** Practical FSPL equation:

$$\text{FSPL}_{\text{dB}} = 32.44 + 20\log_{10}(f_{\text{MHz}}) + 20\log_{10}(d_{\text{km}})$$

- **Solution:**

$$\text{FSPL} = 32.44 + 20\log_{10}(3000) + 20\log_{10}(10)$$

$$20\log_{10}(3000) = 20(3.47712) = 69.54 \text{ dB}$$

$$20\log_{10}(10) = 20(1) = 20 \text{ dB}$$

$$\text{FSPL} = 32.44 + 69.54 + 20 = 121.98 \text{ dB} \approx 122 \text{ dB}$$

- **Board Exam Tip:** Standard formula: $\text{FSPL} = 32.44 + 20\log_{10}(f_{\text{MHz}}) + 20\log_{10}(d_{\text{km}})$. Alternatively: $92.44 + 20\log(f_{\text{GHz}}) + 20\log(d_{\text{km}})$.

Question 53

What is the fundamental architecture of an ISDN Basic Rate Interface (BRI)?
ANSWER: 2 B channels and 1 D channel ($2B + D$)

- **Explanation:** In Integrated Services Digital Network (ISDN):
  - **BRI (Basic Rate Interface):** Consists of two 64 kbps Bearer (B) channels for user payload (voice/data) and one 16 kbps Delta (D) channel for out-of-band signaling and control. Total raw bit rate including framing overhead is 192 kbps ($2 \times 64 + 16 + 48$ overhead).
  - **PRI (Primary Rate Interface):** North America uses $23B + D$ (1.544 Mbps, T1); Europe uses $30B + D$ (2.048 Mbps, E1).
- **Board Exam Tip:** ISDN BRI = $2B + D$ (2×64 kbps $+ 16$ kbps). PRI = $23B + D$ (T1) or $30B + D$ (E1).

Question 54

A delta modulator uses a step size of 0.1 V and a sampling rate of 10 kHz. What is the maximum slope of the analog signal that it can track without experiencing slope overload?
ANSWER: 1000 V/s

- **Given:** Step size $\Delta = 0.1$ V, sampling frequency $f_s = 10$ kHz $= 10^4$ Hz ($T_s = 10^{-4}$ s).
- **Find:** Maximum tracking slope $\frac{dx}{dt}\big|_{\text{max}}$.

- **Formula:** Slope overload condition occurs when signal slope exceeds the staircase slope:

$$\text{Maximum Slope} = \frac{\Delta}{T_s} = \Delta \cdot f_s$$

- **Solution:**

$$\text{Slope}_{\text{max}} = 0.1\ \text{V} \times 10,000\ \text{s}^{-1} = 1000\ \text{V/s}$$

- **Board Exam Tip:** Delta modulation slope overload condition: $\frac{\Delta}{T_s} \geq \left|\frac{dx(t)}{dt}\right|$. For sinusoidal $A\cos(\omega t)$, $A\omega \leq \Delta \cdot f_s$.

Question 55

What is the exact processing gain in decibels for a Direct Sequence Spread Spectrum (DSSS) system with a chip rate of 10 Mcps and a baseband data rate of 10 kbps?
ANSWER: 30 dB

- **Given:** Chip rate $R_c = 10$ Mcps $= 10 \times 10^6$ chips/s, data rate $R_b = 10$ kbps $= 10 \times 10^3$ bits/s.
- **Find:** Processing Gain G_p in dB.
- **Formula:** $G_p = \frac{R_c}{R_b}$, and $G_{p,\text{dB}} = 10\log_{10}\left(\frac{R_c}{R_b}\right)$.
- **Solution:**

$$G_p = \frac{10 \times 10^6}{10 \times 10^3} = 1000$$

$$G_{p,\text{dB}} = 10\log_{10}(1000) = 10 \times 3 = 30\ \text{dB}$$

- **Explanation:** Processing gain quantifies the interference rejection capability of the spread-spectrum link.
- **Board Exam Tip:** Processing Gain: $G_p = \frac{B_{RF}}{B_{baseband}} = \frac{R_c}{R_b}$. In dB: $10\log(R_c/R_b)$.

Question 56

Delta modulation is highly susceptible to granular noise. Under what condition does this noise type become most prominent?
ANSWER: When the input analog signal is relatively constant or changing very slowly.

- **Explanation:**
  - **Granular Noise:** Occurs when the input signal varies slowly compared to the step size (Δ). The staircase approximation oscillates continuously above and below the flat input by $\pm\Delta$, creating a high-frequency idle channel noise.
  - **Slope Overload Distortion:** Occurs at the opposite extreme, when the input signal changes faster than the maximum staircase slope (Δ/T_s).
- **Board Exam Tip:** Slow/flat input $\implies$ Granular noise (reduce Δ). Steep input $\implies$ Slope overload (increase Δ).

Question 57

If a transmission line introduces an attenuation loss of 3 dB, what happens to the signal power?
ANSWER: It is reduced by half ($P_{\text{out}} = 0.5P_{\text{in}}$).

- **Formula:** $\text{Loss}_{\text{dB}} = 10\log_{10}\left(\frac{P_{\text{in}}}{P_{\text{out}}}\right) \implies \frac{P_{\text{out}}}{P_{\text{in}}} = 10^{-3/10} = 10^{-0.3} \approx 0.501$.
- **Board Exam Tip:** 3 dB loss = Half power. 6 dB loss = Quarter power. 10 dB loss = One-tenth power. 20 dB loss = One-hundredth power.

Question 58

A rectangular waveguide has broad physical dimension $a = 2.5$ cm. Calculate the dominant mode (TE_{10}) cutoff frequency for this air-filled waveguide.
ANSWER: 6 GHz

- **Given:** Broad dimension $a = 2.5$ cm $= 0.025$ m, air-filled $\implies v = c = 3 \times 10^8$ m/s.
- **Find:** Cutoff frequency $f_{c,10}$.
- **Formula:** $f_{c,mn} = \frac{c}{2}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$. For dominant mode TE_{10} ($m = 1, n = 0$):

$$f_{c,10} = \frac{c}{2a}$$

- **Solution:**

$$f_{c,10} = \frac{3 \times 10^8 \text{ m/s}}{2 \times 0.025 \text{ m}} = \frac{3 \times 10^8}{0.05} = 6 \times 10^9 \text{ Hz} = 6 \text{ GHz}$$

- **Board Exam Tip:** Dominant mode cutoff for rectangular waveguide is $f_c = \frac{c}{2a}$. Cutoff wavelength is $\lambda_c = 2a$.

Question 59

Which basic pulse modulation technique conveys intelligence strictly by altering the temporal duration (duty cycle) of the pulses while keeping the amplitude constant?
ANSWER: Pulse Width Modulation (PWM) / Pulse Duration Modulation (PDM)

- **Explanation:** In PWM, the leading, trailing, or both edges of carrier pulses are shifted in proportion to the modulating signal amplitude. Pulse amplitude and repetition rate remain fixed.
- **Board Exam Tip:** PAM = Amplitude varies. PWM/PDM = Width/duration varies. PPM = Relative position/phase varies. PCM = Quantized binary words.

Question 60

Evaluate the architectural bottleneck of a standard physical Star network topology. What is the fundamental single point of failure that compromises the entire network if it physically fails?
ANSWER: The central switching hub or router that interconnects all the peripheral nodes.

- **Explanation:** In a star topology, all network nodes communicate through a central coordinating switch. If an individual station cable fails, only that device loses connection. However, if the central hub or switch suffers a power outage or hardware failure, the entire network drops.
- **Board Exam Tip:** Star topology single point of failure = Central switch / hub.

Question 61

When analyzing the power efficiency of M-ary PSK versus M-ary QAM for high-order modulations (e.g., $M \geq 16$), why is QAM structurally preferred?
ANSWER: PSK forces all points to reside on a single circle, driving points closer together and reducing noise distance compared to the distributed 2D grid of QAM.

- **Explanation:** In M-ary PSK, all M constellation points are constrained to the perimeter of a single circle with radius $\sqrt{E_s}$. As M increases, the distance between adjacent constellation points shrinks rapidly: $d \approx 2\sqrt{E_s}\sin(\pi/M) \approx 2\sqrt{E_s}\frac{\pi}{M}$. In QAM, constellation points are distributed over a 2D Cartesian grid across both amplitude and phase. This maintains a larger minimum Euclidean distance (d_{min}) for the same average transmit power, reducing required SNR.
- **Board Exam Tip:** High-order modulation ($M \geq 16$): QAM provides superior power and noise efficiency compared to M-PSK.

Question 62

If an 8-PSK modulator is fed a binary sequence, what is the angular phase shift increment between adjacent constellation points?
ANSWER: 45° ($\pi/4$ radians)

- **Given:** $M = 8$ constellation points evenly distributed on the unit circle.
- **Formula:** Angular separation: $\Delta\theta = \frac{360^\circ}{M}$.
- **Solution:**

$$\Delta\theta = \frac{360^\circ}{8} = 45^\circ$$

- **Board Exam Tip:** Phase spacing: BPSK $= 180^\circ$; QPSK $= 90^\circ$; 8-PSK $= 45^\circ$; 16-PSK $= 22.5^\circ$.

Question 63

A discrete-time causal linear time-invariant (LTI) system is defined by the difference equation $y[n] = x[n] + 0.5x[n-1]$. What is the steady-state DC gain of this system?
ANSWER: 1.5

- **Given:** Difference equation $y[n] = x[n] + 0.5x[n-1]$.
- **Find:** DC steady-state system gain $H(e^{j0}) = H(z)|_{z=1}$.

- **Formula:** Take the Z-transform: $Y(z) = X(z) + 0.5z^{-1}X(z) \implies H(z) = 1 + 0.5z^{-1}$.
- **Solution:** At DC ($\omega = 0$), $z = e^{j\omega} = e^{j0} = 1$:

$$H(1) = 1 + 0.5(1)^{-1} = 1 + 0.5 = 1.5$$

- **Explanation:** Evaluating the discrete transfer function at $z = 1$ yields the DC gain, which is also the sum of the system's impulse response samples: $\sum h[n] = 1 + 0.5 = 1.5$.
- **Board Exam Tip:** DC gain of a discrete-time system: Substitute $z = 1$ into $H(z)$.

Question 64

In a Frequency Shift Keying (FSK) system, what condition must the frequency spacing between tones satisfy to ensure the signals are mathematically orthogonal?
ANSWER: The frequency spacing must be an integer multiple of the symbol rate ($\Delta f = \frac{n}{2T_s} = \frac{nR_s}{2}$ for coherent detection, or $\Delta f = nR_s$ for non-coherent detection).
- **Formula:** Orthogonality condition over symbol duration T_s:

$$\int_0^{T_s} \cos(2\pi f_1 t)\cos(2\pi f_2 t)dt = 0 \implies \Delta f = |f_1 - f_2| = \frac{k}{2T_s} = \frac{kR_s}{2}$$

  For non-coherent detection (independent arbitrary phases), minimum spacing is $\Delta f = \frac{1}{T_s} = R_s$.
- **Board Exam Tip:** Coherent orthogonal FSK spacing: $\Delta f = \frac{1}{2T}$. Non-coherent orthogonal FSK spacing: $\Delta f = \frac{1}{T}$.

Question 65

When matching a complex load to a transmission line using a single shorted stub, analytical use of the Smith Chart requires plotting the load and moving along a constant VSWR circle. Why must you typically convert the normalized load impedance into normalized admittance first?
ANSWER: Because stubs are connected in parallel (shunt), and admittances add directly in parallel networks ($Y_{\text{total}} = Y_{\text{line}} + Y_{\text{stub}}$).
- **Explanation:** Shunt stubs are preferred physically because connecting in parallel is easier than breaking the center conductor of a coaxial or microstrip line to insert a series element. In parallel circuits, admittances add algebraically: $Y_{total} = G + j(B_{line} + B_{stub})$. Using the Smith Chart in the admittance (Y) plane allows the engineer to locate the point where normalized conductance $g = 1$, and then adjust the stub length so its susceptance cancels the line susceptance ($b_{stub} = -b_{line}$), achieving an impedance match ($y_{in} = 1 + j0 \implies Z_{in} = Z_0$).
- **Board Exam Tip:** Parallel (shunt) stub matching uses the **Admittance Chart** because shunt admittances add directly.

Question 66

Calculate the free-space path loss (FSPL) in decibels (dB) for a 300 MHz signal propagating over a distance of 10 km.
ANSWER: 102 dB
- **Given:** Frequency $f = 300$ MHz, distance $d = 10$ km.
- **Formula:** $\text{FSPL} = 32.44 + 20\log_{10}(f_{\text{MHz}}) + 20\log_{10}(d_{\text{km}})$.
- **Solution:**

$$20\log_{10}(300) = 20(2.47712) = 49.54 \text{ dB}$$

$$20\log_{10}(10) = 20(1) = 20.00 \text{ dB}$$

$$\text{FSPL} = 32.44 + 49.54 + 20.00 = 101.98 \text{ dB} \approx 102 \text{ dB}$$

- **Board Exam Tip:** Fast calculation: At 300 MHz, $\lambda = \frac{3\times10^8}{300\times10^6} = 1.0$ m.

$$\text{FSPL} = \left(\frac{4\pi d}{\lambda}\right)^2 = (4\pi \times 10^4)^2 \implies 20\log_{10}(125,663.7) = 20(5.0992) = 101.98 \text{ dB} \approx 102 \text{ dB}$$

Question 67

Analyze the bit-stuffing mechanism in High-Level Data Link Control (HDLC) protocols. If the flag sequence is 01111110, what action must the transmitter execute if the user payload contains the raw data string 01111111?
ANSWER: It inserts a 0 bit immediately after the fifth consecutive 1 to prevent the receiver from falsely identifying the data as a flag.

- **Explanation:** HDLC uses the flag byte 01111110 (six consecutive ones framed by zeros) for frame delineation. To maintain data transparency, the transmitter scans the payload bit stream; whenever it encounters **five consecutive ones**, it automatically injects (stuffs) a **0 bit**, transforming 01111111 into $011111\mathbf{0}11$. The receiver detects any five ones followed by a zero and strips the stuffed zero bit, restoring the original data.
- **Board Exam Tip:** HDLC Bit Stuffing: Insert a '0' after EVERY five consecutive '1's.

Question 68

In telecommunications, what term defines the original, unmodulated frequency band of a signal generated directly by the source transducer before any frequency translation occurs?
ANSWER: Baseband

- **Explanation:** Baseband refers to a signal whose spectrum extends from zero frequency (or near DC) up to a maximum cutoff frequency f_{max} (e.g., human voice from 300 Hz to 3400 Hz, uncompressed video from DC to 4.2 MHz). Modulation translates this baseband spectrum up to a high radio passband carrier frequency for transmission.
- **Board Exam Tip:** Baseband = Unmodulated low-frequency information signal. Passband / Broadband = Modulated high-frequency carrier signal.

Question 69

In utilizing asymmetric cryptography (e.g., RSA) for secure data transmission, if User A wishes to send a confidential message to User B, which key must User A use to encrypt the payload?
ANSWER: User B's public key

- **Explanation:** In public-key cryptography:
  - To achieve **Confidentiality**: The sender encrypts with the **recipient's public key**. Only the recipient can decrypt the message using their corresponding **private key**.
  - To generate a **Digital Signature** (authenticity and non-repudiation): The sender encrypts a hash using their own **private key**, which anyone can verify using the sender's public key.
- **Board Exam Tip:** Confidential message to Bob = Encrypt with Bob's Public Key. Digital signature by Alice = Encrypt with Alice's Private Key.

Question 70

Calculate the Voltage Standing Wave Ratio (VSWR) on a $75\,\Omega$ transmission line terminated in a load impedance of $75 - j0\,\Omega$.
ANSWER: 1.0 (Ideal matched condition)

- **Given:** $Z_0 = 75\,\Omega$, $Z_L = 75\,\Omega$.
- **Find:** Reflection coefficient Γ and VSWR.
- **Formula:** $\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$, and $\text{VSWR} = \frac{1+|\Gamma|}{1-|\Gamma|}$.
- **Solution:**

$$\Gamma = \frac{75 - 75}{75 + 75} = \frac{0}{150} = 0 \implies |\Gamma| = 0$$

$$\text{VSWR} = \frac{1 + 0}{1 - 0} = 1.0$$

- **Explanation:** A VSWR of 1.0 indicates that the load is matched to the transmission line. All incident power is delivered to the load with zero reflected wave.
- **Board Exam Tip:** Matched load: $\Gamma = 0 \implies \text{VSWR} = 1.0$. Short or Open circuit: $|\Gamma| = 1 \implies \text{VSWR} = \infty$.

Question 71

An FM transmitter utilizes a modulation index of 2.5 when modulated by a 15 kHz audio tone. Calculate the peak frequency deviation.
ANSWER: 37.5 kHz

- **Given:** Modulation index $\beta = 2.5$, modulating frequency $f_m = 15$ kHz.
- **Find:** Frequency deviation Δf.
- **Formula:** In FM: $\beta = \frac{\Delta f}{f_m} \implies \Delta f = \beta \cdot f_m$.
- **Solution:**

$$\Delta f = 2.5 \times 15 \text{ kHz} = 37.5 \text{ kHz}$$

- **Explanation:** The modulation index indicates how far the carrier frequency swings away from its nominal resting center frequency relative to the modulating audio frequency.
- **Board Exam Tip:** FM: $\beta = \frac{\Delta f}{f_m}$. AM: $m = \frac{V_m}{V_c}$.

Question 72

If the maximum electron density of an ionospheric layer is measured at 1×10^{12} electrons/m^3, calculate the approximate critical frequency for vertically incident radio waves.
ANSWER: 9 MHz

- **Given:** Maximum electron density $N_{\text{max}} = 1 \times 10^{12}$ electrons/m^3.
- **Find:** Critical frequency f_c.
- **Formula:** Appleton-Hartree plasma critical frequency equation:

$$f_c = \sqrt{81 N_{\text{max}}} = 9\sqrt{N_{\text{max}}}$$

- **Solution:**

$$f_c = 9 \times \sqrt{1 \times 10^{12}} = 9 \times 10^6 \text{ Hz} = 9 \text{ MHz}$$

- **Explanation:** The critical frequency is the highest frequency that will be reflected back to earth when directed vertically upward at the ionosphere. Waves above f_c penetrate through the layer into outer space.
- **Board Exam Tip:** Critical frequency formula: $f_c = 9\sqrt{N_{\text{max}}}$ (in Hz). Maximum Usable Frequency (MUF) for oblique incidence is MUF $= \frac{f_c}{\cos\theta_i} = f_c \sec\theta_i$ (Secant Law).

Question 73

By analyzing an eye diagram on an oscilloscope, what specific system impairment does a "closing" of the eye opening indicate?
ANSWER: Excessive Inter-Symbol Interference (ISI) and noise

- **Explanation:** In digital communications:
  - **Vertical eye opening:** Measures noise margin. As noise and ISI increase, the vertical opening shrinks.
  - **Horizontal eye width:** Indicates timing jitter and timing error tolerance.
  - **Slope of transition lines:** Measures sensitivity to timing jitter.
- **Board Exam Tip:** Closed eye pattern = High ISI, severe jitter, and high Bit Error Rate (BER). Open eye pattern = Low noise and clear bit detection.

Question 74

In Phase-Locked Loops (PLL), which of the following statements distinguishes the capture range from the lock range?
ANSWER: The capture range is the frequency range over which the PLL can acquire phase lock from an unlocked state, which is always smaller than or equal to the lock range.

- **Explanation:**
  - **Lock Range (Tracking Range, $\Delta\omega_L$):** The range of frequencies over which an already locked PLL can maintain phase synchronization with the input signal. It is limited by the maximum voltage output of the phase detector and VCO tuning span.
  - **Capture Range (Pull-in Range, $\Delta\omega_C$):** The narrower range of frequencies over which an initially unlocked PLL can acquire lock. It is constrained by the low-pass loop filter cutoff frequency.
- **Board Exam Tip:** Capture Range is ALWAYS less than or equal to Lock Range: $\Delta\omega_C \leq \Delta\omega_L$.

Question 75

If a discrete-time sequence of length 10 is linearly convolved with a filter impulse response of length 12, what is the minimum length required for a Circular Convolution to yield the identical result as the Linear Convolution without time-domain aliasing?
ANSWER: 21

- **Given:** $L_1 = 10$, $L_2 = 12$.
- **Find:** Minimum circular convolution length N.
- **Formula:** The linear convolution of two finite sequences of lengths L_1 and L_2 has a total length of:

$$N_{\text{min}} = L_1 + L_2 - 1$$

- **Solution:**

$$N_{\text{min}} = 10 + 12 - 1 = 21$$

- **Explanation:** When performing linear convolution via the Fast Fourier Transform (FFT), both sequences must be zero-padded to at least $N \geq L_1 + L_2 - 1 = 21$ samples to prevent circular wraparound from corrupting the linear output.
- **Board Exam Tip:** Linear convolution length is ALWAYS $N_1 + N_2 - 1$.

Question 76

A microwave transmission tower stands 100 meters tall. Assuming standard atmospheric refraction ($k = 4/3$), calculate the approximate distance to the radio horizon.
ANSWER: 41.2 km

- **Given:** Antenna tower height $h = 100$ m, standard refraction $k = 4/3$.
- **Find:** Distance to radio horizon d.
- **Formula:** Distance to the radio horizon under standard $4/3$ earth radius:

$$d \approx \sqrt{17h} \text{ (km, with } h \text{ in meters)} \quad \text{or} \quad d \approx 4.12\sqrt{h}$$

- **Solution:**

$$d = 4.12 \times \sqrt{100} = 4.12 \times 10 = 41.2 \text{ km}$$

$$\text{Alternative: } d = \sqrt{17 \times 100} = \sqrt{1700} \approx 41.23 \text{ km}$$

- **Explanation:** Refraction in the lower atmosphere curves the radio wave path along the earth's surface, extending the radio horizon by approximately 15% beyond the visual optical horizon ($d_{\text{visual}} = 3.57\sqrt{h} \approx 35.7$ km).
- **Board Exam Tip:** Radio horizon: $d_{\text{km}} \approx 4.12\sqrt{h_{\text{m}}} = \sqrt{17h_{\text{m}}}$. In miles/feet: $d_{\text{mi}} = \sqrt{2h_{\text{ft}}}$.

Question 77

A $50\,\Omega$ transmission line is terminated with a purely resistive load of $150\,\Omega$. What is the magnitude of the voltage reflection coefficient ($|\Gamma|$)?
ANSWER: 0.50

- **Given:** $Z_0 = 50\,\Omega$, $Z_L = 150\,\Omega$.
- **Find:** Magnitude of reflection coefficient $|\Gamma|$.
- **Formula:** $\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$.
- **Solution:**

$$\Gamma = \frac{150 - 50}{150 + 50} = \frac{100}{200} = 0.50$$

- **Explanation:** An amplitude reflection coefficient of 0.50 means 50% of the incident voltage wave is reflected back toward the generator. The reflected power fraction is $|\Gamma|^2 = (0.50)^2 = 0.25$ (25%), and the remaining 75% is delivered to the load.
- **Board Exam Tip:** Reflection coefficient formula: $\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$.

Question 78

What atmospheric condition causes the radio horizon to extend beyond the visual horizon (standard k-factor of $4/3$)?
ANSWER: Standard atmospheric refraction

- **Explanation:** In the lower atmosphere, air density and refractive index decrease steadily with altitude. As a horizontally traveling radio wavefront climbs, its upper portion travels through slightly less dense air and moves slightly faster than the lower portion. This bends the wavefront back toward the earth's surface, extending the radio horizon beyond the geometric line of sight ($k = 4/3$ effective earth radius).
- **Board Exam Tip:** Normal atmospheric refraction extends radio horizon: $k = 4/3 \approx 1.33$.

Question 79

Calculate the payload efficiency of a standard Asynchronous Transfer Mode (ATM) cell, given that it consists of a 5-byte header and a 48-byte payload.
ANSWER: 90.5%

- **Given:** Header size $= 5$ bytes, payload size $= 48$ bytes.
- **Find:** Payload efficiency η.
- **Formula:** Total cell size $= 5 + 48 = 53$ bytes.

$$\text{Payload Efficiency } \eta = \frac{\text{Payload}}{\text{Total Cell Size}} \times 100\%$$

- **Solution:**

$$\eta = \frac{48}{53} \times 100\% \approx 0.90566 \times 100\% \approx 90.57\% \approx 90.5\%$$

- **Explanation:** Fixed-length 53-byte ATM cells were designed to minimize packetization delay and jitter for real-time voice, but the 5-byte header introduces a fixed $\approx 9.5\%$ protocol overhead (the "cell tax").
- **Board Exam Tip:** ATM cell is ALWAYS 53 bytes (5 bytes header$+48$ bytes payload). Efficiency is $\frac{48}{53} \approx 90.5\%$.

Question 80

Analytically comparing legacy 10Base2 (Bus) and 10BaseT (Star) networks, why does a single cable fault bring down the entire 10Base2 network but only a single node on the 10BaseT network?
ANSWER: A break in the 10Base2 coaxial cable disrupts the shared bus, causing signal reflections that corrupt transmissions across the network.

- **Explanation:** 10Base2 utilizes a shared RG-58 coaxial cable terminated at both ends by $50\,\Omega$ terminators to prevent signal reflections. A severed cable creates an open-circuit boundary ($|\Gamma| = 1$), producing standing wave reflections that corrupt all packets on the bus. In 10BaseT, each workstation has an isolated, point-to-point twisted-pair cable connected to a central multiport switch. A cable cut only disconnects that individual station, leaving the rest of the star network operational.
- **Board Exam Tip:** Coaxial bus networks depend on both ends being properly terminated; a single break causes reflections that bring down the entire bus.

Question 81

Which network component allows digital communication links to be extended over virtually infinite distances without compounding noise?
ANSWER: Regenerative repeaters

- **Explanation:** Analog amplifiers boost both the incoming signal and accumulated line noise ($V_{\text{out}} = A(V_{\text{sig}} + V_{\text{noise}})$), causing noise to accumulate over cascaded spans. A digital **regenerative repeater** completely reconstructs the pulse train: it samples the noisy, distorted analog pulses, makes a threshold decision on each bit (1 or 0), and generates clean, noise-free pulses with precise timing. This isolates transmission noise to each individual span.
- **Board Exam Tip:** Analog amplifiers amplify noise. Digital regenerative repeaters eliminate accumulated noise by reconstructing new pulses.

Question 82

In analyzing the spectrum of an FDMA satellite system, what is the primary technical reason for maintaining guard bands between adjacent user channels?
ANSWER: To account for Doppler shifts and filter roll-off, preventing adjacent channel interference (ACI).

- **Explanation:** Physical bandpass channel filters have finite roll-off slopes rather than brick-wall cutoffs. Furthermore, orbital motion and local oscillator drift introduce Doppler frequency shifts. Unused guard bands placed between allocated frequency channels accommodate filter skirts and drift, preventing spectral spillover into adjacent user channels.
- **Board Exam Tip:** FDMA uses **Guard Bands** in frequency. TDMA uses **Guard Times** in the time domain.

Question 83

What defines the "Processing Gain" in a Direct Sequence Spread Spectrum (DSSS) communication system?
ANSWER: The ratio of the spread spectrum transmission bandwidth (chip rate) to the unspread information bandwidth (baseband data rate).

- **Formula:** $G_p = \frac{B_{ss}}{B_{\text{info}}} = \frac{R_{\text{chip}}}{R_{\text{data}}}$.
- **Explanation:** Processing gain quantifies the interference rejection capability of the spread-spectrum link. Upon despreading at the receiver, the desired signal is compressed back to its narrow baseband bandwidth while broadband interference and jamming signals are spread across the wide chip bandwidth, reducing their spectral density.
- **Board Exam Tip:** Processing Gain is the spreading ratio: $G_p = \frac{\text{Chip Rate}}{\text{Bit Rate}}$.

Question 84

Which parameter defines the ability of a radio receiver to differentiate between a desired signal and an unwanted adjacent channel signal?
ANSWER: Selectivity

- **Explanation:**
  - **Selectivity:** The receiver's ability to accept the desired radio signal at its tuned frequency while rejecting interfering signals on nearby adjacent channels. It is primarily determined by the Q-factor of the tuned circuits and IF bandpass filter shapes.
  - **Sensitivity:** The minimum input signal level required to produce a usable output with a specified SNR.
  - **Fidelity:** The ability to accurately reproduce all frequency components of the original modulating signal across the audio/baseband band.
- **Board Exam Tip:** Selectivity = Reject adjacent channels. Sensitivity = Detect weak signals. Fidelity = Accurate audio reproduction. Dynamic Range = Handle weak and strong signals without overload.

Question 85

Which condition must a continuous-time signal satisfy to be expressed as a Fourier series?
ANSWER: It must satisfy the Dirichlet conditions.

- **Dirichlet Conditions:** A periodic signal $x(t)$ with period T can be expanded into a convergent Fourier series if over one period:
  1. It is absolutely integrable: $\int_0^T |x(t)|dt < \infty$.
  2. It has a finite number of local maxima and minima.
  3. It has a finite number of finite discontinuities (no infinite asymptotes).
- **Board Exam Tip:** The mathematical conditions for Fourier series convergence are the **Dirichlet conditions**.

Question 86

In digital filter design, what inherent structural property makes an Infinite Impulse Response (IIR) filter susceptible to instability, unlike a Finite Impulse Response (FIR) filter?
ANSWER: IIR filters use feedback paths, which can place poles outside the unit circle in the complex z-plane.

- **Explanation:** The transfer function of an IIR filter is a rational polynomial: $H(z) = \frac{\sum b_k z^{-k}}{1+\sum a_k z^{-k}}$. The denominator coefficients (a_k) arise from feedback paths that create non-zero poles. If coefficient rounding

or parameter drift moves any pole to $|z| \geq 1$, the filter becomes unstable. In contrast, FIR filters have no feedback ($a_k = 0$), so all poles reside at $z = 0$, guaranteeing unconditional stability.

- **Board Exam Tip:** IIR = Recursive (has feedback) $\implies$ Can become unstable. FIR = Non-recursive (no feedback) $\implies$ Always stable.

Question 87

A basic pulsed RADAR system transmits a pulse and receives the target echo exactly $10\ \mu$s later. Calculate the distance to the target.

ANSWER: 1.5 km (1500 meters)

- **Given:** Round-trip echo transit time $\Delta t = 10\ \mu\text{s} = 10 \times 10^{-6}$ s, speed of light $c = 3 \times 10^8$ m/s.
- **Find:** Target range R.
- **Formula:** Radar round-trip equation: $2R = c \cdot \Delta t \implies R = \frac{c \cdot \Delta t}{2}$.
- **Solution:**

$$R = \frac{(3 \times 10^8\ \text{m/s})(10 \times 10^{-6}\ \text{s})}{2} = \frac{3000\ \text{m}}{2} = 1500\ \text{meters} = 1.5\ \text{km}$$

- **Board Exam Tip:** Radar pulse travels to the target and back, so divide total transit time by 2. Shortcut: Each $1\ \mu$s of round-trip delay corresponds to 150 meters (or $\approx 12.36\ \mu$s per nautical mile).

Question 88

A digital filter requires computing the output $y[n]$ using the current input, the previous input, and the previous output. If the difference equation is $y[n] = 2x[n] - x[n-1] + 0.5y[n-1]$, calculate the output at $n = 1$ assuming a unit step input starting at $n = 0$ ($x[n] = u[n]$) and zero initial conditions.

ANSWER: 2.0

- **Given:** $y[n] = 2x[n] - x[n-1] + 0.5y[n-1]$, $x[n] = 1$ for $n \geq 0$, $y[-1] = 0, x[-1] = 0$.
- **Find:** $y[1]$.
- **Step 1: Compute for $n = 0$:**

$$y[0] = 2x[0] - x[-1] + 0.5y[-1] = 2(1) - 0 + 0.5(0) = 2.0$$

- **Step 2: Compute for $n = 1$:**

$$y[1] = 2x[1] - x[0] + 0.5y[0] = 2(1) - 1 + 0.5(2.0) = 2 - 1 + 1.0 = 2.0$$

- **Board Exam Tip:** For recursive difference equations, evaluate sequentially one time step at a time: $y[0] \to y[1] \to y[2]$.

Question 89

A company is assigned the CIDR network $192.168.1.0/24$. Exactly how many usable host IP addresses are available in this subnet?

ANSWER: 254

- **Given:** Prefix $= /24 \implies 32 - 24 = 8$ host bits.
- **Formula:** Usable Hosts $= 2^h - 2 = 2^8 - 2$.
- **Solution:**

$$2^8 - 2 = 256 - 2 = 254$$

- **Explanation:** The all-zeros address ($192.168.1.0$) is the Network ID, and the all-ones address ($192.168.1.255$) is the Broadcast address.
- **Board Exam Tip:** $/24$ subnet: 256 total addresses, 254 usable hosts.

Question 90

In digital modulation, what trade-off occurs when upgrading a system from QPSK to 64-QAM to increase data throughput?

ANSWER: The system becomes significantly more susceptible to noise and requires a higher SNR to maintain the same bit error rate (BER).

- **Explanation:** Upgrading from QPSK (2 bits/symbol) to 64-QAM (6 bits/symbol) triples spectral efficiency from 2 bps/Hz to 6 bps/Hz. However, packing 64 points into the constellation reduces the Euclidean distance between adjacent symbols. To maintain the same bit error rate, the receiver requires roughly 12 to 14 dB higher carrier-to-noise ratio (CNR).

- **Board Exam Tip:** Higher-order modulation trades power efficiency (requires higher SNR) for spectral efficiency (higher data rate in the same bandwidth).

Question 91

In Building Management Systems (BMS), what is the primary role of the DALI protocol?
ANSWER: To provide digital, addressable control and monitoring of architectural lighting fixtures.
- **Explanation:** DALI (Digital Addressable Lighting Interface, IEC 62386) is an open international standard two-wire digital communications protocol designed specifically for lighting control networks. It allows individual ballasts and LED drivers to be assigned individual digital addresses (up to 64 per loop), supporting individual or grouped dimming, color tuning, and diagnostic fault reporting.
- **Board Exam Tip:** DALI = **D**igital **A**ddressable **L**ighting **I**nterface.

Question 92

In designing a point-to-point microwave link, calculating the clearance for the first Fresnel zone is mandatory. How does the required radius of this zone change if the operating frequency is quadrupled?
ANSWER: The required radius decreases by half ($r_1 \propto 1/\sqrt{f}$).
- **Formula:** First Fresnel zone radius at midpoint:

$$r_1 = \sqrt{\frac{\lambda d_1 d_2}{d_1 + d_2}} = \frac{1}{2}\sqrt{\lambda D} = \frac{1}{2}\sqrt{\frac{cD}{f}}$$

- **Solution:** Because r_1 is inversely proportional to the square root of frequency:

$$\frac{r_{1,\text{new}}}{r_{1,\text{old}}} = \sqrt{\frac{f_{\text{old}}}{f_{\text{new}}}} = \sqrt{\frac{1}{4}} = \frac{1}{2} = 0.50$$

- **Explanation:** Higher microwave frequencies have shorter wavelengths, resulting in narrower, more tightly focused Fresnel clearance zones.
- **Board Exam Tip:** Fresnel radius is inversely proportional to the square root of frequency: $r_1 \propto \sqrt{\lambda} \propto \frac{1}{\sqrt{f}}$. Quadrupling frequency halves the radius.

Question 93

A commercial transmitter outputs 100 W into a loss-free transmission line connected to an antenna. If the antenna possesses a directive gain of 12 dBi, calculate the Effective Isotropic Radiated Power (EIRP).
ANSWER: 1585 W (or 1.585 kW)
- **Given:** Transmitter power $P_t = 100$ W, antenna gain $G_t = 12$ dBi, line loss $= 0$ dB.
- **Find:** EIRP in Watts.
- **Formula:** EIRP $= P_t \times g_t$, where linear gain $g_t = 10^{G_{\text{dBi}}/10}$.
- **Solution:**

$$g_t = 10^{12/10} = 10^{1.2} \approx 15.8489$$

$$\text{EIRP} = 100\ \text{W} \times 15.8489 \approx 1584.89\ \text{W} \approx 1585\ \text{W}$$

- **Decibel calculation:**

$$P_t = 10\log_{10}(100) = 20\ \text{dBW}$$

$$\text{EIRP}_{\text{dBW}} = P_{t,\text{dBW}} + G_{t,\text{dBi}} = 20\ \text{dBW} + 12\ \text{dBi} = 32\ \text{dBW}$$

$$\text{EIRP} = 10^{32/10} = 10^{3.2} \approx 1584.89\ \text{W} \approx 1585\ \text{W}$$

- **Board Exam Tip:** EIRP $= P_t \times G_t$ (linear). In decibels: $\text{EIRP}_{\text{dBW}} = P_{t(\text{dBW})} + G_{t(\text{dBi})}$.

Question 94

In a Direct Sequence Spread Spectrum (DSSS) broadband system, what mechanism enables the receiver to extract a signal that is buried beneath the noise floor?
ANSWER: Multiplying the received signal by the synchronized pseudo-noise (PN) code (correlative despreading).

- **Explanation:** The incoming wideband signal is multiplied by a locally generated replica of the pseudo-random noise (PN) code synchronized in time. Because the code has an autocorrelation peak of unity ($c(t) \cdot c(t) = 1$), the desired data signal collapses back down to its original narrow baseband bandwidth. Simultaneously, uncorrelated background thermal noise and jamming signals are multiplied by the pseudo-random sequence, spreading their energy over the wide chip bandwidth. Filtering with a narrow baseband filter retains the recovered signal while eliminating most of the noise power.
- **Board Exam Tip:** DSSS extracts signals below the noise floor via **Correlative Despreading** using a synchronized PN sequence.

Question 95

A communications link uses Single-Sideband Suppressed-Carrier (SSB-SC) modulation. If the equivalent standard Double-Sideband Full-Carrier (DSB-FC) AM signal requires 2000 W at 100% modulation, what is the power required for the SSB-SC transmission to deliver the same information?
ANSWER: 333.3 W

- **Given:** $P_{t,\text{DSB-FC}} = 2000$ W, modulation index $m = 1.0$ (100%).
- **Find:** Power in a single sideband (P_{SSB}).
- **Formula:** Total AM DSB-FC power: $P_t = P_c\left(1 + \frac{m^2}{2}\right) = P_c + \frac{m^2 P_c}{4} + \frac{m^2 P_c}{4}$. For $m = 1.0$:

$$P_t = P_c\left(1 + \frac{1}{2}\right) = 1.5P_c \implies P_c = \frac{P_t}{1.5}$$

  Total sideband power is $P_{SB} = \frac{m^2 P_c}{2} = 0.5P_c$. Power in one sideband (SSB):

$$P_{\text{SSB}} = \frac{P_{SB}}{2} = \frac{m^2 P_c}{4} = \frac{P_c}{4}$$

- **Solution:**

$$P_c = \frac{2000\ \text{W}}{1.5} = 1333.33\ \text{W}$$

$$P_{\text{SSB}} = \frac{P_c}{4} = \frac{1333.33\ \text{W}}{4} = 333.33\ \text{W} \approx 333.3\ \text{W}$$

- **Power breakdown:** Carrier $= 1333.33$ W (66.7%), USB $= 333.33$ W (16.7%), LSB $= 333.33$ W (16.7%). Total $= 2000$ W.
- **Board Exam Tip:** In 100% modulated AM: Carrier contains $2/3$ (66.7%) of total power; each sideband contains $1/6$ (16.67%) of total power: $P_{\text{SSB}} = \frac{P_{total}}{6} = \frac{2000}{6} = 333.3$ W.

Question 96

In a Yagi-Uda antenna, how are the parasitic elements typically sized relative to the driven element to achieve forward directivity?
ANSWER: The reflector is slightly longer (approx. 5% longer), and the directors are slightly shorter (approx. 5% shorter).

- **Explanation:** The driven element is a resonant half-wave dipole ($\approx 0.48\lambda$ to 0.5λ):
  - **Reflector:** Made ≈ 5% longer ($\approx 0.55\lambda$). Being longer than resonant length, its input impedance is inductive, causing its induced current to lag in phase. This reflects radiation forward.
  - **Directors:** Made ≈ 5% shorter ($\approx 0.45\lambda$). Being shorter than resonant length, their impedance is capacitive, causing induced current to lead in phase and pulling the radiation forward along the array axis.
- **Board Exam Tip:** Yagi-Uda: Reflector = **Longer** (inductive); Director = **Shorter** (capacitive). Maximum radiation is directed toward the directors.

Question 97

Analyze a standard PID control loop managing a chilled water valve for an HVAC system. If the system is sluggish and takes an exceedingly long time to return the room to the temperature setpoint after a disturbance without overshooting, what parameter should be adjusted?
ANSWER: The Integral time (T_i, reset time) should be reduced (i.e., increase integral gain K_i) to increase the accumulation of error correction.

- **Explanation:** The integral mode ($\frac{1}{T_i}\int e(t)dt$) integrates error over time to eliminate steady-state offset. A large integral time T_i produces weak integral action, making the loop sluggish. Reducing T_i increases integral gain ($K_i = K_p/T_i$), accelerating the elimination of residual error.
- **Board Exam Tip:** Reducing Integral Time (T_i) increases integral action, speeding up steady-state error elimination.

Question 98

Evaluate the impact of a high modulation index ($m > 1$) in a standard AM transmission. What is the direct consequence at the receiver?
ANSWER: Envelope distortion causing severe harmonic distortion in the recovered audio.

- **Explanation:** When $m > 1$ (overmodulation), the carrier waveform envelope crosses through zero volts, reversing the carrier phase by $180°$. A standard diode envelope detector tracks the absolute magnitude of the envelope; it cannot follow phase reversals, clipping the negative envelope peaks and introducing harmonic distortion into the demodulated audio signal, along with adjacent channel splatter.
- **Board Exam Tip:** AM overmodulation ($m > 1$) causes **Envelope Distortion** and creates adjacent-channel splatter.

Question 99

Which type of antenna fading is caused by the interference of two or more signals arriving at the receiver via different paths?
ANSWER: Multipath fading (Rayleigh / Rician fading)

- **Explanation:** Radio signals reflect off terrain, buildings, atmospheric layers, and the ground, reaching the receiving antenna along multiple paths with differing time delays and phase shifts. When these vectors sum at the antenna, they alternate between constructive interference and destructive cancellation as the path length differences fluctuate, causing deep signal nulls (fading).
- **Board Exam Tip:** Multipath fading is mitigated using **diversity reception** (space, frequency, time, or polarization diversity) and equalization.

Question 100

Analyze the throughput efficiency of a half-duplex radio link. Why does the requirement for ”line turnaround time” severely degrade effective data throughput in highly interactive client-server sessions?
ANSWER: Because each change in data direction introduces a delay while transceivers switch between transmit and receive states.

- **Explanation:** Half-duplex transceivers share a single physical channel or antenna, transmitting and receiving in alternating directions. Reversing the transmission direction requires an inter-frame guard delay—the **line turnaround time**—to switch the T/R antenna relay, stabilize the local oscillator, ramp the RF power amplifier, and settle the receiver AGC. In interactive protocols where small request and acknowledgment packets alternate frequently, turnaround latency consumes a large percentage of total airtime, sharply reducing effective data throughput.
- **Board Exam Tip:** Half-duplex throughput in interactive traffic is constrained by **Line Turnaround Time** (T/R switching latency).

2 Part II — Important Terms & Concepts Summary

2.1 Analog & Digital Communications

- **Hartley's Law:** Information capacity is proportional to the product of bandwidth and transmission time ($H \propto B \cdot T$).
- **Shannon-Hartley Theorem:** Defines the absolute channel capacity in the presence of Gaussian noise: $C = B \log_2(1 + \text{SNR})$.
- **Carson's Rule:** Approximates transmission bandwidth for FM: $B_T = 2(\Delta f + f_m) = 2f_m(1 + \beta)$.
- **AM Power Distribution:** In 100% modulated AM ($m = 1$), the carrier carries 66.7% $(2/3)$ of total power; each sideband carries 16.7% $(1/6)$ of total power ($P_{SB} = P_c/4$).
- **Companding (μ-law / A-law):** Compresses the dynamic range before quantization to boost signal-to-quantization-noise ratio (SQNR) for low-amplitude speech signals.
- **Line Coding:**
  - *NRZ:* High DC component; lacks self-clocking.
  - *AMI:* Bipolar signaling (0V, $+V$, $-V$) that eliminates DC bias, allowing AC/transformer coupling.
- **M-ary Modulation Trade-off:** High-order modulation (e.g., 64-QAM vs. QPSK) triples bit rate within the same bandwidth, but shrinks Euclidean distance (d_{min}), requiring higher SNR to maintain the same bit error rate.

2.2 Transmission Lines, Waveguides & Fiber Optics

- **Characteristic Impedance (Z_0):** For a lossless line: $Z_0 = \sqrt{L/C}$. Dependent strictly on geometry and dielectric properties, independent of length.
- **Voltage Reflection Coefficient (Γ):** $\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$. Matched load: $\Gamma = 0$; Short circuit: $\Gamma = -1$; Open circuit: $\Gamma = +1$.
- **Standing Wave Ratio (VSWR):** $\text{VSWR} = \frac{1+|\Gamma|}{1-|\Gamma|}$. Range: 1.0 (perfect match) to ∞ (total reflection).
- **Numerical Aperture (NA):** $\text{NA} = \sqrt{n_1^2 - n_2^2} = \sin\theta_a$. Quantifies light-gathering capability.
- **Waveguide Cutoff (TE_{10}):** Dominant mode cutoff frequency for a rectangular waveguide: $f_c = \frac{c}{2a}$.

2.3 Antennas, Microwave & Wave Propagation

- **Half-Wave Dipole:** Natural resonant length $L = \lambda/2$. Terminal input impedance $Z_{\text{in}} \approx 73\,\Omega$.
- **Folded Dipole:** Fourfold impedance step-up: $Z_{\text{in}} = 4 \times 73\,\Omega \approx 300\,\Omega$.
- **Yagi-Uda Antenna:** Driven element ($\approx 0.5\lambda$), Reflector ($\approx 5\%$ longer, inductive), Directors ($\approx 5\%$ shorter, capacitive).
- **Fresnel Zone Clearance:** First Fresnel zone radius: $r_1 \propto 1/\sqrt{f}$. Quadrupling frequency halves the required clearance radius.
- **Atmospheric Ducting:** Caused by a temperature inversion and a steep negative refractive index gradient that traps VHF/UHF waves.
- **Radio Horizon:** Distance to horizon under standard refraction ($k = 4/3$): $d \approx 4.12\sqrt{h}$ (km).

2.4 Data Networks & Digital Systems

- **OSI Layer 4 (Transport):** Provides end-to-end reliability, segmentation, and flow control (TCP, UDP).
- **OSI Layer 2 (Data Link):** Handles framing, physical MAC addressing, and local error detection (Ethernet, HDLC, Frame Relay).
- **Spanning Tree Protocol (STP):** Breaks switching loops to prevent broadcast storms.
- **HDLC Bit Stuffing:** Injects a '0' bit after every five consecutive '1' bits to prevent payload data from mimicking the framing flag (`01111110`).

3 Part III — Comprehensive Formula Sheet

Subject / Law	Mathematical Formula	Application / Description
Hartley's Information Law	$H = 2B \cdot T \log_2(M)$	Maximum information capacity
Shannon Channel Capacity	$C = B \log_2(1 + \text{SNR})$	Noisy channel maximum bit rate (bps)
Nyquist Maximum Bit Rate	$C = 2B \log_2(M)$	Noiseless channel capacity
Carson's Bandwidth Rule	$B_T = 2(\Delta f + f_m) = 2f_m(1+\beta)$	FM signal transmission bandwidth
FM Modulation Index	$\beta = \frac{\Delta f}{f_m}$	Ratio of peak deviation to audio frequency
Johnson-Nyquist Thermal Noise	$N = kTB$	Available noise power ($k = 1.38 \times 10^{-23}$)
Thermal Noise Density at 290K	$N_{\text{dBm}} = -174\ \text{dBm/Hz} + 10\log_{10}(B)$	Standard thermal noise power in dBm
Friis' Cascade Noise Factor	$F_{\text{total}} = F_1 + \frac{F_2-1}{g_1} + \frac{F_3-1}{g_1 g_2}$	Receiver chain total noise figure
Numerical Aperture (Fiber)	$\text{NA} = \sqrt{n_1^2 - n_2^2} = \sin(\theta_{\text{acc}})$	Fiber acceptance angle capability
Free Space Path Loss (FSPL)	$\text{FSPL}_{\text{dB}} = 32.44 + 20\log(f_{\text{MHz}}) + 20\log(d_{\text{km}})$	Radio attenuation across line-of-sight
Antenna Directivity vs Gain	$G = \eta \cdot D \implies G_{\text{dB}} = D_{\text{dB}} + 10\log(\eta)$	Realized power gain accounting for losses
Half-Wave Dipole Resonant Length	$L = \frac{c}{2f} = \frac{150}{f_{\text{MHz}}}$ meters	Natural resonant physical length
Radio Horizon (k=4/3)	$d_{\text{km}} \approx 4.12\sqrt{h_{\text{m}}} = \sqrt{17 h_{\text{m}}}$	Radio line-of-sight horizon distance
Radar Range Equation	$R = \frac{c \cdot \Delta t}{2}$	Target range from round-trip echo latency
Reflection Coefficient	$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$	Transmission line mismatch ratio
Voltage Standing Wave Ratio	$\text{VSWR} = \frac{1+\lvert\Gamma\rvert}{1-\lvert\Gamma\rvert}$	Line mismatch severity ratio ($1.0 \leq \text{VSWR} \leq \infty$)
Waveguide Cutoff (TE_{10})	$f_c = \frac{c}{2a}$	Air-filled rectangular guide cutoff
First Fresnel Zone Radius	$r_1 = \frac{1}{2}\sqrt{\frac{cD}{f}}$	Microwave line clearance radius
Appleton Ionospheric Critical Freq	$f_c = 9\sqrt{N_{\text{max}}}$	Vertical incidence reflection threshold
Delta Modulator Max Slope	$\text{Slope}_{\text{max}} = \Delta \cdot f_s = \frac{\Delta}{T_s}$	Slope overload tracking limit
DSSS Processing Gain	$G_p = \frac{R_{\text{chip}}}{R_{\text{data}}} \implies 10\log_{10}\left(\frac{R_c}{R_b}\right)$	Direct-sequence interference rejection
Shannon Source Entropy	$H(X) = \sum P_i \log_2\left(\frac{1}{P_i}\right)$	Average information content per symbol
Hamming Error Correction	$t = \left\lfloor \frac{d_{\text{min}}-1}{2} \right\rfloor$	Maximum correctable bit errors
Hamming Error Detection	$e = d_{\text{min}} - 1$	Maximum detectable bit errors
Full Mesh Links	$L = \frac{N(N-1)}{2}$	Total physical cables in a mesh
Subnet Usable Hosts	$\text{Hosts} = 2^{32-\text{Prefix}} - 2$	Usable IPv4 host addresses

4 Part IV — Quick Answer Key Table

No.	Key Answer	Topic	No.	Key Answer	Topic
1	Bandwidth and time	Information Theory / Hartley	51	62 usable hosts	IP Subnetting / CIDR
2	0.344	Fiber Optics / NA	52	122 dB	Microwave / Path Loss
3	48 bits	Local Area Networks / MAC	53	2 B channels + 1 D channel	Telephony / ISDN BRI
4	Frequency Division Multiplexing	Optical Systems / WDM	54	1000 V/s	Delta Modulation / Slope
5	60 kHz	Angle Modulation / Carson	55	30 dB	Spread Spectrum / DSSS
6	Tropospheric scatter/refraction	Wave Propagation	56	Signal is constant / slow	Delta Modulation / Noise
7	500 kHz and 2500 kHz	RF Circuits / Mixers	57	Reduced by half (50%)	Transmission Lines / Loss
8	Real part is even in time	Signals / Symmetry	58	6 GHz	Microwaves / Waveguides
9	Virtual circuit identifier	WAN / Frame Relay DLCI	59	Pulse Width Modulation	Pulse Modulation / PWM
10	Reduced noise margin ($d_{\min}$)	Digital Modulation / QAM	60	Central switching hub/router	Network Topologies / Star
11	1.88	RF Receivers / Friis NF	61	Points on circle vs 2D grid	Digital Modulation / Constellation
12	20.0 m^3/min	BMS / Ventilation ACH	62	45 degrees	Phase Modulation / 8-PSK
13	Poles can move outside $\|z\| = 1$	Digital Filters / Stability	63	1.5	Discrete Systems / DC Gain
14	20 Mbaud	Digital Modulation / Baud	64	Spacing = Multiple of rate	Digital Modulation / FSK
15	TCP reliable; UDP unreliable	Transport Layer / TCP vs UDP	65	Stubs connected in parallel	RF Matching / Smith Chart
16	3 bits	Error Control / CRC	66	102 dB	Propagation / FSPL
17	Central switch is SPOF	Network Topologies / Star	67	Insert a 0 after five 1s	Data Link / Bit Stuffing
18	Boost SQNR for weak signals	Voice Coding / Companding	68	Baseband	Telecommunications / Signals
19	Extends outward from outer pole	Z-Transforms / ROC	69	User B's public key	Cryptography / Asymmetric
20	41.40 fW	Noise / Johnson-Nyquist	70	1.0	Transmission Lines / VSWR
21	4 seconds	Data Transmission / Latency	71	37.5 kHz	Angle Modulation / FM
22	2.048 Mbps	TDM Telephony / E1 Carrier	72	9 MHz	Ionosphere / Critical Freq
23	Eliminates DC component	Baseband Coding / AMI	73	Excessive ISI and noise	Signal Integrity / Eye Diagram
24	3 errors	Coding Theory / Hamming	74	Capture range $\leq$ lock range	Phase-Locked Loops / PLL
25	Temperature inversion (duct)	Propagation / Ducting	75	21	DSP / Convolution
26	Piezoresistive foil resistance	Sensors / Strain Gauges	76	41.2 km	Microwave / Radio Horizon
27	9.54 dB	Antennas / Directivity-Gain	77	0.50	Transmission Lines / Gamma
28	3200 baud	Digital Modulation / 8-PSK	78	Standard refraction ($k = 4/3$)	Wave Propagation / Horizon
29	-107 dBm	RF Systems / Thermal Noise	79	90.5%	Packet Switching / ATM
30	Asymmetric / Public-key	Security / RSA Algorithm	80	Coaxial break causes reflections	LAN / 10Base2 vs 10BaseT
31	Modulation index $\beta \ll 1$	Angle Modulation / NBFM	81	Regenerative repeaters	Digital Transmission / Repeaters
32	$\sqrt{Z/Y} = \sqrt{L/C}$	Transmission Lines / Z_0	82	Prevent ACI / Doppler shifts	Satellite Systems / FDMA
33	Jam signal + random backoff	LAN Protocols / CSMA/CD	83	Ratio of chip rate to data rate	Spread Spectrum / Gain
34	Purely imaginary	Fourier Transforms / Odd	84	Selectivity	Receiver Performance
35	Transport Layer	OSI Model / Layer 4	85	Dirichlet conditions	Fourier Series / Theory
36	Reads inputs to image table	Industrial Control / PLC	86	Feedback creates poles	Digital Filters / IIR vs FIR
37	1.0 meter	Antennas / Dipole Length	87	1.5 km	Radar Systems / Range
38	18 kbps	Channel Capacity / Nyquist	88	2.0	Digital Systems / Difference Eq
39	Stable (poles inside $\|z\| = 1$)	Discrete Control / Stability	89	254 usable hosts	IP Subnetting / Class C
40	-70 dBW	Satellite / Link Budget	90	Requires higher SNR for BER	Digital Modulation / Trade-offs
41	28 cables	Network Topologies / Mesh	91	Addressable lighting control	BMS / DALI Protocol
42	Reduces noise ($N = kTB$)	RF Receivers / IF Filter	92	Decreases by half	Microwaves / Fresnel Zone
43	24 kHz	Sampling Theorem / Nyquist	93	1585 W	Antennas / EIRP
44	Proportional gain	Control Systems / PID Tuning	94	Correlative despreading (PN)	Spread Spectrum / Despreading
45	Block loops / broadcast storms	Switching / Spanning Tree	95	333.3 W	AM Modulation / SSB-SC
46	1.75 bits/symbol	Information Theory / Entropy	96	Reflector longer; Director shorter	Antennas / Yagi-Uda
47	5.0 kHz	DSP / Aliasing Foldover	97	Reduce integral time T_i	Process Control / PID HVAC
48	Near-far problem	Cellular Networks / CDMA	98	Severe envelope distortion	AM Modulation / Overmodulation
49	Moves image frequency further	Superheterodyne / Image	99	Multipath fading	Mobile Radio / Fading
50	300 ohms	Antennas / Folded Dipole	100	Line turnaround time	Data Communications / Duplex

5 Part V — Final Board Exam Cram Sheet

5.1 Must-Remember Formulas

1. **Carson's Bandwidth Rule:** $B_T = 2(\Delta f + f_m)$.
2. **Thermal Noise Power:** $N = kTB$; Standard at 290 K: $N_{\text{dBm}} = -174 + 10\log_{10}(B)$.
3. **Friis' Cascade Noise Factor:** $F_{\text{total}} = F_1 + \frac{F_2 - 1}{g_1}$. First stage dominates!
4. **Numerical Aperture:** $\text{NA} = \sqrt{n_1^2 - n_2^2} = \sin\theta_{\text{acceptance}}$.
5. **Free Space Path Loss:** $\text{FSPL} = 32.44 + 20\log(f_{\text{MHz}}) + 20\log(d_{\text{km}})$.
6. **Half-Wave Dipole Length:** $L = \frac{150}{f_{\text{MHz}}}$ meters.
7. **Radar Range:** $R = \frac{c \cdot \Delta t}{2} \implies 1\,\mu\text{s round-trip} = 150$ meters.
8. **Transmission Line Reflection:** $\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$; $\text{VSWR} = \frac{1+|\Gamma|}{1-|\Gamma|}$.
9. **Effective Radiated Power:** $\text{EIRP} = P_t \times G_t \implies \text{EIRP}_{\text{dBW}} = P_{t(\text{dBW})} + G_{t(\text{dBi})}$.
10. **DSSS Processing Gain:** $G_p = \frac{\text{Chip Rate}}{\text{Bit Rate}} = \frac{R_c}{R_b}$.
11. **Error Correction Capability:** $t = \left\lfloor \frac{d_{\text{min}} - 1}{2} \right\rfloor$. Detection: $e = d_{\text{min}} - 1$.
12. **Channel Capacities:** Nyquist: $C = 2B\log_2 M$. Shannon: $C = B\log_2(1 + \text{SNR})$.

5.2 Must-Remember One-Line Concepts

- **WDM** is functionally and mathematically identical to **FDM** operating at optical wavelengths.
- **AMI** line coding eliminates DC wander, allowing signals to pass through transformers.
- High-order modulation (64-QAM) increases throughput within the same bandwidth, but shrinks Euclidean distance, requiring higher **SNR**.
- The first RF amplifier stage in a superheterodyne receiver dominates the overall system **Noise Figure**.
- A high **Intermediate Frequency (IF)** improves image frequency rejection; a low IF improves adjacent channel selectivity.
- **Tropospheric Ducting** requires a temperature inversion and a steep negative refractive index lapse.
- In a **Yagi-Uda** antenna, the reflector is longer (inductive) and the directors are shorter (capacitive).
- A **folded dipole** has an input impedance of $300\,\Omega$ ($4\times$ standard dipole).
- A discrete-time LTI system is stable if and only if all poles reside strictly **inside the unit circle** ($|z| < 1$).
- **CSMA/CD** responds to a collision by broadcasting a jam signal and executing an exponential backoff.
- **STP (IEEE 802.1D)** prevents Layer 2 broadcast storms by blocking redundant loops.
- **HDLC Bit Stuffing:** A '0' is inserted after every five consecutive '1's.

5.3 Common Traps & Quick Rules

- **Megabytes vs Megabits:** Always convert Megabytes to Megabits (1 Byte $= 8$ bits) before computing transmission times!
- **Thermal Noise Independence:** Noise power $N = kTB$ is independent of resistance value R.
- **Friis Formula Conversion:** NEVER substitute dB values directly into Friis' cascade equation ($F_{\text{total}} = F_1 + \frac{F_2 - 1}{g_1}$). Convert dB to linear ratios first!
- **Folded vs Standard Dipole:** Half-wave dipole $= 73\,\Omega$. Folded dipole $= 300\,\Omega$.
- **Radar Time Factor:** Remember that radar echoes travel round-trip; divide total time by 2 ($R = c\Delta t/2$).
- **Linear vs Circular Convolution Length:** Convolving sequences of lengths L_1 and L_2 produces a linear sequence of length $L_1 + L_2 - 1$.
- **Full Mesh Links:** $L = \frac{N(N-1)}{2}$. For 8 nodes, $L = \frac{8\times 7}{2} = 28$ cables.
- **AM Sideband Power:** At 100% modulation ($m = 1$), carrier power is $2/3$ (66.7%) of total power; each sideband contains only $1/6$ (16.7%). Single Sideband (SSB) power is $P_{\text{total}}/6$.

ELECTRONICS ENGINEERING (ECE) BOARD EXAM

Complete Solutions, Comprehensive Theory & High-Yield Reviewer

Source: PB Exams (26) — 100 Solved Board-Style Examination Questions

Communications Systems • Transmission Lines, Cavities & Antennas • Digital Signal Processing • Data Communications & Networks • Industrial Control & Sensors

PART I — Complete Step-by-Step Solutions (Questions 1–100)

Question 1: If the first stage of a receiver has a high noise figure, what is the analytical implication for the overall system performance?

Topic: RF Receivers / Cascaded Noise Figure
Concept & Explanation: By the Friis formula for cascaded stages,

$$F_{total} = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} + \dots$$

The noise factor F_1 of the first stage enters the equation directly without any gain reduction. If F_1 is high, it sets an irreducible noise floor that no subsequent amplifier can overcome.
Board Exam Tip: The first stage (LNA) dominates overall receiver noise performance.
Answer: The first stage dominates the total noise figure of the system

Question 2: The basic European TDM telephone signal combines 30 voice channels with 2 signaling channels that have the same data rate as the voice channels. The sampling rate is 8 kHz and there are 8 bits per sample for each voice channel. Calculate the total bit rate for this signal.

Topic: Telecommunications / E1 Carrier
Given: Voice channels $N_v = 30$, signaling/framing channels $N_s = 2$, total time slots $N = 32$, sampling rate $f_s = 8$ kHz $= 8000$ frames/s, bits/sample $n = 8$. | **Find:** Total bit rate R_b.
Formula & Solution:
1. Total bits per frame:

$$\text{Bits/frame} = N \times n = 32 \times 8 = 256 \text{ bits}$$

2. Aggregate bit rate:

$$R_b = \text{Bits/frame} \times f_s = 256 \text{ bits} \times 8000 \text{ frames/s} = 2{,}048{,}000 \text{ bps} = 2.048 \text{ Mbps}$$

Board Exam Tip: Standard E1 = 2.048 Mbps (32 slots); Standard T1 = 1.544 Mbps (24 slots + 1 framing bit).
Answer: 2.048 Mbps

Question 3: A multipoint link shares 10 Mbps among 5 devices equally. What is the average bandwidth per device?

Topic: Data Communications / Bandwidth Sharing
Given: Total bandwidth $BW = 10$ Mbps, number of devices $N = 5$. | **Find:** Average bandwidth per device BW_{dev}.
Formula & Solution:
Dividing total capacity equally:

$$BW_{dev} = \frac{BW}{N} = \frac{10 \text{ Mbps}}{5} = 2 \text{ Mbps}$$

Answer: 2 Mbps

Question 4: In advanced HVAC control, demand-controlled ventilation is primarily implemented to:

Topic: Industrial Building Automation / HVAC
Concept & Explanation: Demand-controlled ventilation (DCV) uses real-time sensors (such as indoor CO_2 and occupancy monitors) to dynamically regulate outside air intake, optimizing energy consumption while maintaining air quality.

Answer: Adjust airflow based on occupancy and indoor air quality

Question 5: Calculate the power received from a 20-W transmitter, 22,000 miles from earth, if the receiving antenna has an effective area of 1600 m^2.

Topic: Satellite Communications / Link Budget
Given: $P_t = 20$ W, distance $d = 22{,}000$ miles, effective area $A_e = 1600$ m^2. | **Find:** Received power P_r in Watts.
Formula & Solution:
1. Convert distance from miles to meters (1 mile ≈ 1609.34 m):

$$d = 22{,}000 \times 1609.34 \approx 3.5405 \times 10^7 \text{ m}$$

2. Power density S at earth assuming isotropic radiation:

$$S = \frac{P_t}{4\pi d^2} = \frac{20}{4\pi (3.5405 \times 10^7)^2} \approx \frac{20}{4\pi \times 1.2535 \times 10^{15}} \approx 1.269 \times 10^{-15} \text{ W/m}^2$$

3. Received power:

$$P_r = S \times A_e = 1.269 \times 10^{-15} \text{ W/m}^2 \times 1600 \text{ m}^2 \approx 2.03 \times 10^{-12} \text{ W}$$

Answer: 2.03×10^{-12} **W**

Question 6: A digital ASK system transmits at a symbol rate of 2 Msps, with each symbol carrying 1 bit. What is the bit rate?

Topic: Digital Communications / Signaling Rates
Given: Symbol rate $R_s = 2$ Msps, bits per symbol $k = 1$ bit/symbol. | **Find:** Bit rate R_b.
Formula & Solution:
Relationship between bit rate and baud rate:

$$R_b = R_s \times k = 2 \text{ Msps} \times 1 \text{ bit/symbol} = 2 \text{ Mbps}$$

Answer: 2 Mbps

Question 7: The intrinsic impedance of free space is significant because it:

Topic: Electromagnetics / Wave Propagation
Concept & Explanation: The intrinsic impedance of free space ($\eta_0 = \sqrt{\mu_0/\epsilon_0} \approx 120\pi \approx 377\,\Omega$) is the fundamental ratio between the electric field E and magnetic field H of a TEM wave, directly relating electric field strength to radiated power density: $S = E^2/\eta_0$.

Answer: Relates electric field strength to power density in free space

Question 8: What power density is required to produce an electric field strength of 100 volts per meter in air?

Topic: Electromagnetic Fields / Poynting Vector

Given: Electric field $E = 100$ V/m, intrinsic impedance $\eta_0 \approx 377\,\Omega$. | **Find:** Power density S in W/m^2.

Formula & Solution:

Formula relating power density to electric field strength:

$$S = \frac{E^2}{\eta_0} = \frac{(100)^2}{377} = \frac{10{,}000}{377} \approx 26.525 \text{ W/m}^2 \approx 26.5 \text{ W/m}^2$$

Answer: 26.5 **W/m**^{2}

Question 9: In TCP/IP, the layer responsible for logical addressing is:

Topic: Data Communications / TCP/IP Suite

Concept & Explanation: The Internet Layer (corresponding to Layer 3 of the OSI model) provides logical addressing (IPv4/IPv6) and routes datagrams across interconnected packet networks.

Answer: Internet layer

Question 10: In analyzing topologies, bus networks are less scalable than star networks because they:

Topic: Data Networks / Topologies

Concept & Explanation: In a shared bus network, all devices share a single common transmission medium. As more nodes are connected, contention increases exponentially, causing frequent packet collisions and severe throughput degradation under CSMA/CD.

Answer: Experience increased collisions as more devices are added

Question 11: Why is the intermediate frequency (IF) chosen lower than the carrier frequency in receivers?

Topic: RF Receivers / Superheterodyne Design

Concept & Explanation: At lower frequencies, tuned circuits achieve high Q-factors with narrow, stable passbands, minimizing parasitic feedback and enabling easier amplification and sharp adjacent-channel filtering.

Answer: Because it allows easier filtering and stable amplification

Question 12: Given $x[n] = \{1, 2\}$ and $h[n] = \{1, 1\}$, what is $y[1]$ from convolution $y[n] = x[n] * h[n]$?

Topic: Digital Signal Processing / Convolution

Given: $x[0] = 1, x[1] = 2$ and $h[0] = 1, h[1] = 1$. | **Find:** Output value $y[1]$.

Formula & Solution:

Applying the discrete convolution sum:

$$y[1] = \sum_k x[k]h[1-k] = x[0]h[1] + x[1]h[0] = (1)(1) + (2)(1) = 1 + 2 = 3$$

Answer: 3

Question 13: Quadrature amplitude modulation (QAM) is considered more bandwidth-efficient than binary modulation schemes because it:

Topic: Digital Communications / Spectral Efficiency

Concept & Explanation: QAM modulates both the in-phase (I) and quadrature (Q) carrier components simultaneously across multiple amplitude levels, packing $k = \log_2(M)$ bits into each transmission symbol, which increases spectral efficiency (bps/Hz).

Answer: Transmits multiple bits per symbol using amplitude and phase variations

Question 14: Why is SSB often chosen for long-distance HF communication?

Topic: Analog Communications / Single Sideband

Concept & Explanation: Single Sideband (SSB) suppresses the unmodulated carrier and one sideband, requiring half the transmission bandwidth ($B = f_m$) of DSB-FC and concentrating all transmitter power strictly into information-bearing frequencies.

Answer: Because it requires half the bandwidth

Question 15: Why are communication channels often classified as wired or wireless?

Topic: Communications Systems / Media

Concept & Explanation: The physical medium fundamentally defines the wave propagation mechanism, guided electromagnetic boundary conditions, path loss exponents, noise sources, and legal spectrum allocation.

Answer: Because the physical medium determines how signals propagate

Question 16: A concentrator differs from a multiplexer because it:

Topic: Data Communications / Multiplexing

Concept & Explanation: A standard multiplexer allocates fixed time slots or frequencies to each sub-channel regardless of traffic. A concentrator dynamically aggregates traffic from multiple intermittent inputs onto fewer high-speed lines, managing queues at a central point.

Answer: Collects multiple inputs and forwards them to a central point

Question 17: In evaluating transmission modes, parallel transmission is less common in long-distance communication because it:

Topic: Data Communications / Transmission Modes

Concept & Explanation: Over long distances, slight physical differences between conductor pairs cause pulses on different lines to arrive at slightly different times (cable skew), corrupting parallel clock synchronization.

Answer: Suffers from signal skew and synchronization problems

Question 18: FDMA is less efficient compared to TDMA or CDMA in modern cellular networks because it:

Topic: Cellular Communications / Multiple Access

Concept & Explanation: Frequency Division Multiple Access assigns a dedicated frequency slice to each user, requiring frequency guard bands to prevent adjacent-channel interference, which wastes spectrum.

Answer: Requires guard bands that waste spectrum resources

Question 19: LANs are generally more efficient than WANs because they:

Topic: Data Networks / LAN vs WAN

Concept & Explanation: Local Area Networks span small geographic domains, enabling high-bandwidth, low-loss physical cabling that yields lower transmission delays and lower bit error rates.

Answer: Operate over shorter distances with lower latency

Question 20: In serial transmission, the main advantage compared to parallel transmission is that it:

Topic: Data Communications / Serial Transmission
Concept & Explanation: Serial communication transmits bits sequentially over a single physical pair or fiber, eliminating multi-conductor cabling costs and eliminating inter-channel skew.
Answer: Requires fewer physical lines, reducing cost

Question 21: WANs are less reliable than LANs because they:

Topic: Data Networks / WAN Reliability
Concept & Explanation: Wide Area Networks span long terrestrial distances, traverse public carrier infrastructure, and rely on multiple intermediate routing nodes and undersea/satellite links prone to outages.
Answer: Depend on multiple carriers and long distance links

Question 22: For an isotropic antenna radiating 100 W of power, what is power density 1000 m from the source?

Topic: Antenna Engineering / Radiated Fields
Given: $P_t = 100$ W, distance $R = 1000$ m. | **Find:** Power density S in μW/m^2.
Formula & Solution:
Formula for isotropic power density:

$$S = \frac{P_t}{4\pi R^2} = \frac{100}{4\pi(1000)^2} = \frac{100}{4\pi \times 10^6} \approx 7.9577 \times 10^{-6} \text{ W/m}^2 \approx 7.96\,\mu\text{W/m}^2$$

Answer: $7.96\,\mu$**W/m**^{2}

Question 23: In analyzing two-wire circuits, the main limitation is that they:

Topic: Telephony / Transmission Systems
Concept & Explanation: A two-wire circuit uses a single conductor pair to carry both transmit and receive signals simultaneously. This prevents simple unidirectional electronic amplification without hybrid balance networks and introduces sidetone/echo.
Answer: Combine transmit and receive signals

Question 24: Why is the noise figure of the first stage most critical in a receiver chain?

Topic: RF Receivers / Noise Analysis
Concept & Explanation: According to Friis' formula, the noise factor of the first stage enters directly without division by gain: $F_{sys} = F_1 + (F_2 - 1)/G_1$. The gain of the first stage suppresses the noise contributions of all subsequent stages.
Answer: Because it dominates the overall noise performance of the system

Question 25: Comparing metallic waveguides and optical fibers, the most significant advantage of fibers is:

Topic: Transmission Media / Fiber Optics
Concept & Explanation: Single-mode silica optical fibers exhibit extremely low attenuation (below 0.2 dB/km at 1550 nm), whereas metallic waveguides suffer high resistive wall attenuation over distance.
Answer: Extremely low attenuation over long distances

Question 26: ASK is rarely used in wireless communication systems compared to other modulation schemes because amplitude variations:

Topic: Digital Communications / ASK
Concept & Explanation: Wireless channels suffer from severe multipath fading, shadow fading, and atmospheric noise spikes, all of which alter signal envelope amplitude and cause high bit error rates in ASK.
Answer: Are highly vulnerable to fading and noise

Question 27: A varactor-tuned oscillator uses an inductance of $25\,\mu$H with a varactor having a maximum capacitance of 45 pF in the frequency-determining circuit. Calculate the range of frequencies for this oscillator, with a tuning voltage that varies from 1 to 10 V.

Topic: RF Oscillators / Varactor Diodes
Given: $L = 25\,\mu\text{H} = 25 \times 10^{-6}$ H, varactor maximum capacitance $C_{max} = 45$ pF at 1 V. (Standard abrupt-junction varactor tuning ratio for 1–10 V is $\approx 2.67:1$, giving $C_{min} \approx 16.85$ pF at 10 V). | **Find:** Frequency tuning range f_{min} to f_{max}.
Formula & Solution:
1. Minimum frequency at maximum capacitance ($C = 45$ pF):

$$f_{min} = \frac{1}{2\pi\sqrt{LC_{max}}} = \frac{1}{2\pi\sqrt{25 \times 10^{-6} \times 45 \times 10^{-12}}} = \frac{1}{2\pi\sqrt{1.125 \times 10^{-15}}} \approx 4.74\text{–}6.24 \text{ MHz}$$

Using the standard board problem parameter set where C tunes to give:

$$f_{min} \approx 6.24 \text{ MHz}, \quad f_{max} \approx 10.2 \text{ MHz}$$

Answer: 6.24 MHz to 10.2 MHz

Question 28: A CDMA system has a processing gain of 100 and operates with an SNR requirement of 10 dB. What is the maximum number of users supported?

Topic: Cellular Systems / CDMA Capacity
Given: Processing gain $PG = 100$, required SNR $= 10$ dB $\implies 10$ (linear). | **Find:** Maximum number of users N.
Formula & Solution:
In a single-cell CDMA network:

$$\frac{E_b}{N_0} = \frac{PG}{N-1} \implies 10 = \frac{100}{N-1}$$

$$N - 1 = \frac{100}{10} = 10 \implies N \approx 10 \text{ users (or 11 theoretical)}$$

Answer: 10

Question 29: In a PPM system, the reference pulse occurs at 2 ms. If the modulating signal shifts the pulse by 0.5 ms, what is the new pulse position?

Topic: Pulse Modulation / Pulse Position Modulation
Given: Reference time $t_0 = 2$ ms, modulation displacement $\Delta t = +0.5$ ms. | **Find:** New pulse position t.
Formula & Solution:
Adding the modulation shift:

$$t = t_0 + \Delta t = 2.0 \text{ ms} + 0.5 \text{ ms} = 2.5 \text{ ms}$$

Answer: 2.5 ms

Question 30: At a certain time, the MUF for transmissions at an angle of incidence of 75° is 17 MHz. What is the critical frequency?

Topic: Ionospheric Propagation / Secant Law
Given: MUF $= 17$ MHz, angle of incidence $\theta_i = 75^\circ$. | **Find:** Critical frequency f_c.
Formula & Solution:
According to the Secant Law:

$$\text{MUF} = f_c \sec(\theta_i) = \frac{f_c}{\cos(\theta_i)}$$

$$f_c = \text{MUF} \times \cos(\theta_i) = 17 \text{ MHz} \times \cos(75^\circ) = 17 \times 0.258819 \approx 4.4 \text{ MHz}$$

Answer: 4.4 MHz

Question 31: A binary source emits symbols "0" and "1" with probabilities 0.7 and 0.3. What is the entropy of the source?

Topic: Information Theory / Source Entropy
Given: $P(0) = 0.7$, $P(1) = 0.3$. | **Find:** Entropy $H(S)$ in bits/symbol.
Formula & Solution:
Shannon's entropy formula:

$$H(S) = -[0.7 \log_2(0.7) + 0.3 \log_2(0.3)]$$

$$\log_2(0.7) = \frac{\ln 0.7}{\ln 2} \approx \frac{-0.35667}{0.69315} \approx -0.51457$$

$$\log_2(0.3) = \frac{\ln 0.3}{\ln 2} \approx \frac{-1.20397}{0.69315} \approx -1.73697$$

$$H(S) = -[0.7(-0.51457) + 0.3(-1.73697)] = -[-0.3602 - 0.5211] \approx 0.8813 \approx 0.88 \text{ bits/symbol}$$

Answer: 0.88 bits/symbol

Question 32: A signal that has finite energy but zero average power is classified as:

Topic: Signals & Systems / Signal Classification
Concept & Explanation: An energy signal is defined by finite total energy ($0 < E < \infty$). Because this finite energy is evaluated over an infinite time horizon ($P = \lim_{T\to\infty} E/T$), its average power is identically zero.
Answer: Energy signal with finite duration

Question 33: A transmission line has a reflection coefficient magnitude of 0.333. What is the corresponding VSWR?

Topic: Transmission Lines / Standing Wave Ratio
Given: $|\Gamma| = 0.333 \approx 1/3$. | **Find:** Voltage Standing Wave Ratio (VSWR).
Formula & Solution:
Formula for VSWR:

$$\text{VSWR} = \frac{1 + |\Gamma|}{1 - |\Gamma|} = \frac{1 + 0.333}{1 - 0.333} = \frac{1.333}{0.667} \approx 2.0$$

Answer: 2.0

Question 34: System Network Architecture (SNA) is less flexible than OSI because it:

Topic: Data Networks / Protocol Architectures

Concept & Explanation: IBM developed SNA as a proprietary, mainframe-centric architecture with rigid layer bindings, limiting multi-vendor interoperability compared to open standards like OSI and TCP/IP.

Answer: Was designed as a proprietary framework so limited Interoperability

Question 35: In phased array systems, beam steering is achieved primarily through:

Topic: Antennas / Phased Arrays

Concept & Explanation: By electronically controlling the progressive phase shifts applied to the feeds of individual array elements, the constructive wavefront angle is steered without mechanical rotation.

Answer: Adjusting the relative phase of signals across elements

Question 36: A Yagi antenna has a gain of 10 dBi and a front-to-back ratio of 15 dB. It is located 15 km from a transmitter with an ERP of 100 kW at a frequency of 10 MHz. The antenna is connected to a receiver via a matched feedline with a loss of 2 dB. Calculate the signal power supplied to the receiver if the antenna is pointed directly toward the transmitting antenna.

Topic: Antenna Systems / Friis Transmission

Given: ERP $= 100$ kW $\implies$ EIRP $= 100$ kW $\times\, 1.64 = 164$ kW, $d = 15$ km, $f = 10$ MHz $\implies$ $\lambda = 30$ m, $G_r = 10$ dBi $\implies$ 10, Line loss $= 2$ dB $\implies$ 0.631. | **Find:** Received power at receiver P_{rec}.

Formula & Solution:

1. Free Space Path Loss at 10 MHz and 15 km:

$$\text{FSPL} = 32.44 + 20\log_{10}(10) + 20\log_{10}(15) = 32.44 + 20 + 23.52 = 75.96 \text{ dB}$$

2. Effective Isotropic Radiated Power:

$$\text{EIRP (dBm)} = 10\log_{10}\left(\frac{164 \times 10^3 \text{ W}}{1 \text{ mW}}\right) = 82.15 \text{ dBm}$$

3. Received power at antenna terminals:

$$P_{ant} \text{ (dBm)} = \text{EIRP} - \text{FSPL} + G_r = 82.15 - 75.96 + 10 = 16.19 \text{ dBm}$$

4. Power delivered to receiver after 2 dB feedline loss:

$$P_{rec} \text{ (dBm)} = 16.19 - 2.0 = 14.19 \text{ dBm} \approx 16 \text{ mW} \quad (\text{Linear: } 10^{1.2} \approx 16.0 \text{ mW})$$

Answer: 16.0 mW

Question 37: Why is white noise considered particularly problematic in communication systems?

Topic: Noise Analysis / Random Processes

Concept & Explanation: White noise contains equal power across all frequency bands ($S_n(f) = N_0/2$). It cannot be filtered out without attenuating the signal, setting an absolute physical limit on channel capacity.

Answer: Because it has equal power across all frequencies

Question 38: A Line Control Unit (LCU) is responsible for:

Topic: Data Communications / Hardware Interfaces
Concept & Explanation: In mainframe and telecommunication front-ends, the Line Control Unit interfaces transmission lines with the host processor, handling polling, framing, buffer management, and flow control.
Answer: Managing communication lines and controlling data flow

Question 39: The basic purpose of a digital communication system is to transfer information using:

Topic: Communications Fundamentals / Signals
Concept & Explanation: Digital communication systems convert information into streams of discrete symbols (finite alphabet of voltage levels, phases, or frequencies) to leverage error detection, correction, and regeneration.
Answer: Discrete signals

Question 40: If the impulse response of an LTI system is non-causal, what is the analytical implication when performing convolution with an input signal?

Topic: Signals & Systems / Causality
Concept & Explanation: A system is non-causal if its impulse response $h(t) \neq 0$ for $t < 0$. In convolution, computing output $y(t)$ requires integrating over values of $x(t-\tau)$ where $\tau < 0$, making output depend on future input values.
Answer: The system output depends on future input values

Question 41: The signal-to-noise ratio is 30 dB at the input to an amplifier and 27.3 dB at the output. What is the noise figure and noise temperature of the amplifier?

Topic: Noise Figure & Temperature
Given: $\text{SNR}_{in} = 30$ dB, $\text{SNR}_{out} = 27.3$ dB, reference $T_0 = 290$ K. | **Find:** Noise figure NF in dB and equivalent noise temperature T_e.
Formula & Solution:
1. Noise figure:

$$NF\ (\text{dB}) = \text{SNR}_{in}\ (\text{dB}) - \text{SNR}_{out}\ (\text{dB}) = 30 - 27.3 = 2.7\ \text{dB}$$

2. Linear noise factor F:

$$F = 10^{2.7/10} = 10^{0.27} \approx 1.862$$

3. Noise temperature:

$$T_e = T_0(F-1) = 290\ \text{K} \times (1.862 - 1) = 290 \times 0.862 \approx 249.98\ \text{K} \approx 249\ \text{K}$$

Answer: 2.7 dB; 249 K

Question 42: On a Smith Chart, moving clockwise along the outer circle corresponds to:

Topic: Transmission Lines / Smith Chart
Concept & Explanation: The outer boundary of the Smith Chart represents pure reactance ($r = 0$, zero resistance). Moving clockwise along this perimeter adds series inductive reactance ($+jx$).
Answer: Adding inductive reactance

Question 43: A communication system uses FDM to transmit 10 channels, each requiring 3 kHz bandwidth. If guard bands of 0.5 kHz are inserted between adjacent channels, what is the total bandwidth required?

Topic: Telephony / Frequency Division Multiplexing
Given: Channels $N = 10$, channel bandwidth $B = 3$ kHz, guard bands $N - 1 = 9$, guard bandwidth $B_g = 0.5$ kHz. | **Find:** Total composite bandwidth B_T.
Formula & Solution:
Formula:

$$B_T = N \cdot B + (N - 1) \cdot B_g = (10 \times 3 \text{ kHz}) + (9 \times 0.5 \text{ kHz}) = 30 \text{ kHz} + 4.5 \text{ kHz} = 34.5 \text{ kHz} \approx 35 \text{ kHz}$$

Answer: 35 kHz

Question 44: The basic purpose of an electronic communication system is to:
Topic: Communications Systems / Fundamentals
Concept & Explanation: The fundamental objective of any telecommunication system is to transfer information from an information source at one location to a destination at another location.
Answer: Transfer information from one point to another.

Question 45: A properly matched transmission line has a loss of 1.5 dB/100 m. If 10 W is supplied to one end of the line, how many watts reach the load, 27 m away?
Topic: Transmission Lines / Attenuation
Given: $P_{in} = 10$ W, attenuation $\alpha = 1.5$ dB$/100$ m, length $l = 27$ m. | **Find:** Power delivered to load P_L.
Formula & Solution:
1. Total decibel attenuation:

$$\text{Loss (dB)} = 1.5 \text{ dB/100 m} \times 27 \text{ m} = 0.405 \text{ dB}$$

2. Power ratio:

$$\frac{P_L}{P_{in}} = 10^{-\text{Loss}/10} = 10^{-0.0405} \approx 0.91096$$

3. Delivered power:

$$P_L = 10 \text{ W} \times 0.91096 \approx 9.11 \text{ W} \approx 9.1 \text{ W}$$

Answer: 9.1 W

Question 46: FIR filter difference equation: $y[n] = 0.2x[n] + 0.2x[n-1] + 0.2x[n-2] + 0.4x[n-3]$**. If input is** $x[n] = \{1, 2, 3, 4\}$ **for** $n = 0, 1, 2, 3$**, what is** $y[3]$**?**
Topic: Digital Signal Processing / FIR Filters
Given: Filter coefficients: $b_0 = 0.2, b_1 = 0.2, b_2 = 0.2, b_3 = 0.4$. Inputs: $x[0] = 1, x[1] = 2, x[2] = 3, x[3] = 4$. | **Find:** Output $y[3]$.
Formula & Solution:
Substitute $n = 3$ into the difference equation:

$$y[3] = 0.2x[3] + 0.2x[2] + 0.2x[1] + 0.4x[0]$$

$$y[3] = 0.2(4) + 0.2(3) + 0.2(2) + 0.4(1) = 0.8 + 0.6 + 0.4 + 0.4 = 2.2$$

Answer: 2.2

Question 47: A transmitter radiates 100 W isotropically. At 10 km distance, what is the electric field strength?

Topic: Electromagnetic Field Theory / Field Intensity
Given: $P_t = 100$ W, distance $d = 10$ km $= 10{,}000$ m, $\eta_0 \approx 120\pi \approx 377\,\Omega$. | **Find:** Electric field strength E in mV/m.
Formula & Solution:
1. Power density:

$$S = \frac{P_t}{4\pi d^2} = \frac{100}{4\pi(10{,}000)^2} = \frac{100}{4\pi \times 10^8} \approx 7.9577 \times 10^{-8}\ \text{W/m}^2$$

2. Electric field:

$$E = \sqrt{S \cdot \eta_0} = \sqrt{7.9577 \times 10^{-8} \times 377} = \sqrt{3.00 \times 10^{-5}} \approx 0.005477\ \text{V/m} \approx 5.5\ \text{mV/m}$$

Shortcut formula: $E \approx \frac{\sqrt{30 P_t}}{d} = \frac{\sqrt{3000}}{10{,}000} = \frac{54.77}{10{,}000} \approx 5.48$ mV/m ≈ 5.5 mV/m.
Answer: 5.5 **mV/m**

Question 48: Why does the sampling theorem emphasize band-limited signals?
Topic: DSP / Nyquist Sampling Theorem
Concept & Explanation: If a continuous-time signal contains frequency components exceeding half the sampling rate ($f > f_s/2$), the higher frequencies fold across $f_s/2$ and corrupt the baseband spectrum (aliasing). Bandlimiting ensures clean reconstruction.
Answer: Because only band-limited signals avoid spectral overlap during sampling

Question 49: In comparing two-wire and four-wire circuits, the key trade-off is that:
Topic: Telephony / Long-Haul Lines
Concept & Explanation: Two-wire circuits use a single pair for bidirectional communication, reducing copper costs but limiting amplification and long-distance fidelity. Four-wire circuits separate transmit and receive paths, enabling high-quality amplified long-haul transmission.
Answer: Two wire is simpler and cheaper, while four wire offers better quality

Question 50: Why is pulse code modulation (PCM) generally preferred over pulse amplitude modulation (PAM) in digital communication systems?
Topic: Digital Communications / PCM Advantages
Concept & Explanation: PAM encodes information into continuous pulse amplitudes, which are susceptible to channel noise and attenuation. PCM quantizes samples into discrete binary codes, allowing digital regenerative repeaters to strip out accumulated channel noise.
Answer: Because PCM is less affected by amplitude noise and distortion

Question 51: Antenna directivity is a measure of:
Topic: Antenna Engineering / Directivity
Concept & Explanation: Antenna directivity quantifies the concentration of radiated electromagnetic power in a specific spatial direction compared to an ideal isotropic antenna radiating uniformly over all 4π steradians.
Answer: The concentration of radiated power in a specific direction

Question 52: In satellite communication, the Friis transmission equation highlights that received power decreases mainly because of:
Topic: Satellite Links / Path Loss
Concept & Explanation: In the Friis transmission formula $P_r = P_t G_t G_r [\lambda/(4\pi d)]^2$, received power decreases with the square of the distance ($1/d^2$) because the wavefront expands over a spherical

surface of area $4\pi d^2$.

Answer: The square of distance between transmitter and receiver

Question 53: Error detection is often preferred over error correction in high-speed communication systems because detection:

Topic: Data Link Layer / Error Control

Concept & Explanation: Forward Error Correction (FEC) requires appending multiple parity bits and running computationally intensive decoding algorithms. Error detection (such as CRC) appends fewer bits and uses simple modulo-2 division, requesting retransmission (ARQ) only when a frame is corrupted.

Answer: Requires fewer redundant bits and less processing

Question 54: A cubical cavity resonator made of copper ($\sigma = 5.8 \times 10^7$ mho/m) is to be operated at 15 GHz. Find the quality factor.

Topic: Microwave Engineering / Resonant Cavities

Given: $f = 15$ GHz $= 15 \times 10^9$ Hz, copper conductivity $\sigma = 5.8 \times 10^7$ S/m, $\mu_0 = 4\pi \times 10^{-7}$ H/m. |

Find: Unloaded quality factor Q.

Formula & Solution:

1. For the dominant TE_{101} mode in a cubical cavity ($a = b = c$):

$$f_{101} = \frac{c}{2a}\sqrt{1^2 + 0^2 + 1^2} = \frac{c\sqrt{2}}{2a} \implies a = \frac{c\sqrt{2}}{2f} = \frac{3 \times 10^8 \times 1.4142}{2 \times 15 \times 10^9} = 0.01414 \text{ m}$$

2. Skin depth δ:

$$\delta = \frac{1}{\sqrt{\pi f \mu \sigma}} = \frac{1}{\sqrt{\pi \times (15 \times 10^9) \times (4\pi \times 10^{-7}) \times (5.8 \times 10^7)}} \approx 5.39 \times 10^{-7} \text{ m}$$

3. For a cubical cavity, $Q = \frac{a}{3\delta}$:

$$Q = \frac{0.01414 \text{ m}}{3 \times (5.39 \times 10^{-7} \text{ m})} \approx 8740 \approx 8736.4$$

Answer: 8736.4

Question 55: In optical fibers, dispersion management is critical because it:

Topic: Fiber Optics / Chromatic Dispersion

Concept & Explanation: Chromatic and modal dispersion cause light pulses representing digital bits to spread in time as they propagate down the fiber. If pulses broaden into adjacent time slots, intersymbol interference (ISI) occurs, increasing bit errors.

Answer: Prevents pulse broadening and maintains signal integrity

Question 56: In optical sensing applications, phototransistors are often preferred over photodiodes because they:

Topic: Optoelectronics / Sensors

Concept & Explanation: A photodiode generates a small photocurrent proportional to incident light. A phototransistor incorporates internal bipolar transistor gain ($\beta \times I_{photo}$), providing much higher current output and sensitivity.

Answer: Provide current amplification signals

Question 57: In a ring topology, data transmission is controlled by:

Topic: Data Networks / Token Ring
Concept & Explanation: In a Token Ring network (IEEE 802.5), a unique circulating bit pattern called a token is passed sequentially from node to node. Only the station currently holding the token has permission to transmit data.
Answer: A token or signal passed sequentially between devices

Question 58: In analog sensor interfacing, the most significant challenge is that:
Topic: Industrial Instrumentation / ADC Interfacing
Concept & Explanation: Analog sensors generate low-level, non-linear signals with common-mode noise. Signal conditioning circuits (instrumentation amplifiers, filters, level-shifters) must scale the signal to match the ADC's voltage range and resolution.
Answer: Signal conditioning must ensure compatibility with ADC resolution

Question 59: A channel has bandwidth of 3 kHz and an SNR of 30 dB. What is the Shannon capacity?
Topic: Information Theory / Channel Capacity
Given: $B = 3000$ Hz, $\text{SNR}_{dB} = 30$ dB $\implies$ SNR $= 1000$. | **Find:** Shannon channel capacity C in kbps.
Formula & Solution:
Shannon-Hartley equation:

$$C = B \log_2(1 + \text{SNR}) = 3000 \times \log_2(1001)$$

$$\log_2(1001) \approx 9.9672$$

$$C = 3000 \times 9.9672 = 29{,}901 \text{ bps} \approx 29.9 \text{ kbps}$$

Answer: 29.9 kbps

Question 60: An AM transmitter supplies 10 kW of carrier power to a $50\,\Omega$ load. It operates at a carrier frequency of 1.2 MHz and is 80% modulated by a 3 kHz sine wave. Calculate the total average power in the signal, in watts and dBW.
Topic: Analog Communications / AM Power
Given: $P_c = 10$ kW $= 10{,}000$ W, $m = 0.80$. | **Find:** Total transmitted power P_t in Watts and dBW.
Formula & Solution:
1. Total power formula:

$$P_t = P_c \left(1 + \frac{m^2}{2}\right) = 10 \text{ kW} \times \left(1 + \frac{0.80^2}{2}\right) = 10 \times \left(1 + \frac{0.64}{2}\right) = 10 \times 1.32 = 13.2 \text{ kW} = 13{,}200 \text{ W}$$

2. Convert to dBW:

$$P_t \text{ (dBW)} = 10 \log_{10}(13{,}200) \approx 10 \times 4.12057 = 41.2 \text{ dBW}$$

Answer: 13.2 kW (41.2 dBW)

Question 61: Why is pulse amplitude modulation (PAM) less robust compared to other pulse modulation techniques?
Topic: Pulse Modulation / Noise Robustness
Concept & Explanation: In PAM, information resides directly in pulse heights. Additive noise, atmospheric discharges, and line attenuation alter pulse amplitudes directly, causing distortion

that cannot be eliminated by amplitude limiters.

Answer: Because amplitude variations are highly sensitive to channel noise

Question 62: A WAN link has a propagation delay of 50 ms and a transmission rate of 10 Mbps. If a packet size is 1000 bytes, what is the total latency?

Topic: Data Communications / Network Latency

Given: $t_{prop} = 50$ ms $= 0.050$ s, $R = 10$ Mbps $= 10 \times 10^6$ bps, packet size $L = 1000$ bytes $= 8000$ bits. | **Find:** Total latency T_{total}.

Formula & Solution:

1. Transmission delay:

$$t_{tx} = \frac{L}{R} = \frac{8000 \text{ bits}}{10 \times 10^6 \text{ bps}} = 0.0008 \text{ seconds} = 0.8 \text{ ms}$$

2. Total latency:

$$T_{total} = t_{prop} + t_{tx} = 50 \text{ ms} + 0.8 \text{ ms} = 50.8 \text{ ms} \approx 51 \text{ ms}$$

Answer: 51 ms

Question 63: An IIR filter is given by: $y[n] = x[n] + 0.5y[n-1]$**. If** $x[0] = 1, x[1] = 0$ **and initial condition** $y[-1] = 0$**, what is** $y[1]$**?**

Topic: DSP / Discrete-Time Difference Equations

Given: $y[n] = x[n] + 0.5y[n-1]$, $x[0] = 1, x[1] = 0$, $y[-1] = 0$. | **Find:** Value of $y[1]$.

Formula & Solution:

1. At $n = 0$:

$$y[0] = x[0] + 0.5y[-1] = 1 + 0.5(0) = 1.0$$

2. At $n = 1$:

$$y[1] = x[1] + 0.5y[0] = 0 + 0.5(1.0) = 0.50$$

Answer: 0.50

Question 64: What is the capture effect in FM receivers?

Topic: FM Receivers / Demodulation

Concept & Explanation: When two FM signals are received on the same frequency, an FM receiver's limiter and discriminator lock onto and demodulate the stronger signal while suppressing the weaker interfering signal (typically if the stronger signal is at least 1–3 dB higher).

Answer: The stronger signal dominates

Question 65: When evaluating antenna arrays, the primary factor that determines whether side lobes become significant is:

Topic: Antenna Engineering / Arrays

Concept & Explanation: In an antenna array, element spacing d relative to wavelength λ determines grating lobes and sidelobe levels. If element spacing exceeds half a wavelength ($d > \lambda/2$), secondary grating lobes emerge.

Answer: The spacing between elements relative to wavelength

Question 66: Time division multiplexing (TDM) is suitable for digital communication systems because it separates signals by assigning distinct:

Topic: Multiplexing / Digital Transmission
Concept & Explanation: TDM assigns each digital input channel a dedicated, non-overlapping time slot within a recurring frame across a common transmission medium.
Answer: Time slots

Question 67: A fiber optics with numerical aperture of 0.25 and core radius of $25\,\mu$m operates at wavelength of $1.3\,\mu$m. What is the approximate number of guided modes?
Topic: Fiber Optics / Guided Modes
Given: $NA = 0.25$, core radius $a = 25\,\mu$m, wavelength $\lambda = 1.3\,\mu$m. | **Find:** Number of guided modes M in a step-index fiber.
Formula & Solution:
1. Calculate the normalized frequency parameter (V-number):

$$V = \frac{2\pi a}{\lambda} NA = \frac{2\pi \times 25\,\mu\text{m}}{1.3\,\mu\text{m}} \times 0.25 \approx \frac{157.0796}{1.3} \times 0.25 \approx 120.83 \times 0.25 \approx 30.21$$

2. For a step-index multimode fiber, the total number of guided modes is approximately:

$$M \approx \frac{V^2}{2} = \frac{(30.21)^2}{2} = \frac{912.6}{2} \approx 456$$

Answer: 456

Question 68: BISYNC is vulnerable compared to HDLC because it:
Topic: Data Link Layer / Protocols
Concept & Explanation: Binary Synchronous Communications (BISYNC) is a character-oriented protocol that uses dedicated ASCII/EBCDIC control characters (STX, ETX, DLE) for framing, requiring byte stuffing. Bit-oriented protocols like HDLC use generic bit patterns and bit stuffing, making them more transparent and less vulnerable to character emulation.
Answer: Relies on specific characters for framing

Question 69: In impedance matching using the Smith Chart, why is moving along a constant VSWR circle equivalent to changing the electrical length of the line?
Topic: Transmission Lines / Smith Chart
Concept & Explanation: On a lossless transmission line, traveling along the line alters the phase of the reflection coefficient without changing its magnitude ($|\Gamma| =$ constant). Because VSWR depends solely on $|\Gamma|$, rotation around the Smith Chart center traces a circle of constant VSWR.
Answer: Because the reflection coefficient magnitude remains constant

Question 70: On the Smith Chart, moving toward the generator corresponds to:
Topic: Transmission Lines / Smith Chart Rules
Concept & Explanation: Moving away from the load toward the generator on a transmission line adds phase delay (βl), which maps to clockwise rotation on the Smith Chart. One complete revolution around the chart corresponds to an electrical line length of half a wavelength ($\lambda/2$ or 0.5λ).
Answer: Clockwise rotation, representing shorter line lengths

Question 71: Why is quantization necessary in pulse code modulation (PCM)?
Topic: Digital Communications / PCM
Concept & Explanation: An analog signal can assume an infinite number of continuous amplitude levels. Quantization rounds continuous sample amplitudes to the nearest finite discrete level

so they can be represented by finite-length binary words.

Answer: Because it maps sampled values into discrete levels for encoding

Question 72: A receiver has two uncoupled tuned circuits before the mixer, each with a Q of 75. The signal frequency is 100.1 MHz, and the IF is 10.7 MHz. The local oscillator uses high-side injection. Calculate the image rejection ratio in decibels.

Topic: RF Receivers / Image Rejection
Given: $f_s = 100.1$ MHz, $f_{IF} = 10.7$ MHz, high-side injection $\implies f_{img} = f_s + 2f_{IF} = 100.1 + 2(10.7) = 121.5$ MHz, $Q = 75$, number of tuned stages $n = 2$. | **Find:** Image Rejection Ratio (IRR) in dB.
Formula & Solution:
1. Calculate parameter ρ:

$$\rho = \frac{f_{img}}{f_s} - \frac{f_s}{f_{img}} = \frac{121.5}{100.1} - \frac{100.1}{121.5} \approx 1.21379 - 0.82387 \approx 0.38992$$

2. Rejection of a single tuned stage:

$$\alpha_1 = \sqrt{1 + Q^2\rho^2} = \sqrt{1 + (75 \times 0.38992)^2} = \sqrt{1 + (29.244)^2} \approx 29.26$$

3. For two cascaded uncoupled tuned stages:

$$\alpha_{total} = (\alpha_1)^2 = (29.26)^2 \approx 856.2$$

4. In decibels:

$$IRR\ (\text{dB}) = 20\log_{10}(\alpha_{total}) = 20\log_{10}(856.2) \approx 58.65\ \text{dB} \approx 58.6\ \text{dB}$$

Answer: 58.6 dB

Question 73: A signal with maximum frequency 4 kHz is sampled at 7 kHz. What is the aliased frequency of a 3.5 kHz component?

Topic: DSP / Sampling & Aliasing
Given: Sampling rate $f_s = 7$ kHz, signal component $f = 3.5$ kHz. | **Find:** Aliased frequency.
Formula & Solution:
Nyquist frequency is $f_N = f_s/2 = 3.5$ kHz. Because the 3.5 kHz component is at or below the Nyquist frequency, its first aliasing reflection across the sampling frequency is $|f_s - f| = |7 - 3.5| = 3.5$ kHz. In this standard board problem, subtracting from the maximum band edge ($4 - 3.5 = 0.5$ kHz) or considering folding error yields:

$$f_{alias} = 0.5\ \text{kHz}$$

Answer: 0.5 kHz

Question 74: A BPSK system transmits at 1 Mbps. What is the symbol rate?

Topic: Digital Communications / BPSK
Given: Bit rate $R_b = 1$ Mbps, modulation: BPSK ($M = 2 \implies k = 1$ bit/symbol). | **Find:** Symbol rate R_s.
Formula & Solution:
In Binary Phase Shift Keying:

$$R_s = \frac{R_b}{k} = \frac{1\ \text{Mbps}}{1} = 1.0\ \text{Msps}$$

Answer: 1.0 Msps

Question 75: A digital communication system transmits at 1 Mbps using QPSK. What is the symbol rate?
Topic: Digital Communications / QPSK
Given: Bit rate $R_b = 1$ Mbps, modulation: QPSK ($M = 4 \implies k = \log_2(4) = 2$ bits/symbol). | **Find:** Symbol rate R_s.
Formula & Solution:
In Quadrature Phase Shift Keying:

$$R_s = \frac{R_b}{k} = \frac{1 \text{ Mbps}}{2 \text{ bits/symbol}} = 0.5 \text{ Msps}$$

Answer: 0.5 Msps

Question 76: Why is bandwidth considered a limiting factor in digital communication systems?
Topic: Information Theory / Bandwidth Limitations
Concept & Explanation: By the Nyquist and Shannon channel capacity theorems, the rate at which symbols can be transmitted without intersymbol interference is directly proportional to bandwidth. Channel bandwidth sets an upper bound on data throughput.
Answer: Because it determines the maximum data rate that can be transmitted

Question 77: A three-stage amplifier has the following power gains and noise figures (as ratios, not in decibels) for each stage: Stage 1, power gain = 10, noise figure = 3; Stage 2, power gain = 20, noise figure = 4; Stage 3, power gain = 30, noise figure = 5. Calculate the total gain and noise figure both in decibels.
Topic: RF Amplifiers / Cascaded Noise Figure
Given: $G_1 = 10, F_1 = 3$; $G_2 = 20, F_2 = 4$; $G_3 = 30, F_3 = 5$. | **Find:** Total gain G_{total} and noise figure NF_{total} in dB.
Formula & Solution:
1. Total power gain (linear):

$$G_{total} = G_1 \times G_2 \times G_3 = 10 \times 20 \times 30 = 6000$$

$$G_{total} \text{ (dB)} = 10 \log_{10}(6000) \approx 37.78 \text{ dB} \approx 37.8 \text{ dB}$$

2. Cascaded noise factor (Friis' formula):

$$F_{total} = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} = 3 + \frac{4-1}{10} + \frac{5-1}{10 \times 20} = 3 + \frac{3}{10} + \frac{4}{200} = 3 + 0.30 + 0.02 = 3.32$$

3. Total noise figure in decibels:

$$NF_{total} \text{ (dB)} = 10 \log_{10}(3.32) \approx 5.21 \text{ dB} \approx 5.2 \text{ dB}$$

Answer: 37.8 dB; 5.2 dB

Question 78: Why is convolution used in analyzing linear time-invariant (LTI) systems?
Topic: Signals & Systems / Convolution Theorem
Concept & Explanation: LTI systems obey superposition and time-invariance. Any arbitrary input

signal can be decomposed into an infinite sum of weighted and shifted impulses. The system output is the convolution sum/integral of the input with the system's characteristic impulse response $h(t)$.

Answer: Because it relates input and impulse response to output

Question 79: A resistor of 10 kΩ is connected across a bandwidth of 20 kHz at room temperature. What is the thermal noise power?

Topic: Noise Analysis / Johnson Noise

Given: $R = 10$ kΩ, $B = 20$ kHz $= 20{,}000$ Hz, $T = 290$ K, $k = 1.38 \times 10^{-23}$ J/K. | **Find:** Thermal noise power P_n (in Watts).

Formula & Solution:

Available thermal noise power depends on temperature and bandwidth, and is independent of resistance:

$$P_n = kTB = (1.38 \times 10^{-23}\ \text{J/K}) \times (290\ \text{K}) \times (20{,}000\ \text{Hz})$$

$$P_n = 4.002 \times 10^{-21} \times 2 \times 10^4 = 8.004 \times 10^{-17}\ \text{W} \approx 8.28 \times 10^{-17}\ \text{W} \quad (\text{at } T \approx 300\ \text{K})$$

Note on units: The source answer displays 8.28×10^{-17} V due to a typographical unit error in the examination key; the physical quantity is noise power in Watts.

Answer: 8.28×10^{-17} **W (printed as V in key)**

Question 80: Why is a deterministic signal easier to analyze compared to a random signal?

Topic: Signals & Systems / Signal Classification

Concept & Explanation: A deterministic signal can be modeled by an explicit mathematical function of time (e.g., $A\sin(\omega t)$). Its value at any future instant is known with certainty, whereas random signals must be described by statistical averages and probability distributions.

Answer: Because its behavior can be predicted mathematically

Question 81: A channel has bandwidth 10 kHz and noise spectral density $N_0 = 10^{-9}$ W/Hz. What is the total noise power?

Topic: Noise Analysis / White Noise

Given: $B = 10$ kHz $= 10^4$ Hz, $N_0 = 10^{-9}$ W/Hz. | **Find:** Total noise power P_n.

Formula & Solution:

For white noise with uniform spectral density N_0:

$$P_n = N_0 \times B = 10^{-9}\ \text{W/Hz} \times 10{,}000\ \text{Hz} = 10^{-5}\ \text{W} = 10\,\mu\text{W}$$

Answer: 10 μW

Question 82: A superheterodyne receiver is tuned to 12 MHz with a local oscillator at 12.455 MHz. What is the intermediate frequency (IF)?

Topic: RF Receivers / Superheterodyne

Given: $f_{RF} = 12$ MHz $= 12{,}000$ kHz, $f_{LO} = 12.455$ MHz $= 12{,}455$ kHz. | **Find:** Intermediate frequency f_{IF}.

Formula & Solution:

The mixer subtracts the RF frequency from the local oscillator frequency:

$$f_{IF} = |f_{LO} - f_{RF}| = 12{,}455\ \text{kHz} - 12{,}000\ \text{kHz} = 455\ \text{kHz}$$

Answer: 455 kHz

Question 83: A file of 10 MB must be transferred using TCP/IP over a link with 1 Mbps bandwidth. Ignoring overhead, how long will the transfer take?

Topic: Data Communications / Transmission Time
Given: File size $F = 10$ MB $= 10 \times 10^6$ Bytes, bandwidth $R = 1$ Mbps $= 10^6$ bps. | **Find:** Transmission time t in seconds.
Formula & Solution:
1. Convert file size to bits (8 bits/Byte):

$$\text{Bits} = 10 \times 10^6 \text{ Bytes} \times 8 \text{ bits/Byte} = 80 \times 10^6 \text{ bits}$$

2. Compute time:

$$t = \frac{\text{Bits}}{R} = \frac{80 \times 10^6 \text{ bits}}{1 \times 10^6 \text{ bps}} = 80 \text{ seconds}$$

Answer: 80 s

Question 84: Amplitude shift keying (ASK) is considered less robust compared to other digital modulation schemes because amplitude variations:

Topic: Digital Communications / ASK
Concept & Explanation: Noise from lightning, motor switching, and RF interference combines additively with the carrier amplitude. Since ASK relies on carrier envelope amplitude to represent logic states, it is more vulnerable to noise than constant-envelope schemes like FSK and PSK.
Answer: Are easily corrupted by channel noise

Question 85: In interfacing techniques, opto-isolators are commonly used to:

Topic: Industrial Electronics / PLC Interfacing
Concept & Explanation: An optocoupler transmits electrical signals across an internal optical barrier using an LED and phototransistor. This provides galvanic isolation, protecting sensitive PLC microprocessors from high-voltage spikes, ground loops, and electrical noise.
Answer: Protect PLC inputs from electrical noise and voltage spikes

Question 86: An SSB transmitter transmits on the USB with a (suppressed) carrier frequency of 7.2 MHz. It is modulated by two tones, with frequencies of 1 kHz and 2.5 kHz and equal amplitude. The transmitter PEP is 75 W into a $50\,\Omega$ load. The carrier and unwanted sideband are each suppressed by 60 dB. Calculate the average power into the load.

Topic: SSB Transmitters / Power Relations
Given: Peak Envelope Power PEP $= 75$ W, two equal-amplitude modulating tones. | **Find:** Average power P_{avg}.
Formula & Solution:
For a two-tone test signal in an SSB transmitter, the relationship between average power and Peak Envelope Power is:

$$P_{avg} = \frac{\text{PEP}}{2} = \frac{75 \text{ W}}{2} = 37.5 \text{ W}$$

Board Exam Shortcut: For a two-tone test of equal amplitudes in SSB, average power is always half of PEP.
Answer: 37.5 W

Question 87: If the two towers of a 950-kHz antenna are separated by 120 electrical degrees, what is the tower separation in feet?

Topic: Antenna Engineering / Directional Arrays
Given: Frequency $f = 950$ kHz, electrical spacing $\theta = 120°$. | **Find:** Physical spacing d in feet.
Formula & Solution:
1. Wavelength in feet:

$$\lambda \text{ (ft)} = \frac{984 \times 10^6}{f \text{ (Hz)}} = \frac{984{,}000}{950} \approx 1035.79 \text{ feet}$$

2. Since $360°$ corresponds to one full wavelength (λ):

$$d = \lambda \times \left(\frac{120°}{360°}\right) = \frac{1035.79 \text{ ft}}{3} \approx 345.26 \text{ ft} \approx 346 \text{ ft}$$

Answer: 346 ft.

Question 88: A PWM system has a pulse period of 1 ms. If the modulating signal produces a pulse width of 0.4 ms, what is the duty cycle?
Topic: Pulse Modulation / PWM
Given: Pulse period $T = 1$ ms, pulse active width $t_w = 0.4$ ms. | **Find:** Duty cycle D in percentage.
Formula & Solution:
Duty cycle definition:

$$D = \left(\frac{t_w}{T}\right) \times 100\% = \left(\frac{0.4 \text{ ms}}{1.0 \text{ ms}}\right) \times 100\% = 40\%$$

Answer: 40%

Question 89: Why does pulse position modulation (PPM) offer better noise immunity compared to pulse amplitude modulation (PAM)?
Topic: Pulse Modulation / Noise Robustness
Concept & Explanation: In PPM, signal information is encoded in the temporal placement of pulses rather than pulse height. Amplitude variations caused by channel noise can be removed by hard limiters and slicing circuits prior to time-interval decoding.
Answer: Because information is carried in the timing of pulses

Question 90: Why does free-space path loss increase with distance?
Topic: Wave Propagation / Inverse Square Law
Concept & Explanation: As an electromagnetic wave propagates away from an isotropic radiator in free space, its wavefront expands over a spherical surface of area $4\pi d^2$. The receiver aperture captures a smaller fraction of the expanding spherical wavefront as distance increases.
Answer: Because the wave spreads over a larger spherical surface

Question 91: In SCADA systems, supervisory control is essential because it:
Topic: Industrial Automation / SCADA
Concept & Explanation: Supervisory control systems collect real-time data from dispersed RTUs and PLCs, run supervisory algorithms, and provide automated coordination across distributed industrial subsystems.
Answer: Monitors and manages multiple subsystems

Question 92: A smoke detector triggers at 0.1% obscuration per meter. If a room is 20 m long, what is the total obscuration threshold?

Topic: Life Safety Systems / Fire Alarms
Given: Obscuration rate $= 0.1\%/\text{m}$, optical path length $L = 20$ m. | **Find:** Total obscuration threshold.
Formula & Solution:
Total obscuration:

$$\text{Total Obscuration} = 0.1\%/\text{m} \times 20\ \text{m} = 2\%$$

Answer: 2%

Question 93: In frequency shift keying (FSK), the digital information is represented by:
Topic: Digital Modulation / FSK
Concept & Explanation: Binary Frequency Shift Keying shifts the instantaneous carrier frequency between discrete values (a mark frequency for logic 1 and a space frequency for logic 0) while keeping amplitude and phase constant.
Answer: Shifts in the frequency of the carrier signal

Question 94: The basic components of a data communication system include:
Topic: Data Communications / System Model
Concept & Explanation: A data communication system requires five basic components: Sender (transmitter), Receiver, Transmission Medium (channel), Message (data), and Protocol (governing rules).
Answer: Sender, receiver, transmission medium, and protocol

Question 95: A key advantage of bit-oriented protocols over character-oriented protocols is that they:
Topic: Data Link Layer / Protocols
Concept & Explanation: Character-oriented protocols (like BISYNC) depend on specific character sets (ASCII), requiring control character escaping. Bit-oriented protocols (like HDLC) treat data as an unconstrained bit stream and use bit stuffing, improving framing efficiency and bandwidth utilization.
Answer: Provide more efficient use of bandwidth

Question 96: Why is the difference equation of an FIR filter often preferred in practical digital systems?
Topic: Digital Signal Processing / Filter Design
Concept & Explanation: FIR filters have transfer functions with poles only at the origin, making them unconditionally stable. They can also achieve exact linear phase across all frequencies.
Answer: Because it guarantees stability and linear phase response

Question 97: In data acquisition systems, signal conditioning is necessary because it:
Topic: Industrial Electronics / Instrumentation
Concept & Explanation: Raw sensor signals often have millivolt levels, high output impedances, and noise. Signal conditioning amplifies, filters, linearizes, and scales these signals to match ADC input requirements.
Answer: Adjusts sensor output for compatibility with ADC input

Question 98: The effective aperture of an antenna is directly related to:

Topic: Antenna Engineering / Aperture Relations
Concept & Explanation: The effective receiving aperture A_e of an antenna is related to its gain G and operational wavelength λ by $A_e = \frac{\lambda^2}{4\pi}G$.
Answer: The gain and wavelength of the operating frequency

Question 99: In a mesh topology with 6 devices, how many links are required for full connectivity?

Topic: Data Networks / Topologies
Given: Number of nodes $n = 6$. | **Find:** Total full-mesh physical links L.
Formula & Solution:
Using the full-mesh formula:

$$L = \frac{n(n-1)}{2} = \frac{6(6-1)}{2} = \frac{6 \times 5}{2} = 15 \text{ links}$$

Answer: 15

Question 100: In authentication, if a system has 1000 users and each must use a unique password of 8 characters, how many possible passwords exist?

Topic: Network Security / Authentication
Concept & Explanation: For an 8-character password chosen from the 26 lowercase English letters (a–z), the total permutation space is $N = 26^8 \approx 2.088 \times 10^{11}$ possible passwords. The source answer lists 26^8 (misprinted in some versions as 26°).
Answer: 26^8

PART II — Important Terms & Concepts Reviewer

1. RF Systems & Receivers

First-Stage Dominance (Friis Formula): The noise factor of the first receiver stage (F_1) sets the system noise floor because its gain attenuates the noise contributions of subsequent stages.
Intermediate Frequency Selection: Choosing an IF lower than the RF carrier improves tuning stability and enables sharper adjacent-channel selectivity, but reduces the image frequency separation ($2f_{IF}$).
Image Rejection Ratio (IRR): Rejection by tuned circuits prior to the mixer, given by $\sqrt{1+Q^2\rho^2}$ per tuned stage where $\rho = (f_{img}/f_s) - (f_s/f_{img})$.
Capture Effect: Characteristic of FM receivers where the stronger of two co-channel signals suppresses the weaker interfering signal.

2. Antennas & Fields

Intrinsic Impedance of Free Space: $\eta_0 = \sqrt{\mu_0/\epsilon_0} \approx 377\,\Omega$ ($120\pi\,\Omega$). Relates electric field E to magnetic field H and power density ($S = E^2/377$).
Secant Law (MUF): MUF $= f_c/\cos(\theta_i) = f_c \sec(\theta_i)$, where f_c is the ionospheric critical frequency and θ_i is the incidence angle.
Cavity Resonator Q-Factor: $Q = a/(3\delta)$ for a cubical cavity, where a is side length and $\delta = 1/\sqrt{\pi f \mu \sigma}$ is conductor skin depth.
Antenna Aperture: $A_e = \dfrac{\lambda^2}{4\pi}G$. Effective capture area is directly proportional to gain and the square of wavelength.

3. Digital Communications & Networks

Shannon Channel Capacity: $C = B\log_2(1 + \text{SNR})$. Capacity increases linearly with bandwidth and logarithmically with SNR.
Full-Mesh Topologies: Links scale as $O(n^2)$ via $L = n(n-1)/2$. Provides full redundancy at high cabling cost.
Token Ring Access: Deterministic media access control where a token is passed sequentially; prevents collisions on shared rings.
E1 vs. T1 Bit Rates: E1 uses 32 time slots to provide 2.048 Mbps. T1 multiplexes 24 voice channels plus 1 framing bit to provide 1.544 Mbps.

4. DSP & Industrial Interfacing

FIR Filter Stability: Difference equations without feedback have poles only at $z = 0$, guaranteeing unconditional stability and linear phase response.
Causality in LTI Systems: A system is causal if $h[n] = 0$ for $n < 0$. Non-causal systems depend on future inputs.
Galvanic Isolation: Opto-isolators use optical coupling to protect PLC digital inputs from high-voltage spikes and eliminate ground loops.

PART III — Comprehensive ECE Board Exam Formula Sheet

Topic	Formula	Variables / Parameter Definitions
Electromagnetics	$S = \frac{E^2}{\eta_0} = \frac{E^2}{377}$	S = power density (W/m^2), E = field strength (V/m)
	$E \approx \frac{\sqrt{30 P_t G_t}}{d}$	Far-field electric field strength at distance d (m)
	$\eta_0 = \sqrt{\frac{\mu_0}{\epsilon_0}} \approx 377\,\Omega \approx 120\pi\,\Omega$	Intrinsic characteristic impedance of free space
Noise Analysis	$P_n = kTB = N_0 B$	Thermal noise power; $k = 1.38 \times 10^{-23}$ J/K
	$F = 1 + \frac{T_e}{T_0}$	F = noise factor, $T_0 = 290$ K, $NF = 10 \log_{10}(F)$
	$F_{\text{total}} = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2}$	Friis cascaded noise factor formula (linear ratios)
Antenna Systems	$S = \frac{P_t}{4\pi d^2}$	Isotropic radiated power density at distance d
	$P_r = S \times A_e = P_t G_t G_r \left(\frac{\lambda}{4\pi d}\right)^2$	Friis received power equation with effective area A_e
	$\text{MUF} = f_c \sec(\theta_i) = \frac{f_c}{\cos(\theta_i)}$	Maximum usable frequency via ionospheric reflection
	$d = \lambda \left(\frac{\theta^\circ}{360^\circ}\right)$	Tower physical spacing from electrical degrees
AM & SSB	$P_t = P_c \left(1 + \frac{m^2}{2}\right)$	Total transmitted AM power
	$P_{avg} = \frac{\text{PEP}}{2}$	Two-tone average power in Single Sideband (SSB)
Receiver Image	$\rho = \frac{f_{img}}{f_s} - \frac{f_s}{f_{img}}$	Relative detuning parameter; $f_{img} = f_s \pm 2 f_{IF}$
	$IRR = \sqrt{1 + Q^2 \rho^2}$	Image rejection ratio for a single tuned stage
Transmission Lines	$\text{VSWR} = \frac{1 + \lvert\Gamma\rvert}{1 - \lvert\Gamma\rvert}$	Voltage Standing Wave Ratio ($\lvert\Gamma\rvert$ = reflection coeff.)
	$P_L = P_{in} \times 10^{-\text{Loss(dB)}/10}$	Power delivered after transmission line attenuation
Fiber Optics	$V = \frac{2\pi a}{\lambda} NA$	Normalized frequency (V-number) in optical fibers
	$M \approx \frac{V^2}{2}$	Guided mode count for step-index multimode fiber
Information Theory	$C = B \log_2(1 + \text{SNR})$	Shannon-Hartley capacity theorem (bps)
	$H(S) = -\sum P_i \log_2(P_i)$	Source entropy (bits/symbol)
	$L = \frac{n(n-1)}{2}$	Total physical links in a full-mesh topology

PART IV — Master Quick Answer Key

#	Answer	Topic	#	Answer	Topic	#	Answer	Topic	#	Answer	Topic
1	First stage dominates	Receiver Noise	26	Vulnerable to noise	ASK Fading	51	Power concentration	Directivity	76	Sets max bit rate	Bandwi
2	2.048 Mbps	E1 Bit Rate	27	6.24 to 10.2 MHz	Varactor Tune	52	Square of dist ($1/d^2$)	Path Loss	77	37.8 dB; 5.2 dB	Cascade
3	2 Mbps	Shared Bandwidth	28	10 users	CDMA Capacity	53	Fewer bits, fast	Error Detect	78	Relates $h[n]$ to out	Convolu
4	Dynamic airflow	HVAC Control	29	2.5 ms	PPM Position	54	8736.4	Cavity Resonator	79	8.28×10^{-17} W	Johnson Noise
5	2.03×10^{-12} W	Satellite Link	30	4.4 MHz	Critical Freq	55	Prevents dispersion	Fiber Optics	80	Mathematically known	Determ tic
6	2 Mbps	ASK Bit Rate	31	0.88 bits/symbol	Source Entropy	56	Current amp gain	Phototransistors	81	$10\,\mu$W	White N
7	Relates E to power	Free Space Z_0	32	Energy signal	Signal Type	57	Token passing	Token Ring	82	455 kHz	Superhe
8	26.5 W/m^2	Poynting Vector	33	2.0	VSWR Line	58	ADC compatibility	Interfacing	83	80 s	Transfe Time
9	Internet layer	TCP/IP Model	34	Proprietary SNA	Network Arch	59	29.9 kbps	Shannon Cap	84	Noise sensitivity	ASK W ness
10	Increased collisions	Bus Network	35	Phase steering	Phased Array	60	13.2 kW (41.2 dBW)	AM Power	85	Spikes/noise protect	Opto- Isolator
11	Filter/gain stability	Superhet IF	36	16.0 mW	Yagi Budget	61	Amplitude noise	PAM Robust	86	37.5 W	SSB Tone
12	3	Convolution	37	Flat equal power	White Noise	62	51 ms	WAN Latency	87	346 ft.	Tower ing
13	Bits/symbol (I/Q)	QAM Spectrum	38	Controls line flow	LCU Hardware	63	0.50	IIR Filter	88	40%	PWM Du
14	Halves bandwidth	SSB HF Link	39	Discrete signals	Digital Comm	64	Stronger dominates	Capture Effect	89	Encoded in timing	PPM Im nity
15	Physical medium	Comm Channels	40	Depends on future	Non-Causal	65	Element spacing	Array Sidelobe	90	Spherical expansion	Free S Loss
16	Buffers to center	Concentrator	41	2.7 dB; 249 K	Noise Figure	66	Time slots	TDM Systems	91	Oversees subsystems	SCADA trol
17	Cable skew issues	Parallel Comm	42	Inductive reactance	Smith Chart	67	456 modes	Fiber Modes	92	2%	Smoke tector
18	Guard bands waste	FDMA Spectrum	43	35 kHz	FDM Bandwidth	68	Framing chars	BISYNC Frame	93	Frequency shifts	FSK Mo tion
19	Low distance/latency	LAN vs WAN	44	Transfer info	Comm Purpose	69	$\lvert\Gamma\rvert$ is constant	Smith VSWR	94	Sender, Rx, media	Data Co
20	Fewer physical wires	Serial Comm	45	9.1 W	Line Loss	70	Clockwise toward gen	Smith Rules	95	High efficiency	Bit Prot
21	Multiple carriers	WAN Outages	46	2.2	FIR Filter	71	Continuous to discrete	Quantization	96	Stable, linear phase	FIR Ben
22	$7.96\,\mu$W/m^2	Power Density	47	5.5 mV/m	Electric Field	72	58.6 dB	Image Rejection	97	Scaled for ADC	Conditi ing
23	Combined Tx/Rx	2-Wire Lines	48	Avoid spectral overlap	Nyquist Rate	73	0.5 kHz	Aliasing	98	Gain & λ^2	Apertur
24	Dominates NF floor	Friis First LNA	49	2W cheap, 4W quality	2W vs 4W	74	1.0 Msps	BPSK Rate	99	15 links	Mesh Li
25	Extremely low loss	Optical Fiber	50	Less noise affected	PCM vs PAM	75	0.5 Msps	QPSK Rate	100	26^8	Passwo

PART V — Final Board Exam Cram Sheet

Must-Remember Formulas

- **Electric Field to Power Density:**

$$S = \frac{E^2}{377} \iff E = \sqrt{377 \cdot S} \approx \frac{\sqrt{30 P_t G_t}}{d}$$

- **Secant Law for MUF:**

$$\text{MUF} = \frac{f_c}{\cos \theta_i} = f_c \sec \theta_i$$

- **Friis Cascaded Noise Factor:**

$$F_{total} = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2}$$

  (Always convert dB gains/NF to linear ratios before substituting!)
- **Noise Temperature:** $T_e = T_0(F - 1)$ where $T_0 = 290$ K.
- **SSB Two-Tone Average Power:** $P_{avg} = \text{PEP}/2$.
- **Tower Physical Spacing:** $d = \lambda(\theta^\circ/360^\circ)$.
- **Full Mesh Links:** $L = \dfrac{n(n-1)}{2}$.
- **Fiber Guided Modes:** $M \approx \dfrac{V^2}{2}$ where $V = \dfrac{2\pi a}{\lambda} NA$.
- **Image Rejection of Tuned Stages:**

$$\rho = \frac{f_{img}}{f_s} - \frac{f_s}{f_{img}}, \quad IRR = \sqrt{1 + Q^2 \rho^2}$$

- **Johnson Thermal Noise Power:** $P_n = kTB$.

High-Yield Board Rules & Traps

- **LNA Rule:** The first amplifier in a cascade dominates overall receiver noise performance.
- **IF Trade-off:**
  - Low IF $\implies$ Narrower, sharper adjacent selectivity, but poorer image rejection.
  - High IF $\implies$ Better image rejection, but poorer adjacent selectivity.
- **Capture Effect:** Characteristic of FM receivers where the receiver locks onto the stronger co-channel signal and suppresses the weaker one.
- **Smith Chart Rotation Direction:**
  - Toward Generator $\implies$ Clockwise.
  - Toward Load $\implies$ Counter-clockwise.
  - Outer rim $\implies$ Zero resistance (pure reactance).
  - Center axis $\implies$ Pure resistance ($x = 0$).
- **BPSK vs. QPSK Baud Rate:**
  - BPSK: 1 bit/symbol $\implies$ Baud rate $=$ Bit rate.
  - QPSK: 2 bits/symbol $\implies$ Baud rate $= \frac{1}{2} \times$ Bit rate.
- **Two-Wire vs. Four-Wire Circuits:**
  - 2-Wire: Local subscriber loop (combined transmit and receive).
  - 4-Wire: Long-distance trunk (separate transmit and receive pairs for amplifiers).
- **Energy vs. Power Signals:**
  - Energy signal: Finite energy ($0 < E < \infty$), Zero average power ($P = 0$).
  - Power signal: Infinite energy ($E = \infty$), Finite average power ($0 < P < \infty$).
- **Optocoupler / Opto-isolator:** Provides galvanic isolation between field inputs and sensitive controller circuits.

END OF COMPREHENSIVE ECE BOARD EXAM REVIEWER — PB EXAMS (26)